Pain, Agitation/Sedation, Delirium, Immobility, Sleep Disruption, and Neuromuscular Blockade

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Learning Objectives

1. Develop a management strategy for the prevention and treatment of pain, agitation/sedation, and delirium, immobility, and sleep disruption (PADIS) in an intensive care unit (ICU) patient with various comorbidities.
2. Discuss relevant pharmacokinetic and pharmacodynamic considerations of PADIS medications as they pertain to disturbances in critical care physiology.
3. Identify relevant adverse effects, drug interactions, and drug withdrawal syndromes in the management of PADIS.
4. Evaluate patients in the ICU for PADIS using a validated screening tool.
5. Construct a plan for the management of delirium.
6. Identify the long-term effects of critical illness in adult ICU patients.
7. Create a management strategy for PADIS-related medications that are continued beyond ICU discharge.
8. Describe a treatment and monitoring plan for critically ill patients receiving neuromuscular blockade.

Abbreviations in This Chapter

ARDS	Acute respiratory distress syndrome
BPS	Behavioral Pain Scale
CAM-ICU	Confusion assessment method for the intensive care unit
CPOT	Critical-Care Pain Observation Tool
GABA	γ-Aminobutyric acid
ICDSC	Intensive Care Delirium Screening Checklist
ICP	Intracranial pressure
ICU	Intensive care unit
NMBA	Neuromuscular blocking agent
PAD	Pain, agitation, and delirium
PADIS	Pain, agitation/sedation, delirium, immobility, and sleep disruption
PRIS	Propofol-related infusion syndrome
RASS	Richmond Agitation Sedation Scale
SAS	Sedation-Agitation Scale
SAT	Spontaneous awakening trial
SBT	Spontaneous breathing trial
SCCM	Society of Critical Care Medicine
TOF	Train of four

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. P.J. has been receiving propofol 50–60 mcg/kg/minute and fentanyl 75–100 mcg/hour for 4 days. She has no significant medical history. Laboratory results today show that transaminases have increased to 5 times baseline, lactate is increased to 5 mmol/L, and triglyceride concentration is 450 mg/dL. No new medications have been added. Given these laboratory values, which complication would be most appropriate to address?
   A. Deep venous thrombosis (DVT).
   B. Critical illness-induced polyneuropathy.
   C. Intensive care unit (ICU) delirium.
   D. Propofol-related infusion syndrome (PRIS).

2. T.I. is a 35-year-old man admitted to the ICU for severe alcohol withdrawal. His medical history is otherwise unknown. Laboratory values are within normal limits on admission. He has been receiving a lorazepam infusion 6–8 mg/hour for active alcohol withdrawal. On day 4, his blood pressure is 130/75 mm Hg, oxygen saturation is 98% on 2 L of oxygen, blood urea nitrogen (BUN) is 50 mg/dL, and serum creatinine (SCr) is 2.0 mg/dL; he has a new anion gap of 20 mEq/L, an osmolar gap of 18 mmol/L H_2O, and a fractional excretion of sodium of 0.2. Which is the most likely cause for his clinical presentation?
   A. Acute respiratory distress syndrome (ARDS).
   B. Propylene glycol toxicity.
   C. Delirium tremens.
   D. Acute tubular necrosis.

3. R.B. is a 25-year-old man admitted to the ICU for acute pancreatitis and sepsis. He is intermittently agitated on hydromorphone 2 mg/hour and midazolam 6 mg/hour (Richmond Agitation-Sedation Scale [RASS] score of –1 to +2) and is not oxygenating adequately after adjustments on the ventilator. The physician would like to initiate therapeutic paralysis. Which is the next best step in the treatment of this patient?

A. Spontaneous awakening and breathing trials.
B. Cisatracurium infusion.
C. Intermittent vecuronium.
D. Sedate the patient to a "deeply sedated" clinical state.

4. P.V. is a 70-year-old woman (weight 50 kg, decreased from 60 kg 2 months ago) admitted to the ICU in ARDS. She has a history of cirrhosis and is currently fluid overloaded (net positive 5 L). She has been on a continuous infusion of fentanyl and propofol for 5 days. Which pharmacologic factor would best be considered with respect to her analgesics or sedatives?
   A. Risk of PRIS in patients with ARDS.
   B. Unpredictable clearance of fentanyl.
   C. Enzymatic induction of fentanyl by propofol.
   D. Hypocalcaemia secondary to extended use of propofol.

5. L.B. is a 38-year-old woman intubated in the neurosurgery ICU for 72 hours receiving propofol. The nurse is requesting medications for "severe agitation and hallucinations." Her heart rate and blood pressure have steadily increased since admission, and a chart review reveals years of chronic pain while receiving oxycodone and tramadol at home. Her laboratory values are normal, but she is not tolerating enteral-route medications. Which is the most appropriate recommendation at this time?
   A. Quetiapine as needed for agitation.
   B. Fentanyl infusion.
   C. Lorazepam as needed for agitation.
   D. Hydromorphone patient-controlled analgesia.

6. S.P. has just been intubated in the ICU and is in severe alcohol withdrawal. He has a history of frequent delirium tremens and alcohol withdrawal seizures. Which medication is most appropriate to begin initial management of pain, agitation, and delirium (PAD) in this patient?
   A. Dexmedetomidine.
   B. Phenytoin.
   C. Fentanyl.
   D. Midazolam.

7. H.F., a 65-year-old man admitted to the ICU from home for aspiration pneumonia requiring intubation, is initiated on levofloxacin and metronidazole. Other medications include fentanyl and dexmedetomidine infusions as well as amiodarone, a home medication. His last RASS was -2), and he has intermittent agitation. Vital signs and laboratory values are normal, and corrected QT (QTc) is 550 milliseconds. Which is the most appropriate recommendation at this time?
   A. Add quetiapine for agitation, and monitor QTc.
   B. Start nonpharmacologic management of agitation/delirium.
   C. Discontinue dexmedetomidine because of his prolonged QTc.
   D. Give lorazepam as needed for agitation.

8. S.V., a 70-year-old woman with a history of hypertension, is transferred from the floor to the ICU for worsening pneumonia and new-onset hypoactive delirium. She has been nil per os (NPO) since admission 3 days prior. She remains febrile (temperature 102°F [38.89°C]) with decreased urine output; other vital signs and laboratory values are within normal limits. Her medications include ceftriaxone, heparin, and hydrochlorothiazide. Which best represents the most important set of considerations regarding her delirium?
   A. Dementia and sleep disorder.
   B. Undetected alcohol withdrawal.
   C. Adrenal insufficiency.
   D. Dehydration and untreated infection.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1 -5
   b. Task B: 2
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1 -6
   b. Task B: 1, 4, 13, 14, 16
   c. Task C: 1, 4
3. Domain 3: Professional Practice
   a. Task A: 4
   b. Task B: 1, 4, 6

I. PAIN, AGITATION/SEDATION, DELIRIUM, IMMOBILITY, AND SLEEP DISRUPTION (PADIS) IN THE INTENSIVE CARE UNIT

A. Background
   1. The Society of Critical Care Medicine (SCCM) published updated guidelines for the management of PADIS in adult ICU patients in 2018 (Crit Care Med 2018;46:e825-e873). These guidelines, together with recently published research, help guide ICU clinicians in the challenging task of optimizing patient comfort and outcomes while avoiding the complications of under- or oversedation. The PADIS guidelines were written by 32 international experts, four methodologists, and four critical illness survivors who met virtually monthly and annually at annual SCCM congresses. Rigorous research has developed our understanding of the assessment tools and medications used for PADIS, the prevention and treatment methods used for PADIS, and the long-term effects of the ICU environment on patients and caregivers. Recommendations for specific ICU populations such as burn, neurologic, neurosurgical (including traumatic brain injury), and cardiac populations may need specialized consideration.
   2. The PADIS guidelines add to the 2013 pain, agitation, and delirium (PAD) guidelines by:
      a. Adding two critical care topics: rehabilitation/mobilization and sleep disruption
      b. Including patients as collaborators and coauthors
      c. Inviting an international panel of experts from high-income countries to incorporate more diverse practices and expertise from the global critical care community
   3. Each section of the 2018 PADIS guidelines was represented by content experts, methodologists, and ICU survivors. Population, Intervention, Comparison, and Outcome and nonactionable descriptive questions were developed by each section. Members of the guideline group voted on their ranking, and patients voted on their perceived importance. According to the Grading of Recommendations Assessment, Development, and Evaluation principles, every Population, Intervention, Comparison, and Outcomes question was evaluated by members of each section to determine the best level of evidence, assess quality, and determine recommendations as "strong," "conditional," or "good" practice. Evidence gaps and clinical caveats were explicitly identified.

B. Pharmacy Intervention – Pharmacists provide unique and valuable insight into the management of PADIS in the ICU. Much of the management for PADIS involves medications with complex pharmacologic profiles, allowing many opportunities for pharmacy expertise on the critical care team. As the management of PADIS in the ICU continues to evolve, pharmacists should seek avenues for contributing to the critical care community through development of hospital protocols and assessing for quality improvement; providing education for medical, pharmacy, and nursing colleagues; and/or doing research on pertinent questions surrounding the management of PADIS in the ICU (Crit Care Med. 2020;48:e813-e834).

II. PAIN IN THE INTENSIVE CARE UNIT

A. Introduction
   1. More than half of ICU survivors report severe pain as the most traumatic memory of their ICU stay.
   2. Both short- and long-term negative sequelae are related to uncontrolled pain in the ICU. Assessing pain in the ICU is challenging, particularly in patients who cannot effectively communicate. If patients cannot adequately communicate their degree of pain but retain motor activity, medications should be titrated according to validated behavioral pain scales.

B. Incidence and Causes of Pain: Pain may occur in any type of ICU patient, and considerations for pain management often require an individualized approach to optimize treatment. The interdisciplinary team should complete a comprehensive review of all variables such as acute and chronic pain, routine nursing care that may cause discomfort, and procedural-based pain.
   1. Common causes of pain in the ICU include, but are not limited to, acute trauma, injury or burns, postoperative pain, exacerbation of chronic pain, heart disease, ischemia, acute or chronic underlying disease state pain such as cancer pain, pancreatitis, or other abdominal pathology.
   2. Less discernible causes of pain may include those from either routine nursing care or the provision of life-sustaining measures: presence of an endotracheal tube and endotracheal tube suctioning, wound care, tube or Foley insertion, immobility, bed repositioning, bathing, medication administration, and physical and occupational therapy. Other examples of painful invasive procedures include intravenous line placement, endoscopy and bronchoscopy, chest tube placement or removal, paracentesis, lumbar puncture, biopsies, and fracture reductions.

C. Short- and Long-term Consequences of Pain in the ICU
   1. Acute pain can invoke a stress response, resulting in a hypercatabolic state, decreased tissue perfusion, and impaired wound healing. Uncontrolled pain decreases a patient's immune response to infection by suppressing natural killer cell activity and neutrophil function.
   2. Long-term studies (12 months post-ICU stay) report detrimental physiologic and psychological function in patients who recall significant pain during their hospitalization, particularly in patients admitted with a traumatic injury (Lancet Respir Med 2014;2:369-79).
      a. Health-related quality of life is decreased in up to 20% of patients.
      b. Chronic pain is reported in up to 40% of patients.
      c. Posttraumatic stress disorder is reported in 5%–20% of patients.

D. Assessment of Pain
   1. The gold standard for assessing pain remains the patient's self-report of pain. Several scenarios in the ICU make the self-reporting of pain challenging for clinicians (e.g., mechanical ventilation, presence of sedation and/or delirium). SCCM guidelines currently recommends two validated behavioral pain scales to be done in a repetitive and routine manner: the Behavioral Pain Scale (BPS) (Table 1) and the Critical-Care Pain Observation Tool (CPOT) (Table 2).
      a. Assessment scales should be used routinely in all ICU patients. Most nursing protocols assess pain every 4–6 hours while the patient is awake. In addition, it is important to reassess the degree of pain within about 30 minutes to 1 hour after administering an "as-needed" pain medication to determine the appropriateness of the pain medication or dose.
      b. Pain scores should be documented in the medical chart and then used to help formulate daily titrations in pain medications.
      c. Patients should be treated within 30 minutes of a "significant pain" score. A BPS greater than 5 or a CPOT score of 3 or greater, or a numeric rating scale score of 4 or greater is indicative of pain.
   2. The use of vital signs alone is not recommended for assessing pain in the ICU patient. Abnormal vital signs such as tachycardia and hypertension are appropriate for use as a prompt to further investigate the need for pain control.
   3. Further research is needed to determine the effectiveness of a preprocedural pain assessment tool and the ways in which this assessment will affect analgesic administration. A study by Puntillo et al. in 2014 found the procedures most likely to double the patient's pain intensity score (from preprocedure to during-procedure scoring) were chest tube removal, wound drain removal, and arterial line insertion. This study found that higher-intensity pain and pain distress before the procedure were associated with a high risk of increased pain during the procedure (Am J Respir Crit Care Med 2014;189:39-47).

4. Kanji et al. found that the CPOT is a valid pain assessment in noncomatose, delirious adult ICU patients who are not able to reliably self-report the absence or presence of pain (Crit Care Med 2016;44:943-7).

Table 1. Behavioral Pain Scale (BPS)[a]

Item	Description	Score
Facial expression	Relaxed	1
	Partly tightened (e.g., brow lowering)	2
	Fully tightened (e.g., eyelid closing)	3
	Grimacing	4
Upper limbs	No movement	1
	Partly bent	2
	Fully bent with finger flexion	3
	Permanently retracted	4
Compliance with ventilation	Tolerating movement	1
	Coughing but tolerating ventilation most of the time	2
	Fighting ventilator	3
	Unable to control ventilation	4

[a]A BPS score > 5 indicates significant pain.

Adapted with permission from: Lippincott Williams & Wilkins/Wolters Kluwer Health. Payen J, Bru O, Bosson J, et al. Assessing pain in critically ill sedated patients by using a behavioral pain scale. Crit Care Med 2001;29:2258-63.

Table 2. Critical-Care Pain Observation Tool (CPOT)[a]

Indicator	Description	Score	
Facial expression	No muscular tension observed	Relaxed, neutral	0
	Presence of frowning, brow lowering, orbit tightening, and levator contraction	Tense	1
	All of the above facial movements plus eyelids tightly closed	Grimacing	2
Body movements	Does not move at all (does not necessarily mean absence of pain)	Absence of movement	0
	Slow, cautious movements; touching or rubbing the pain site; seeking attention through movements	Protection	1
	Pulling tube, trying to sit up, moving limbs/ thrashing, not following commands, striking at staff, trying to climb out of bed	Restlessness	2
Muscle tension Evaluation by passive flexion and extension of upper extremities	No resistance to passive movements	Relaxed	0
	Resistance to passive movements	Tense, rigid	1
	Strong resistance to passive movements, inability to complete them	Very tense or rigid	2
Compliance with the ventilator (intubated patients)	Alarms not activated, easy ventilation	Tolerating ventilator or movement	0
	Alarms stop spontaneously	Coughing but tolerating	1
	Asynchrony: Blocking ventilation, alarms often activated	Fighting ventilator	2

Table 2. Critical-Care Pain Observation Tool (CPOT)[a] *(continued)*

Indicator	Description	Score	
OR Vocalization (extubated patients)	Talking in normal tone or no sound	Talking in normal tone or no sound	0
	Sighing, moaning	Sighing, moaning	1
	Crying out, sobbing	Crying out, sobbing	2
Total, range			0–8

[a]A CPOT score ≥ 3 indicates significant pain.

Adapted with permission from: Lippincott Williams and Wilkins/Wolters Kluwer Health. Gelinas C, Fillion L, Puntillo K, et al. Validation of the Critical-Care Pain Observation Tool in adult patients. Am J Crit Care 2006;15:420-7.

E. Prevention and Treatment of Pain in the ICU
   1. In a patient whose pain is inadequately controlled in the ICU, intravenous opioids are considered first-line treatment for nonneuropathic pain. The PADIS guidelines suggest acetaminophen as an adjunct to an opioid to decrease pain intensity and opioid consumption for pain management in critically ill patients.
   2. Preprocedural pain management should be considered in all ICU patients. One study reported that up to 60% of patients did not receive preprocedural systemic pain medication for common procedures and wound care in the ICU, although 89% of patients received a topical anesthetic for central venous catheter placement (Am J Crit Care 2002;11:415-29).
      a. The PADIS guidelines suggest using an opioid at the lowest effective dose for procedural pain management in critically ill patients undergoing a procedure. The American Society for Pain Management Nursing (ASPMN) published recommendations for preprocedural pain management in 2011. The ASPMN recognizes both the psychological and the physical elements of procedural pain and agrees with combining nonpharmacologic and pharmacologic methods. Examples of nonpharmacologic options recommended by ASPMN include relaxation and breathing techniques, imagery, massage, music, thermal measures, and positioning (Pain Manag Nurs 2011;12:95-111). The PADIS guidelines recommend not using inhaled volatile anesthetics for procedural pain management in critically ill adults. The PADIS guidelines also suggest using a nonsteroidal anti-inflammatory drug (NSAID) administered intravenously, orally, or rectally as an alternative to opioids for pain management during discrete and infrequent procedures in critically ill adults. Use of an NSAID topical gel for procedural pain management in critically ill adults is not recommended by the PADIS guidelines.
      b. Preemptive analgesia for chest tube removal is recommended together with nonpharmacologic relaxation techniques. The PADIS guidelines suggest neither local analgesia nor nitrous oxide for pain management during chest tube removal in critically ill patients.
   3. Postoperative thoracic epidural anesthesia/analgesia is recommended for patients undergoing abdominal aortic aneurysm treatment. Thoracic epidural anesthesia is "suggested" for traumatic rib fractures in the ICU.
   4. Nonpharmacologic therapy for pain
      a. The PADIS guidelines suggest offering cold therapy for procedural pain management in critically ill adults.
      b. The PADIS guidelines suggest offering relaxation techniques for procedural pain management in critically ill adults. Music therapy is suggested for both nonprocedural and procedural pain in critically ill adults.

5. Pharmacotherapy for pain
    a. Intravenous opioids on an as-needed, scheduled, or continuous infusion basis are recommended to treat pain in the ICU. The pharmacokinetics of different opioids may vary; thus, opioids should be chosen according to patient comorbidities and individual needs (Table 3). Fentanyl is the most commonly used intravenous opioids in American adult ICUs.

Table 3. Opiates Commonly Used in the ICU

Drug	Metabolic/Drug Interaction Considerations	Usual CI Starting Dose[a]	Drug-Specific Adverse Effects[b]	Drug Accumulation Factors
Fentanyl	3A4 major substrate	12.5–25 mcg/hr; 0.35–0.5 mcg/kg loading dose follwed by 12.5–25 mcg/hr	Muscle rigidity	Hepatic failure; high volume of distribution; high lipophilicity; unpredictable clearance (long context-sensitive half-time) with prolonged infusion
Morphine	Glucuronidation	1–2 mg/hr	Hypotension, bradycardia from histamine release	Hepatic failure; active metabolite (3- and 6-morphine glucuronide) accumulates in renal failure
Hydromorphone	Glucuronidation	0.25–0.5 mg/hr	Overdose effects from dosing errors (high-potency opiate)	Hepatic failure
Methadone	3A4 and 2B6 major substrates	N/A	QTc prolongation, serotonin syndrome	Long half-life; hepatic and renal failure will delay clearance
Remifentanil	Blood and tissue esterases	Loading dose: 1.5 mcg/kg CI: 0.5–15 mcg/kg/hr	Chest wall rigidity; rebound pain on discontinuation	

[a]Usual starting dose in the ICU for pain management in an opiate-naive patient.

[b]Other common significant adverse effects for all opiates to be considered: Constipation, respiratory depression, bradycardia, hypotension, altered mental status.

CI = continuous infusion.

    i. General mechanism of action of opiates: Bind to mu-opioid receptors in the central nervous system (CNS)
    ii. Commonly used intravenous opioids in the ICU: Fentanyl, morphine, hydromorphone, remifentanil, and methadone
    iii. Tolerance: May quickly develop to all opiates, particularly when given as a continuous infusion. If switching to a different intravenous or oral opiate, equianalgesic dosing may be difficult to estimate, and low starting doses should be considered.
    iv. Significant adverse effects: Decreased respiratory drive: This may be a desired effect in some ICU scenarios; however, a depressed respiratory drive is a critical negative implication during the ventilator weaning process; decreased blood pressure and heart rate, constipation, gastrointestinal (GI) intolerance, altered sensorium.

v. GI intolerance and constipation: A bowel regimen should be initiated on day 1 unless contraindicated, with assessment for efficacy every 24–48 hours. GI intolerance in the ICU can result in increased time on mechanical ventilation, delayed time to attaining nutritional goals, and prolonged ICU stay. Constipation may also contribute to agitation.
vi. Altered mental status: Opiates may induce a sedative effect as well as an altered sensorium in some patients. Unless contraindicated, clinicians should consider tapering the opiate dose in an altered patient who has adequate pain control.
vii. Patient-controlled analgesia: In alert and clinically stable patients, use of patient-controlled analgesia may be considered to titrate to the patient's perceived level of pain. Patient-controlled analgesia may also be useful on discontinuation of continuous infusion opiates.

b. Fentanyl
   i. Pharmacokinetics: Hepatic metabolism, cytochrome P450 (CYP) 3A4 substrate. Quick onset and short duration of action; lacks a pharmacologically active metabolite. Highly lipophilic, high volume of distribution and protein binding; maintains a three-compartment model; continuous infusion dosing may lead to prolonged and unpredictable clearance (prolonged context-sensitive half-time).
   ii. Many dosage forms: Injectable (intravenous, intramuscular, intrathecal, epidural), transdermal, transmucosal, nasal spray. Different dosage forms should not be converted on a 1:1 mcg basis; use specific manufacturer recommendations if converting. Injectable form of fentanyl is most commonly used in the ICU setting. The fentanyl patch is not generally appropriate for use in the ICU because of its latent onset (about 12 hours) and erratic/increased absorption in a febrile patient.
   iii. Adverse effects: Respiratory depression, bradycardia, hypotension, CNS depression, constipation, ileus, risk of serotonin syndrome when used with other serotonergic agents

c. Morphine
   i. Pharmacokinetics: Hepatic metabolism by glucuronidation to two major active metabolites, morphine-3-glucuronide (45%–55%) and morphine-6-glucuronide (10%–15%). The glucuronide metabolites of morphine are both renally eliminated; accumulation can occur with the chronic use of morphine or in patients with decreased renal function. Morphine-3-glucuronide does not have analgesic activity, but adverse effects may include seizure activity or agitation. Morphine-6-glucuronide does have analgesic activity by the mu-receptor and may cause additive sedation and respiratory depression if accumulation occurs. Continuous morphine infusions are rarely used for analgesia in the ICU setting because of the concerns with the active metabolites.
   ii. Dosage forms: Injectable (intravenous, subcutaneous, intrathecal, epidural) and oral. Intravenous-to-oral conversion is not a 1:1 mg ratio.
   iii. Adverse effects: Histamine release may cause significant hypotension; bradycardia, respiratory depression, CNS depression, constipation, ileus.

d. Hydromorphone
   i. Pharmacokinetics: Hepatic metabolism by glucuronidation to an inactive but potentially neurotoxic metabolite. Low volume of distribution, highly water soluble, and relatively low protein binding.
   ii. Dosage forms: Injectable (intravenous, subcutaneous) and oral; intravenous-to-oral conversion is not a 1:1 mg ratio.
   iii. Adverse effects: CNS alterations (e.g., abnormal dreams, aggressive behavior, altered thinking), respiratory depression, hypotension, constipation

e. Remifentanil (Ultiva)
   i. Pharmacokinetics: Clearance by blood and tissue esterase; clearance not dependent on organ function. Fast onset and short duration of action with little to no accumulation. High volume of distribution, high protein binding.
   ii. Research primarily done in Europe; limited reported use in U.S. adult ICUs for ongoing analgesic use.
   iii. Dosage form: Injectable only
   iv. Adverse effects: Respiratory depression, hypotension, bradycardia, constipation
   v. Rebound pain: Quick offset (5–10 minutes) may lead to rebound pain and withdrawal symptoms, and additional pain medication may be needed if remifentanil is interrupted or discontinued.
   vi. Benefit in adult ICUs: Decreased time on mechanical ventilation with short-term use (72 hours or less)

f. Methadone
   i. Pharmacokinetics: Phase I hepatic metabolism to inactive metabolites. Many drug interactions: major substrate of CYP 2B6 and 3A4. Moderate inhibitor of CYP2D6, weak inhibitor of CYP3A4. Longer-acting opiate with variable duration of action (12–48 hours); may accumulate quickly in patients with hepatic failure or patients receiving hemodialysis. Animal studies have found that the d-isomer of methadone works as both a partial mu-agonist and an N-methyl-d-aspartate receptor antagonist (the l-isomer is a full mu-agonist). These properties of the d-isomer are thought to decrease the tolerance effect to other opioids. Methadone is currently marketed as the racemic mixture. On initiating oral methadone, steady state and peak analgesic effect may not be reached for 3–5 days; oversedation and respiratory depression may occur if titrated too quickly.
   ii. Dosage forms: Injectable (intravenous, intramuscular, subcutaneous) and oral. Intravenous-to-oral conversion is not a 1:1.3 mg ratio.
   iii. Adverse effects: Dose-dependent QTc prolongation, altered mental status, respiratory depression, confusion, dizziness, arrhythmias, constipation, risk of serotonin syndrome when used with other serotonergic agents

6. Non-opioid adjunctive pain medications should be considered in combination with opioids to reduce opioid requirements. Clinically stable patients may tolerate a conversion from opiates to non-opiate medications.
   a. Local and regional anesthetics such as bupivacaine. The PADIS guidelines do not recommend the use of intravenous lidocaine as an adjunct to opioid therapy for pain management in critically ill adults.
   b. Acetaminophen (Tylenol, Ofirmev): The PADIS guidelines suggest using acetaminophen as an adjunct to opioids to decrease pain intensity and opioid consumption.
      i. Total daily acetaminophen doses should be considered from all acetaminophen combination products, with a maximum total daily dose of 4 g. Decreased total daily dosing should be considered in patients with significant liver disease or those older than 60 years of age.
      ii. Intravenous acetaminophen: Dose reduction recommended if the creatinine clearance (CrCl) is 30 mL/minute/1.73 m^2 or less or with continuous renal replacement therapy (every 8 hours); contraindicated in severe hepatic disease. The cost of the intravenous formulation of acetaminophen is considerably higher than that of the oral or rectal formulations. The intravenous formulation can also cause hypotension.
   c. Intravenous or oral NSAIDs: Ibuprofen, ketorolac. The PADIS guidelines suggest not routinely using a COX-1 selective NSAID as an adjunct to opioid therapy for pain management in critically ill adults. Use with caution in critically ill patients with renal or hepatic dysfunction. May increase the risk of acute renal failure, bleeding, or GI adverse effects.

d. Ketamine (Ketalar) has been used for analgesia and sedation in the ICU, primarily in the pediatric population. The PADIS guidelines suggest using low-dose ketamine (1–2 mcg/kg/minute) as an adjunct to opioid therapy when seeking to reduce opioid consumption in postsurgical adults admitted to the ICU. Published data for the use of ketamine in adults for analgesia and/or sedation are limited to case reviews, and long-term cognitive effects of ketamine are not known. Data from animal studies suggest a significant decline in cognitive function after continued use of ketamine.
   i. Called a "dissociative anesthetic," providing analgesic activity at subanesthetic doses. It is a schedule III controlled substance and works primarily as an N-methyl-d-aspartate receptor antagonist. Ketamine is void of the constipation, respiratory depression, and hypotensive effects that plague the opiate class.
   ii. May decrease dose requirements of concurrently administered opioids
   iii. Other uses include rapid sequence intubation, refractory pain syndromes, cancer pain, neuropathic pain, asthma (bronchodilatory effects), refractory seizure activity, and depression.
   iv. Dosing range is varied; usual starting dose for analgesia or sedation is 0.1 mg/kg/hour. Reviews of ketamine use in adult ICUs report a dosing range of 0.1–2.5 mg/kg/hour and a range in duration of 3 hours to 9 days.
   v. Significant adverse effects: Mild to severe emergence reactions (e.g., confusion, excitement, irrational behavior, hallucinations, delirium) in around 12% of patients, enhanced skeletal muscle tone, tachycardia, hypertension, hypotension

e. The PADIS guidelines suggest not offering cybertherapy or hypnosis for pain management in critically ill adults.
f. The PADIS guidelines suggest offering massage for pain management in critically ill adults.
g. The PADIS guidelines suggest offering music therapy to relieve both non-procedural and procedural pain in critically ill adults.

7. The PADIS guidelines suggest using a neuropathic pain medication (e.g., gabapentin, carbamazepine, pregabalin) with opioids for neuropathic pain management in critically ill patients. The PADIS guidelines recommend using neuropathic pain medications with opioids for pain management in ICU adults after cardiovascular surgery. There is a potential for significant adverse effects and drug interactions, requiring close monitoring and follow-up. If the patient is discharged home on an anticonvulsant for neuropathic pain, follow-up should be documented and the primary care provider notified.
   a. Gabapentin (Neurontin)
      i. Suggested starting dose range: 300–600 mg/day divided two or three times daily; requires renal adjustment. The target dose is 900–3600 mg/day in three divided doses.
      ii. Pharmacokinetics: Renally excreted, dose adjusted for reduced CrCl
      iii. Adverse effects: May be severe, including CNS depression, delirium, paresthesias, and asthenias
   b. Carbamazepine (Tegretol)
      i. Suggested starting dose range: 50–100 mg twice daily; use with caution in patients with hepatic impairment, and adjust for a CrCl less than 10 mL/minute/1.73 m^2 or with hemodialysis. The target dose is 100–200 mg every 4–6 hours; the maximum dose is 1200 mg/day.
      ii. Pharmacokinetics: Strong inducer of many CYP enzymes, substrate of CYP3A4. Closely monitor for drug interactions.
      iii. Adverse effects: Somnolence, severe skin reactions (e.g., Stevens-Johnson syndrome, toxic epidermal necrolysis), pancytopenia, syndrome of inappropriate antidiuretic hormone
   c. Pregabalin (Lyrica)
      i. 50 mg three times daily; may be increased in 1 week depending on tolerability and effect; maximum dose: 100 mg three times daily.
      ii. Pharmacokinetics: 90% excretion in urine

iii. Adverse effects: Peripheral edema, dizziness, drowsiness, headache, fatigue, weight gain, xerostomia, visual field loss, and blurred vision

F. Analgosedation Method in the ICU: This method of sedation advocates the use of opiate medications before prescribing an anxiolytic/hypnotic medication to provide patient comfort in the ICU unless anxiolytics are otherwise indicated. The PADIS guidelines recommend analgosedation (analgesia is used before a sedative to reach the sedative goal) or analgesia-based sedation (an opioid is used instead of a sedative to reach the sedative goal). Providing pain relief early in the ICU stay may decrease the agitation associated with pain and/or general discomfort while minimizing the use of alternative medications commonly used for agitation (e.g., benzodiazepines). The guidelines recognize that current data using analgosedation are primarily limited to open-label trials, using remifentanil as the analgesic, and mostly conducted in Europe, where critical care staffing and management practices differ from those in the United States. Despite these limitations, it remains notable that studies using the analgosedation method found a significant decrease in benzodiazepine dosage requirements when opiates were the primary medications used for discomfort and agitation. This is a positive step in decreasing the untoward adverse effects of the benzodiazepine class of sedatives. There is a potential for high cumulative doses of opiates with the analgosedation method, necessitating daily monitoring of their adverse effects (e.g., respiratory depression, altered mental status, GI slowing). Opioid use in the ICU also increases the daily risk of delirium development in a dose dependent manner.

G. Data: NONSEDA is a randomized, clinical parallel-group, multinational superiority study comparing no sedation (bolus doses of morphine) and light sedation (propofol for 48 hours then midazolam titrated to a RASS of −2 to −3). There was no difference in 90-day mortality between groups (no sedation 148 [42.4%] vs. light sedation 130 [37%]; 95% CI −2.2 to 12.2; p=0.65). However, 27% of the no-sedation group crossed over to receive sedation during the first 24 hours after randomization.

Patient Case

Questions 1 and 2 pertain to the following case.
T.O. is a 70-year-old man just admitted to the ICU with multiple fractures after a motor vehicle accident. His medical history includes hypertension. He is now agitated after intubation. His laboratory values are normal, and his vital signs include blood pressure 175/95 mm Hg and heart rate 110 beats/minute.

1. Which grouping of initial sedatives is most appropriate at this time?
   A. Fentanyl infusion and midazolam infusion
   B. Propofol infusion and fentanyl as needed
   C. Midazolam as needed and fentanyl as needed
   D. Fentanyl infusion and propofol infusion, if needed

2. After 2 weeks in the ICU, T.O. is being prepared for chest tube removal. He is currently receiving a fentanyl drip with adequate pain control. Which is the best pain management regimen for chest tube removal?
   A. Give intravenous acetaminophen 15 minutes before chest tube removal.
   B. Make no change in pain treatment because his current pain regimen is adequate.
   C. Increase his pain medication infusion dose by 50% the morning of his chest tube removal.
   D. Give fentanyl 50 mcg injectable 15 minutes before chest tube removal.

III. AGITATION IN THE INTENSIVE CARE UNIT

A. Agitation in the ICU – Maintaining patient comfort for the duration of an ICU stay can be extremely challenging, requiring significant resources and daily discipline from the nursing, medical, and pharmacy team. Ongoing research has improved our understanding of the consequences of either under- or overtreating agitation in the ICU, and clinicians should continue to apply this knowledge to their daily selection and titration of medications. Treatment of a patient who presents with agitation must always begin with attempts to identify and correct the etiology of the agitation. Common causes of agitation in the ICU include pain, delirium, hypoxia, hypoglycemia, dehydration, and drug or alcohol withdrawal. Close inspection of significant patient variables will also help determine the appropriate sedative:
   1. Pain control
   2. Substance abuse and smoking history
   3. Neurologic function: Baseline and acute mental status, history of seizure activity, dementia, psychiatric history
   4. Clinical variables: Blood pressure, heart rate, respiratory rate
   5. Comorbidities (baseline and acute): Cardiac, renal, hepatic, gastric, pulmonary, pancreatic
   6. Home medication use: Any medication from which a patient could withdraw: Benzodiazepines, opioids, nicotine, antidepressants, other γ-aminobutyric acid (GABA) receptor agonists

B. Common Medications for the Treatment of Agitation – Include propofol, dexmedetomidine, and benzodiazepines (usually lorazepam and midazolam) (Table 4). Benzodiazepines are first-line agents for status epilepticus, alcohol withdrawal, benzodiazepine dependence or withdrawal, and are useful for deep sedation or amnesia and with the use of neuromuscular blockade. Other indications for benzodiazepines may exist, which must be scrutinized throughout the ICU stay.

Table 4. Sedatives for Patients on Mechanical Ventilation in the ICU

Drug	Onset and Duration	Precautions for Use	CYP Substrate (major)	Usual Dose	Significant Adverse Effects
Propofol	Onset: 1 min Duration: short term: 0.5–1 hr; long term > 7 days: variable; 25–50 hr has been observed (depends on depth and time on sedation)	Hypotension, bradycardia, hepatic/ renal failure, pancreatitis	2B6	5–50 mcg/kg/ min; 0.3–3 mg/ kg/hr	Hypotension, pancreatitis, respiratory depression, bradycardia, PRIS
Dexmedetomidine	Onset: 5–10 min (with LD) 1–2 hr (without LD) Duration: 1–2 hr	Hepatic failure; symptomatic bradycardia	2A6	LD: 0.5–1 mcg/ kg (optional) MD: 0.2–0.7 mcg/kg/hr	Hypo/hyper-tension, bradycardia, pyrexia
Lorazepam	Onset: 5–20 min Duration: 4–8 hr; prolonged with continuous infusion	Delirium, renal failure	N/A	Intermittent: 1–4 mg IV every 4–6 hr	Oversedation, propylene glycol toxicity
Midazolam	Onset: 3–5 min Duration: 2–6 hr, prolonged with continuous infusion	Hepatic failure, end-stage renal failure or dialysis, delirium	3A4 (active metabolite)	0.02–0.1 mg/ kg/hr	Oversedation

CI = continuous infusion; IV = intravenously; LD = loading dose; MD = maintenance dose; N/A = not applicable; PRIS = propofol-related infusion syndrome.

C. SCCM provides the following statement in the PADIS guidelines regarding sedation in the ICU: "We suggest using either propofol or dexmedetomidine over benzodiazepines [midazolam or lorazepam] for sedation in critically ill, mechanically ventilated adults" and those after cardiac surgery. SCCM states that "benzodiazepine use may be a risk factor for the development of delirium in adult ICU patients." Two randomized studies evaluated the differences in clinical outcomes while adult ICU patients were receiving sedation with either a benzodiazepine or a non-benzodiazepine strategy. Heterogeneity occurred among the findings of these two studies (see No. 1 and No. 2 below), which may be partly because of differences in study design.
   1. The MENDS study compared the sedative effects of lorazepam and dexmedetomidine in medical and surgical adult ICU patients (n=103). Dexmedetomidine had more "delirium-free + coma-free" days than lorazepam (7 vs. 3 days, p=0.01), and the prevalence of "delirium or coma" was lower in the dexmedetomidine group (87% vs. 98%, p=0.03). The assessment of "delirium-free + coma-free" was a more appropriate outcome to evaluate than "delirium without coma," given that delirium cannot be assessed in patients with coma. More patients were within 1 point of their RASS goal with dexmedetomidine (67%) than with lorazepam (55%), p=0.008, but there was no difference in mechanical ventilator–free days, ICU length of stay, or 28-day mortality. This study has been criticized because both groups received continuous infusions of sedatives without additional bolusing, whereas in clinical practice, most practitioners would bolus lorazepam before increasing the infusion rate. In addition, patients were not required to have a spontaneous awakening trial (SAT). (JAMA 2007;298:2644-53).
   2. The SEDCOM study compared the sedative effects of midazolam with those of dexmedetomidine in medical and surgical adult ICU patients (n=366). The prevalence of delirium was lower in the dexmedetomidine group (54%) than in the midazolam group (76.6%), p<0.001. Median time to extubation was shorter in the dexmedetomidine group (3.7 days) than in the midazolam group (5.6 days), p<0.01; however, the times in target sedation range, ICU length of stay, and mortality were no different between the two groups. This study did allow bolus dosing of study drugs, and patients were required to have an SAT if safety criteria were met (JAMA 2009;301:489-99).

D. Clinical Outcome Differences Among Sedative Agents in Medical and Surgical ICU Patients: A meta-analysis, "Benzodiazepine versus Nonbenzodiazepine-Based Sedation for Mechanically Ventilated Critically Ill Adults," reviewed trials from 1996 to 2013 (Crit Care Med 2013;41:S30-8):
   1. Studies from this review contained the following criteria: (1) randomized controlled parallel-group design; (2) medical and surgical adult ICU patients on mechanical ventilation receiving intravenous sedation; (3) patients receiving a non-benzodiazepine (propofol 1% or dexmedetomidine) compared with a benzodiazepine (lorazepam or midazolam); and (4) patients having predefined outcomes. Excluded cardiac surgery and obstetric patients.
   2. Four primary outcomes from six randomized trials were reported in the review (1235 patients):
      a. ICU length of stay (all six studies reported): ICU length of stay was longer in a benzodiazepine strategy than in a non–benzodiazepine-based strategy (mean difference 1.6 days; 95% confidence interval [CI], 0.72–2.5; p=0.0005).
      b. Duration of mechanical ventilation (four studies reported): Longer duration of mechanical ventilation in a benzodiazepine-based strategy than in a non–benzodiazepine-based strategy (mean difference 1.9 days; 95% CI, 1.7–2.09; p=0.00001)
      c. Delirium prevalence (two studies reported): No difference in delirium prevalence between a benzodiazepine and a non–benzodiazepine-based strategy (relative risk [RR] 0.98; 95% CI, 0.76–1.27; p=0.94)
      d. Short-term (45 days or less) all-cause mortality (four studies reported): No difference in risk of death between a benzodiazepine and a non–benzodiazepine-based strategy (RR 0.98; 95% CI, 0.76–1.27, p=0.94)

E. Propofol
   1. SCCM suggests using a non-benzodiazepine (propofol or dexmedetomidine) for sedation to improve clinical outcomes in mechanically ventilated patients. Throughout the past decade, propofol has been increasingly used worldwide for sedation in the ICU. Propofol's short duration of action, lack of accumulation, and relatively clean adverse effect profile at low to moderate doses makes it an appealing alternative.
   2. Mechanism of action: General anesthetic by potentiation of the $GABA_A$ receptor; may inhibit *N*-methyl-D-aspartate receptor activity at high doses. Propofol decreases cardiac β-adrenergic responsiveness and attenuates β-adrenergic signal transduction in cardiac myocytes, resulting in direct cardiac depressive effects.
   3. Pharmacokinetics: Hepatic conjugation; clearance may be prolonged (from minutes to hours) in patients with severe hepatic impairment or cirrhosis or with long-term infusions as it redistributes from fat and muscle to plasma. Highly lipophilic pharmacokinetics and a large volume of distribution lead to extensive tissue distribution. Propofol maintains a three-compartment linear model: plasma, rapidly equilibrating tissues (e.g., major organs), and slowly equilibrating tissues (e.g., fat deposits). Substrate of CYP 2B6, 2C9, 2C19, and 3A4; pharmacokinetic studies of healthy volunteers show a 25% increase in propofol plasma concentrations when given with midazolam, a weak CYP 3A4 and 2C9 inhibitor.
   4. Lipid formulation considerations: Standard propofol is a 1% (10 mg/mL) lipid emulsion containing 1.1 kcal/mL (0.1 g of fat per 1 mL of propofol); this should be accounted for when calculating nutritional intake (e.g., propofol at 50 mcg/kg/minute in a 70-kg patient would provide around 500 calories per day contributed by fat). Propofol contains 0.005% disodium edetate (EDTA) to decrease the rate of microorganism growth, which is known to chelate trace metals, including zinc. Zinc supplementation should be considered in patients at high risk of zinc deficiency (sepsis, burns, large-volume diarrhea) if propofol is used for more than 5 days. Strict aseptic technique must be followed when handling propofol; manufacturers recommend discarding propofol bottles and changing intravenous tubing every 12 hours to decrease the risk of contamination. Propofol is not contraindicated in patients with egg or soybean allergies as historically thought. Propofol contains egg yolks, and most patients are allergic to egg whites. Propofol contains soybean oil not soybean-derived proteins, which have been linked to allergic reactions.
   5. Dosing range for ICU sedation: Usual starting dose 5–10 mcg/kg/minute, titrated every 5–10 minutes to goal sedative effect. Abrupt discontinuation of propofol is not recommended because of its rapid clearance (5–10 minutes).
   6. Data: In a multicenter European trial (PRODEX), Jakob et al. compared propofol (n=249) with dexmedetomidine (n=251) for sedation in prolonged mechanical ventilation. Patients in both groups were treated with daily sedation interruption trials and spontaneous breathing trials (SBTs), and pain was treated with fentanyl boluses. Proportion of time in target RASS (0, −3) without rescue therapy was the same in the propofol group (65%) as in the dexmedetomidine group (65%). There was no difference in median time on mechanical ventilation in propofol (5 days) versus dexmedetomidine (4 days), p=0.24. Patients' ability to communicate discomfort was better in the dexmedetomidine group. Rates of hypotension and bradycardia were similar between the two groups. Critical illness polyneuropathy was more common in the propofol group (n=11) than in the dexmedetomidine group (n=2), p<0.02. The composite outcome of agitation, anxiety, and delirium occurred in propofol (29%) versus dexmedetomidine (18%) (p=0.008) (JAMA 2012;307:1151-60).
   7. Adverse effects: Bradycardia and hypotension (may be more common or severe in patients with cardiac dysfunction, intravascular volume depletion, or low systemic vascular resistance); respiratory depression, hypertriglyceridemia (most institutions use a threshold of 800–1000 as a cutoff of when to stop propofol), pancreatitis with or without hypertriglyceridemia, PRIS

8. PRIS: This is a rare but life-threatening complication of propofol, usually occurring at doses greater than 50 mcg/kg/minute for 48 hours or more. The mechanism of PRIS may include alterations in hepatic metabolism of the lipid emulsion, leading to an accumulation of ketone bodies and lactate and/or disruptions in the mitochondrial respiratory chain and inhibition of oxidative phosphorylation. Patients with urea cycle disorders may experience alterations in propofol metabolism within 24–48 hours of propofol use. Consider avoiding in patients with acute liver failure, or pancreatitis, because the symptoms of PRIS may be difficult to distinguish from the underlying disease state abnormalities. PRIS carries a high mortality rate, and propofol should be discontinued immediately if symptoms are present.
    a. Clinical characteristics of PRIS: Metabolic acidosis, acute renal failure, cardiovascular collapse, cardiac arrhythmias including Brugada-like syndrome, rhabdomyolysis, myoglobinemia, myoglobinuria, hyperkalemia, hypertriglyceridemia, elevated creatine kinase concentrations
    b. Risk factors for PRIS or other adverse effects of propofol: Neurologic injury, sepsis, use of vasoactive medications, high-dose propofol, acute liver failure

F. Dexmedetomidine
1. SCCM suggests that in adult ICU patients with delirium unrelated to alcohol or benzodiazepine withdrawal, dexmedetomidine infusions rather than benzodiazepine infusions should be administered for sedation to reduce the duration of delirium.
2. Mechanism of action: Highly selective and dose-dependent α_2-adrenoceptor agonist in the CNS. Dexmedetomidine provides a hypnotic and sedative effect by inhibition of norepinephrine release from the locus coeruleus; dexmedetomidine also produces a weak antinociceptive effect by way of inhibition of neuronal transmission through presynaptic C-fibers and release of substance P, and hyperpolarization of postsynaptic α receptors in the dorsal horn of the spinal column. Dexmedetomidine does not directly affect respiratory drive; therefore, intubation is not required with use. Dexmedetomidine is considered a weak sedative and would not be appropriate for use when deep sedation is required (e.g., in a patient requiring neuromuscular blockade). Both anterograde amnesia and retrograde amnesia have been described in some patients (20%–50%) in adult and pediatric studies. A benzodiazepine may be required if full amnesia is desired because of the clinical scenario.
3. Pharmacokinetics: Hepatic by glucuronidation and renal excretion. Onset with loading dose 15–20 minutes; onset without loading dose greater than 20 minutes to 1 hour; terminal half-life = 3 hours (may be significantly prolonged in hepatic impairment). Highly protein bound 94%.
4. Clinical effects: Sedation and weak opiate-sparing antinociceptive effects.
5. Dosing for ICU sedation: Optional loading dose 0.5–1 mcg/kg intravenously for 10 minutes, followed by 0.2–0.7 mcg/kg/hour. The loading dose may initially cause severe tachycardia and hypertension, but it can then quickly lead to significant bradycardia and/or hypotension secondary to receptor saturation. Because of these untoward hemodynamic effects, the loading dose is rarely administered in clinical ICU practice on initiation of dexmedetomidine, and the drip is usually initiated at 0.2–0.4 mcg/kg/hour. For the maintenance infusion dose, randomized trials have safely used dexmedetomidine at higher than manufacturer-recommended doses, up to 1.5 mcg/kg/hour. Clinical efficacy with doses greater than 1.5 mcg/kg/hour remains unclear. Other routes of administration have been described for dexmedetomidine, including intramuscular, subcutaneous, epidural, and intranasal.
6. Duration of use: Although the package insert recommends a therapy of 24 hours or less, randomized trials have used dexmedetomidine for up to 5–7 days; thus, ICU clinicians often administer dexmedetomidine for longer than 24 hours. Safety beyond 7 days of use has not been well established.
7. Data: The Dexmedetomidine to Lessen ICU Agitation (DahLIA) study was a double-blind placebo-controlled, parallel-group randomized clinical trial in 15 ICUs in Australia and New Zealand in which 39 patients were randomized to dexmedetomidine and 32 patients to placebo. At 7 days, dexmedetomidine increased ventilator-free hours compared with placebo (median, 144.8 hours vs. 127.5 hours, 95% CI

4 to 33.2 hours, p=0.01). Patients in the dexmedetomidine group had decreased time to extubation compared with placebo (median 21.9 hours vs. 44.3 hours, 95% CI 5.3 to 3.1 hours, p<0.001). An accelerated resolution of delirium was found in the dexmedetomidine group compared with placebo (median, 23 hours vs. 40 hours, 95% CI, 3 to 28 hours, p=0.01). However, propofol use was common in both groups after randomization (72% in the dexmedetomidine group vs. 88% in the placebo group) (median cumulative dose 980 mg (IQR 280–3050) vs. 5390 mg (IQR 1880–10,803), p<0.001).

8. Data: The DESIRE trial was a open-label, multicenter randomized clinical trial conducted in eight ICUs in Japan in which 100 patients with sepsis were randomized to dexmedetomidine and 101 patients with sepsis were randomized to placebo. Mortality at 28 days was not different between the two groups (23% in the dexmedetomidine group vs. 31% in the placebo group; HR 0.69; 95% CI, 0.38–1.22; p=0.20). Ventilator-free days over 28 days were not different between groups (median 20 days (IQR 5–24) in the dexmedetomidine group vs. 18 days (IQR 0.5–23) in the control group (p=0.20).
9. Data: SPICE III was an open-label, randomized controlled trial comparing dexmedetomidine as primary sedation with usual care (propofol, midazolam, or other sedation) in patients receiving less than 12 hours of mechanical ventilation who are expected to be on the ventilator for at least one additional day. The target RASS goal was -2 to +1. Death at 90 days occurred in 569 of 1956 (29.1%) of the usual care group and 566 of 1948 (29.1%) of the dexmedetomidine group (adjusted risk difference 0.0 percentage points; 95% CI, –2.9 to 2.8). In the dexmedetomidine group, 64% of patients additionally received propofol, 3% received midazolam, and 7% received both during the first 2 days after randomization. Sixty percent of patients received propofol, 12% received midazolam, and 20% received both in the usual care group. Given the many sedatives administered in both groups, application of the study results is difficult. In the dexmedetomidine group, bradycardia and hypotension occurred in 5.1% and 2.7% of patients. A post hoc analysis of SPICE III included 703 mechanically ventilated patients. Over the first 5 ICU days, more patients in the dexmedetomidine group had a temperature of 38.3°C or greater (43.3% vs 32.7%, p=0.004) and 39.0°C or greater (19.4% vs 12.5%, p=0.013). A post hoc subgroup analysis of SPICE III compared the effect of dexmedetomidine versus usual care on vasopressor requirements in patients with septic shock. This subroup included 83 patients, of which, 44 (53%) received dexmedetomidine and 39 (47%) usual care. In the first 48 hours, median NEq dose was 0.03 [0.01, 0.07] μg/kg/min in the dexmedetomidine group and 0.04 [0.01, 0.16] μg/kg/min in the usual care group (p = 0.17). Patients in the dexmedetomidine group had a lower NEq/MAP ratio, meaning lower vasopressor requirements to maintain the target MAP.
10. Data: MENDS 2 is multicenter, double-blind trial comparing dexmedetomidine to propofol. No difference was found between dexmedetomdine and propofol in the number of days alive and without delirium of coma (adjusted median 10.7 vs 10.8 days; odds ratio, 0.96; 95% CI 0.74–1.26), ventilator free days (adjusted median, 23.07 vs. 24 days, odds ratio 0.98; 95% CI, 0.63–1.51), or death at 90 days 38% vs. 39%; hazard ratio, 1.06; 95% CI 0.74–1.52).
11. Data: Dexmedetomidine was compared with other sedatives in critically ill mechanically ventilated adults. A total of 77 randomized trials were included. Compared with other sedatives, dexmedetomidine reduced the risk of delirium (RR 0.67; 95% CI, 0.55–0.81; moderate certainty), duration of mechanical ventilation (mean difference [MD] -1.8 hours; 95% CI, -2.89 to -0.71; low certainty), and ICU length of stay (MD -0.32 days; 95% CI, -0.42 to -0.22; low certainty). The risk of bradycardia (RR 2.39; 95% CI, 1.82–3.13; moderate certainty) and hypotension (RR 1.32; 95% CI, 1.07–1.63; low certainty) was increased with dexmedetomidine.
12. Data: A rapid practice guideline was published looking at use of dexmedetomidine for sedation in mechanically ventilated adult patients in the ICU. It suggested use of dexmedetomidine over other sedative agents if the reduction in delirium was more highly valued than the potential for hypotension and bradycardia.
13. Adverse effects: Tachycardia, pyrexia bradycardia, hypertension, hypotension, dry mouth. Should generally be avoided in patients with acute decompensated heart failure or advanced heart block

14. Other potential uses in the ICU: Procedural sedation, palliative care pain and anxiety control, adjunct to opiates for sickle cell crisis, adjunct to benzodiazepines or propofol for alcohol withdrawal, bridge to extubation while tapering off longer-acting sedatives and/or opiates, to provide sedation and anxiolysis during noninvasive mechanical ventilation

G. Lorazepam (Ativan)
   1. Pharmacokinetics: Benzodiazepine that binds to the postsynaptic $GABA_A$ receptor, undergoes hepatic clearance by conjugation to inactive compounds; moderate to high volume of distribution and high protein binding. Onset of action is 15–30 minutes, slower than more lipophilic benzodiazepines (e.g., midazolam). Duration of action of intermittent dosing is 4–8 hours. As a continuous infusion, clearance of lorazepam decreases in an unpredictable fashion, and prolonged sedation may occur.
   2. Effects: Anxiolysis/sedation, anticonvulsant, muscle relaxant. Maintains anterograde amnesia properties; however, studies report that patients who received benzodiazepines in the ICU may maintain delusional versus factual memories.
   3. Dosing range: 1–4 mg every 4–6 hours intermittent dosing is recommended before using continuous infusion; accumulation and prolonged awakening times with continuous infusion of lorazepam may occur because of prolonged duration of action.
   4. Data: Carson et al. studied the number of days on mechanical ventilation in medical ICU patients receiving intermittent lorazepam (n=64) compared with continuous infusion propofol (n=68); each group underwent daily interruption of sedation if the fraction of inspired oxygen (Fio_2) was less than 80%. Median time on mechanical ventilation was 9 days in the lorazepam group versus 4.4 days in the propofol group, p=0.006; ICU length of stay was 12.7 days in the lorazepam group versus 8.6 days in the propofol group (p=0.05); no difference in hospital mortality. Delirium was not assessed in this study (Crit Care Med 2006;34:1326-32).
   5. Propylene glycol toxicity: Because of its insolubility, injectable lorazepam is diluted in propylene glycol. Propylene glycol toxicity can occur with lorazepam infusions for more than 48 hours, particularly at doses of 6–8 mg/hour or greater, and can manifest as new-onset renal failure, respiratory failure, metabolic acidosis, and altered mental status. Most hospitals cannot measure quantitative levels of propylene glycol; therefore, surrogate markers of propylene glycol toxicity such as an elevated osmolar gap (greater than 10 mOsm/kg) and elevated anion gap with new metabolic acidosis are recommended for monitoring. If these metabolic abnormalities are present while on a lorazepam infusion, lorazepam should be discontinued.
   6. Other adverse effects: Paradoxical agitation, confusion, prolonged duration of sedative action, respiratory depression, hypotension, bradycardia

H. Midazolam (Versed)
   1. Pharmacokinetics: Benzodiazepine that binds to the postsynaptic $GABA_A$ receptor; undergoes phase I hepatic metabolism to an active glucuronidated metabolite, α_1-hydroxymidazolam, which is then renally excreted. A short- to medium-acting benzodiazepine in patients with normal renal and hepatic function. CYP3A4 substrate. Midazolam is highly lipophilic, has a large volume of distribution, and is highly protein bound.
   2. Clearance: Clearance of midazolam or its metabolite is significantly altered if either hepatic (primary drug accumulation) or renal (active metabolite α-hydroxymidazolam accumulation) functions are significantly impaired. Continuous renal replacement therapy partly clears the active metabolite but does not effectively clear the parent compound and is therefore not recommended as a method for definitive midazolam clearance (Am J Kidney Dis 2005;45:360-71). High lipophilicity and a large volume of distribution may lead to significant drug accumulation and a depot effect in the ICU patient. In general, clearances of midazolam infusions have wide interpatient variability in the ICU, and emergence times may be significantly prolonged.

3. Effects: Anxiolysis/sedation, anticonvulsant, muscle relaxant. Maintains anterograde amnesia properties.
4. Dosing range: 1–4 mg every 2–4 hours intermittently or as needed should be considered before initiating continuous infusion. Older adult patients may tolerate only 1–2 mg per dose.
5. Data: In a multicenter European trial (MIDEX), Jakob et al. compared midazolam (n=251) with dexmedetomidine (n=249) for sedation in prolonged mechanical ventilation. Patients in both groups were treated with daily sedation interruption trials and SBTs, and pain was treated with fentanyl boluses. There was no difference in the primary outcome: proportion of time in target RASS (0, -3) without rescue therapy in the midazolam group (56%) versus the dexmedetomidine group (60%). Median time on mechanical ventilation was lower in the dexmedetomidine group (5 days) than in the midazolam group (6.8 days), p=0.03. Patients in the dexmedetomidine group were more arousable, more cooperative, and better able to communicate discomfort or pain to clinical staff than were patients in the midazolam group. Hypotension occurred more often in the dexmedetomidine group (20.6%) than in the midazolam group (11.6%; p=0.007); and bradycardia was more common in the dexmedetomidine group (14.2%) than in the midazolam group (5.2%; p<0.001). The two treatment groups showed no difference in neurocognitive adverse events after 48 hours of follow-up, including agitation, anxiety, and delirium (JAMA 2012;307:1151-60).
6. Adverse effects: Paradoxical agitation and prolonged duration of sedative action, respiratory depression, hypotension, bradycardia

I. Titration of Sedation in the ICU
1. Titration of medications using a sedation protocol to a goal level of sedation is arguably one of the most salient clinical practice standards within ICU care. This titration practice has consistently been shown to decrease time on mechanical ventilation, decrease ICU length of stay, and decrease rates of tracheostomy, which may all result in faster physical and cognitive rehabilitation time.
   a. The goal level of sedation should be reassessed daily, documented, and communicated clearly to the nursing and medical staff because the goal may change throughout a patient's ICU stay.
   b. Level of sedation should be assessed every 2–4 hours throughout the day and evening. Consider assessing every 4 hours during nighttime sleeping hours to minimize sleep interruption.
   c. The two validated sedation scales currently recommended by the PADIS guidelines are the RASS (Table 5) and the Riker Sedation-Agitation Scale (Table 6).
2. Recommended methods for titration of medications include either (1) titration to a "light" versus "deep" level of sedation, unless clinically contraindicated, or (2) daily SAT. The PADIS guidelines suggest using light sedation (vs. deep sedation) in critically ill, mechanically ventilated adults. Light sedation is defined as a RASS of –2 to +1 or an SAS of 3 or 4, per the PADIS Guidelines. However, a pilot study from 2022 suggested that a RASS of –1 and an SAS of 4 indicate light sedation.
   a. Patients should receive sedation only if required.
   b. Sedatives should be titrated to allow patient responsiveness and awareness, as shown by patients' ability to purposefully respond to a combination of any three of the following actions on request: open eyes, maintain eye contact, squeeze hand, stick out tongue, and wiggle toes. A numerical score on a sedation scale may not fully represent a patient's ability to follow purposeful commands.
3. Data: In a multicenter, prospective, longitudinal Australian and New Zealand Study, Shehabi et al. evaluated 251 medical/surgical patients who were ventilated and sedated for 24 hours or more. Within 4 hours of initiating ventilation, deep sedation occurred in 191 patients (76.1%) and in 171 patients (68%) at 48 hours. Early deep sedation was an independent predictor of time to extubation (HR 0.90; 95% CI, 0.87–0.94; p<0.001), hospital death (HR 1.11, 95% CI, 1.02–1.20; p=0.01), and 180-day mortality (HR 1.08, 95% CI, 1.01–1.16; p=0.026) (JAMA 2012;307:1151-60).

4. In a single-center, nested cohort study within the Awakening and Breathing Controlled (ABC) randomized trial, Seymour et al. evaluated 140 patients for whom hourly doses of benzodiazepines and propofol during the daytime (7 a.m. to 11 a.m.) and nighttime (11 p.m. to 7 a.m.) for 5 days were measured. Greater daytime benzodiazepine doses were independently associated with failed SBT and extubation and subsequent delirium in adjusted models ($p<0.02$ for all). A failed SBT ($p<0.01$) and delirium ($p=0.05$) were associated with nighttime increases in benzodiazepine doses (Crit Care Med 2012;40:2788-96).

J. SAT Paired with SBT: The daily coordination of an SAT completed before an SBT is a method of weaning sedation before attempts at breathing trials in order to maximize a patient's chances of weaning from mechanical ventilation. This pairing of an SAT before an SBT is becoming recognized as an important component to ICU care and management of sedation. Important safety screens are incorporated into the daily SAT because studies have shown that the SAT is not appropriate for all ICU patients. If a patient does not pass the safety screen and does not undergo the SAT, this should not preclude the appropriate titration of sedatives to a goal level of sedation throughout the remainder of the day:
   1. SAT safety screen (criteria may vary; published trial protocols have had variations): If any are present, discontinue the protocol and repeat in 12–24 hours or according to hospital protocol:
      a. Current RASS greater than 2; or goal for deeper sedation (e.g., RASS -3 to -5)
      b. Active seizures requiring a continuous infusion of a sedative to control
      c. Active alcohol withdrawal requiring a continuous infusion of a sedative to control
      d. Fio_2 of 70% or greater (these criteria are not consistently present among published trial protocols)
      e. Neuromuscular blockade
      f. Myocardial ischemia in previous 24 hours or ongoing myocardial ischemia
      g. Intracranial pressure (ICP) greater than 20 mm Hg or need for control of ICP
   2. If pass SAT safety screen, begin SAT: Hold continuous sedative and analgesic infusions. Bolus opioids are recommended for breakthrough pain. Continuous opioid infusions allowed to continue while stopping sedatives if presence of active pain. If the patient "passes" the SAT, continue to the SBT safety screen.
   3. SAT failure (if any are present, discontinue the protocol, and repeat in 12–24 hours or according to hospital protocol):
      a. Anxiety/agitation/pain present (e.g., RASS greater than +1 for 5 minutes or more)
      b. Respiratory rate greater than 35 breaths/minute for 5 minutes or more
      c. Oxygen saturation less than 88% for 5 minutes or more
      d. ICP greater than 20 mm Hg
      e. Acute cardiac ischemia or arrhythmia
      f. Respiratory or cardiac distress (e.g., heart rate increase of 20 beats/minute or greater, heart rate less than 55 beats/minute, use of accessory muscles, abdominal paradox, diaphoresis, or dyspnea)
   4. If SAT fails: Consider giving patient bolus opioids first (up to three doses in 1 hour) before restarting infusion. Reinitiate sedation infusion, if necessary, at half the previous dose and titrate to goal. Determine the reasons for SAT failure. Repeat SAT steps in 12–24 hours or according to hospital protocol.
   5. SBT safety screen (if any are present, discontinue the protocol; repeat in 12–24 hours or according to hospital protocol):
      a. Agitation
      b. Oxygen saturation less than 88%, Fio_2 greater than 50%
      c. PEEP (positive end-expiratory pressure) of 7 cm H_2O or greater
      d. Myocardial ischemia in previous 24 hours
      e. Increasing vasopressor requirements
      f. Lack of inspiratory efforts

6. SBT: If a patient tolerates the SBT for 30 to 120 minutes, consider extubation.
7. SBT failure:
    a. Respiratory rate greater than 35 breaths/minute (for more than 5 minutes) or less than 8 breaths/minute
    b. Oxygen saturation less than 88% for more than 5 minutes
    c. ICP greater than 20 mm Hg, mental status change
    d. Acute cardiac ischemia or arrhythmia
    e. Respiratory distress (use of accessory muscles, abdominal paradox, diaphoresis, and dyspnea)
8. If SBT fails: Place the patient on prior ventilator settings. Repeat bundle in 12–24 hours or according to hospital protocol.

K. Sedation Protocol Compared with the Paired SAT-SBT Protocol: Studies comparing a standard sedation protocol with daily pairing of a SAT with SBT have shown decreased days on mechanical ventilation, days in the ICU, and decreased rates of delirium when the SAT is paired with the SBT.
    1. The ABC trial included 336 mechanically ventilated patients from four tertiary care hospitals. Patients were randomized to patient-targeted sedation protocol plus the SBT ("usual care" control group) or to daily SAT paired with the SBT (intervention group). Both groups were deeply sedated on enrollment (RASS -4), and both groups had been admitted for 2.2 days before enrollment. In the intervention group, patients who passed the safety screen underwent an SAT: sedatives and analgesics used for sedation were discontinued, and analgesics used for active pain were continued. Patients "passed" their SAT if they opened their eyes to command or tolerated being off sedation for at least 4 hours without meeting failure criteria. The mean ventilator-free days was 11.6 days in the usual care control group versus 14.7 days in the SAT plus SBT group (p=0.02). The time to discharge was 12.9 days in the control group versus 9.1 days in the intervention group (p=0.01). Self-extubations were higher in the intervention group, but there was no difference in self-extubations requiring reintubation between groups. Rates of delirium assessed by the confusion assessment method for the intensive care unit (CAM-ICU) were no different between groups (74% vs. 71%).
    2. The first study evaluating the ABCDE (Awakening and Breathing Coordination, Delirium Monitoring/Management, and Early Mobility) Bundle compared clinical outcomes in patients before (n=146) and after (n=150) bundle-protocol implementation; 187 patients were on mechanical ventilation. The bundle protocol consisted of a daily-paired SAT/SBT, delirium screening with the CAM-ICU every 8 hours, and an early mobility protocol. The "before" bundle patients were enrolled from February to October 2011; the "after" bundle patients were enrolled from October 2011 to April 2012. There were some differences in patient type on admission, including more elective admissions in the post-bundle group (39 vs. 30), more cardiothoracic surgery patients in the post-bundle group (20 vs. 6), more surgical patients in the pre-bundle group (21 vs. 11), and more patients coming from an outside hospital in the pre-bundle group (9 vs. 1). The post-bundle group had more median ventilator-free days (24 vs. 21 days, p=0.04), less delirium at any time (49 vs. 62%, p=0.03), and shorter percentage of ICU days with delirium (33.3 vs. 50 %, p=0.003) (Crit Care Med 2014;42:1024-36).
    3. The SLEAP investigators from the Canadian Critical Care Trials Group studied the outcomes of patients receiving a daily sedation protocol alone versus patients receiving a daily sedation protocol plus a daily sedation interruption (Crit Care Med 2015;43:557-66; Crit Care Med 2015;43:2180-90; JAMA 2012;308:1985-92). From January 2008 to July 2011, 430 patients were enrolled from 16 tertiary care medical and surgical ICUs. Only opiate and benzodiazepine infusions were allowed in the study. According to the sedation-alone protocol, the RASS goal was -3 to 0, and the Sedation-Agitation Scale (SAS) goal was 3 or 4. Nurses assessed sedation levels on an hourly basis and titrated medications every 15–30 minutes to achieve sedation goals. If patients were oversedated in either group (SAS 1 or 2; RASS −4 or −5), infusions were discontinued. According to the sedation protocol with daily sedation

interruption, nurses stopped benzodiazepine and opiate infusions once a day and assessed hourly for wakefulness (e.g., a light SAS or RASS score, plus ability to follow at least three commands).

a. Clinical outcomes (published 2012): There was no difference in the primary outcome of time to successful extubation between the two groups (7 days in both groups). There was a significant difference in time to extubation in the prespecified surgical/trauma group between sedation protocol with daily sedation interruption and sedation protocol alone (6 vs. 13 days; hazard ratio [HR] 2.55; 95% CI, 1.40–5.44). No difference in time to extubation was detected between groups among medical ICU patients (9 vs. 8 days; HR 0.92; 95% CI, 0.72–1.18). However, significantly lower daily doses of both benzodiazepines and opiates boluses and continuous infusions were used in the sedation protocol–alone group than in the sedation protocol plus interruption group. Although the sedation protocol suggested to target light sedation, the actual mean RASS/SAS was not reported for either group making it unclear how deeply sedated either group was.
b. Delirium outcomes (published 2015): Delirium by the Intensive Care Delirium Screening Checklist (ICDSC) was diagnosed in 53.8% of patients in the study; there was no difference in delirium in the sedation protocol–alone group versus the protocol plus daily sedation interruption group. Patients who had delirium had a longer duration of mechanical ventilation, longer ICU and hospital stay, longer use of restraints, higher rates of tracheostomy, and higher incidence of unintentional device removal. Patients with delirium received almost the twice the mean dose of midazolam equivalents/patient/day (104 mg vs. 57 mg), higher fentanyl equivalents/patient/day (1497 mcg vs. 1150 mcg), more frequent use of anticholinergics (18 vs. 8.6%), and more frequent use of trazadone or zopiclone (17.7 vs. 9.8%) than did patients who were not delirious. Patients who developed delirium had a higher incidence of alcohol and cigarette use than did patients who did not develop delirium.
c. Recall in ICU survivors (published 2015): The SLEAP investigator study did patient interviews on days 3, 28, and 90 post-ICU discharge to determine differences in recall between the sedation-alone protocol group and the protocol plus daily sedation interruption group. There were no differences in type of recall between the sedation strategies. Delusional memories were common at day 28 (70% of patients) but were unrelated to the presence of delirium or the total dose of benzodiazepines or opiates. Patients with no recall had received lower total doses of benzodiazepines than had patients with recall. Emotional memories such as panic and confusion declined over time.

Table 5. Richmond Agitation-Sedation Scale (RASS)

Score	Term	Description
+4	Combative	Overtly combative or violent; immediate danger to staff
+3	Very agitated	Pulls on or removes tube(s) or has aggressive behavior toward staff
+2	Agitated	Frequently nonpurposeful movement or patient-ventilator dyssynchrony
+1	Restless	Anxious or apprehensive, but movements not aggressive or vigorous
0	Alert and calm	
−1	Drowsy	Not fully alert but has sustained (> 10 s) awakening, with eye contact, to voice
−2	Light sedation	Briefly (< 10 s) awakens with eye contact to voice
−3	Moderate sedation	Any movement (but no eye contact) to voice
−4	Deep sedation	No response to voice, but any movement to physical stimulation
−5	Unarousable	No response to voice or physical stimulation

Table 5. Richmond Agitation-Sedation Scale (RASS) *(continued)*

Score	Term	Description
Procedure 1. Observe patient. Is patient alert and calm (score 0)? Does patient have behavior that is consistent with restlessness or agitation (score +1 to +4 using the criteria listed above, under *Description*)? 2. If patient is not alert, in a loud speaking voice, state the patient's name and direct the patient to open eyes and look at speaker. Repeat once if necessary. Can prompt patient to continue looking at speaker. Patient has eye opening and eye contact, which is sustained for more than 10 seconds (score −1). Patient has eye opening and eye contact, but this is not sustained for 10 seconds (score −2). Patient has any movement in response to voice, excluding eye contact (score −3). 3. If patient does not respond to voice, physically stimulate patient by shaking shoulder and then rubbing sternum if there is no response to shaking shoulder. Patient has any movement to physical stimulation (score −4). Patient has no response to voice or physical stimulation (score −5).		

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Table 6. Riker Sedation-Agitation Scale

7	Dangerous agitation	Pulling at ETT, trying to remove catheters, climbing over bedrail, striking at staff, thrashing side to side
6	Very agitated	Does not calm despite frequent verbal reminding of limits, requires physical restraints, biting ETT
5	Agitated	Anxious or mildly agitated, trying to sit up, calms down to verbal instructions
4	Calm and cooperative	Calm, awakens easily, follows commands
3	Sedated	Difficult to arouse, awakens to verbal stimuli or gentle shaking but drifts off again, follows simple commands
2	Very sedated	Arouses to physical stimuli but does not communicate or follow commands, may move spontaneously
1	Unarousable	Minimal or no response to noxious stimuli, does not communicate or follow commands

ETT = endotracheal tube.

Adapted with copyright permission from: Lippincott Williams and Wilkins/Wolters Kluwer Health. Simmons LE, Riker RR, Prato BS, et al. Assessing sedation during intensive care unit mechanical ventilation with the Bispectral Index and the Sedation Agitation Scale. Crit Care Med 1999;27:1499-504.

L. Acute Withdrawal Syndrome of Long-term Analgesia and/or Sedation in the ICU
   1. Patients who have been receiving high doses of continuous infusion sedation and/or analgesia in the ICU for an extended period may be at risk of sedative or analgesia withdrawal as dose tapering begins. In a retrospective review of adult trauma/surgical ICU patients, 32% of patients experienced either sedative or opiate withdrawal soon after discontinuing these medications. The patients in this study had been in the ICU for 20 days or more and were receiving higher mean daily analgesic and sedative doses than were the non-withdrawal patients (fentanyl 6.4 mg vs. 1.4 mg; lorazepam 38 mg vs. 11 mg). The withdrawal patients in this study were also more likely to have received an NMBA (Crit Care Med 1998;26:676-84).
   2. Data: An international, multicenter, observational, point prevalence study in 229 adult ICUs showed that a small number of ICUs have policies or protocols for opioid or sedative weaning or iatrogenic withdrawal syndrome. Even when these protocols exist, they are rarely used.

3. The risk factors and incidence of sedation or analgesia withdrawal in adult ICU patients have not been well characterized; however, these are important considerations in the long-term ICU patient receiving high doses of these medications. Use of longer-acting agents given orally or by feeding tube has been described to assist in the transition off long-term continuous infusions. The medical indication and dosing plan for using oral medications to taper off continuous infusions should be clearly documented in the medical chart on patient discharge from the ICU. Clonidine is a potential consideration in patients to aid in dexmedetomidine withdrawal.

Patient Cases

3. A 48-year-old man with cirrhosis and now hepatorenal syndrome was intubated for respiratory distress. He has been receiving midazolam 1 mg/hour and fentanyl 75 mcg/hour for 2 days; his RASS (−4 to −5) and CPOT has been 1 for 24 hours. Oxygen requirements have decreased, and vital signs are normal. Which is the most appropriate change in his medications?
   A. Decrease midazolam; give as-needed lorazepam for agitated RASS score.
   B. Discontinue midazolam and fentanyl; give as-needed fentanyl for CPOT of 3 or greater.
   C. Discontinue midazolam; initiate propofol drip.
   D. Change midazolam to dexmedetomidine drip.

4. T.L. is a 55-year-old woman intubated for respiratory distress for severe pneumonia. She is receiving fentanyl 50 mcg/hour and dexmedetomidine 1.0 mcg/kg/hour. Her home medications are confirmed to include esomeprazole 20 mg daily, lorazepam 1 mg three times daily, and citalopram 10 mg daily. The nurse reports intermittent agitation with tachycardia and a negative pain score. Which is the most appropriate recommendation?
   A. Increase fentanyl drip for agitated RASS score.
   B. Reinitiate lorazepam and citalopram.
   C. Give fentanyl boluses as needed for agitation.
   D. Increase dexmedetomidine.

IV. DELIRIUM IN THE INTENSIVE CARE UNIT

A. Delirium is an acute and fluctuating disturbance in consciousness resulting in the inability to receive, process, store, or recall information. In the ICU, delirium may present as hyperactive (agitated and restless), hypoactive (flat affect, apathy, lethargy, decreased responsiveness), or mixed hyper/hypoactive states. Most common in the ICU are mixed and hypoactive states of delirium. Two screening tools are currently recommended by the PAD guidelines: (1) the CAM-ICU and (2) the ICDSC. Both the CAM-ICU and the ICDSC require a RASS (-2) or a SAS (3) or more alert to be completed.
   1. The CAM-ICU assesses four features: (1) acute change or fluctuation in mental status from baseline, (2) inattention, (3) altered level of consciousness, and (4) disorganized thinking. If features 1 *and* 2 plus feature 3 *or* 4 are present, the patient is considered positive for delirium. Detailed training is available at www.icudelirium.org.

2. The ICDSC consists of eight items, evaluated during an 8- to 24-hour period. The eight symptoms are level of consciousness, inattention, disorientation, hallucinations-delusions-psychosis, psychomotor agitation or retardation, inappropriate speech or mood, sleep-wake cycle disturbances, and symptom fluctuation. A point is given for any symptom that is present during the previous 24 hours; a score of 4 or higher indicates the presence of delirium.

B. Background – 30%–80% of ICU patients reportedly develop delirium, depending on the severity of illness and the diagnostic method, yet assessment for delirium is still not routine in most U.S. ICUs. During a patient's hospitalization, the presence of delirium is associated with difficulty in weaning mechanical ventilation and longer duration of mechanical ventilation, increased use of physical and chemical restraints, longer duration of ICU stay, and additional stress to family and friends who may not understand the course of delirium. Delirium is also associated with up to a 3-fold increase in mortality, increase in cognitive decline, delay in cognitive recovery, and increased likelihood of being discharged to a nursing home. Two studies found that a longer duration of delirium was independently associated with worse activity of daily living scores and worse cognitive impairment scores at 3 and 12 months post-ICU discharge (Crit Care Med 2014;42:369-77; N Engl J Med 2013;369:1306-16). A retrospective study reported increased difficulty in the weaning of mechanical ventilation when delirium was detected in patients during the first spontaneous weaning trial compared with in patients who did not have delirium (Respirology Nov 2015;1-8).
   1. The underlying pathophysiology of delirium is not well understood; however, it may involve a complex set of factors:
      a. Cerebral hypoperfusion and alterations in cerebral blood flow
      b. Degradation of the blood-brain barrier, causing influx of inflammatory cytokines and microvascular thrombosis
      c. Depletion in central neurotransmitters (e.g., dopamine, norepinephrine, serotonin)
      d. Depletion in acetylcholine
      e. Medication withdrawal
   2. Risk factors for delirium: A systematic review of studies from 2001 to 2013 described 11 variables identified as risk factors for developing delirium in the ICU, extracting from only a strong or moderate level of evidence (Crit Care Med 2015;43:40-7):
      a. Age
      b. Preexisting dementia
      c. History of baseline hypertension
      d. Sedative-associated coma
      e. APACHE II (Acute Physiology and Chronic Health Evaluation II) score
      f. Delirium on the previous day
      g. Emergency surgery
      h. Mechanical ventilation
      i. Organ failure
      j. (Poly)trauma
      k. Metabolic acidosis
   3. Other reported risk factors or precipitants:
      a. Infection
      b. Dehydration or malnutrition
      c. Sleep deprivation
      d. Centrally acting medications (benzodiazepines, opiates, anticholinergics)
      e. Lack of exposure to sunlight
      f. Lack of personal interaction
      g. Physical restraints or insertion of catheters or tubes

4. Medication-induced altered mental status – Although the development of delirium is considered multifactorial, any patient who presents with a change in mental status should have his or her medications and doses immediately scrutinized as part of the initial workup for delirium. Several classes of medications have long been recognized for their effects on mental status and cognitive function, in or out of the ICU. These medications have the potential to affect a patient's level of consciousness or course of delirium at any point in the patient's hospital stay. Anticholinergics, benzodiazepines, opiates, antipsychotics, antispasmodics, anticonvulsants, corticosteroids, and others should be used with caution in a hospitalized patient, with close monitoring of the patient's cognitive adverse effects. Because renal and hepatic function may fluctuate throughout an ICU stay and affect the clearance of these medications, doses must be thoughtfully titrated. Research on the degree of impact these medications have on the overall course of sedation and delirium in the ICU is difficult to characterize, and research is undergoing. Given the multifactorial nature of delirium in the ICU, clinicians should be leery of solely assigning blame to medications but should remain vigilant when assessing the need and doses of the aforementioned medications.
    a. Benzodiazepines: The PADIS guidelines state "benzodiazepine use may be a risk factor for the development of delirium in adult ICU patients." Of interest, research directed at finding an independent association between the use of benzodiazepines and the development of delirium in the ICU including a meta-analysis, a randomized trial, and a systematic review has yielded mixed results (Crit Care Med 2015;43:40-7; Crit Care Med 2015;43:557-66; Crit Care Med 2013;41:S30-8). Investigators used more appropriate statistical analysis for a fluctuating illness such as delirium (e.g., Markov monitoring was used to determine the probability of a daily transition to delirium while assuming this was independent of the patient history beyond the prior day) in addition to more frequent delirium monitoring to examine the use of benzodiazepines and the transition from an awake state without delirium to delirium, or from coma to delirium by the next day (Intensive Care Med 2015;41:2130-7). This study found that a midazolam equivalent dose of just 5 mg/day increased the odds of developing delirium the next day by 4%, and the use of benzodiazepine infusions was an independent risk factor for the transition to delirium in the study population. However, there was a large difference in the daily dose of benzodiazepines between the continuous infusion (daily median dose 99 mg) compared with the intermittent dosing group (daily median dose 4.1 mg), bringing into question whether the risk factor for the transition to delirium was indeed the cumulative dose and not the method of administration. As the data evolve, the scrutiny of benzodiazepine use should persist for ICU clinicians. Routine strategies to preferentially use a non-benzodiazepine sedative (e.g., propofol or dexmedetomidine) and to avoid benzodiazepine infusions unless clinically indicated should be used in addition to diligent performance of daily SATs to discontinue use as soon as possible.
    b. Anticholinergics: These medications are known for their sedating and altering effects on mentation and should be avoided or used with extreme caution in the ICU setting. One proposed mechanism for delirium is a decline in acetylcholine concentrations; therefore, any medication that may further inhibit the activity of acetylcholine could worsen the patient's mental status. A prospective cohort study of 1112 critically ill patients was conducted to determine whether anticholinergic exposure increased the probability of transitioning to delirium. On 6% of ICU days, transition from "awake and without delirium" to "delirium" occurred. A nonsignificant increase in the probability of transitioning to delirium the following day resulted from a 1-unit increase in the Anticholinergic Drug Scale (odds ratio 1.05; 95% CI, 0.99–1.10). The dose of the medication was not evaluated to determine the effects on the transition to delirium, and it was not evaluated if patients were already delirious and remained delirious while receiving anticholinergic medications (Crit Care Med 2015;43:1846-52).

c. Systemic corticosteroids: The neuropsychiatric effects of systemic steroids in various clinical settings have been described for more than 50 years. Common symptoms include mania, depression, mood lability, anxiety, insomnia, delirium, and psychosis. The incidence of these symptoms varies greatly depending on the clinical setting, dose of steroid, and patient's underlying medical history. Reported risk factors for neuropsychiatric effects secondary to steroids include a daily prednisone dose equivalent to 40 mg or greater, hypoalbuminemia, underlying psychiatric disorder, and blood-brain barrier damage (Curr Opin Organ Transplant 2014;19:201-8). Although steroids are commonly used in the ICU for various indications and at various doses, significant research directed at the neuropsychiatric effects of steroids in the critically ill population is lacking. In a secondary analysis of a multicenter observational study of adult medical and surgical ICU patients with acute lung injury (n=330), Schreiber et al. found a significant and independent association between the use of systemic corticosteroids and the transition to delirium from a non-comatose, non-delirious state within 24 hours of corticosteroid administration (odds ratio [OR] 1.52 [1.05–2.21], p=0.03). Delirium was documented on one or more days in 83% of patients, with a median duration of 7 days. There was no significant association in prednisone-equivalent dose and transition to delirium. Schreiber et al. recognize that a direct causal relationship could not be determined between corticosteroid use and delirium from this observational study; however, they believe that the study adds valuable data toward our understanding of risk factors for delirium in the ICU (Crit Care Med 2014;42:1480-6). A second study that investigated steroids and transition to delirium in a mixed medical and surgical ICU population (n=1112) found no association between steroid use and a transition to delirium. The median prednisone equivalent dose was 50 mg (Crit Care Med 2015;43:e585-8).

5. Outcomes of sedation-related versus illness-related delirium: A single-center study using propofol and fentanyl timed its CAM-ICU assessments before and after a daily sedation interruption protocol. Rapidly reversible delirium was defined as delirium while patients were receiving sedation that resolved within 2 hours after performing an SAT. This type of delirium was rare (12% of the 102 patients), but these patients has a prognosis that was similar to patients who did not have delirium. Most patients (75%) had persistent delirium, delirium that did not resolve with cessation of sedatives, a higher risk of death, and longer length of stay (Am J Respir Crit Care Med 2014;189:658-65). Patients can have both sedation- and illness-related delirium, and additional research in this area is needed to clarify the differences in short- and long-term outcomes.

C. Monitoring for Delirium: SCCM recommends assessing critically ill patients for delirium with a validated tool such as either the CAM-ICU or the ICDSC. The PAD guidelines summarized their review of five delirium assessment scales used for adult ICU patients. The two scales with the highest psychometric (e.g., validity and reliability) scores were the CAM-ICU and the ICDSC. Both scales were designed for patients in the ICU either on or off mechanical ventilation, and both showed high sensitivity and specificity when tested against the American Psychiatric Association's criteria for delirium.

1. Delirium should be assessed at least every 8-12 hours and documented in the medical chart; results should be discussed with the medical team. Because these assessment scales cannot distinguish between sedation- and disease-related causes of delirium, delirium assessments should ideally be timed both before and after SATs with appropriate time allowed for drug clearance (www.icudelirium.org, Am J Respir Crit Care Med 2014;189:658-65). If this timing is not feasible and a patient screens positive for delirium while receiving ongoing analgesia or sedation, an SAT should be conducted if the patient passes the safety screen to assist in ruling out a medication-induced cause of delirium.

2. If a patient's delirium score is positive, the medical team should correct possible etiologies (e.g., decrease sedative doses, if safe), decrease ongoing risk factors, address inciting factors (e.g., metabolic derangements, infection, withdrawal), and try nonpharmacologic treatment and preventive measures when appropriate.

D. Prevention of Delirium: With a lack of data supporting the use of pharmacologic agents to prevent delirium, the PADIS guidelines suggest not using haloperidol, an atypical antipsychotic, dexmedetomidine, a statin, or ketamine to prevent delirium in critically ill patients. Instead, the recommendations are focused on nonpharmacologic prevention methods when feasible, particularly for patients at high risk of delirium. Preventive efforts may help avert 30%–40% of new-onset delirium cases, particularly in older adults. Recommended nonpharmacologic strategies by the PADIS guidelines include:
   1. Performing rehabilitation or mobilization in critically ill adults
   2. Using a multicomponent intervention to reduce or shorten delirium (e.g., reorientation, cognitive stimulation, use of clocks); improve sleep (e.g., minimize light and noise); improve wakefulness (i.e., reduce sedation); reduce immobility (e.g., early rehabilitation/mobilization); and reduce hearing and/or visual impairment (e.g., enable use of devices such as hearing aids or eyeglasses).
   3. Decreasing the use of benzodiazepines and anticholinergics in patients at risk of delirium; use the lowest effective doses of any sedating medication (e.g., opiates, antipsychotics).
   4. REDUCE was a randomized, double-blind, placebo-controlled study of 1789 critically ill patients randomized to prophylactic haloperidol 1 mg, haloperidol 2 mg, or placebo. The 1-mg haloperidol group was prematurely stopped because of futility (JAMA 2018;319:680-90). No difference occurred in the median days that patients survived in 28 days in the 2-mg haloperidol group compared with 28 days in the placebo group (95% CI, 0-0; p=0.93) with a hazard ratio of 1.003 (95% CI, 0.78–1.30; p=0.82). None of the 15 secondary outcomes were statistically different between the three groups. These outcomes included delirium incidence (mean difference 1.5%; 95% CI, −3.6% to 6.7%), delirium- and coma-free days (mean difference 0 days; 95% CI, 0–0 days), and duration of mechanical ventilation, ICU, and hospital length of stay (mean difference 0 days; 95% CI, 0–0 days for all three measures). Adverse events did not differ between groups.

E. Sleep in the ICU: Uninterrupted sleep (ideally 4 hours or more) is vital for an adequate immune response to illness, to maintain normal metabolic and hormonal balance, and to help decrease delirium and/or agitation. Disturbances in the ICU such as multiple alarms and frequent physical interruptions (e.g., examination, turning, laboratory tests, medication administration) make it challenging for patients to maintain the slow-wave sleep cycle needed for optimal immune function. Sleep research in the ICU is ongoing, and more information will be forthcoming regarding its effects in the critically ill patient. The PADIS guidelines suggest not routinely using physiologic sleep monitoring clinically in critically ill adults. In addition, the PADIS guidelines suggest not using aromatherapy, acupressure, or music at night to improve sleep in critically ill adults. However the PADIS guidelines do suggest using noise and light reduction strategies to improve sleep in critically ill adults. The PADIS guidelines make no recommendation regarding the use of melatonin to improve sleep or regarding the use of dexmedetomidine at night to improve sleep. The PADIS guidelines suggest not using propofol to improve sleep in critically ill adults. In addition, the PADIS guidelines suggest using a sleep-promoting, multicomponent protocol in critically ill adults.
   1. To avoid waking patients at night, pharmacists should ensure that medications are scheduled during the daytime and evening hours, if possible—particularly orally or subcutaneously administered medications.
   2. Sleep protocols should seek to cluster patient care activities (e.g., vital sign checks, radiology tests, laboratory checks, sedation assessments) around nighttime sleeping hours unless clinically indicated in a specific patient population.

3. A two-center, double-blind randomized trial randomized 100 critically ill adults without delirium to nocturnal dexmedetomidine or placebo (Am J Respir Crit Care Med 2018;197:1147-56). During the ICU stay, nocturnal dexmedetomidine was associated with a greater proportion of patients who remained delirium free (dexmedetomidine (40 of 50 patients [80%]) than with placebo (27 of 50 patients [54%]; relative risk 0.44; 95% CI, 0.23–0.82; p=0.006). Adverse events did not differ between the two groups.

F. Treatment of Delirium: The cause of delirium may be multifactorial, and identifying and correcting the underlying etiology is the first step in management. Patients can also progress to alcohol withdrawal or withdrawal from other chronic medications/substances and present with hyperactive delirium. The PADIS guidelines suggest not using an atypical antipsychotic, haloperidol, or a statin to treat subsyndromal delirium or delirium (N Engl J Med 2018;379:2506-16). The Modifying the Impact of the ICU-Associated Neurological Dysfunction-USA (MIND USA) Study is a multicenter, randomized, placebo-controlled study of 566 patients showing that haloperidol and ziprasidone did not reduce delirium, time on the ventilator, ICU or hospital length of stay, or death compared with placebo. Arrhythmias, parkinsonism (extrapyramidal symptoms), neuroleptic malignant syndrome, study drug discontinuation, and other safety concerns were extremely low across all three groups. Additionally, a planned secondary analysis of the MIND USA study showed that neither haloperidol nor ziprasidone significantly increased QTc intervals compared with placebo. The AID-ICU trial was a multicenter, randomized, placebo-controlled trial of 963 patients with delirium randomized to haloperidol or placebo. The mean number of days alive and out of the hospital was 35.8 (95% CI, 32.9–38.6) in the haloperidol group and 32.9 (95% CI, 29.9–35.8) in the placebo group with an adjusted mean difference of 2.9 days (95% CI, -1.2 to 7; p=0.22). Antipsychotics remain viable for the short-term control of agitation (e.g., alcohol or drug withdrawal) or severe anxiety with the need to avoid respiratory suppression (e.g., heart failure, COPD, or asthma). If an antipsychotic is initiated, low starting doses should be considered, and daily review of drug interactions, adverse effects, dosing titration, and need for the antipsychotic should be completed. In addition, a strategy for discontinuation or outpatient follow-up should be documented to help avoid inadvertent continuation beyond the hospital environment (Table 7). Serious adverse effects are associated with the use of any antipsychotic; effects such as arrhythmias, serotonin syndrome, neuroleptic malignant syndrome, extrapyramidal symptoms, and oversedation should be closely monitored on a daily basis. Dose ranges for atypical antipsychotics for ICU delirium are not well described. The American Geriatrics Society 2015 Beers Criteria for medication use in older adults includes the following recommendation: "Avoid antipsychotics for behavioral problems of dementia or delirium unless nonpharmacological options have failed or are not possible AND the older adult is threatening substantial harm to self or others." If the ICU team decides to use antipsychotics in older adults, lower starting doses should be considered, together with daily review of drug interactions and adverse effects. The PADIS guidelines suggest using dexmedetomidine for delirium in mechanically ventilated adults when agitation precludes weaning/extubation. The PADIS guidelines suggest not using bright light therapy to reduce delirium in critically ill adults.

G. Inadvertent Continuation of Antipsychotics Beyond ICU Discharge: The PADIS guidelines discuss the use of adjunctive medications for ICU patients (e.g., antipsychotics, gabapentin, carbamazepine). Although their use in the ICU may be appropriate, there is a potential for inadvertent continuation of these medications on hospital discharge if a treatment plan is not clear in the medical record. This has been a well-documented problem with other medications initiated in the ICU (e.g., histamine receptor blockers, proton pump inhibitors), and studies have been published describing the continuation of newly prescribed antipsychotics from the ICU and hospital, even when an indication for continuation was not documented (J Crit Care 2015;30:814-6; J Crit Care 2016;33:119-24). Continued use of these medications beyond the hospital stay could lead to serious adverse effects, drug interactions, and significant drug cost as well as a presumption of a psychiatric or neuromuscular disorder associated with these drugs. Communication to the next direct patient care provider is crucial to appropriately direct the next steps in medication reconciliation.

Table 7. Antipsychotics[a]

Drug	CYP Substrate (major)	Usual Starting Dose	Significant Adverse Effects[b]	Formulations
Haloperidol	3A4, 2D6	1–2 mg older adults; 2–4 mg if history of psychiatric disorders	Anticholinergic: * Sedation: * EPS: ** NMS: *	PO, IM, IV
Olanzapine	1A2	5 mg	Anticholinergic: ** Sedation: ** EPS: * NMS: * Neuromuscular weakness	PO, disintegrating tablet, IV, IM
Quetiapine	3A4	12.5–25 mg	Anticholinergic: ** Sedation: ** NMS: * Orthostatic hypotension: **	PO
Risperidone	2D6	0.5–1 mg	Anticholinergic: * Sedation * EPS: ** NMS: * Orthostatic hypotension: ** Cardiac conduction abnormalities	PO, disintegrating tablet
Ziprasidone	1A2 (minor) 3A4 (minor)	20 mg PO; 10 mg IM	Anticholinergic: * Sedation: * EPS: * NMS: *	Oral, IM

NOTE: * = lower risk; ** = medium-higher risk

[a]Not all medications listed are FDA label approved for use in delirium; not all are recommended by SCCM for the treatment of delirium in the ICU.

[b]Adverse effects other than QTc prolongation. Documented QTc prolongation incidence: IV haloperidol = ziprasidone > risperidone > olanzapine = quetiapine.

EPS = extrapyramidal symptoms; IM = intramuscular(ly); IV = intravenous(ly); NMS = neuroleptic malignant syndrome; PO = oral(ly)

1. Quetiapine (Seroquel): A randomized, placebo-controlled pilot trial compared the efficacy and safety of scheduled quetiapine with placebo for the treatment of delirium in ICU patients during a 10-day study (Crit Care Med 2010;38:419-27). Significant exclusions were as follows: patients with end-stage liver disease, those with alcohol withdrawal, those with a QTc greater than 500, and those receiving concomitant QTc-prolonging agents. This small pilot study (n=36), in which the placebo group was administered as-needed intravenous haloperidol, found that quetiapine was associated with a shorter time to first resolution of delirium, reduced duration of delirium, and less agitation than placebo. Mortality and ICU length of stay were not different from placebo.
    a. Pharmacokinetics: Hepatically metabolized to one active and two inactive metabolites. Metabolites renally cleared. Many drug interactions, CYP3A4 (major) and CYP2D6 (minor) substrates. Peak plasma concentrations for oral about 1½ hours (immediate release).
    b. Initial dose range for ICU delirium: 50 mg one to three times daily. Consider lower starting doses for older adult patients because of sedating effects. The Devlin study initiated 50 mg every 12 hours and titrated to a maximum dose of 200 mg every 12 hours.
    c. Adverse effects (early onset): Sedation, orthostatic hypotension, extrapyramidal symptoms, QTc prolongation

2. Olanzapine (Zyprexa): Available in oral, orally disintegrating, and intramuscular (immediate and extended release) dosage forms. Intramuscular administration may result in plasma concentrations 5 times those of oral administration. The immediate release intramuscular formulation can also be administered intravenously. The U.S. Food and Drug Administration (FDA) warns that the use of intramuscular olanzapine has resulted in unexplained deaths; use of intramuscular olanzapine with benzodiazepines may result in significant oxygen desaturation.
    a. Pharmacokinetics: Metabolized by glucuronidation and CYP 1A2, 2D6 oxidation. Clearance is significantly increased (around 40%) in smokers and decreased in females (around 30%). Many drug interactions, CYP1A2 (major) and CYP2D6 (minor) substrates. Weak inhibitor of several CYP isoenzymes. Peak plasma concentrations for oral: About 6 hours.
    b. Suggested starting dose for ICU delirium: 5 mg orally once daily
    c. Adverse effects (early onset): Drowsiness, extrapyramidal symptoms, neuromuscular weakness, serotonin syndrome. High doses may cause cardiac arrhythmias, cardiopulmonary arrest, and extreme sedation to coma-like states.
3. Risperidone (Risperdal): Available in oral and oral dispersible tablets (M-tabs) and intramuscular injection dosage forms
    a. Pharmacokinetics: Hepatically metabolized to active metabolites, renally cleared. Many drug interactions, CYP2D6 (major) and CYP3A4 (minor) substrates and P-glycoprotein. Peak plasma concentrations for oral about 1 hour.
    b. Suggested starting dose for ICU delirium: 0.25–0.5 mg once or twice daily
    c. Adverse effects (early onset): Cardiac arrhythmias, anticholinergic effects, extrapyramidal symptoms
4. Ziprasidone (Geodon): Studied in a multicenter, randomized, placebo-controlled pilot trial of mechanically ventilated patients to test the hypothesis that antipsychotics would improve days alive without delirium or coma in the ICU (MIND trial). Medical and surgical adult ICU patients (n=101) from six tertiary-care centers in the United States on mechanical ventilation who had an abnormal level of consciousness or were receiving analgesia/sedative medications were randomly assigned to receive haloperidol, ziprasidone, or placebo every 6 hours for up to 14 days during a 21-day study. During the study, no difference was found in median days alive without delirium or coma between the haloperidol (14 days), ziprasidone (15 days), and placebo (12.5 days) groups, p=0.66. The study also found no difference in ventilator-free days, hospital length of stay, or mortality among the three groups (Crit Care Med 2010;38:428-37).
    a. Ziprasidone is available in oral and intramuscular dosage forms.
    b. Pharmacokinetics: Hepatic by glutathione and aldehyde oxidase. Minor substrates of CYP 1A2, 3A4. Peak plasma concentrations for oral about 6 hours; intramuscular about 1 hour.
    c. Suggested starting dose for ICU delirium: 20 mg twice daily (oral)
    d. Adverse effects (early onset): Somnolence, extrapyramidal symptoms, dizziness, orthostatic hypotension

V. ABCDEF BUNDLE

A. Incorporating multiple concomitant patient care interventions into one consolidated bundle may be an effective strategy to improve clinical outcomes in critically ill patients. SCCM recommends implementing the "ABCDEF" bundle to align and coordinate care using an interprofessional approach (e.g., physician, nursing, pharmacy, respiratory therapy, physical and occupational therapy). The following practice principles are applied to the bundle:
   1. A: Assess, prevent, and manage pain
   2. B: Both SATs and SBTs
   3. C: Choice of analgesia and sedation
   4. D: Delirium: Assess, prevent, and manage
   5. E: Early mobility and exercise
   6. F: Family engagement and empowerment.

B. Assess, Prevent, and Manage Pain ("A" of the bundle): Asking the patient to self-report pain or to use the CPOT or BPS if the patient is nonverbal. Preventing pain by recognizing patients with known sources of pain (i.e., rib fractures) and scheduling analgesics when indicated. Managing pain by ordering the most appropriate pharmacologic agent on the basis of the source of pain and renal and liver function.

C. Daily coordination of the SAT with the SBT ("B" of the bundle) versus usual care with the SBT has been shown to significantly decrease the time on mechanical ventilation and ICU length of stay in randomized studies. This was reviewed earlier in the chapter in the Agitation section.

D. Choice of Sedation ("C" of the bundle): Use a multidisciplinary approach, including focused pharmacy input, to choose a sedative according to individual patient needs, hemodynamic stability, and organ function (e.g., hepatic, renal, cardiac, pulmonary, pancreatic).

E. Delirium Assessment, Prevention, and Management ("D" of the bundle): Regularly assess for delirium using the CAM-ICU or the ICDSC every 8–12 hours. Use delirium preventive measures in all patients when safe to do so.

F. Early Mobility ("E" of the bundle): Perform a mobility safety screen, and implement a daily mobility protocol.

G. Family Engagement and Empowerment ("F" of the bundle): Good communication with the family is critical at every step of the patient's clinical course, and empowering the family to be part of the team to ensure best care is adhered to diligently will improve many aspects of the patient's experience.

H. In a prospective, multicenter, cohort study from 68 academic, community, and federal ICUs during a 20-month collection period, performance of the complete ABCDEF bundle was associated with a lower likelihood of death within 7 days (HR 0.32; CI, 0.17–0.62), next-day mechanical ventilation (OR 0.28; CI, 0.22–0.36), coma (OR 0.35; CI, 0.22–0.56), delirium (OR 0.60; CI, 0.49–0.72), physical restraint use (OR 0.37; CI, 0.30–0.46), ICU readmission (OR 0.54; CI, 0.37–0.79), and discharge to a facility other than home (OR 0.64; CI, 0.51–0.80). There was a dose response in between higher proportional bundle performance and improvement in each clinical outcome ($p<0.002$). Pain was more commonly reported as bundle performance increased ($p=0.0001$), probably because more patients were awake.

VI. POST-INTENSIVE CARE SYNDROME

A. Advances in critical care have decreased mortality and resulted in an increased likelihood of surviving critical illness. Post-intensive care syndrome (PICS) describes new or worsening impairments in physical, cognitive, or mental health status after critical illness and persisting beyond acute care hospitalization. Physical impairments include both pulmonary dysfunction and neuromuscular weakness. Impairments in memory and executive functioning are examples of cognitive dysfunction. Mental health impairments include depression, posttraumatic stress disorder, and anxiety. Medication management including glucose control, use of the ABCDEF bundle, and review of medication lists at every transition of care are important roles of the pharmacist to prevent PICS. Family members can also have PICS, termed PICS-F, in which these individuals have depression, posttraumatic stress disorder, anxiety, and prolonged grief. SCCM has taken progressive steps to help clinicians and families recognize the prolonged PICS and PICS-F through THRIVE. The establishment of an ICU follow-up clinic is one proposed method to manage long-term complications of patients with PICS and family members of PICS-F through optimization of physical, cognitive, and mental health; improved coordination of care; and reduction in health care use. The pharmacist should be considered a key member of the PICS clinic team who performs complete medication management on all patients seen in the clinic.
   1. Physical impairment: A prospective, longitudinal study of 109 survivors with ARDS was conducted. Median (Interquartile Range) Total Lung Capacity (TVC) was 92% (77-97%), 92% (82–101%), and 95% (81–103%) of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. Forced expiratory volume in 1 second (FEV_1) was 75% (58–92%), 85% (69–98%), and 86% (74–100%) of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. Six-minute walk test was 49%, 64%, and 66% of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. These 109 survivors of ARDS were further analyzed annually up to 5 years after ICU discharge. TVC was 94% (84–108%), 93% (78–107%), 92% (79–104%), and 94% (78–105%) of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge. FEV_1 was 87% (75–99%), 79% (66–97%), 85% (68–98%), and 83% (69–98%) of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge. Six-minute walk test was 68%, 67%, 71%, and 76% of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge.
   2. Cognitive impairment: A large, multicenter, prospective observational cohort study of 821 adult medical ICU and surgical ICU patients (called the BRAIN-ICU study) estimated the prevalence of long-term cognitive impairment after critical illness secondary to respiratory failure, cardiogenic shock, or septic shock (N Engl J Med 2013;369:1306-16). Delirium was the strongest independent predictor of cognitive impairment in the 50% of patients after critical illness. A Repeatable Battery for Neuropsychological Status (RBANS) score similar to Alzheimer disease (2 standard deviations below the population mean) was found in 26% of patients, and a score similar to moderate traumatic brain injury (1.5 standard deviations below the population mean) was found in 40% of patients 3 months after discharge. Both young and older adults, with and without comorbidities, experienced these deficits, which persisted at 12 months in 24% and 34% of these individuals having RBANS scores similar to Alzheimer disease and moderate traumatic brain injury, respectively.
   3. Mental health impairments: A multicenter, prospective observational cohort study of 821 adult medical ICU and surgical ICU patients estimate the prevalence of depression, posttraumatic stress disorder (PTSD), and functional disability after critical illness secondary to respiratory failure, cardiogenic shock, or septic shock. Depression was found in 149 patients (37%) and 116 patients (33%) at 3 and 6 months after ICU discharge, respectively. Disabilities in basic activities of daily living were found in 139 patients (32%) and 102 patients (23%) at 3 and 6 months after ICU discharge, respectively. Disabilities in instrumental activities of daily living were found in 108 patients (26%) and 87 patients (23%) at 3 and 6 months after ICU discharge, respectively. PTSD was found in 27 patients (7%) at 3 and 6 months after ICU discharge.

4. A prospective, observational feasibility study was conducted at an academic hospital between July 2012 and December 2015 (Crit Care 2018;46:141-8). Patients were identified if at high risk of PICS; if so, post-ICU care was offered. Sixty-two patients were seen in the clinic. Median time from hospital discharge to ICU recovery center visit was 29 days. Cognitive impairment was identified in 64% of patients. Anxiety and depression were identified in 37% and 27% of patients. One-third of the patients were unable to ambulate independently. The median 6-minute walk distance was 56% of predicted. Only seven of the previously working patients (15%) had returned to work. Referral services and case management were provided 142 times. The median number of pharmacy interventions per patient was four.

B. Medication Management
   1. Glucose dysregulation: A retrospective study of 74 patients with ARDS found that a blood glucose value of 153.5 mg/dL resulted in a 2.9 greater chance of developing cognitive impairment. A second retrospective case-control study of 37 surgical ICU patients with at least one episode of hypoglycemia found that cognitive impairment was higher in the hypoglycemic group ($p<0.01$). Intensive insulin therapy (maintaining blood glucose levels between 80 and 100 mg/dL) in surgical ICU patients decreased neuropathy from 51.9% to 28.7%, and the prevalence of critical illness polyneuropathy (CIP) and critical illness myopathy (CIM) from 49% to 25% in surgical ICU patients ($p<0.0001$) (Neurology 2005;64:1348-53). Intensive insulin therapy also decreased CIP and CIM from 51% to 39% in the medical ICU ($p=0.02$) in patients who had an ICU stay of at least 1 week. The percentage of patients needing mechanical ventilation for at least 2 weeks was reduced from 42% to 32% in the surgical ICU ($p=0.04$) and from 47% to 35% in the medical ICU ($p=0.01$). Subsequently, NICE-SUGAR showed increased mortality in the intensive insulin group (81 to 108 mg/dL) (27.5%) versus conventional glucose control (less than 180 mg/dL) (24.5%) ($p=0.02$). The SCCM guidelines for the use of an insulin infusion for the management of hyperglycemia in critically ill patients suggests that a blood glucose of 150 mg/dL or greater initiates interventions to maintain blood glucose less than 180 mgdL and to avoid hypoglycemia based off the results of NICE-SUGAR.
   2. Continuation of inappropriate medications: The frequency of prescribed potentially inappropriate medications (PIMs) and actually inappropriate medications (AIMs) was evaluated in a single-center study of 120 older adult ICU survivors. PIMs were defined as potentially harmful on the basis of prior studies and pharmacologic effects. PIMs could further be classified as AIMs if the benefit of the drug was considered less than the harm. The 2003 Beers Criteria and medication safety data published since 2003 were used to identify medications (Arch Intern Med. 2011;171:1032-4). Medications were identified at five points during the hospital stay: admission, ward admission, ICU admission, ICU discharge, and hospital discharge. The most common categories of PIMs identified at hospital discharge were opioids, anticholinergic medications, antidepressants, and drugs causing orthostasis. Thirty-six percent of these PIMs were considered AIMs. The PIM categories at hospital discharge with the highest positive predictive values for being AIMs were anticholinergics (55%), nonbenzodiazepine hypnotics (67%), benzodiazepines (67%), atypical antipsychotics (71%), and muscle relaxants (100%). In multivariate analysis, the number of discharge PIMs was independently predicted by the number of preadmission PIMs ($p<0.001$), discharge to somewhere other than home ($p=0.03$), and discharge from a surgical service ($p<0.001$). Almost two-thirds of AIMs were initiated in the ICU.
   3. Not restarting home medications: A large population-based Canadian cohort study of 396,380 patients evaluated records of hospital and outpatient medications prescribed from at least one of five of the following groups: (1) statins, (2) antiplatelet/anticoagulant agents, (3) levothyroxine, (4) respiratory inhalers, and (5) gastric acid–suppressing drugs. Patients were divided into three groups: hospitalization with an ICU admission, hospitalization without ICU admission, and nonhospitalized patients who served as the control group. Compared with control patients, those admitted to a hospital without an ICU stay were more likely to have medications discontinued among all five of the medication groups.

Patients admitted to a hospital with an ICU stay were also more likely to have medications discontinued among all five of the medication groups than were control patients. Except for respiratory inhalers, there was a higher risk of medication discontinuation in all medication groups in patients hospitalized with an ICU admission than in patients hospitalized without an ICU admission. The composite outcome of death, hospitalization, and emergency department visit up to 1 year after hospital discharge in all study patients was higher in patients in whom a statin or antiplatelet or anticoagulant was discontinued (JAMA. 2011;306:840-7).

4. A prospective, observational cohort study was conducted of all outpatient appointments of a tertiary care hospital's post-ICU clinic between July 2012 and December 2015. The pharmacist completed medication reconciliation, interview, counseling, and resultant interventions during the post-ICU clinic appointment. The pharmacist did a full medication review in 56 of the patients (90%). All 56 patients had at least one pharmacy intervention. Medications were discontinued at the clinic appointment for 22 of the patients (39%). New medications were initiated in 18 of the patients (32%). An adverse drug event was identified in nine of the patients (18%). Adverse drug event preventive measures were implemented in 18 patients (32%). Thirteen patients (23%) had an influenza vaccination administered. Two patients (4%) received a pneumococcal vaccination (Ann Pharmacother 2018;52:713-23).
5. A prospective, observational study was conducted in 12 ICUs/post-ICU clinics between September 2019 and July 2021. A full medication review was performed in 472 patients by a pharmacist at the post-ICU clinic.. In 397 (84%) patients, pharmacy interventions were made. Medications were stopped in 124 (26%) and started in 91 (19%) patients. The number of patients that had a dose decreased was 51 (11%). The number of patients that had a dose increased was 43 (9%). Adverse drug event (ADE) preventive measures were initiated in 115 (24%) patients. Identification of ADEs occurred in 69 (15%) patients. In 30 (6%) patients, medication interactions were identified.

Patient Cases

5. T.L. (from question 4) was extubated 24 hours ago, is currently receiving dexmedetomidine 0.2 mcg/kg/hour, and has received two doses of fentanyl 25 mcg over 24 hours for pain. She is alert and calm with intermittent periods of agitation. Her pain score is now negative, and she is newly positive for delirium by CAM-ICU. Her laboratory values and vital signs are normal. Which would best be recommended for the management of delirium?

   A. Continue dexmedetomidine, and start quetiapine for delirium.
   B. Discontinue dexmedetomidine, and increase maintenance fluids for dehydration.
   C. Discontinue dexmedetomidine, and order patient mobility as tolerated.
   D. Continue dexmedetomidine, and schedule oxycodone sustained release every 12 hours.

6. P.V. is a 70-year-old woman intubated for severe respiratory failure (Fio_2 80%) and refractory shock from methicillin-resistant *Staphylococcus aureus* pneumonia, for which she was administered antibiotics, vasopressors, and steroids. She is on day 5 of mechanical ventilation (Fio_2 50%) and has been off vasopressors for 48 hours. The nurse describes PAD, but the patient denies pain. Medications include vancomycin 1000 mg daily, heparin 5000 units subcutaneously every 12 hours, hydrocortisone 50 mg every 6 hours, and fentanyl 75 mcg/hour. Which is the most appropriate recommendation at this time?

   A. Increase fentanyl, and add midazolam for agitation.
   B. Decrease fentanyl, and discontinue hydrocortisone.
   C. Decrease fentanyl, and add haloperidol for delirium.
   D. Increase fentanyl, and change vancomycin to linezolid.

VII. NEUROMUSCULAR BLOCKADE IN THE INTENSIVE CARE UNIT

A. The most recent SCCM guidelines for the sustained use of neuromuscular blockade in the ICU were published in 2016. Surveys have reported a dramatic decrease in the use of NMBAs during the past 20 years, from around 80% to 15% in patients on mechanical ventilation (Crit Care Med 2016;44:2079-103). This change in practice may be secondary to a better understanding of the serious adverse effects of prolonged paralysis, together with accepted standards of care for modes of mechanical ventilation in patients with ARDS.

B. Clinical Scenarios for the Use of NMBAs in the ICU May Include:
   1. Rapid sequence intubation
   2. ARDS
   3. Status asthmaticus
   4. Elevated ICP
   5. Elevated intra-abdominal pressure
   6. Targeted temperature management after cardiac arrest

C. Acute Respiratory Distress Syndrome
   1. Cisatracurium has been the most-studied NMBA for ARDS since 2000, primarily as short-term treatment and in severe cases of ARDS. In 2010, a randomized placebo-controlled trial (n=340) found that short-term fixed-dose cisatracurium (48 hours) significantly improved 90-day survival, increased ventilator-free days, increased organ dysfunction–free days, and decreased barotrauma in patients with severe ARDS (Pao_2/Fio_2 less than 120 mm Hg). The investigators found no difference in neuromuscular weakness compared with placebo. Other cisatracurium studies have shown improvements in oxygenation and a reduction in inflammatory mediators.
   2. In retrospective studies, the use of NMBAs in ARDS was associated with a prolonged duration of mechanical ventilation, prolonged ICU length of stay, and increased mortality.
   3. Protective mechanisms of NMBAs in severe ARDS: Researchers have proposed mechanisms by which an NMBA may protect the lung against further injury in severe ARDS. These mechanisms are not completely understood, but they may help explain the beneficial effects of NMBAs in early, severe ARDS:
      a. Provide improved adaptation to the ventilator through increased thoracopulmonary compliance.
      b. Increase functional residual capacity, and decrease intrapulmonary shunt.
      c. Provide uniform distribution of pulmonary perfusion and pressures, favoring the perfusion of ventilated areas.
      d. Limit over-distention of high-compliance lung regions and recruit areas of smaller compliance.
      e. Decrease muscular oxygen consumption by decreasing ventilator asynchrony.
      f. Decrease production of proinflammatory cytokines in lungs and blood.
      g. Provide protective role against ventilator-induced trauma, including decreased incidence of pneumothoraces.
   4. The 2016 SCCM NMBA guidelines suggest that a NMBA be administered by continuous intravenous infusion early in the course of ARDS for patients with a Pao_2/Fio_2 of less than 150 mm Hg (weak recommendation). Use of NMBAs in ARDS remains controversial. Short-term use of cisatracurium (48 hours or less) when used early may be beneficial for severe ARDS (Pao_2/Fio_2 less than 120 mm Hg). The Reevaluation of Systemic Early Neuromuscular Blockade (ROSE) PETAL was a randomized, parallel-design study comparing cisatracurium with placebo for 48 hours on the effects of 90-day mortality. It was published after the guidelines and did not support using this early in course of ARDS.

D. Targeted Temperature Management After Cardiac Arrest
   1. The 2016 SCCM NMBA guidelines make no recommendation on the routine use of NMBAs for patients undergoing therapeutic hypothermia after cardiac arrest (insufficient evidence). They recommend use of a protocol that includes guidance on NMBA administration in patients undergoing therapeutic hypothermia (good practice statement).
   2. NMBAs have been used to prevent or treat shivering during therapeutic hypothermia.
   3. The optimal combination and dosing of sedatives and paralytics have not been well established because the metabolism of these drugs is significantly slowed during hypothermia, and potency may be decreased. NMBAs have been used in both a bolus and a continuous infusion fashion during therapeutic hypothermia.

E. Sedation During NMBA: The 2016 SCCM NMBA guidelines recommend that optimal clinical practice require administering analgesic and sedative drugs before and during neuromuscular blockade with the goal of achieving deep sedation (good practice statement). They also suggest that patients receiving a continuous infusion of NMBA should receive a structured regimen of physiotherapy (weak recommendation). It is critical that patients be in a sedated, non-agitated, and pain-free state before initiating an NMBA. Once the patient becomes paralyzed from the NMBA, the ability to accurately assess mental status or pain is ostensibly challenging and often unattainable. The deeper the degree of paralysis, the higher the risk of drug accumulation because nurses cannot routinely complete sedation interruption or taper to a lighter level of sedation. Common scenarios that slow the clearance of sedatives (e.g., hepatic and renal failure or a hypothermic state) can add to the likelihood of increased drug exposure and delayed awakening times once the paralytic and sedatives are discontinued. This risk of drug accumulation underscores the importance of a daily assessment for need of paralysis and frequent tapering of NMBA dosing once it is safe for the patient.

F. Two Classes of NMBAs According to Mechanism of Action: Depolarizing and Nondepolarizing:
   1. Depolarizing NMBAs: Bind and activate acetylcholine receptors, causing persistent depolarization, which then renders muscle fibers resistant to further cholinergic stimulation. Succinylcholine is the only available depolarizing NMBA. Because of its quick onset and short duration, it is commonly the drug of choice for urgent or emergency intubation.
      a. Pharmacokinetics: Hydrolyzed by plasma pseudocholinesterase
      b. Usual dose: 0.05–1.5 mg/kg intravenously or intramuscularly
      c. Onset intravenously: 30–60 seconds; intramuscularly: 2–3 minutes
      d. Duration intravenously: 4–6 minutes; intramuscularly: 10–30 minutes
      e. Should not be used in patients with a history of malignant hyperthermia, hyperkalemia or risk factors for hyperkalemia (stroke, paralysis, or spinal, crush, or burn injuries after 24 hours), glaucoma, or penetrating eye injuries.
      f. Adverse effects: Arrhythmias, bradycardia or tachycardia, hyperkalemia, rhabdomyolysis
   2. Nondepolarizing NMBAs: Nicotinic receptor antagonists (competitive), blocking the action of acetylcholine at the neuromuscular junction. Divided into aminosteroid group (pancuronium, vecuronium, and rocuronium) and benzyl isoquinolinium group (atracurium, cisatracurium, doxacurium, and mivacurium).
      a. Pancuronium: Long-acting aminosteroid; intermittent or scheduled bolus may be preferred to continuous infusion because of accumulation and variable clearance. Older NMBA, not used much in the United States.
         i. Pharmacokinetics: Hepatically metabolized (30%–50%) and renally cleared as unchanged drug (50%–70%). Accumulation and prolonged duration of paralysis will occur with varying degrees of hepatic and/or renal dysfunction. Duration about 60–120 minutes.
         ii. Adverse effects: Vagolytic activity, sympathetic stimulation, bradycardia, prolonged effect

b. Vecuronium: Intermediate-acting aminosteroid; often used as a continuous infusion
   i. Pharmacokinetics: Hepatically metabolized (30%–50%); cleared renally (20%–30%), with fecal excretion. Has an active metabolite, around half the activity of parent compound. Duration 30 minutes after bolus intubation dose.
   ii. Adverse effects: Vagolytic activity at higher doses, prolonged weakness
   iii. Can be reversed with sugammadex by forming a complex that decreases the amount of vecuronium available to bind to nicotinic receptors in the neuromuscular junction.

c. Rocuronium: An intermediate-acting aminosteroid; considered a suitable alternative to succinylcholine for rapid sequence intubation (dose: 0.6–1.2 mg/kg) because of its rapid onset of action (60–90 seconds). Duration 30–40 minutes.
   i. Pharmacokinetics: Primarily hepatically metabolized, minimal renal excretion. No active metabolite. Prolonged effects have been observed in patients with hepatic or renal failure.
   ii. Adverse effects: Vagolytic activity at higher doses, bradycardia
   iii. Can be reversed with sugammadex by forming a complex that decreases the amount of rocuronium available to bind to nicotinic receptors in the neuromuscular junction.

d. Atracurium: Intermediate-acting benzyl isoquinolinium; a mixture of 10 stereoisomers (contains 15% cisatracurium)
   i. Pharmacokinetics: Undergoes Hofmann elimination to form the toxic metabolite laudanosine at high levels. Laudanosine is a cerebral stimulant that may precipitate seizure activity, clearance dependent on liver and kidney function. Duration of atracurium 20–40 minutes.
   ii. Adverse effects: Histamine release may cause cardiovascular adverse effects and bronchospasm; laudanosine accumulation may cause seizure activity.

e. Cisatracurium: An intermediate-acting benzyl isoquinolinium. Differences compared with atracurium: It is only one isomer, has a slower onset at normal bolus doses, no histamine release.
   i. Pharmacokinetics: Undergoes Hofmann elimination, forms laudanosine but at much lower levels than atracurium. Renal and hepatic dysfunction do not alter cisatracurium clearance. Duration 30–60 minutes.
   ii. Adverse effects: Prolonged weakness with continued use

G. Drug Interactions with NMBAs: Certain medications may decrease the activity of NMBAs, whereas others can enhance or prolong the paralytic action.
   1. Drugs decreasing the activity of NMBAs:
      a. Calcium: Antagonizes the effect of magnesium on neuromuscular blockade
      b. Carbamazepine: Competitor of acetylcholine receptor
      c. Phenytoin: Depressed postsynaptic response to acetylcholine
      d. Ranitidine: Unknown mechanism
      e. Theophylline: Unknown mechanism
   2. Drugs prolonging the activity of NMBAs:
      a. Antibiotics: Aminoglycosides, clindamycin, tetracyclines, vancomycin. Decreases prejunctional acetylcholine release with decreased postjunctional acetylcholine receptor sensitivity; blocks acetylcholine receptor.
      b. Cardiac medications: β-Blockers, calcium channel blockers, procainamide, quinidine, and furosemide. Decreases prejunctional acetylcholine release.
      c. Immunosuppressants: Steroids (decrease end plate sensitivity to acetylcholine), cyclosporine (inhibits metabolism of certain NMBAs)

H. Choice of NMBA: Intermediate- to longer-acting agents such as vecuronium may be tried in bolus fashion initially before continuous infusion, particularly if organ dysfunction is present. The duration of paralysis for NMBAs cleared by Hofmann degradation may be more reliable when used as a continuous infusion because their clearance is not dependent on renal or hepatic function.

I. Train-of-Four (TOF) Monitoring and Dose Titration
   1. The SCCM NMBA guidelines make no recommendation concerning the use of electroencephalogram-derived parameters (e.g., Bispectral Index [BIS], E-entropy, Cerebral State Index, and Patient State Index) as a measure of sedation during continuous NMBA administration (insufficient evidence). They suggest against the use of peripheral nerve stimulation (PNS) with TOF alone for monitoring the depth of neuromuscular blockade in patients receiving continuous NMBA infusions (very low quality evidence). They make no recommendation on the use of PNS to monitor the degree of block in patients undergoing therapeutic hypothermia (insufficient evidence). Typically, the goal of using an NMBA is to improve patient-ventilator synchrony and increase oxygenation. This may be achieved with varying degrees of paralysis and may not necessitate 100% block.
   2. Monitoring the depth of neuromuscular blockade by peripheral nerve stimulators (e.g., TOF), together with measured oxygenation parameters, helps find the "lowest effective paralytic dose" and allows quicker recovery of spontaneous neuromuscular transmission once the NMBA is discontinued. Some clinicians do not believe that TOF monitoring is necessary and believe that using the clinical values alone is sufficient to determine NMBA dosing.
   3. TOF delivers four supramaximal electrical impulses every 0.5 seconds to the ulnar, facial, or posterior tibial nerve. Response to the impulse is then measured by muscle twitches visualized from the associated innervated muscles (thumb or eye). Goals of paralysis can usually be reached with 2 or 3 of 4 twitches; 0 of 4 twitches indicates complete neuromuscular blockade, usually necessitating a decrease in NMBA dose. Oxygenation goals may be reached even with 4 of 4 twitches, indicating that the NMBA dose is effective and an increase is not warranted.
   4. A baseline electrical current should be established before initiating an NMBA to determine how much electrical current is needed to produce a twitch. Usually 10–20 mA (amperage) is sufficient. The conduction of the electrical impulse may be dampened because of peripheral edema, loss of electrode adhesion, incorrect electrode placement, and hypothermia, which can lead to inaccurate readings. These factors should be reassessed with each use of the TOF.

J. Complications of NMBAs
   1. Prolonged weakness: Several case reports associate the use of NMBAs and prolonged weakness, which could include myopathy, polyneuropathy, or neuromyopathy. Other risk factors may include concomitant use of corticosteroids, persistent hyperglycemia, and type of NMBA used. However, data are inconsistent and not controlled, and further studies are needed to clarify specific risk factors for prolonged weakness associated with NMBAs. Following a trend in creatine kinase concentration every 48–72 hours may help assess the presence of myopathy secondary to paralysis and prolonged immobilization. A creatine kinase concentration should not be solely relied on for the presence of myopathy, and daily determination of the need for the NMBA should still be considered, even with a normal creatine kinase.
   2. Corneal abrasions: Paralysis eliminates the ability of the eyes to close and blink, increasing the risk of corneal ulcerations and infection. Prophylactic eye protection must be used in all patients on NMBAs (e.g., lubricating eye ointments or eye covers).
   3. Thrombosis: Caused partly by immobility, patients receiving an NMBA may be up to 8 times more likely to have a DVT than those not on an NMBA. Prophylaxis for a DVT must be provided for all patients on an NMBA.

4. Awareness: Recent case reports document patient awareness during paralysis in the ICU. These patients report weird dreams, fear, resistance of restraints, thoughts of life and death, and pain. Patients must be deeply sedated before initiating an NMBA (Ann Emerg Med 2021;77:532-44).
5. Resistance to paralysis and/or potentiation: Certain disease states may produce an up-regulation in acetylcholine skeletal muscle receptors, leading to higher-than-normal doses of the NMBA (e.g., muscle trauma, muscle atrophy, burns). Acid-base disorders, electrolyte imbalances, and adrenal insufficiency may also cause unpredictable alterations in dosing requirements.
6. Anaphylaxis: Allergic reactions can occur after the first dose of an NMBA because the ammonium ions in NMBAs are commonly found in the household environment and in household products. If an allergic reaction is suspected, skin prick testing for the NMBA against a control can be done within 6 weeks of the reaction.

Patient Cases

7. A 55-year-old man intubated for severe ARDS (Pao_2/Fio_2 ratio less than 100 mm Hg) is receiving fentanyl 200 mcg/hour, midazolam 8 mg/hour, and propofol 40 mcg/kg/minute. He is deeply sedated but remains hypoxic and dyssynchronous with the ventilator after several changes in mechanical ventilation settings. Which is the most appropriate consideration at this time?
   A. Start scheduled lorazepam every 6 hours.
   B. Add quetiapine 50 mg every 8 hours.
   C. Change propofol to dexmedetomidine.
   D. Start a cisatracurium infusion.

8. A 70-year-old woman who is day 2 in the ICU is receiving a neuromuscular blocking agent (NMBA) and is sedated for severe ARDS. The TOF over 24 hours is 2 of 4 twitches at an amplitude of 10 mA. Arterial blood gas is pH 7.38, Pco_2 40 mm Hg, Po_2 91 mm Hg, and bicarbonate 24 mEq/L on 50% inspired oxygen and 10 cm H_2O PEEP; the patient is synchronous with the ventilator, and other clinical markers are stable. Which changes in management would be best to recommend?
   A. Decrease stimulator amplitude to decrease pain from excessive electrical current.
   B. Increase stimulator amplitude to test for more frequent twitches.
   C. Decrease the NMBA dose because the patient is clinically stable.
   D. Increase the NMBA dose until the TOF induces fewer twitches.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

This patient has a clear indication for intravenous pain medication from his recent trauma and multiple fractures. An "as-needed" opiate would likely not keep up with his pain control needs (Answers B and C are incorrect). His age and history of hypertension place him at risk of delirium; therefore, a benzodiazepine is not the best initial choice for this patient. A fentanyl infusion for pain with a propofol infusion if needed for sedation is the most appropriate answer (Answer A is incorrect; Answer D is correct).

2. **Answer: D**

Chest tube removal is specifically cited in the PADIS guidelines as an indication for both preemptive analgesia and nonpharmacologic relaxation techniques. This is given a "strong" recommendation, determining that the benefits outweigh the risks of preemptive therapy (Answer D is correct). Acetaminophen given just before the procedure will most likely not adequately treat pain associated with chest tube removal (Answer A is incorrect). Increasing an opiate infusion several hours before a bedside procedure can expose the patient to substantially higher amounts of drug than needed and cause delayed awakening times or other significant adverse effects from opiates (Answer C is incorrect). Extensive studies of appropriate preemptive analgesia for chest tube removal have not been completed; however, administering an opiate appropriately timed before manipulation of a chest tube is an accepted standard of therapy (Answer B is incorrect).

3. **Answer: B**

This patient has end-stage liver failure and acute renal failure; he will therefore not predictably clear midazolam or fentanyl infusions. With a RASS of −4 to −5, indicating no meaningful responsiveness to stimuli, all sedatives should be held if the patient is otherwise clinically stable to allow time for clearance of medications. If the patient was to develop pain based on a CPOT of 3 or more, then intermittent fentanyl is appropriate (Answer B is correct). A decrease in sedative dose or changing to a different sedative is not needed at this time based on the deeply sedated RASS score and would only further delay awakening time. Intermittent doses of benzodiazepines should avoided. (Answers A, C, and D are incorrect).

4. **Answer: B**

Withdrawal from certain home medications may occur if these medications are not reinitiated within a few days of admission. The onset of withdrawal symptoms will vary depending on the half-life of each medication. Symptoms may include agitation, anxiety, psychosis, insomnia, hypertension, and tachycardia and can occur with medications such as opiates, GABA receptor agonists, antiepileptics, antidepressants, and antipsychotics. A pharmacist can assist the medical team by obtaining a thorough medication history and assessment of home medication adherence to help identify drug withdrawal symptoms. Reinitiating these medications can be considered, unless contraindicated because of the clinical scenario (e.g., drug-drug interactions, drug-disease state interactions). The Agitation and Sedation section of the PADIS guidelines discusses identifying and treating the etiology of agitation before adding other medications; reinitiating the benzodiazepine and antidepressant to treat withdrawal symptoms is the most appropriate answer (Answer B is correct). Neither fentanyl nor dexmedetomidine would treat withdrawal from a benzodiazepine or antidepressant (Answers A, C, and D are incorrect).

5. **Answer: C**

The PADIS guidelines stress using nonpharmacologic means to manage delirium when it is safe for the patient. Strong evidence for using dexmedetomidine to treat delirium is still not available and the MIND USA Study has shown that anti-psychotics are ineffective at treating delirium. This patient's presentation of "alert and calm with intermittent periods of agitation" is a common scenario, and initial therapy should focus on reorienting and getting the patient interactive and mobile (Answers A and D are incorrect; Answer C is correct). Dehydration is a common cause of agitation, and it should be addressed; however, with normal laboratory values and vital signs, this patient is unlikely dehydrated at this time (Answer B is incorrect).

6. **Answer: B**

In the general population, systemic corticosteroids are known to cause many neuropsychiatric events, including hyperactivity and agitation; in a recent study of adult ICU patients with acute lung injury, only age and use of systemic corticosteroids in the preceding 24 hours were

independently associated with the transition to delirium from a non-delirious state (Answer B is correct). Benzodiazepines have a sedating effect and may calm an acutely agitated patient; however, they would not be recommended in this patient because they could worsen her confusion or delirium (Answer A is incorrect). The PADIS guidelines state that no evidence supports the use of haloperidol to reduce the duration of delirium (Answer C is incorrect). Vancomycin is not currently recognized as a cause of delirium; therefore, changing to linezolid is not indicated (Answer D is incorrect).

7. **Answer: D**

The midazolam dose is high, and the patient is "deeply sedated"; therefore, adding another benzodiazepine will likely not improve this patient's clinical status. Quetiapine has no indication for general sedation in a critically ill patient, and it should not be a consideration for sedation in this patient with severe ARDS. Dexmedetomidine is considered a weak sedative with no effect on respiratory drive; therefore, it would likely not improve this patient's ventilator dyssynchrony and hypoxia (Answers A–C are incorrect). At this stage in the patient's clinical course, it is reasonable to consider an NMDA. In a 2010 study of cisatracurium versus placebo for 48 hours in early ARDS, the cisatracurium group had more days free of mechanical ventilation and decreased mortality (30% vs. 44%) at 90 days for the subgroup of patients with severe ARDS (Pao_2/Fio_2 ratio less than 120 mm Hg). The incidence of pneumothorax was lower in the cisatracurium group than in the placebo group (4% vs. 11%). There were more days free of organ failure (non-lung) in the cisatracurium group than in the placebo group (15.8 vs. 12.2 days) in the first 28 days. A recent meta-analysis concluded that using short-term cisatracurium in patients with severe ARDS decreases mortality and time on mechanical ventilation compared with placebo. The risk of prolonged neuromuscular weakness was not found in these studies; however, use beyond 48 hours may increase this risk (Answer D is correct).

8. **Answer: C**

The TOF method of assessment is primarily used to help determine the degree of neuromuscular blockade and should not be used to titrate the dose of the NMBA. The patient's clinical status and laboratory values are the true determinants for dose adjustment of the NMBA. Patients may be at their clinical goal with a TOF of 2 or 3 twitches of 4. This is the ideal scenario, and it will predict a faster reversal of neuromuscular blockade (Answer C is correct). A TOF of 0 or 1 of 4 twitches predicts a significantly slower neuromuscular recovery time, and clinicians should try to decrease the NMBA as soon as the patient is clinically stable by laboratory values and ventilator management (Answer D is incorrect). A baseline electrical current intensity (amperage) should be established before the onset of neuromuscular blockade and should not be changed during paralysis unless a new baseline is indicated (Answer A is incorrect). As the electrical intensity (amperage) is established, an increase in the amperage is not indicated during infusion of the NMBA in order to increase the number of twitches. A decrease in the dose of NMBA would be indicated if an increase in the number of twitches were the clinical goal (Answer B is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**

Propofol infusion syndrome is a well-documented and complex set of adverse events, potentially resulting in multiorgan failure. An elevation in lactate, creatine kinase, transaminases, SCr, and triglycerides and the presence of a metabolic acidosis are some of the abnormalities that should concern the critical care provider for the presence of PRIS (Answer D is correct). Both DVT and critical illness polyneuropathy are serious concerns in the ICU patient; however, the abnormalities in the case are not representative of these complications (Answers A and B are incorrect). There are currently no known abnormal laboratory values to help determine whether delirium is present in an ICU patient (Answer C is incorrect).

2. **Answer: B**

This patient is at risk of propylene glycol toxicity after receiving a lorazepam drip for more than 48 hours. Lorazepam is dissolved in propylene glycol, an alcohol that can induce an osmolar gap and metabolic lactic acidosis, particularly in patients with significant hepatic or renal failure. Quantitative propylene glycol levels may be unavailable; therefore, surrogate markers such as an abnormal osmolar gap (greater than 10 mmol) and metabolic acidosis may indicate propylene glycol toxicity and a need to discontinue lorazepam. Although lorazepam drips are not routinely used for general sedation in adult ICUs, they may be used for other indications (e.g., severe EtOH [ethyl alcohol] or benzodiazepine withdrawal), and clinicians should remain aware of this serious complication (Answer B is correct). With an oxygen saturation of 98% on 2 L of oxygen, this patient does not meet the predefined criteria of ARDS (Answer A is incorrect). Encephalopathy will not cause a metabolic acidosis; therefore, an ammonia concentration would not be helpful at this stage (Answer C is incorrect). With a low fractional excretion of sodium and a high BUN/SCr ratio, the patient's laboratory values are indicative of a pre-renal concern versus acute tubular necrosis (Answer D is incorrect).

3. **Answer: D**

It is inappropriate to initiate an NMBA in a patient who has a sedation score indicating "agitation." This implies that the patient may potentially detect pain or discomfort while paralyzed (Answers B and C are incorrect). The goal should be to achieve a deeply sedated and/or non-agitated state before initiating an NMBA in an effort to avoid any patient discomfort that may be undetected during paralysis (Answer D is correct). The SAT would be inappropriate in someone who is rated "agitated" on the sedation scale or in a patient requiring escalating doses of sedation (Answer A is incorrect).

4. **Answer: B**

The pharmacokinetics/dynamics of prolonged fentanyl infusions have not been well described in the adult ICU population. Most data for fentanyl are derived from short-term infusions or boluses in healthy volunteers and in animal models. Fentanyl is hepatically metabolized primarily by the CYP3A4 enzyme, and decreased clearance of fentanyl has been described in patients with significant liver disease. Other properties of fentanyl (e.g., high volume of distribution, high protein binding, and high lipophilicity) may contribute to unpredictable clearance and a prolonged context-sensitive half-time for patients in acute renal failure or in patients who have inadequate nutritional status (Answer B is correct). Propofol is a CYP3A4 inhibitor; therefore, it should not induce the metabolism of fentanyl (Answer C is incorrect). Propofol is known to chelate trace elements and increase urinary loss of zinc when used for more than 5 days; propofol has not been shown to cause hypocalcemia (Answer D is incorrect). Disease states identified as risk factors for PRIS may include sepsis, acute liver failure, and history of pancreatitis; ARDS is not currently a documented risk factor (Answer A is incorrect).

5. **Answer: B**

Withdrawal from certain home medications may occur if these medications are not reinitiated within a few days of admission. The onset of withdrawal symptoms will vary depending on the half-life of each medication. Symptoms may include agitation, anxiety, psychosis, insomnia, hypertension, and tachycardia and can occur with medications such as opiates, GABA receptor agonists, antiepileptics, antidepressants, and antipsychotics. A pharmacist can assist the medical team by obtaining a thorough medication history and assessment of home medication compliance to help identify drug withdrawal symptoms. Reinitiating these medications

can be considered unless contraindicated because of the clinical scenario (e.g., drug-drug interactions, drug-disease state interactions). The Agitation and Sedation section of the PADIS guidelines discusses identifying and treating the etiology of agitation before adding other medications; fentanyl should treat this patient's chronic pain and treat opiate/tramadol withdrawal (Answer B is correct). Adding quetiapine or lorazepam for agitation would not address or treat the underlying etiology of potential opiate/tramadol withdrawal (Answers A and C are incorrect). Patient-controlled analgesia with an opiate would help treat opiate withdrawal; however, this patient is not alert enough to use it (Answer D is incorrect).

6. **Answer: D**

Up to 30%–60% of ICU patients may go through alcohol withdrawal on cessation of alcohol use. The presence of alcohol withdrawal in ICU patients may prolong their ICU stay, increase hospital costs significantly, and lead to other complications during the hospital stay. Early and aggressive symptom-triggered management with a benzodiazepine, particularly in a patient with a history of alcohol withdrawal, is a key element in management. There is no otherwise published protocol for alcohol withdrawal in ICU patients. In this case, although dexmedetomidine is useful as an adjunctive agent to help decrease sympathetic storm and agitation secondary to alcohol withdrawal, it is not currently recommended for use as a single agent for alcohol withdrawal (Answer A is incorrect). Opiates do not treat alcohol withdrawal (Answer C is incorrect). Phenytoin is a known antiepileptic for use in epilepsy; however, it has not been shown to be effective in preventing or treating alcohol withdrawal (Answer B is incorrect). Benzodiazepines are the drugs of choice for alcohol withdrawal seizures; therefore, midazolam is the most appropriate drug for this patient, whose medical history is significant for recurrent alcohol withdrawal seizures (Answer D is correct).

7. **Answer: B**

This patient is at risk of delirium given the patient's age and being critically ill in the ICU. Avoiding medications that may cause or worsen delirium, such as benzodiazepines, is a strong recommendation in the PAD guidelines (Answer D is incorrect). With a QTc of 550 milliseconds, increasing the quetiapine dose for agitation could further increase the QTc and put this patient at high risk of cardiac arrhythmias. Starting quetiapine would not be the best option (Answer A is incorrect). Starting non-pharmacologic management of agitation/delirium is the best option (Answer B is correct). Dexmedetomidine does not prolong the QTc and should be continued in the setting of agitation (Answer C is incorrect).

8. **Answer: D**

Dementia is a progressive and chronic state of cognitive impairment that worsens over weeks to months; this differs from the acute onset characteristic of ICU delirium (Answer A is incorrect). Alcohol withdrawal should always be a consideration for patients who become altered or agitated in the ICU. This patient's vital signs are normal, and she is presenting in a hypoactive state; these findings are not typical for acute alcohol withdrawal, and alternative causes for her decline in mental status need to be considered (Answer B is incorrect). Adrenal insufficiency in the ICU usually presents with abnormal laboratory values and/or hemodynamic instability. This patient's laboratory values and hemodynamics are reported as normal; therefore, adrenal insufficiency is unlikely (Answer C is incorrect). Both iatrogenic and non-iatrogenic causes for delirium are well documented in the literature. Dehydration and infection are common causes of delirium, particularly in the older adult population. This patient has several reasons for being dehydrated in the hospital: NPO status, persistent fevers, and taking hydrochlorothiazide. Untreated infection or lack of source control is also a concern with the presence of persistent fever, even while the patient is taking antibiotics (Answer D is correct).

Pulmonary Disorders I

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Pulmonary Disorders I

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Learning Objectives

1. Evaluate the role of pharmacologic management options in acute respiratory distress syndrome.
2. Review clinical practice guidelines pertaining to acute respiratory distress syndrome.
3. Recommend an evidence-based approach for non-pharmacologic therapy in managing critically ill patients with acute respiratory distress syndrome.
4. Evaluate key variables and commonly used modes for treatment with mechanical ventilation.
5. Assess appropriateness of drug therapy for endotracheal intubation, including agents for premedication, induction, and neuromuscular blockade.

Abbreviations in This Chapter

AC/VC	Assist control/volume control
ARDS	Acute respiratory distress syndrome
COVID-19	Coronavirus disease 2019
CVP	Central venous pressure
ECMO	Extracorporeal membrane oxygenation
ICU	Intensive care unit
MAP	Mean arterial pressure
MV	Mechanical ventilation
NMBA	Neuromuscular blocking agent
PEEP	Positive end-expiratory pressure
PS	Pressure support
RSI	Rapid sequence intubation
SIMV	Synchronized intermittent mandatory ventilation

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. Which best describes the category of acute respiratory distress syndrome (ARDS) that most benefits from prone positioning and cisatracurium administration?
   A. Acute lung injury.
   B. Moderate to severe.
   C. Mild to moderate.
   D. Mild.

2. Several clinical trials have been conducted in moderate to severe ARDS. Which of the following partial pressure of arterial oxygen (Pa_{O_2})/ fraction of inspired oxygen (F_{IO_2}) parameters best describes this ARDS subpopulation?
   A. 100–200 mm Hg.
   B. Greater than 150 mm Hg.
   C. Less than 150 mm Hg.
   D. 50–150 mm Hg.

3. Which of the following parameters best describes "low tidal volumes" for invasive mechanical ventilation in ARDS?
   A. 6 mL/kg of total body weight.
   B. 4 mL/kg of ideal body weight.
   C. 7 mL/kg of total body weight.
   D. 5 mL/kg of adjusted body weight.

4. A 65-year-old man presents to the emergency department with severe shortness of breath, tachypnea, altered mental status, and diaphoresis. A chest radiograph reveals diffuse, bilateral opacities. His vital signs are as follows: blood pressure 94/54 mm Hg, respiratory rate 26 breaths/minute, heart rate 120 beats/minute, pain score 2/10, and temperature 105.8°F (41.0°C). The patient's wife states that his symptoms began about 2 days ago and gradually worsened over the past day. The patient is transferred to the ICU, where he is intubated. His arterial blood gas values are pH 7.30, partial pressure of arterial carbon dioxide (Pa_{CO_2}) 50 mm Hg, Pa_{O_2} 50 mm Hg, and oxygen saturation (Sa_{O_2}) 85% while receiving F_{IO_2} 100%. Which is the best therapy plan for the next 24 hours?
   A. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a Richmond Agitation-Sedation Scale (RASS) score of –4; and prone positioning.
   B. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of –5 and cisatracurium administration to limit plateau pressures; and prone positioning.

C. Empiric antibiotic therapy, aggressive diuresis (goal central venous pressure less than 4 mm Hg), and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of −4; and prone positioning.
D. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of −4; and supine positioning.

5. P.D. is a 67-year-old woman (height 160 cm; total body weight 68 kg) admitted to the ICU with ARDS pneumonia. Her mechanical ventilation parameters are as follows: tidal volume 280 mL, respiratory rate 20 breaths/minute, positive end-expiratory pressure (PEEP) 12 cm H_2O, and F_{IO_2} 80% with plateau pressure of 22 mm Hg. Her mean arterial pressure (MAP) is currently 68 mm Hg with norepinephrine discontinued 4 hours ago and CVP ranging between 8 and 14 cm H_2O over the last 2 hours. Also, her total urine output over the past 24 hours is 1150 mL. The ICU team is discussing fluid management using the FACTT Lite protocol. Which of the following options is the most appropriate strategy?
   A. Administer furosemide 40 mg IV once and re-evaluate in 4 hours.
   B. Administer furosemide 1mg/hour continuous infusion and re-evaluate in 8 hours.
   C. Administer 0.9% sodium chloride 500 mL over 1 hour and re-evaluate in 4 hours.
   D. Monitor only and re-evaluate in 8 hours.

6. Which of the following statements most appropriately describes published clinical trials of corticosteroids in ARDS?
   A. Dexamethasone demonstrated improved clinical outcomes among ARDS patients on lung-protective strategies.
   B. Dexamethasone initiated after 7 days of ARDS onset was associated with worse clinical outcomes.
   C. Methylprednisolone demonstrated improved clinical outcomes among ARDS patients on lung-protective strategies.
   D. Methylprednisolone initiated after 7 days of ARDS onset was associated with worse clinical outcomes.

7. Which order of medication administration would be most appropriate for a 34-year-old woman with no significant medical history receiving rapid sequence intubation (RSI)?
   A. Rocuronium, etomidate, midazolam.
   B. Fentanyl, succinylcholine, propofol.
   C. Atropine, rocuronium, etomidate.
   D. Fentanyl, etomidate, succinylcholine.

8. A 78-year-old man (weight 70 kg) presents to the ICU after being intubated for acute respiratory failure. His ventilator settings are as follows: assist control/volume control mode, tidal volume 700 mL (10 mL/kg), respiratory rate 20 breaths/minute, F_{IO_2} 50%, PEEP 5 cm H_2O, and pressure support 10 cm H_2O. The first arterial blood gas values are pH 7.25, Pa_{CO_2} 65 mm Hg, HCO_3 26 mEq/L, Pa_{O_2} 65 mm Hg, and Sa_{O_2} 90%. His MAP is 72 mm Hg without requiring vasoactive support, with a serum creatinine of 1.2 mg/dL and urinary output of 350 mL over the past 6 hours. Which approach would be best for this patient's initial treatment?
   A. High tidal volume mechanical ventilation (MV) and conservative fluid management.
   B. Low tidal volume MV and conservative fluid management.
   C. High tidal volume MV and liberal fluid management.
   D. Low tidal volume MV and liberal fluid management.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
    a. Task A: 1, 2, 4, 6
2. Domain 2: Therapeutics and Patient Management
    a. Task A: 1, 3, 6, 7
    b. Task B: 1, 13
    b. Task C: 1
3. Domain 3: Personal Practice
    a. Task B: 1

I. ACUTE RESPIRATORY DISTRESS SYNDROME

A. Definition and Epidemiology
   1. First described in 1967; several definitions have evolved over the years
   2. Berlin ARDS definition was established in 2012 to address issues with previous definitions (Table 1) (JAMA 2012;307:2526-33).

Table 1. Berlin ARDS Definition

Variable	Criteria
Timing	Onset within 1 wk of a known clinical insult or new or worsening respiratory symptoms
Chest imaging	Bilateral opacities not fully explained by effusions, lobar/lung collapse, or nodules
Origin of edema	Respiratory failure not fully explained by cardiac failure or fluid overload Need objective assessment (ECHO) to exclude hydrostatic edema if no risk factors present
Oxygenation	
Mild	200 mm Hg < Pa_{O_2}/F_{IO_2} ≤ 300 mm Hg with PEEP or CPAP ≥ 5 cm H_2O
Moderate	100 mm Hg < Pa_{O_2}/F_{IO_2} ≤ 200 mm Hg with PEEP ≥ 5 cm H_2O
Severe	Pa_{O_2}/F_{IO_2} ≤ 100 mm Hg with PEEP ≥ 5 cm H_2O

ARDS = acute respiratory distress syndrome; CPAP = continuous positive airway pressure; ECHO = echocardiogram; F_{IO_2} = fraction of inspired oxygen; Pa_{O_2} = partial pressure of oxygen; PEEP = positive end-expiratory pressure..

   3. ARDS represents 10.4% of all ICU admissions and 23.4% of patients requiring mechanical ventilation (MV) (JAMA 2016;315:788-800).
      a. Mild ARDS accounts for 30%, moderate for 46.6%, and severe for 23.4%.
      b. Hospital mortality is 34.9% for mild ARDS, 40.3% for moderate ARDS, and 46.1% for severe ARDS.
      c. Multisystem organ failure is the leading cause of death in patients with ARDS, with the number of extrapulmonary organ failures correlating with an incremental increase in mortality (Intensive Care Med 2011;37:1932-41).

B. Cause and Pathophysiology
   1. Direct and indirect causes of lung injury
      a. Direct: Pneumonia, aspiration, trauma
      b. Indirect: Sepsis, transfusion injury, pancreatitis, burn injury, trauma
   2. Pathogenesis
      a. The hallmark clinical manifestation of ARDS is hypoxemia from alveolar collapse and edema.
      b. The exudative phase is characterized by increased permeability from the alveolar epithelium and capillary endothelial complex damage, leading to diffuse alveolar edema with fluid and cellular debris (e.g., neutrophils, cytokines, platelets). In addition, type II cells responsible for surfactant production are damaged. The peak incidence of this phase typically occurs within the first week after the initial insult.
      c. The fibroproliferative phase marks either recovery or progression of ARDS. Patients may partly or fully recover pulmonary function from drainage of the alveolar fluid, type II cellular repair, and improvement in the integrity of the endothelium-epithelium complex. However, patients whose conditions are progressively worsening may develop significant alveolar, interstitial, and capillary fibrosis. This phase typically manifests later in the course of ARDS (i.e., more than 7 days after the initial insult).

C. Management Strategies
   1. MV
      a. Lung-protective strategies using low tidal volume ventilation are considered the standard of care and cornerstone in managing ARDS.
      b. The landmark multicenter trial by the Acute Respiratory Distress Syndrome Network (ARDSNet) showed a survival benefit over conventional ventilation using higher tidal volumes (12 mL/kg) (N Engl J Med 2000;342:1301-8). Clinical practice guidelines recommend limiting tidal volumes (4–8 mL/kg of predicted body weight) (Am J Respir Crit Care Med 2017;195:1253-63; Intensive Care Med 2023;49:727-759) and inspiratory pressures (plateau pressure less than 30 cm H_2O) (Am Respir Crit Care Med 2017;195:1253-63). Although optimizing higher PEEP values may be beneficial (Am J Respir Crit Care Med 2017;195:1253-63), individualizing PEEP in ARDS remains unclear (Intensive Care Med 2023;49:727-59). Although predicted body weight and ideal body weight are not identical, clinicians often apply ideal body weight for tidal volumes for easy of use and familiarity.
   2. Prone positioning
      a. Prone over supine positioning improves gas exchange and may reduce ventilator-induced lung injury. Compression atelectasis attributed to the weight of the heart, ventral lungs, and abdominal viscera may be exacerbated in the supine position in patients with ARDS. The gravitational effects by placing a patient in the prone position improve ventilation/perfusion matching and end-expiratory lung volume by more homogeneous tidal volume delivery (Chest 2017;151:215-24; Am J Respir Crit Care Med 2017;195:1253-63).
      b. The PROSEVA study group represents the most recently published randomized clinical trial comparing prone and supine positioning in ARDS. In this trial, prone positioning for at least 16 hours/day compared with supine positioning in early ARDS decreased 28-day (adjusted hazard ratio [HR] 0.42; 95% confidence interval [CI], 0.26–0.66) and 90-day (adjusted HR 0.48; 95% CI, 0.32–0.72) mortality (N Engl J Med 2013;368:2159-68). Significantly more ventilator-free days at days 28 and 90 were observed with proning. In addition, 90-day extubation rates were significantly higher in patients in the prone position.
      c. Several meta-analyses have suggested an overall mortality benefit associated with prone positioning over supine in ARDS, though this finding was inconsistent (Intensive Care Med 2014;40:332-41; Crit Care 2014;18:R109; Crit Care Med 2014;42:1252-62; Ann Am Thorac Soc 2017;14(suppl 4):S280-S8; J Thorac Dis 2015;7:356-67; Am J Respir Crit Care Med 2021;203:1366-77). However, most meta-analyses have shown a survival benefit with proning in patients with moderate to severe ARDS (i.e., Pa_{O_2}/F_{IO_2} less than 150 mm Hg) compared with mild hypoxemia (i.e., Pa_{O_2}/F_{IO_2} greater than 200 mm Hg) (Crit Care 2014;18:R109; Crit Care Med 2014;42:1252-62; Ann Am Thorac Soc 2017;14(suppl 4):(S280-S8). In addition, mortality was reduced in patients with ARDS with prone positioning and concurrent lung-protective MV strategies compared with pronation and high tidal volume ventilation (Intensive Care Med 2014;40:332-41; Crit Care Med 2014;42:1252-62; J Thorac Dis 2015;7:356-67; Am J Respir Crit Care Med 2021;203:1366-77). Prone positioning in moderate and severe ARDS has been associated with the greatest reduction in mortality compared with other strategies, including ECMO (Am J Respir Crit Care Med 2021;203:1366-77).
      d. Implementation of prone positioning requires robust planning and resources (J Pharm Pract 2019;32:347-60). Contraindicated criteria of use should be established to mitigate any adverse events (e.g., spinal cord injury, elevated intracranial pressure [ICP]). Health care providers should be trained on appropriate movement of patients between supine and prone positioning to ensure securing of central lines as well as the endotracheal tube. Institutions should also consider the need for manual maneuvering of patients compared with specialized rotating beds. One important patient safety consideration is rapid health care staff access to patients in specialized beds during medical emergencies (e.g., cardiopulmonary arrest, self-extubation).

e. Recently published clinical practice guidelines strongly recommend prone positioning in moderate to severe ARDS (Table 2) (Am J Respir Crit Care Med 2017;195:1253-63; BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69; Intensive Care Med 2023;49:727-59). Despite overwhelming support for this therapy in ARDS, several important questions remain, including the optimal duration and timing of use pertaining to ARDS onset. Patients with ARDS should remain in the prone position for more than 12 hours each day, with consideration for longer daily durations (i.e., at least 16 hours per day). Prone positioning therapy should be considered early in the course of ICU admission in patients with moderate to severe ARDS.

Table 2. Clinical Practice Guideline Recommendations for Prone Positioning in ARDS

Source	Summary	GRADE Recommendation
Am J Respir Crit Care Med 2017;195:1253-63	Suggest prone positioning > 12 hr/day	Strong recommendation; moderate-high level of evidence
BMJ Open Respir Res 2019;6:e000420	• NOT recommended for all levels of ARDS severity • Recommended for at least 12 hr/day in moderate to severe ARDS (i.e., Pa_{O_2}/F_{IO_2} < 150 mm Hg)	Strongly in favor
Ann Intensive Care 2019;9:69	Should be used in moderate to severe ARDS (Pa_{O_2}/F_{IO_2} < 150 mm Hg) for at least 16 consecutive hr	High level of evidence
Intensive Care Med 2023;49:727-59	Recommend using early after intubation in moderate to severe ARDS (Pa_{O_2}/F_{IO_2} < 150 mm Hg for at least 16 consecutive hours)	Strong recommendation; high level of evidence

3. Extracorporeal membrane oxygenation (ECMO)
   a. The CESAR trial showed that treatment at an ECMO referral center improved 6-month survival (Lancet 2009;374:1351-63).
      i. ARDS with a Murray score (i.e., acute lung injury score on the basis of chest radiography, hypoxemia score, peak end-expiratory pressure, and static compliance) of 3 or greater, or hypercapnia and a pH less than 7.20
      ii. Not all patients randomized to the ECMO referral center received ECMO (22 of 90 patients in the experimental group did not receive ECMO).
      iii. Potential for confounding because of experiential bias – Clinicians at an ECMO treatment center may have more experience in treating ARDS.
   b. The Respiratory ECMO Survival Prediction (RESP) score was developed from a registry of patients receiving ECMO (Am J Respir Crit Care Med 2014;189:1374-82).
      i. The RESP score predicts patient survival after ECMO initiation.
      ii. Does not help quantify the decision of whether to initiate ECMO or determine whether ECMO treatment will improve survival in a specific patient
   c. In the EOLIA trial, early use of ECMO compared with conventional management in severe ARDS (e.g., Pa_{O_2}/F_{IO_2} less than 80 mm Hg) was not associated with reduced 60-day mortality rates (relative risk [RR] 0.76; 95% CI, 0.55–1.04, p=0.09) (N Engl J Med 2018;378:1965-75).
      i. This trial was prematurely discontinued because of predefined futility criteria, which resulted in not achieving power.
      ii. The high crossover rate of 28% of controls into the ECMO group may have introduced bias.

d. The 2019 clinical practice guidelines state that ECMO should be considered in patients with severe ARDS or more specifically with Pa_{O_2}/F_{IO_2} ratios less than 80 mm Hg and/or in patients with elevated plateau pressures despite optimizing PEEP, neuromuscular blocking agents (NMBAs), and prone positioning (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69). The 2023 clinical practice guidelines recommend ECMO use in severe ARDS (as defined in EOLIA trial eligibility criteria) at an established ECMO center (strong recommendation; moderate level of evidence in favor) (Intensive Care Med 2023;49:727-59).

e. Concomitant veno-venous ECMO and prone positioning
   i. A meta-analysis (n=13 studies including only 1 randomized clinical trial) evaluated the clinical impact of prone positioning with veno-venous ECMO in ARDS compared with veno-venous ECMO alone (Intensive Care Med 2022;48:270-80).
   ii. Pooled data showed combination veno-venous ECMO and prone positioning 28-day survival was significantly improved (n=10 studies; RR 1.31; 95% CI, 1.21–1.41). Also, combination therapy was also associated with significantly improved ICU and in-hospital survival as well as overall 60- and 90-day survival.
   iii. A priori subgroup analyses found improved 28-day survival with concomitant prone positioning and veno-venous ECMO in both patient cohorts with COVID-19 ARDS patients (n=6 studies; RR 1.32; 95% CI, 1.15–1.50) and ARDS patients without COVID-19 (n=4 studies; RR 1.30; 95% CI, 1.19–1.43).
   iv. Pooled data also showed significant improvements in ventilator-free days at day 28 (n=9 studies; RR –1.29; 95% CI, –2.39 to –0.19). with prone positioning and veno-venous ECMO over ECMO alone in ARDS patients
   v. No major complications were reported with either study group.

4. Fluid management
   a. The FACTT trial compared optimal fluid management strategies – conservative (CVP less than 4 mm Hg) and liberal (CVP 10–14 mm Hg) – in patients with ARDS and hemodynamic stability (not requiring vasopressors or MAP greater than 60 mm Hg) (N Engl J Med 2006;354:2564-75). Diuretics were withheld in patients with shock but were administered according to study protocol once patients had established hemodynamic stability (discontinuation of vasopressors or MAP greater than 60 mm Hg). Although 60-day mortality did not differ (p=0.30), the conservative compared with the liberal strategy was associated with increased ventilator-free days (14.6 ± 0.5 vs. 12.1 ± 0.5, $p<0.001$) and ICU-free days (28 days) (13.4 ± 0.4 vs. 11.2 ± 0.4, $p<0.001$).
   b. Fluid strategies were compared (conservative, liberal, and simplified conservative) in a retrospective comparison among protocols pertaining to ARDS (Crit Care Med 2015;43:288-295). The FACTT Lite protocol provides fluid management recommendations pertaining to the administration of furosemide or fluids as well as monitoring without intervention-based CVP, MAP, urinary output, and pulmonary artery occlusion pressure (optional) (Table 3). No significant differences were found between the FACTT Lite and the FACTT conservative strategies for ventilator-free days (14.9 vs. 14.6, respectively; p=0.61), ICU-free days (14.4 vs. 13.4, respectively; p=0.054), or death at 60 days (22% vs. 25%, respectively; p=0.15). Compared with FACTT liberal, the FACTT Lite approach had improved outcomes (ventilator-free and ICU days and 60-day mortality).
   c. A meta-analysis evaluated the safety and efficacy of conservative versus liberal fluid strategies in ICU patients with either ARDS or sepsis (Intensive Care Med 2017;43:155-70). No differences in mortality were found between study groups in the ARDS subgroup (n=5 studies; RR 0.91; 95% CI, 0.77–1.07) or the mixed ARDS and sepsis subgroup (n=2 studies; RR 0.95; 95% CI, 0.80–1.14).

Table 3. FACTT Lite Simplified Conservative Fluid Management Approach[a]

CVP (Recommended)	Pulmonary Artery Occlusion Pressure (optional)	MAP ≥ 60 mm Hg AND No Vasoactive Support ≥ 12 Hr	
		Urinary Output < 0.5 mL/kg/hr	Urinary Output ≥ 0.5 mL/kg/hr
> 8	> 12	Administer furosemide and reevaluate in 1 hr	Administer furosemide and reevaluate in 4 hr
4–8	8–12	Administer fluid bolus and reevaluate in 1 hr	Administer furosemide and reevaluate in 4 hr
< 4	< 8	Administer fluid bolus and reevaluate in 1 hr	Monitor and reevaluate in 4 hr

[a]Initial recommended furosemide dosing = 20-mg bolus or 3-mg/hr infusion. Recommended 2-fold incremental increase in subsequent furosemide doses up to a maximum of 160-mg bolus/24-mg/hr infusion rate or until goal CVP/pulmonary artery occlusion pressure achieved. See clinical trial publication for further details.
CVP = central venous pressure; MAP = mean arterial pressure.

d. The RADAR-2 trial also evaluated the conservative and usual care fluid strategies in critically ill patients (n=180) (Intensive Care Med 2022;48:190-200). Although this randomized clinical trial was not exclusively conducted in ARDS patients, subgroup analysis in ARDS was planned a priori. In the ARDS subgroup, no differences in mortality, mechanical ventilation duration, or ICU length of stay were observed between treatment arms. However, it must be noted that the sample size of ARDS patients in the intervention (n=19) and usual care (n=11) study groups was very small, which may limit clinical interpretation for this patient population.

5. NMBAs
   a. Early administration (within 48 hours of ARDS onset) of cisatracurium over 48 hours has consistently improved oxygenation without increasing the risk of neuromuscular weakness.
   b. The ACURASYS trial showed decreased 90-day adjusted mortality with cisatracurium compared with placebo (HR 0.68; 95% CI, 0.48–0.98; p=0.04). Subgroup analysis suggested the greatest mortality benefit in patients with ARDS having a Pa_{O_2}/F_{IO_2} less than 120 mm Hg. Cisatracurium was also associated with significantly more ventilator-free days (up to 28 and 90 days) and organ failure–free days (up to 28 days). Cisatracurium did not significantly increase the risk of ICU-acquired neuromuscular weakness (N Engl J Med 2010;363:1107-16; Crit Care Med 2017;45:446-53).
   c. The ROSE trial showed no significant differences in 90-day mortality between cisatracurium (42.5%) and placebo (42.8%) (between-group difference -0.3 percentage points; 95% CI, -6.4 to 5.9; p=0.93). No differences between study groups were found for secondary end points at day 28, including in-hospital mortality, days free of MV, and days not in the ICU or hospital (N Engl J Med 2019;380:1997-2008).
   d. The optimal strategy for NMBA dosing and monitoring in ARDS remains debatable. The ACURASYS and ROSE trials had similar dosing strategies of a cisatracurium 15-mg bolus followed by continuous infusion at 37.5 mg/hour over 48 hours without titration or train-of-four monitoring. Additional boluses were allowed in patients with elevated plateau pressures (i.e., greater than 30 cm H_2O), given concerns for increased risk of ventilator-induced lung injury.
   e. The 2019 clinical guidelines recommend that early and short-course NMBAs be considered in patients with moderate-severe ARDS (Pa_{O_2}/F_{IO_2} less than 150 mm Hg) (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69). However, these guidelines were published before the results of the ROSE trial. However, the 2023 clinical practice guidelines recommend against routine use of NMBAs in ARDS (strong recommendation, moderate level of evidence) (Intensive Care Med 2023;49:727-59).

6. Corticosteroids
    a. Proposed rationale for corticosteroid use in ARDS was to prevent fibroproliferation with subsequent alveolar fibrosis from a proinflammatory response.

Table 4. Clinical Practice Guideline Recommendations for Corticosteroids in ARDS

Source	Summary	GRADE Recommendation
Crit Care Med 2017;45:2078-88	Suggest use in early moderate to severe ARDS (Pa_{O_2}/Fi_{O_2} < 200 mm Hg AND within 14 days of onset)	Conditional recommendation; moderate quality of evidence
BMJ Open Respir Res 2019;6:e000420	Limited quality data, resulting in inability to make a recommendation; rather, states that further research is warranted	Research recommendation
Ann Intensive Care 2019;9:69	Not reported	Not reported

    b. Steroids may improve oxygenation, but randomized trials found no definitive mortality benefit. Although the optimal timing of use remains unknown, consideration may be given in patients with early (less than 7 days) and late (7 days or more), whereas initiation beyond 14 days after onset was associated with higher death rates in the LaSRS trial. (N Engl J Med 2006;354:1671-84; JAMA 1998;280:159-65; Crit Care Med 2017;45:2078-88). However, increased death risk in patients initiated on steroids after 14 days of ARDS onset may be misleading. It should be noted this specific cohort had significant baseline characteristic imbalances as well as atypically low death rates in the control group for 60-day (8%) and 180-day (12%) mortality (N Engl J Med 2006;354:1671-84). Furthermore, no significant differences in mortality were found between steroid and control groups in the greater than 14 days cohort after adjusting for important baseline covariates (Crit Care 2007;11:425).
    c. Published guidelines provide minimal direction for clinical practice decision-making regarding the best approach for corticosteroid use in ARDS (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69; Crit Care Med 2017;45:2078-88). The 2017 guidelines suggest use in selected patients with ARDS, whereas neither of the 2019 guidelines provides any recommendations because of the paucity of quality data (Table 4).
    d. The DEXA-ARDS clinical trial was the first randomized, placebo-controlled clinical trial investigating dexamethasone for patients with early moderate to severe ARDS (Lancet Respir Med 2020;8:267-76). The corticosteroid group consisted of dexamethasone 20 mg intravenously daily over the first 5 days, followed by 10 mg intravenously daily for up to 10 days or upon extubation (if occurring prior to day 10). Dexamethasone was associated with increased mean ventilator-free days over placebo at 1 month (12.3 ± 9.9 and 7.5 ± 9.9 days, respectively; $p<0.0001$). Other secondary outcomes, including mortality (all-cause, ICU, in-hospital) and MV duration (among all ICU survivors at day 60), significantly favored the dexamethasone group. Of note, hyperglycemia, secondary infections, and barotrauma were similar between study groups. The novelty of this trial compared with previously published corticosteroid studies was the concurrent use of contemporary lung-protective MV strategies. Of importance, this clinical trial was released after publication of the clinical practice guidelines mentioned earlier.
7. Corticosteroids in COVID-19
    a. Currently, clinical practice guidelines strongly recommend the use of dexamethasone in ICU patients with COVID-19 (Table 5).

Table 5. Clinical Practice Guideline Recommendations for Corticosteroids in Critically Ill Patients with COVID-19

Source	Summary	GRADE Recommendation
www.covid19treatmentguidelines.nih.gov Last updated October 10, 2023	Recommends dexamethasone for most patients requiring mechanical ventilation or ECMO	Strong recommendation
www.idsociety.org/COVID19guidelines Last updated September 25, 2020	Recommends dexamethasone rather than no dexamethasone	Strong recommendation, moderate certainty of evidence
Crit Care Med 2021;49:e219-234	• Recommends using a short course of systemic corticosteroids over not using corticosteroids • Suggests specifically using dexamethasone over other corticosteroid derivatives	• Strong recommendation, moderate level of evidence • Weak recommendation, very low level of evidence

b. The landmark RECOVERY clinical trial compared dexamethasone 6 mg daily up to 10 days total administered either orally or intravenously compared to usual care alone in 6425 hospitalized patients with COVID-19 (N Engl J Med 2021;384:693-704). Among 1007 patients in the prespecified subgroup analysis requiring invasive MV, the 28-day morality rate was significantly lower in the dexamethasone group (29.3%) than in the usual care group (41.4%) (RR 0.64; 95% CI, 0.51–0.81). In addition, the probability of hospital discharge within 28 days among those receiving invasive MV significantly improved in patients receiving dexamethasone over usual care (RR 1.45; 95% CI, 1.13–1.85). Moreover, MV cessation was more likely with dexamethasone than with usual care (RR 1.47; 95% CI, 1.20–1.78). Of note, use of dexamethasone lowered overall mortality among all hospitalized patients compared with usual care. However, a survival benefit was only observed among patients with COVID-19 requiring oxygen support (both invasive and noninvasive) over those without any respiratory support.

c. The CoDEX clinical trial randomized 299 adult patients with COVID-19 with moderate to severe ARDS to either dexamethasone (20 mg intravenously daily over 5 days followed by 10 mg intravenously daily for another 5 days or until ICU discharge) or usual care (JAMA 2020;6;324:1307-16). The primary end point was ventilator-free days during the initial 28 days consisting of the number of days alive and liberation from MV for 48 hours or more. The mean number of ventilator-free days in the dexamethasone and usual care groups was 6.6 days and 4.0 days (adjusted difference of 2.26 days; 95% CI, 0.2–4.38; p=0.04), respectively. No differences were observed between study groups for secondary aims, including 28-day mortality, ICU-free days, overall MV days, or risk of adverse events. A limitation of this trial is that it was terminated early because of the RECOVERY trial findings publication. This may have introduced bias despite showing a positive benefit associated with dexamethasone.

d. The WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group conducted a meta-analysis of seven randomized clinical trials involving 1703 critically ill patients with COVID-19. It should be noted most patients included in pooled analysis did not meet a strict definition of ARDS but were considered critically ill. This meta-analysis evaluated the impact of systemic corticosteroids, including dexamethasone, methylprednisolone, or hydrocortisone, on 28-day mortality (JAMA 2020;324:1330-41). Overall pooled data among all trials showed that systemic corticosteroids significantly reduced the risk of all-cause mortality (summary OR 0.66; 95% CI, 0.53–0.82; $p<0.001$) using a fixed-effects model. However, no association was observed between corticosteroids and controls using a random-effects meta-analysis approach (OR 0.70; 95% CI, 0.48–1.01). Subgroup analysis showed the survival benefit favored dexamethasone over no steroids according to pooled data from three clinical trials. However, of note, this benefit with dexamethasone was largely driven by the RECOVERY trial consisting of 57% of the overall pooled data. The remaining clinical trials consisting of hydrocortisone (n=3) and methylprednisolone (n=1) showed that these agents did not affect survival according to subgroup analysis. No increased risk of adverse events was found between corticosteroid and control groups.

e. The optimal dexamethasone dosing strategy remains unknown, though most clinicians have adopted the 6-mg/day approach used in the RECOVERY trial. Of note, higher dexamethasone dosing strategies (20 mg/day) have not shown additional beneficial effects compared with lower doses (6 mg daily) (JAMA 2020;324:1330-41).

8. Inhaled pulmonary vasodilators

a. Inhaled nitric oxide and inhaled epoprostenol have improved gas exchange, despite their lack of effect on clinical outcomes (length of stay and mortality). Limited data suggest these agents are equally efficacious. A meta-analysis of randomized clinical trials comparing inhaled nitric oxide and placebo showed no effect on mortality in patients with ARDS (RR 1.10; 95% CI, 0.94–1.29), patients with a baseline Pa_{O_2}/F_{IO_2} of 100 mm Hg or less (RR 1.01; 95% CI, 0.78–1.32), or patients with a baseline Pa_{O_2}/F_{IO_2} greater than 100 mm Hg (RR 0.89; 95% CI, 0.89–1.42) (Crit Care Med 2014;42:404-12).

b. Institutions have used inhaled epoprostenol as a more cost-effective option over inhaled nitric oxide. Direct comparisons between these studies suggest safety and efficacy were similar. However, inhaled epoprostenol has been associated with significant cost savings over inhaled nitric oxide. Major limitations of widespread inhaled nitric oxide use for ARDS include high costs and dedicated equipment for drug delivery. Given that cost is the primary differentiating factor between these two agents, many institutions have transitioned from inhaled nitric oxide to inhaled epoprostenol. Implementation of inhaled epoprostenol delivery systems and processes requires continuous education and pharmacovigilance to mitigate the risk of medication errors and preventable adverse drug events.

c. The 2019 clinical guidelines did not evaluate inhaled epoprostenol; thus, they do not provide any recommendations on its use in ARDS. However, one of the 2019 guidelines states not to use inhaled epoprostenol in ARDS because of weak quality of evidence (BMJ Open Respir Res 2019;6:e000420), whereas the other provides an expert opinion statement for consideration in severe ARDS for patients currently receiving lung-protective ventilation and proning, but before ECMO (Ann Intensive Care 2019;9:69).

d. Despite the paucity of data for inhaled epoprostenol, it is commonly used in the ICU for ARDS as a bridge to more invasive management strategies (i.e., ECMO) and/or as salvage therapy. Although the optimal dosing regimen remains unknown, the most commonly used doses include a weight-based strategy (typically 10–50 ng/kg/minute titrated to effect) (J Pharm Pract 2019;32:347-60). A small prospective study evaluated a dose-response relationship of weight-based inhaled epoprostenol in patients with ARDS, titrating from 0 to 50 ng/kg/minute by increments of 10 ng/kg/minute every 30 minutes (Chest 2000;117:819-27). A significant increase in the absolute median Pa_{O_2}/F_{IO_2} was observed with the 50 ng/kg/minute weight-based dose compared with baseline (202.2 vs. 187.2 mm Hg, respectively; $p<0.008$). However, no significant difference was observed with the median Pa_{O_2}/F_{IO_2} (183.2 mm Hg) associated with the 10 ng/kg/minute regimen compared with baseline. Furthermore, no statistically significant difference was found between the 10 and 50 ng/kg/minute weight-based regimens regarding oxygenation indices. A major caveat of this study was small sample size (n=9), and the patient population was not reflective of contemporary clinical use of inhaled epoprostenol in moderate to severe ARDS, given that only three patients had a baseline Pa_{O_2}/F_{IO_2} less than 150 mm Hg. Recently, one study observed a similar effect on oxygenation using fixed-dose inhaled epoprostenol without titration compared with inhaled nitric oxide (J Intensive Care Med 2021;36:466-76).

e. One study directly compared weight-based and fixed-dose inhaled epoprostenol in ARDS patients (n=294) (Am J Health System Pharm 2023;80:S11-22). The mean Pa_{O_2}/F_{IO_2} increase from baseline to 4 hours after inhaled epoprostenol initiation was significantly higher in the the fixed-dose group compared with the weight-based dosing approach (81.1 ± 106.0 vs. 41.0 ± 72.5 mm Hg, respectively; p=0.0015). Also, responder rates at 4 hours after inhaled epoprostenol initiation was significantly higher with fixed-dosing over weight-based dosing (69.9% vs. 30.1%; p=0.02). Clinical outcomes were comparable between the two dosing groups.

Patient Cases

1. A 56-year-old man is admitted to the ICU with ARDS after experiencing increasing dyspnea during the past 24 hours. His medical history is significant for alcoholism and hypertension. Results of the initial arterial blood gas are as follows: pH 7.24, Pa_{CO_2} 58 mm Hg, HCO_3 24 mEq/L, Pa_{O_2} 50 mm Hg, and Sa_{O_2} 84% while receiving MV AC mode with F_{IO_2} 100%. Chest radiography reveals diffuse bilateral infiltrates. The patient has blood pressure 120/40 mm Hg (MAP 67 mm Hg), heart rate 142 beats/minute, and CVP 8 mm Hg while receiving a norepinephrine (10 mcg/minute) infusion after intravenous fluid resuscitation. Norepinephrine has now been weaned off while maintaining a MAP of 67 mm Hg. Ceftriaxone 1 g intravenously every 24 hours and levofloxacin 750 mg intravenously every 24 hours have been initiated for community-acquired pneumonia. Which is the best therapeutic plan for the patient's ARDS?
   A. Continue fluid resuscitation to maintain a CVP of 10–14 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; prone positioning; and sedative administration to target deep sedation and cisatracurium infusion.
   B. Begin diuresis to target a CVP less than 4 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; supine positioning; and sedative administration to target deep sedation and cisatracurium infusion.
   C. Begin diuresis to target a CVP less than 4 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; supine positioning; and sedative administration to target deep sedation.
   D. Discontinue fluid resuscitation and begin diuresis to target a CVP less than 4 mm Hg while maintaining a MAP greater than 65 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; prone positioning; and sedative administration to target deep sedation and cisatracurium infusion.

Patient Cases *(continued)*

2. A 70-year-old woman (height 63 inches, weight 65 kg) is transferred to your ICU from an outside hospital after outside hospital admission for hypoxic respiratory failure. She had been treated for ARDS at the outside hospital for 3 days before her son requested hospital transfer. On admission, the patient is receiving MV with the following settings: SIMV mode, tidal volume 600 mL (12 mL/kg), respiratory rate 12 breaths/minute, PS 10 cm H_2O, and PEEP 10 cm H_2O. Which is the best therapy for her ARDS?
   A. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; supine positioning.
   B. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; prone positioning; cisatracurium administration.
   C. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; supine positioning; cisatracurium administration.
   D. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; prone positioning.

3. A 63-year-old woman (height 165 cm; total body weight 64 kg) with a medical history of insulin-dependent diabetes and chronic kidney disease was admitted to the general ward 2 days ago for acute COVID-19 pneumonia requiring oxygen therapy via nasal cannula. She was started on dexamethasone 6 mg oral daily. Today, she transferred to the intensive care unit following endotracheal intubation because of worsening oxygenation and meeting severe ARDS criteria. Lung-protective MV is initiated at tidal volume 340 mL, respiratory rate 22 breaths/minute, PEEP 12 cm H_2O, and F_{IO_2} 100%. Her arterial blood gas values are as follows: pH 7.28, P_{aCO_2} 46 mm Hg, P_{aO_2} 60 mm Hg, and HCO_3 24 mEq/L with S_{aO_2} of 92%. Which is most appropriate therapeutic plan regarding her corticosteroids?
   A. Discontinue dexamethasone 6 mg oral daily.
   B. Change dexamethasone 6 mg oral daily to methylprednisolone 40 mg intravenous four times daily.
   C. Change dexamethasone 6 mg oral daily to dexamethasone 6 mg intravenous daily.
   D. Change dexamethasone 6 mg oral daily to dexamethasone 20 mg intravenous daily.

4. K.J. is a 56-year-old man (height 72 inches; total body weight 120 kg) admitted to the ICU for septic shock requiring vasoactive support and ARDS pneumonia. Lung-protective MV is initiated as follows: tidal volume 600 mL, respiratory rate 24 breaths/minute, PEEP 14 cm H_2O, and F_{IO_2} 100% with plateau pressure of 24 mm Hg. His arterial blood gas values are pH 7.28, P_{aCO_2} 46 mm Hg, P_{aO_2} 62 mm Hg, and HCO_3 24 mEq/L. Which of the following treatment strategies may be most appropriate at this time to improve oxygenation?
   A. Inhaled nitric oxide.
   B. ECMO.
   C. Inhaled epoprostenol.
   D. Cistracurium.

5. D.J. is a 49-year-old man with septic shock requiring vasoactive support and ARDS with MV at the following settings: tidal volume 400 mL, respiratory rate 26 breaths/minute, PEEP 20 cm H_2O, and F_{IO_2} 100%, resulting in a plateau pressure of 40 mm Hg. The patient's arterial blood gas values after mechanical ventilation initiation are as follows: pH 7.45, $PaCO_2$ 34 mm Hg, P_{aO_2} 110 mm Hg, and HCO_3 24 mmol/L with S_{aO_2} 88%. Which of the following strategies is most appropriate at this time?
   A. Cisatracurium, furosemide, and prone positioning.
   B. Cistatracurium, furosemide, and ECMO.
   C. Prone positioning, dexamethasone, and cisatracurium.
   D. Prone positioning, dexamethasone, and ECMO.

II. INTUBATION

A. Endotracheal Intubation
   1. Provides access for suctioning of tracheobronchial secretions, maintains a patent airway, and allows administration of medications
   2. Indications include airway protection, facilitation of ventilation and oxygenation, assurance of airway patency, and anesthesia and surgery.
   3. Orotracheal intubation is preferred for elective and emergency cases.
   4. Nasotracheal intubation is beneficial for patients undergoing maxillofacial surgery or dental procedures and for patients with limited mouth opening. May be associated with increased risk of bleeding and sinusitis and should be avoided in patients with severe nasal or midface trauma
   5. Complications include insertion trauma, gastric aspiration, hypoxemia, laryngospasm, esophageal intubation, right main bronchus intubation, cardiac arrhythmias, and hemodynamic impairment

B. Rapid Sequence Intubation (RSI)
   1. Rapid and sequential administration of a rapid-acting induction agent and a NMBA to facilitate intubation and decrease the risk of aspiration
   2. Series of seven distinct steps: preparation, preoxygenation, pretreatment, induction either prior or with paralysis, protection and positioning, placement of the tube in the trachea, and management after intubation

C. Clinical Practice Guideline on Rapid Sequence Intubation
   1. These recently published guidelines were developed to provide evidence-based recommendations on pharmacologic and nonpharmacologic topics related to RSI (Critical Care Medicine 2023;51:1411-30).
   2. Overall, 10 clinically relevant guideline statements involving specific aspects of RSI were finalized; however, a summary of medication-related aspects of RSI are summarized below (Table 6).

Table 6. Clinical Practice Guideline Rapid Sequence Intubation Medication-Related Recommendations

RSI Step	Summary	GRADE Recommendation
Pretreatment	Not reported	Not reported
Induction	Suggest sedative-hypnotic induction agent when neuromuscular blocking agent will be used	Best practice statement (level of evidence not graded)
	No difference suggested between etomidate and other induction agents regarding outcomes (survival, hypotension risk, or need for vasoactive support)	Conditional (moderate level of evidence)
	Suggest against the use of corticosteroids following etomidate administration	Conditional (low level of evidence)
Paralysis	Recommend administering neuromuscular blocking agent following induction	Strong (low level of evidence)
	Suggest administering either rocuronium or succinylcholine (if no contraindications)	Conditional (low level of evidence)

RSI = rapid sequence intubation.

D. Pretreatment
   1. Occurs before an induction agent or NMBA is administered
   2. Pretreatment attenuates the sympathetic and parasympathetic responses (catecholamine release, hypertension, tachycardia, potentially increased ICP in patients with impaired cerebral autoregulation) to laryngoscopy.
   3. Fentanyl or lidocaine can be used as a pretreatment medication (Table 7).
   4. Atropine and defasciculating doses of nondepolarizing NMBAs are not recommended for routine use in RSI for adult patients.

Table 7. Pretreatment Agents

Agent	Dose	Onset	Duration	Advantages	Disadvantages
Fentanyl (Sublimaze)	IV: 1–3 mcg/kg	< 30 s	0.5–1 hr	• Blunts hypertensive response from intubation • Recommended over other opioids because of its rapid onset and short duration of action	• Chest wall rigidity (doses > 100 mcg/kg) • Hypotension, bradycardia, and respiratory depression
Agent	**Dose**	**Onset**	**Duration**	**Advantages**	**Disadvantages**
Lidocaine (Xylocaine)	IV: 1.5 mg/kg	45–90 s	10–20 min	• May prevent increase in ICP through blunting of cough reflex • May reduce bronchospasm in patients with reactive airway disease	• Contraindicated in patients with an amide anesthetic allergy, bradycardia, or severe heart block

ICP = intracranial pressure; IV = intravenous(ly).

E. Induction Agents
   1. Given as rapid intravenous push immediately before the paralyzing agent to help achieve optimal conditions for intubation
   2. Agents should provide rapid loss of consciousness, analgesia, amnesia, and stable hemodynamics.
   3. Agents used for induction during RSI include barbiturates, benzodiazepines (midazolam), etomidate, ketamine, and propofol (Table 8).
   4. Barbiturates
      a. Thiopental is no longer available in the United States.
      b. Methohexital is rarely used because of its adverse effect profile, including respiratory depression, hypotension, and histamine release.
   5. Etomidate
      a. A nonbarbiturate, imidazole derivative with a rapid onset of action and a very short elimination half-life
      b. Enhances the effects of γ-aminobutyric acid, thereby blocking neuroexcitation and inducing unconsciousness (does not provide analgesia)
      c. Transiently inhibits the conversion of cholesterol to cortisol by inhibiting 11ß-hydroxylase, leading to transient adrenal suppression
         i. No convincing or consistent evidence suggests that etomidate is associated with an increased risk of death (Intensive Care Med 2011;37:901-10; Chest 2015;147:335-46).
         ii. Large randomized, prospective, adequately powered studies are needed to clarify the clinical significance of etomidate's effects in patients at risk of adrenal insufficiency.

Table 8. Induction Agents

Agent	Dose	Onset	Duration	Advantages	Disadvantages
Etomidate (Amidate)	IV: 0.2–0.6 mg/kg	10–20 s	4–10 min	• Minimal cardiovascular effects • Decreased ICP with minimal effects on cerebral perfusion	• Myoclonic jerks • Transient decrease in cortisol production
Ketamine (Ketalar)	IV: 1–2 mg/kg IM: 4–10 mg/kg	IV: 10–15 s IM: 3–4 min	IV: 5–15 min IM: 12–25 min	• Catecholamine reuptake inhibition (transient increase in blood pressure and heart rate) • Respiration and airway reflexes maintained • Does not increase ICP • Relieves bronchospasm • Has both amnestic and analgesic effects	• Negative inotropic/chronotropic effects (in catecholamine-depleted patients) • Emergence delirium, nightmares, and hallucinations (premedication with a benzodiazepine does not reduce the incidence)
Midazolam (Versed)	IV or IM: 0.2–0.3 mg/kg	60–90 s	1–4 hr	• Recommended over other benzodiazepines because of its relatively faster onset of action	• Compared with other induction agents, slower onset and longer duration • Dose-dependent respiratory depression and hypotension
Propofol (Diprivan)	IV: 1.5–2.5 mg/kg	15–45 s	3–10 min	• Decreases ICP; however, may also decrease CPP • Mild bronchodilating effects • Drug of choice in pregnancy	• Hypotension and bradycardia • Negative inotropic effects

CPP = cerebral perfusion pressure; IM = intramuscular(ly).

F. NMBAs
   1. Block impulse transmission at the neuromuscular junction, resulting in skeletal muscle paralysis
   2. Used immediately after induction to help achieve optimal conditions for intubation
   3. Have no sedative, analgesic, or amnestic properties
   4. Problematic in patients with a difficult or failed airway
   5. Depolarizing NMBAs (Table 9)
      a. Succinylcholine: Noncompetitively binds to acetylcholine receptors, leading to sustained depolarization of the neuromuscular junction and prevention of muscle contraction
      b. Preferred agent for RSI
   6. Nondepolarizing NMBAs (Table 9)
      a. Rocuronium and vecuronium: Competitive antagonists of acetylcholine at the neuromuscular junction, leading to the prevention of muscle contraction
      b. Intermediate-acting nondepolarizing agents are alternatives when succinylcholine is contraindicated.
      c. Usually have a slower onset of action and longer duration of action
   7. Reversal of nondepolarizing NMBAs with a failed airway
      a. An acetylcholinesterase inhibitor (neostigmine or pyridostigmine) in combination with an anticholinergic (atropine or glycopyrrolate)
      b. Sugammadex (Bridion) binds the aminosteroid class of nondepolarizing NMBAs (vecuronium and rocuronium).

Table 9. Common Neuromuscular Blocking Agents

Agent	Dose	Onset	Duration	Cautions
Succinylcholine (Anectine)	IV: 1–2 mg/kg IM: 3–4 mg/kg (max 150 mg)	IV: 1 min IM: 2–3 min	IV: 3–5 min IM: 10–30 min	• Prolonged effects in pseudocholinesterase deficiency • Hyperkalemia or patients at risk of hyperkalemia (prolonged immobilization, crush injuries, myopathies, burns, muscular dystrophy, stroke, and spinal cord injuries) • Malignant hyperthermia • Bradycardia/hypotension with repeated doses • Mild increase in ICP
Rocuronium (Zemuron)	IV: 0.6–1.2 mg/kg	1–2 min	30–60 min	• Moderate increase in duration with liver dysfunction, minimal increase in duration with renal dysfunction
Vecuronium (Norcuron)	IV: 0.08–0.1 mg/kg	2–3 min	20–60 min	• Prolonged duration in renal and liver dysfunction

G. Management After Intubation
   1. Provide continued sedation/analgesia as needed if an intermediate-acting NMBA was used and assists in adequate oxygenation and ventilation
   2. Minimize long-term use of analgesics and sedatives
   3. Maintain head-of-bed elevation at 30–45 degrees
   4. Mouth and eye care
   5. Bowel regimen
   6. Stress ulcer and deep venous thrombosis prophylaxis

Patient Cases

6. A 55-year-old man (weight 75 kg) is admitted to the burn ICU after a 65% total body surface area burn to the abdomen, back, and lower extremities from a house fire. He is unconscious and unable to protect his airway. His medical history is significant for hypertension and hyperlipidemia. He is currently receiving high-dose norepinephrine and vasopressin to maintain a MAP of 65 mm Hg. His current laboratory test results show the following: sodium (Na) 130 mEq/L, potassium (K) 5.9 mEq/L, chloride (Cl) 122 mEq/L, carbon dioxide (CO_2) 15 mg/dL, blood urea nitrogen (BUN) 10 mg/dL, and serum creatinine (SCr) 1.3 mg/dL. Which group of medications would be most appropriate for RSI?
   E. Fentanyl, propofol, rocuronium.
   F. Fentanyl, ketamine, succinylcholine.
   G. Fentanyl, etomidate, rocuronium.
   H. Fentanyl, propofol, succinylcholine.

7. A 39-year-old homeless man (weight 70 kg) was admitted to the neurosciences ICU with a traumatic head injury after falling off a 3-ft ladder while intoxicated. Imaging reveals a subdural hematoma. The team decides to intubate this patient. His current laboratory values are as follows: Na 133 mEq/L, K 4.5 mEq/L, Cl 97 mEq/L, CO_2 28 mg/dL, BUN 13 mg/dL, SCr 0.7 mg/dL, and glucose 140 mg/dL. Which induction medication would be most appropriate to use for RSI?
   I. Propofol 90 mg intravenous push.
   J. Ketamine 100 mg intravenous push.
   K. Midazolam 15 mg intravenous push.
   L. Etomidate 150 mg intravenous push.

MECHANICAL VENTILATION

A. Critical to Understanding How MV Works: A fundamental knowledge of acid-base disorders and normal respiratory physiology

B. Two Essential Categories of Respiratory Failure: Hypercapnic and hypoxemic. Derangements in Pa_{O_2} or Pa_{CO_2} will help determine the cause of respiratory failure. (Table 10 provides the context for normal oxygenation and ventilation values.)

C. Modes
   1. Assist control (AC) ventilation
      a. Volume control (VC)
         i. The patient receives a predetermined respiratory rate and tidal volume, with additional patient-initiated breaths provided at the preset tidal volume. Patient-initiated respiration generates a negative pressure within the ventilator circuit, which is sensed by the ventilator, and a full tidal volume breath is provided.
         ii. Potential for ventilator dyssynchrony, "double-stacking," and respiratory alkalosis
         iii. Mode used in the ARDSNet trial of tidal volume strategy to limit spontaneous tidal volumes (N Engl J Med 2000;342:1301-8)
      b. Pressure control
         i. The patient will receive a breath at a fixed rate until a predetermined peak pressure limit is reached. The tidal volume is variable and limited by the peak pressure limit.
         ii. Not ideal for patients with low minute ventilation and may lead to hypoventilation and further hypoxia
   2. Synchronized intermittent mandatory ventilation (SIMV): Patients will receive a predetermined respiratory rate and tidal volume plus additional spontaneous, self-generated breaths at whatever tidal volume they can generate. Not ideal for the treatment of ARDS, given patients' ability to exceed the present tidal volume for spontaneous breaths in excess of 6 mL/kg
   3. PS ventilation
      a. Usually used as a weaning mode of MV from a more intensive mode of MV (i.e., AC ventilation)
      b. The patient initiates each breath with assistance from the ventilator in the form of a preset pressure value. The ventilator is set to provide a correct amount of pressure to assist with each inspiratory effort. The tidal volume and respiratory rate depend on the patient.

D. Ventilator Variables
   1. F_{IO_2}
      a. Amount of oxygen that is delivered with each breath, from 21% to 100% at sea level
      b. F_{IO_2} is generally tapered to provide the minimal oxygenation needed to meet patient needs. Concerns for oxygen toxicity with high F_{IO_2} requirements over prolonged period
   2. Tidal volume
      a. Volume of air inspired in a breath (delivered by MV or spontaneously)
      b. Set according to oxygenation and ventilation requirements. Patients with ARDS are treated with a low tidal volume strategy.
   3. Respiratory rate
      a. Set to provide a minimal number of breaths from the ventilator at the set tidal volume
      b. Titrated by minute ventilation, Pa_{CO_2}, and pH. Minute ventilation (liters per minute) = tidal volume (liters) × respiratory rate (breaths/minute).

4. Flow rate
   a. Velocity of air delivered
   b. Velocity is greatest initially upon inspiration and decelerates toward the end of the inspiratory effort.
5. PEEP
   a. Positive pressure in the alveoli during expiration
   b. Pressure at the end of expiration promotes alveoli recruitment while minimizing the risk of collapse in ARDS.
   c. Titrated with F_{IO_2} to provide the minimum necessary support to meet the patient's oxygenation requirements

Table 10. Normal Respiratory Physiology Values

Value	Normal Range
Tidal volume (mL/kg)	5–10
Respiratory rate (breaths/min)	12–20
Minute ventilation (L/min)	5–10
Pa_{CO_2} (mm Hg)	35–45
Pa_{O_2} (mm Hg)	80–100
Sa_{O_2} (%)	95–100

Patient Case

8. A 74-year-old woman (height 63 inches, weight 65 kg) is transferred to your ICU from an outside hospital after admission for hypoxic respiratory failure. She had been treated at the outside hospital for ARDS for 3 days before her son requested transfer. When the patient arrives in your ICU, she is receiving MV with the following settings: SIMV mode, tidal volume 600 mL (12 mL/kg), respiratory rate 12 breaths/minute, PS 10 cm H_2O, and PEEP 10 cm H_2O. Which is the best ventilator plan for treating her ARDS?

   M. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.
   N. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.
   O. PS mode, PS 10 cm H_2O, PEEP 5 cm H_2O.
   P. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 10 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.

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Intubation

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Mechanical Ventilation

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

The patient has early severe ARDS (for less than 48 hours and Pa_{O_2}/F_{IO_2} less than 150 mm Hg) and hemodynamic stability (post-resuscitation MAP greater than 65 mm Hg). According to the findings of an ARDSNet-sponsored multicenter trial, he would qualify for conservative fluid management (CVP less than 4 mm Hg). In addition, vasopressors should be discontinued and diuresis initiated to achieve a target CVP less than 4 mm Hg (Answer A is incorrect). Given the timing and the severity of the patient's ARDS, he should be placed in the prone position (Answers B and C are incorrect). In addition, this patient would qualify for a cisatracurium infusion. Given the timing and severity of his ARDS, he qualifies to receive a lung-protective ventilation strategy (tidal volume 4–6 mL/kg of ideal body weight) and diuresis to a CVP less than 4 mm Hg (hemodynamically stable if weaned off vasopressors) while placed in the prone position and administered a cisatracurium infusion (Answer D is correct).

2. **Answer: A**

The patient presents to your ICU after 3 days of care. Therefore, she does not currently meet the criteria for being administered cisatracurium or placed in the prone position (Answers B–D are incorrect). Currently, the applicable therapy to apply is a lung-protective ventilation strategy (tidal volume 4–6 mL/kg of ideal body weight) (Answer A is correct).

3. **Answer: C**

The patient was appropriately started on corticosteroids for COVID-19 pneumonia during the same hospital admission prior to developing ARDS and subsequently transferred to the ICU. Dexamethasone was associated with favorable results among COVID-19 hospitalized patient requiring oxygen support (both noninvasive and invasive mechanical ventilation) as well as critically ill patients with moderate to severe ARDS; therefore, continuing dexamethasone is appropriate (Answer A is incorrect). Methylprednisolone has not been associated with favorable clinical outcomes among COVID-19 or moderate to severe ARDS patients (Answer B is incorrect). Although the DEXA-ARDS trial found improved clinical outcomes associated with the dexamethasone 20 mg daily (followed by 10 mg daily) regimen in ARDS patients, it is important to recognize this patient population did not consist of COVID-19. Furthermore, a meta-analysis evaluating various corticosteroid regimens failed to find any benefit of higher dexamethasone dosing strategies (i.e., 20 mg daily) compared with lower doses (6 mg daily) (Answer C is correct; Answer D is incorrect).

4. **Answer: B**

Although selective pulmonary vasodilating agents may modestly improve oxygenation parameters at best, they have not shown to improve clinical outcomes. Their role in therapy is uncertain, but it may be mostly considered salvage therapy after failing more proven strategies (Answers A and C are incorrect). The role of neuromuscular blocking agents in ARDS remain controversial. However, the ROSE clinical trial failed to corroborate any improved clinical outcomes previously demonstrated (ACURASYS trial). Cisatracurium may not be appropriate at this time since plateau pressures remain acceptable despite using high PEEP (Answer D is incorrect). ECMO use in ARDS also remains controversial. However, severe ARDS may be a compelling reason to initiate ECMO compared with alternative supportive therapies (Answer B is correct).

5. **Answer: C**

ECMO may not be the most appropriate strategy at this time considering the patient has moderate to severe ARDS with corresponding Pa_{O_2} of 110 mm Hg on lung protective mechanical ventilation alone (Answers B and D are incorrect). Furosemide may not be the most appropriate strategy because the patient is currently requiring vasoactive support (Answers A and B and incorrect). Cisatracurium may be indicated for this patient considering the plateau pressure is elevated as well as prone positioning to help improve oxygenation with suboptimal parameters on mechanical ventilation along; furthermore, corticosteroids may also be indicated given ARDS severity and timing (Answer C is correct).

6. **Answer: C**

Propofol, ketamine, and etomidate can all be used for induction. Propofol may worsen hypotension; therefore, because this patient is already receiving vasoactive medications to maintain his blood pressure,

propofol would not be ideal (Answer A is incorrect). Succinylcholine causes the up-regulation of acetylcholine receptors, predisposing the muscle fibers to release excess potassium because they are depolarized, which leads to significant dysrhythmias or cardiac arrest (Answers B and D are incorrect). Appropriate induction and neuromuscular blockade in this patient would be to administer etomidate (indicated for hemodynamically unstable patients), fentanyl (providing adequate analgesia), and rocuronium (does not increase serum potassium) (Answer C is correct).

7. **Answer: B**

Although propofol may promptly lower ICP, it can also induce hypotension and thus decrease cerebral perfusion pressure (Answer A is incorrect). Midazolam could be used as an induction agent; however, it is not the best agent for RSI because of its delayed onset of action (Answer C is incorrect). Etomidate may be considered in the setting of increased ICP; however, the dose in this case is too high (Answer D is incorrect). Ketamine not only decreases ICP but also prevents fluctuations in ICP (Answer B is correct).

8. **Answer: A**

After recognizing that the patient has ARDS, a lung-protective ventilation strategy should be implemented (tidal volume 4–6 mL/kg). Choosing a PS or SIMV mode would allow the patient to initiate spontaneous breaths in excess of the goal tidal volume (Answers B–D are incorrect). In the ARDSNet study of tidal volume strategy, the AC mode was most commonly used to promote the application of low tidal volumes (Answer A is correct).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: B**

According to the Berlin ARDS definition, the category of acute lung injury was removed in favor of categorizing the severity of ARDS (PaO_2/FIO_2 less than 200 mm Hg) (Answer A is incorrect). Because of the relative difference in mortality rates, mild and moderate ARDS cases are less likely to benefit from therapeutic interventions, given the number needed to treat to show an effective intervention (Answers C and D are incorrect). In the trials evaluating prone positioning and cisatracurium, patients with severe ARDS were the most likely to benefit. Although the criteria used for severe ARDS in these studies differed from those used in the Berlin definition (both studies were initiated before publication of the Berlin ARDS definition), a post hoc analysis shows a survival benefit in favor of the group with the highest mortality rate (Answer B is correct).

2. **Answer: C**

The Belin definition defines ARDS severity with PaO_2/FIO_2 as one element of criteria. Comprehension of subpopulations of ARDS is essential to apply published clinical data among various therapies. Although not universal defined, the most commonly reported inclusion criteria among landmark ARDS clinical trials (ROSE, ACURASYS, DEXA-ARDS, etc.) was moderate to severe ARDS criteria was less than 150 mm Hg (Answer C is correct; Answers A and B are incorrect). Also, these trials did not define a lower limit of ARDS severity (e.g., 50–150 mm Hg) as inclusion criteria (Answer D is incorrect).

3. **Answer: B**

The landmark ARDSNet trial evaluating lung protective mechanical ventilation strategies defined low tidal volumes ranging from 4 to 8 mL/kg using predicted body weight (also known as ideal body weight (Answer B is correct). Total body weight may overestimate ventilation requirements in adult patients with obesity because lung volume is unrelated to increasing total body weight (i.e., lung size remains relatively constant irrespective of adult body mass) (Answers A, C, and D are incorrect).

4. **Answer: B**

This patient has both ARDS and septic shock. In addition, he has likely had ARDS for less than 48 hours; therefore, the time to initiation of several treatments is essential. The patient is actively in shock (as evidenced by his blood pressure), thus making a fluid-conservative strategy (CVP less than 4 mm Hg) impossible (Answer C is incorrect). Because the time to presentation is less than 48 hours and the patient has severe ARDS, he meets the criteria for cisatracurium administration and prone positioning, and a treatment plan should include these two therapies (Answers A and D are incorrect). A therapy plan should include shock resuscitation (fluid-liberal strategy, CVP 10–14 mm Hg), lung-protective ventilation (tidal volume 4–8 mL/kg of ideal body weight), prone positioning, and cisatracurium administration (Answer B is correct).

5. **Answer: D**

The FACTT Lite approach attempts to simplify fluid management for implementation among ARDS patients in clinical practice. The parameters used to determine clinical course of action (remove fluid from diuretic administration, administer fluid boluses, or monitor without administration of either therapy) include MAP, urine output, and CVP (pulmonary arterial occlusion pressure may be optional based on institutional practices for invasive monitoring). Administering fluids in this patient may not be the most appropriate approach because MAP, CVP, and urine output do not suggest hypovolemia (Answer C is incorrect). Although continuous infusion furosemide may be an option within this published protocol, it is important to note both the suggested rate of infusion and time to re-evaluate and inconsistent with the FACTT Lite protocol. Re-evaluation should be relatively short irrespective of the clinical decision-making (i.e., 1 or 4 hours) (Answer B is incorrect). It is important to emphasize conservative management approaches defined in this protocol require the patient achieve a MAP of 60 mm Hg without any vasoactive support for at least 12 hours. The short time period since vasopressor support cessation may be too aggressive to administer without demonstrating clinical stable hemodynamic pressures (Answer A is incorrect). The most appropriate approach would be monitoring until adequate MAP has been achieved without vasoactive support for at least 12 hours (Answer D is correct).

6. **Answer: A**

The DEXA-ARDS clinical trial has been the only contemporary trial evaluating corticosteroids (dexamethasone) among ARDS patients already receiving lung-protective mechanical ventilation strategies (Answer A is correct). Despite the numerous ARDS trials evaluating methylprednisolone, none have concurrently evaluated with lung-protective strategies among all study patients (Answer C is correct). Despite the optimal timing of corticosteroid use in ARDS remains controversial, dexamethasone was started "early" after ARDS onset in the DEXA-ARDS trial (Answer B is incorrect). The LaSRS clinical trial found increased risk of death in the steroid group compared with controls among patients enrolled greater than 14 days after ARDS onset. Unfortunately, this unadjusted subgroup analysis may have biased these findings. Subsequent re-evaluation of these results found no differences in mortality among in this subgroup. Furthermore, "late" (greater than 5–7 days) corticosteroid use in ARDS has not consistently demonstrated worse outcomes (Answer D is incorrect).

7. **Answer: D**

Neuromuscular blocking agents should always be administered after induction agents (Answers A, B, and C are incorrect). In addition, atropine is not routinely recommended (Answer C is incorrect) for RSI in adult patients. Atropine should be kept nearby for patients who are at an increased risk of bradycardia during RSI (use of β-blockers, calcium channel blockers, digoxin, or amiodarone). Induction agents (and pretreatment medications) should be administered before NMBAs (Answer D is correct).

8. **Answer: B**

The patient has moderate to severe ARDS. The initial MV strategy is using low tidal volumes (Answers A and C are incorrect). Considering the patient is hemodynamically stable without requiring vasoactive support, conservative fluid management approaches are warranted (Answer B is correct). Hemodynamically unstable patients may be candidates for liberal fluid management strategies (Answer D is incorrect).

Pulmonary Disorders II

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Pulmonary Disorders II

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Learning Objectives

1. Design a treatment plan for a cystic fibrosis (CF) exacerbation.
2. Compare treatment strategies for pulmonary hypertension (PH) using the clinical classifications of PH.
3. Describe classifications and risk factors for increased severity of asthma exacerbation and treatment plans for patients with acute respiratory failure caused by asthma exacerbation.
4. Recognize evidence-based treatment options for acute exacerbations of chronic obstructive pulmonary disease (COPD).

Abbreviations in This Chapter

ABG	Arterial blood gas
AC/VC	Assist control/volume control
BNP	B-type natriuretic peptide
CF	Cystic fibrosis
CFTR	Cystic fibrosis transmembrane conductance regulator
COPD	Chronic obstructive pulmonary disease
CRP	C-reactive protein
CTEPH	Chronic thromboembolic pulmonary hypertension
CVP	Central venous pressure
ECOPD	Exacerbation of chronic obstructive pulmonary disease
ED	Emergency department
ET	Endothelin
ERA	Endothelin receptor antagonist
FEV_1	Forced expiratory volume in the first second of expiration
FIO_2	Fraction of inspired oxygen
ICU	Intensive care unit
MDI	Metered dose inhaler
mPAP	Mean pulmonary artery pressure
MV	Mechanical ventilation
$PaCO_2$	Partial pressure of carbon dioxide
PaO_2	Partial pressure of oxygen
PAH	Pulmonary artery hypertension
PCWP	Pulmonary capillary wedge pressure
PDE5i	Phosphodiesterase type 5 inhibitor
PEEP	Positive end-expiratory pressure
PH	Pulmonary hypertension
PVR	Pulmonary vascular resistance
RV	Right ventricular
SABA	Short-acting β-agonist
SaO_2	Oxygen saturation
SpO_2	Functional oxygen saturation
6MWD	6-minute walk distance

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. A 21-year-old woman (height 62 inches, weight 50 kg) with a medical history significant for cystic fibrosis (CF) is admitted to the intensive care unit (ICU) with acute respiratory failure requiring mechanical ventilation (MV). After intubation, her arterial blood gas (ABG) results are as follows: pH 7.27, $PaCO_2$ 45 mm Hg, bicarbonate 22 mEq/L, PaO_2 55 mm Hg, and oxygen saturation (SaO_2) 88%. Her ventilator settings are as follows: assist control/volume control (AC/VC) mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, fraction of inspired oxygen 60%, pressure support 5 cm H_2O, and positive end-expiratory pressure (PEEP) 5 cm H_2O. Her blood pressure is 110/70 mm Hg and heart rate is 95 beats/minute. Which is the best holistic therapy plan?
   A. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 150 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a central venous pressure (CVP) goal of 10–14 mm Hg; dornase alfa and hypertonic saline 7% nebulization.
   B. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 500 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a CVP goal of 10–14 mm Hg; dornase alfa and hypertonic saline 7% nebulization.
   C. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 150 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a CVP goal of 10–14 mm Hg.
   D. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 500 mg intravenously every 24 hours; diuresis to maintain a CVP goal of less than 4 mm Hg while mean arterial pressure is greater than 65 mm Hg; dornase alfa and hypertonic saline 7% nebulization.

2. Which adjunctive measure for a CF exacerbation has the highest quality of evidence for improving patient outcomes?
   A. Normal saline (0.9%) nebulization.
   B. Corticosteroid administration.
   C. Concurrent administration of intravenous and inhaled antibiotics.
   D. Aggressive chest physical therapy.

3. A 55-year-old woman with pulmonary artery hypertension (PAH) is admitted to the ICU for severe respiratory failure. She reports increased work of breathing for the past 5 days and full adherence to her PAH medication regimen, which includes macitentan 10 mg daily and sildenafil 40 mg three times daily. Her current vital signs are as follows: blood pressure 76/60 mm Hg, heart rate 140 beats/minute, respiratory rate 30 breaths/minute, and 85% Sao_2 on 6 L of nasal cannula. Right heart catheterization reveals the following: mean pulmonary artery pressure (mPAP) 50 mm Hg, right atrial pressure 25 mm Hg, cardiac index 1.9 L/minute/m^2, and pulmonary capillary wedge pressure (PCWP) 16 mm Hg. A transthoracic echocardiogram reveals an ejection fraction of 60% with severe right ventricular (RV) dilatation. Which regimen would be most appropriate for this patient?
   A. Dopamine infusion.
   B. Epoprostenol infusion.
   C. Phenylephrine infusion.
   D. Furosemide intravenous push.

4. Which best describes one of the primary treatment goals for the management of PAH?
   A. Achieve and maintain WHO functional class (FC) III or IV.
   B. Preserve 6-minute walk distance (6MWD) to greater than 300 m.
   C. Preserve RV size and function (right atrial pressure less than 8 mm Hg).
   D. Normalize B-type natriuretic peptide (BNP) (less than 200 ng/L).

5. A 42-year-old man presents to the emergency department (ED) with anxiety and shortness of breath. Auscultation reveals audible wheezing. He has difficulty speaking in full sentences. He has used his albuterol metered dose inhaler (MDI) at home for the past several hours without symptom resolution. His forced expiratory volume in the first second of expiration (FEV_1) is 35% of predicted. Which best classifies this patient's asthma exacerbation?
   A. Mild.
   B. Moderate.
   C. Severe.
   D. Life threatening.

6. A 55-year-old man presents to the ED with a 2-week history of progressive shortness of breath, wheezing, and dyspnea at rest. He is confused and speaking in short phrases but can indicate that his inhaled bronchodilator has failed to improve his symptoms for the past 2 days. He is afebrile with heart rate 116 beats/minute, respiratory rate 32 breaths/minute, and Sao_2 86% on room air. He is placed on supplemental oxygen and administered albuterol nebulization and methylprednisolone 125 mg intravenously × 1. One hour later, his oxygenation has not improved. Which adjunctive therapy would be best to add to his current regimen?
   A. Antimicrobial agents directed at community-acquired pathogens.
   B. Magnesium sulfate 2 g intravenously over 20 minutes.
   C. Theophylline 4.6 mg/kg intravenous loading dose; then obtain serum concentration in 30 minutes to guide further dosing.
   D. Acetylcysteine 20% 3 mL, nebulized every 6 hours.

7. A 66-year-old man presents to the ICU with acute respiratory failure from a chronic obstructive pulmonary disease exacerbation (ECOPD). He has had no exacerbations in the past 2 years. His home medications include albuterol hydrofluoroalkane 2 puffs four times daily as needed and tiotropium 1 puff once daily. The patient denies any recent sick contacts or changes in sputum production. He has no known drug allergies. He is placed on 3 L of nasal cannula (Sao_2 97%) and inhaled albuterol and ipratropium by nebulization. Which other therapy would be most appropriate for this patient?
   A. Azithromycin 500 mg intravenously daily.
   B. Prednisone 40 mg orally daily.
   C. Levofloxacin 500 mg orally daily.
   D. Methylprednisolone 125 mg intravenously every 6 hours.

8. A 65-year-old Black man (height 177.8 cm, weight 90 kg, SCr 1.5 mg/dL) presents with a medical history of end-stage renal disease on dialysis, COPD, hypertension, and hyperlipidemia. He is admitted from home to the ICU with respiratory failure. His symptoms before admission included shortness of breath, cough, increased sputum production, purulence, and fever. His respiratory viral panel is positive for rhinovirus. His Sao_2 is 85% on room air; he is placed on high-flow nasal cannula (HFNC) 40 L/minute, fraction of inspired oxygen (Fio_2) 60%. The patient's work of breathing decreases, and after 15 minutes, his Sao_2 is 97%. He is continued on the same HFNC settings. He has been administered methylprednisolone 60 mg intravenously once and azithromycin 500 mg intravenously once. Which best depicts the current intervention for this patient that has demonstrated an increase in mortality?
   A. Corticosteroid dose higher than prednisone 40 mg oral equivalent.
   B. Titration of oxygen to peripheral Sao_2.
   C. Azithromycin dose greater than 250 mg.
   D. Maintaining Sao_2 greater than 92%.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1-4
   b. Task B: 2, 3
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1–3, 6, 8
   b. Task B: 1, 2, 12–14
   c. Task C: 1, 4
3. Domain 3: Professional Practice
   a. Task A: 1
   b. Task B: 1

I. CYSTIC FIBROSIS

A. Cystic Fibrosis (CF) – Chronic disease process that affects many organs, including the pancreas, liver, and intestine, but primarily the lung (Chest 2004;125:1-39)
   1. CF is a recessive disorder caused by a mutation in the cystic fibrosis transmembrane conductance regulator (CFTR). CFTR controls the movement of chloride and bicarbonate across the cell membranes of many organs throughout the body (e.g., lungs, gastrointestinal tract, pancreas) (Chest 2018;154:383-93).
   2. Dysfunction of the CFTR within the airways also affects the ability of the airway membranes to absorb sodium and fluid through the epithelial sodium channel (ENaC), which creates a viscous mucus (Chest 2018;154:383-93). The pH of this mucus is more acidic, which impairs ciliary clearance and the function of phagocytic cells (Int J Mol Sci 2022;23:3513).
   3. Chronic therapy has been developed to target specific CFTR mutations, and such therapies may be seen in the ICU as home medications. Several of these agents have important drug-drug interactions as either substrates or inducers of CYP3A4 (Chest 2018;154:383-93).
   4. Social Determinants of Health in CF (Expert Rev Respir Med 2016;10:967-77)
      a. The term *socioeconomic status*, defined as access to education, material goods and services, social support, and position in society, contributes to differences in outcomes among patients with CF through varying exposure to allergens, quality of nutrition, chronic stress levels, and ability to self-manage disease, as well as access to medical care.
      b. Patients of a lower socioeconomic status have higher morbidity and mortality rates. Their disease progression related to lung function and nutrition are adversely affected as well.

B. Pulmonary Exacerbations of CF
   1. Definitions and significance
      a. There is no universally accepted definition for pulmonary exacerbations of CF (Respir Care 2020;65:233-51).
      b. Acute exacerbations include symptoms of increased cough, sputum production, shortness of breath, weight loss, and a decline in lung function.
      c. Pulmonary exacerbations of CF can have a significant impact on quality of life and disease progression. About 25% of patients with an exacerbation will not completely recover lung function at three months (Am J Respir Crit Care Med 2010;182:627-32).
   2. Common bacterial organisms (most common to least common)
      a. *Staphylococcus aureus* is the most common organism isolated from birth to teenage years.
      b. Methicillin resistant *Staphyloccocus aureus* (MRSA) becomes more common in the 11–30 years of age range.
      c. *Pseudomonas aeruginosa* has a prevalence rate of around 15% from birth to teenage years, but rapidly increases in prevalence after 17 years of age. Greater than 60% of adults with CF in the United States are infected with *P. aeruginosa*.
      d. *Haemophilus influenzae*
      e. *Stenotrophomonas maltophilia*
      f. *Achromobacter xylosoxidans*
      g. *Burkholderia cepacia complex*
      h. *Nontuberculous mycobacteria*
   3. Antibiotic therapy for pulmonary exacerbations of CF
      a. Antibiotic dosing is challenging because of the altered pharmacokinetics in patients with CF. Patients with CF can be distinguished by their large volume of distribution and augmented renal clearance.

b. Notably, most pulmonary exacerbations are not due to the acquisition of a new microorganism, and patients with CF commonly have routine cultures collected throughout their care. Using the patient's previous susceptibility pattern or restarting a previous successful antibiotic regimen is common practice.
   i. For patients with history of or suspected infection with *Pseudomonas aeruginosa*, initial treatment with two antipseudomonal agents is recommended, despite limited evidence that this practice improves outcomes (Am J Respir Crit Care Med 2009;180:802-8; Cochrane Database Syst Rev 2021;6:CD002007).
   ii. Two drugs from different classes are recommended rather than double β-lactams, with most data supporting a β-lactam in combination with an aminoglycoside, although toxicity risk and patient-specific factors should be taken into account.
   iii. Aggressive dosing of β-lactam antibiotics is recommended to optimize time above the minimum inhibitory concentration, primarily based on pharmacokinetic data. Current evidence is insufficient to recommend continuous infusion β-lactams for all patients with CF, but this practice should be considered on a case-by-case basis (Pharmacotherapy 2023;43:740-77; Respir Care 2009;54:522-37).
   iv. Extended-interval aminoglycoside dosing (e.g., once-daily dosing of tobramycin 10 mg/kg, targeting a peak concentration of 20–30 mg/L and a trough concentration of less than 1 mg/L) is as effective as conventional dosing, with possibly less nephrotoxicity (Lancet 2005;365:573-8; Cochrane Database Syst Rev 2017;3:CD002009).

c. Evidence is currently lacking for or against the simultaneous administration of inhaled and intravenous antibiotics for acute CF exacerbations (Table 1) (Am J Respir Crit Care Med 2009;180:802-8; Cochrane Database Syst Rev 2022;8:CD008319.

Table 1. Chronic Antimicrobial Therapy

Therapy	Clinical benefit(s)
Inhaled tobramycin[a]	• Improves lung function • Improves quality of life • Reduces exacerbations
Inhaled aztreonam[a]	• Improves lung function • Improves quality of life
Oral azithromycin[a]	• Improves lung function • Reduces exacerbations
Oral azithromycin[b]	• Reduces exacerbations

[a]Patients with history of *Pseudomonas aeruginosa*.

[b]Patients without history of *P. aeruginosa*.

Information from: Ann Am Thorac Soc 2014;11:1640-50

4. Duration of therapy
   a. A 10- to 14-day course of antibiotics is common practice, and durations are primarily driven by clinical response, either symptomatology or FEV_1 measurements.
   b. The Standardized Treatment of Pulmonary Exacerbation (STOP) trial reflected this precedent with a mean duration of intravenous antibiotics in adults of 16.2 days (J Cyst Fibros 2017;16:600-6).
   c. The STOP2 trial excluded ICU patients, but identified that for patients who improved within 7–10 days, 10 days of antibiotics was noninferior to 14 days. Patients without improvement should complete a 14-day course, but treating past 14 days was not associated with improved outcomes (Am J Respir Crit Care Med 2021;204:1295-305).

5. Exacerbation treatment outcomes
    a. Providers and practice will vary in defining when treatment is complete for pulmonary exacerbation. The STOP study identified the most common markers of success for providers including improvement in FEV_1 and patient-reported symptoms (J Cyst Fibros 2017;16:600-6).
    b. Though intravenous antibiotic therapy is known to improve lung function during an exacerbation, eradication of organisms in culture is neither required nor expected (Expert Opin Drug Metab Toxicol 2021;17:53-68).

C. Adjunctive therapy for prevention or treatment of pulmonary exacerbations of CF
1. Airway clearance techniques are a cornerstone of management of lung disease in CF.
    a. The type of therapy will be tailored to a patient's specific needs and preferences and the use of these interventions will likely increase during treatment of a pulmonary exacerbation (Respir Care 2009;54:522-37).
    b. High frequency chest wall oscillation, which involves an inflatable vest attached to a machine, is the most commonly used airway clearance technique (after infancy).
    c. Positive expiratory pressure/oscillating positive expiratory pressure involves breathing through a mask or mouthpiece to provide back pressure to the airways during expiration as gas also builds up behind the mucus by collateral ventilation (Cochrane Database Syst Rev 2019;11:CD003147).
    d. Despite a low level of evidence, physical exercise is recommended as a possible airway clearance therapy and is frequently used as either primary or secondary clearance for patients who are teenagers or older.
2. Nutrition
    a. Providing adequate nutrition according to patient-specific factors during acute exacerbations is key to maintaining metabolic function and promoting optimal outcomes.
    b. Administer pancreatic enzymes to treat exocrine dysfunction.
3. Continuation of patient's current chronic airway clearance therapy
    a. It is recommended to continue nebulized airway clearance therapies during an exacerbation (Table 2); there is limited data to describe the impact of this practice.

Table 2. Chronic Airway Clearance Therapy

Therapy	Mechanism	Clinical benefit(s)
Nebulized dornase alfa[a]	• Enzymatic cleavage of neutrophilic DNA • Reduces mucus viscosity	• Improves lung function • Improves quality of life • Reduces exacerbations
Nebulized hypertonic saline[a]	Osmotically active expectorant	
Nebulized mannitol[b]	Osmotically active expectorant	• Improves lung function

[a]Mogayzel PJ Jr, Naureckas ET, Robinson KA, et al. Cystic fibrosis guidelines: chronic medications for maintenance of lung health. Am J Respir Crit Care Med 2013;187:680-9.

[b]Flume PA, Amelina E, Daines CL, et al. Efficacy and safety of inhaled dry-powder mannitol in adults with cystic fibrosis: an international, randomized controlled study. J Cyst Fibros 2021;20:1003-9.

    b. The only airway clearance therapy that has been studied specifically during pulmonary exacerbations is hypertonic saline. This study of 132 patients, randomized to either 7% hypertonic saline or control, failed to detect a difference in the primary outcome of hospital length of stay, but did find patients in the treatment group were more likely to return to baseline FEV_1 and have significantly improved symptoms at discharge (Thorax 2016;71:141-7).
4. Data remains unclear on the role of continuing patient's chronic antimicrobial regimens during an exacerbation and practice will vary.

5. Corticosteroids
    a. Currently, there is insufficient evidence for administering corticosteroids (Am J Respir Crit Care Med 2009;180:802-8).
    b. However, current practice appears at odds with the 2009 guidelines. Though data from several small studies have been mixed, the STOP trial observed 21% of patients were treated with corticosteroids during a pulmonary exacerbation (J Cyst Fibros 2017;16:600-6).
    c. Attempts to clarify the role of corticosteroids are ongoing. The Prednisone in Cystic Fibrosis Pulmonary Exacerbations (PIPE) trial (currently recruiting) plans to compare 7 days of oral prednisone with placebo in patients with pulmonary exacerbation on intravenous antibiotics and who have not recovered their FEV_1 after 7 days of antibiotics. Study is still recruiting; currently unknown timeline for completion (NCT03070522).

Patient Case

Questions 1 and 2 pertain to the following case.
A 20-year-old woman is admitted to the ICU for an acute CF exacerbation. Before admission, her condition had been stable. She had maintained her ideal body weight and had just completed a home regimen of suppressive antibiotics. The patient requires MV for the management of hypoxic respiratory failure. She is initiated on AC/VC mode with a lung-protective strategy (4–8 mL/kg tidal volume).

1. Which of the following would be best to manage her acute CF exacerbation?
    A. Tobramycin nebulization; hypertonic saline 7% nebulization; and tube feedings to target a hypocaloric goal during her acute illness (15 kcal/kg/day).
    B. Ceftriaxone 2 g intravenously every 24 hours; tobramycin nebulization; normal saline 0.9% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).
    C. Piperacillin/tazobactam 4.5 g intravenously every 6 hours; tobramycin 10 mg/kg intravenously once daily; hypertonic saline 7% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).
    D. Cefepime 2 g intravenously every 8 hours extended infusion; tobramycin 10 mg/kg intravenously every 8 hours; hypertonic saline 7% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).

2. After treatment with intravenous antibiotics, aggressive chest physical therapy, and her maintenance CF medications for 5 days, this patient improved and was extubated. She was moved out of the ICU on day 6 of her hospitalization. Which duration of antibiotics would be best to recommend for this patient?
    A. 7 days.
    B. 10 days.
    C. 14 days.
    D. Until cultures are negative.

II. PULMONARY HYPERTENSION

A. Pathophysiology and Definitions
   1. Pulmonary hypertension (PH) is a chronic, life-threatening disease characterized by vascular changes, including vasoconstriction, cellular proliferation, and thrombosis.
      a. Thromboxane A2 (potent vasoconstrictor) concentrations are increased.
      b. Prostacyclin (potent vasodilator, inhibitor of platelet aggregation, and antiproliferative properties) concentrations are decreased.
      c. Endothelin-1 (ET-1; potent vasoconstrictor, mitogenic properties on pulmonary artery smooth muscle cells) concentrations are increased.
      d. Nitric oxide concentrations (vasodilator, inhibitor of platelet activation, and inhibitor of vascular smooth muscle cell proliferation) are decreased.
   2. Clinical Classification of PH by the World Health Organization (WHO) (Chest 2019;155:565-86; Eur Heart J 2022;43:3618-731:
      a. Group 1 (pulmonary artery hypertension [PAH]): Idiopathic, heritable, or drug/toxin-induced PAH, PAH associated with various diseases, PAH with features of venous/capillary (PVOD/PCH) involvement, or persistent newborn PH
      b. Group 2: PH caused by left heart disease
      c. Group 3: PH caused by chronic lung disease and/or hypoxia
      d. Group 4: PH caused by chronic thromboembolic pulmonary hypertension (CTEPH) or other pulmonary artery obstructions
      e. Group 5: PH with unclear multifactorial mechanisms
   3. PH is defined as a mean pulmonary artery pressure (mPAP) greater than 20 mmHg at rest as measured by right heart catheterization (Eur Respir J 2019;53:1801913; Chest 2019;155:565-86).
      a. Hemodynamic definitions of PH (Eur Heart J 2022;43:3618-731)
         i. Groups 1, 3, 4, and 5: Precapillary PH defined as mPAP greater than 20 mmHg, pulmonary capillary wedge pressure (PCWP) less than or equal to 15 mm Hg, and pulmonary vascular resistance (PVR) greater than 2 Wood units
         ii. Groups 2 and 5: Isolated postcapillary PH defined as mPAP greater than 20 mmHg, PCWP greater than 15 mm Hg, and PVR less than or equal to 2 Wood units.
         iii. Groups 2 and 5: Combined pre- and postcapillary PH defined as mPAP greater than 20 mmHg, pulmonary capillary wedge pressure greater than 15 mm Hg, and PVR greater than 2 Wood units.

B. Diagnosis (Eur Heart J 2022;43:3618-731)
   1. Initial symptoms are nonspecific
      a. Common symptoms: Fatigue, shortness of breath with exertion, chest pain, and syncope
      b. Less common presentations: Dry cough, nausea, and vomiting
      c. Once PH diagnosis is confirmed, patient's WHO Functional Class will be assessed using these symptoms (Table 3).
   2. Due to the nonspecific pattern of symptoms, multiple tests will likely be needed to rule out other causes of such symptoms. This list is not complete, and additional tests may be performed based on patient-specific factors.
      a. Electrocardiogram
      b. Chest radiograph
      c. Pulmonary function tests
      d. Arterial blood gas
      e. Echocardiogram is used to predict likelihood of PH based on specific findings

i. Peak tricuspid regurgitation velocity (pTRV) is a key factor in suggesting the probability of PH, but it is only a suggestive factor and should not be used independently to predict PH. Guidelines differ in their recommendation for threshold of pTRV to suggest PH, with newer guidelines using > 2.8 m/second.
ii. Various measurements in the ventricles, pulmonary artery, inferior vena cava, and right atrium contribute to the determination of PH diagnosis.
iii. Echocardiogram should not be used independently for decision making about treatment for PH.

f. Ventilation/perfusion scan can be used to identify possible group 4 PH
g. Computed tomography (CT) or CT pulmonary angiography
h. Cardiac magnetic resonance imaging can provide noninvasive prognostic assessment of the right heart initially and at follow up.
i. Right heart catheterization is required to confirm diagnosis of PH
j. Perform vasoreactivity testing if appropriate

C. General management of PAH (group 1 PH)
1. Determine WHO Functional Class (FC): Defines patient's baseline functional status and is monitored throughout the disease to guide medication therapy. It is one of the primary predictors of survival throughout the patient's course of disease. (Table 3)

Table 3. WHO Functional Class Assessment[a]

Class	Definition
I	No symptoms (dyspnea, fatigue, syncope, chest pain) with normal activities
II	Symptoms with strenuous normal daily activities that slightly limit functional status and activity level; no symptoms at rest
III	Symptoms of dyspnea, fatigue, syncope, and chest pain with normal daily activities that severely limit functional status and activity level; no symptoms at rest
IV	Symptoms at rest; cannot perform normal daily activities without symptoms

[a]Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

2. Treatment goals for PH
a. Achieve and maintain WHO FC I or II.
b. Preserve 6-minute walk distance (6MWD) to greater than 440 m.
c. Preserve right ventricular (RV) size and function (right atrial pressure less than 8 mm Hg and cardiac index greater than or equal to 2.5 L/minute/m^2).
d. Normalize B-type natriuretic peptide (BNP) to less than 50 ng/L.
e. Maintain peak oxygen consumption greater than 15 mL/minute/kg and ventilatory equivalent for carbon dioxide less than 36 L/minute during cardiopulmonary exercise testing.
3. Supportive therapy
a. Physical activity and rehabilitation under the direction of a PH provider for patients in stable condition may improve quality of life and 6MWD
b. Oxygen: Maintain Sao_2 of 92% or greater and Pao_2 of 60 mm Hg or greater.
c. Diuretics should be used for the symptomatic management of RV dysfunction and signs of fluid overload; choice of diuretic is variable.
d. Digoxin increases cardiac output; consider in patients who develop atrial tachyarrhythmias although data specifically in PAH are not available.
e. Anticoagulation is not generally recommended for patients with PAH. Consider on an individualized basis if appropriate.

f. Aggressive treatment of sleep apnea and hypertension
g. Angiotensin-converting enzyme inhibitors, angiotensin-receptor blockers, angiotensin-receptor/neprilysin inhibitors, sodium-glucose cotransporter-2 inhibitors, β-blockers, or ivabradine are not recommended in patients with PAH unless indicated for a different comorbidity.
h. Treatment of iron deficiency anemia is indicated, defined by serum ferritin less than 100 mcg/L, or serum ferritin 100–299 mcg/L, and transferrin saturation less than 20%.
i. Patients with PAH should be vaccinated against influenza, *Streptococcus pneumoniae* and SARS-CoV-2 at minimum.

4. Special populations
   a. Pregnancy
      i. Despite advances in treatment and management, for patients with PAH who become pregnant, birthing parent and neonatal mortality rates are high.
      ii. People of childbearing potential should be provided with clear contraceptive recommendations. Some oral contraceptives have significant drug–drug interactions with PAH therapy, which should be considered and discussed with the patient.
   b. Surgery
      i. Surgical procedures pose a risk for patients with PAH. Risk increases with severity of PAH.
      ii. Decisions to perform surgery should be made by a multidisciplinary team familiar with the risks and benefits of both the procedure and PAH.

D. Specific treatment options for group 1 PAH
1. Vasodilator therapy with calcium channel blockers (CCBs) (diltiazem, amlodipine, nifedipine)
   a. Patients who should undergo vasoreactivity testing include those with idiopathic PAH, heritable PAH, and PAH secondary to toxins or drugs.
   b. Considered first-line treatment for patients who have a positive response to acute vasoreactivity testing (reduced mPAP by 10 mm Hg or more to an mPAP of 40 mm Hg or less with increased or unchanged cardiac output) (N Engl J Med 1992;327:76-81)
   c. CCBs improve mPAP, PVR, WHO FC, and survival in patients with PAH and a positive vasoreactivity test (Circulation 1987;76:135-41; N Engl J Med 1992;327:76-81).
   d. Patients are not considered candidates for CCB therapy if they have RV dysfunction, depressed cardiac output, or WHO FC IV symptoms.
   e. *Marked hemodynamic improvement* in patients on CCB therapy is defined as mPAP greater than 30 mm Hg and a PVR less than 4 Wood units. If patients do not reach this threshold of response, CCB therapy can be continued or stopped according to patient-specific factors; some patients may experience deterioration when CCBs are withdrawn.
2. Treatment-naïve patients with PAH (nonresponders to vasoreactivity testing)
   a. Drug therapy should be evaluated on the basis of the patient-specific factors below (Chest 2019;155:565-86; Eur Heart J 2022;43:3618-731):
      i. Mortality risk assessment using the 3-strata model (low, intermediate, high), as shown in Table 4
      ii. Presence of cardiopulmonary comorbidities that are associated with increased risk of LV diastolic dysfunction (obesity, hypertension, diabetes, coronary heart disease)
      iii. Tolerance of combination therapy
      iv. Ability and willingness to manage parenteral prostacyclins
      v. Access to resources
   b. Initial therapy in patients without cardiopulmonary comorbidities
      i. Low- or intermediate-risk strata (Table 4): combination therapy with an endothelin receptor antagonist (ERA) and a phosphodiesterase type 5 inhibitor (PDE5i)
      ii. High-risk strata: combination therapy with ERA, PDE5i, and parenteral prostacyclin

Table 4. Three-Strata Risk Assessment for Patients with Pulmonary Artery Hypertension

Multivariable Assessment of Mortality Risk to Guide Initial Therapy			
Assessment	**3-Risk Strata**		
Estimation of 1-year mortality	**Low Risk < 5%**	**Intermediate Risk 5%–20%**	**High Risk > 20%**
Signs of right heart failure	No	No	Yes
Progression of symptoms and clinical manifestations	No	Slow	Rapid
Syncope	No	Occasional during strenuous exercise	Repeated with little or normal physical activity
WHO-FC	I, II	III	IV
6-Minute walk distance	> 440 m	165–440 m	< 165 m
Cardiopulmonary exercise testing	Peak VO_2 > 15 mL/min/kg (> 65% predicted) VE/VCO_2 slope < 36	Peak VO_2 11–15 mL/min/kg (35%–65% predicted) VE/VCO_2 slope 36–44	Peak VO_2 < 11 mL/min/kg (< 35% predicted) VE/VCO_2 slope > 44
BNP or NT-proBNP	BNP < 50 ng/L NT-proBNP < 300 ng/L	BNP 50–800 ng/L NT-proBNP 300–1100 ng/L	NP > 800 ng/L NT-proBNP > 1100 ng/L
Echocardiography parameters	RA area < 18 cm^2 TAPSE/sPAP > 0.32 mm/mm Hg No pericardial effusion	RA area 18–26 cm^2 TAPSE/sPAP 0.19–0.32 mm/mmHg Minimal pericardial effusion	RA area > 26 cm^2 TAPSE/sPAP < 0.19 mm/mm Hg Moderate/large pericardial effusion
Cardiac magnetic resonance imaging	RVEF >54% SVI >40 mL/m^2 RVESVI <42 mL/m^2	RVEF 37–54% SVI 26–40 mL/m^2 RVESVI 42–54 mL/m^2	RVEF < 37% SVI < 26 mL/m^2 RVESVI > 54 mL/m^2
Hemodynamics	RAP < 8 mm Hg CI ≥ 2.5 L/min/m^2 SVI > 38 mL/m^2 SvO_2 > 65%	RAP 8–14 mm Hg CI 2.0–2.4 L/min/m^2 SVI 31–38 mL/m^2 SvO_2 60–65%	RAP >14 mm Hg CI < 2.0 L/min/m^2 SVI < 31 mL/m^2 SvO_2 < 60%

BNP, B-type natriuretic peptide; CI, cardiac index; NT-proBNP, N-terminal pro-brain natriuretic peptide; RA, right atrium; RAP, right atrial pressure; sPAP, systolic pulmonary artery pressure; SvO_2, mixed venous oxygen saturation; RVESVI, right ventricular end-systolic volume index; RVEF, right ventricular ejection fraction; SVI, stroke volume index; TAPSE, tricuspid annular plane systolic excursion; VE/VCO_2, ventilatory equivalents for carbon dioxide; VO_2, oxygen uptake; WHO-FC, World Health Organization functional class.

Adapted from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

c. Initial and subsequent therapy in patients with cardiopulmonary comorbidities: because these patients are under-represented in clinical studies, no evidence-based recommendations can be provided. Although these patients are often treated similarly in practice, they have more treatment-related adverse effects or tolerability issues with combination therapy.
   i. All risk strata: oral monotherapy with PDE5i or ERA
   ii. Follow-up therapeutic changes should be based on patient-specific characteristics

3. Subsequent therapeutic options are based on the four strata risk assessment (Table 5). All 3 variables are recommended to be included in each follow-up assessment for better accuracy at predicting mortality. If only 2 variables are available, it is preferred to include 1 biomarker (BNP or NT-proBNP) and 1 functional marker (either WHO-FC or 6MWD).
   a. Low-risk strata: Continue current therapy
   b. Intermediate/Low-risk strata
      i. Add prostacyclin receptor antagonist (PRA)—or—
      ii. Switch from PDE5i to soluble guanylate cyclase stimulator (sGCs)

Table 5. Four-Strata Risk Assessment For Patients With Pulmonary Artery Hypertension

Multivariable Assessment of Mortality Risk to Guide Follow-Up Therapy				
Assessment	**4-Risk Strata**			
Prognostic determinants	**Low**	**Intermediate/Low**	**Intermediate/High**	**High**
Points[a]	1	2	3	4
WHO-FC	I or II	—	III	IV
6-Minute walk distance	> 440 m	320–440 m	165–319 m	< 165 m
BNP or NT-proBNP	< 50 ng/L < 300 ng/L	50–199 ng/L 300–649 ng/L	200–800 ng/L 650–1100 ng/L	> 800 ng/L > 1100 ng/L

[a]Divide the sum of all points by the number of variables and round to the next whole number.

BNP, B-type natriuretic peptide; NT-proBNP, N-terminal pro-brain natriuretic peptide; WHO-FC, World Health Organization functional class.

Adapted from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

Table 6. Endothelin Receptor Antagonists (oral)

	Bosentan	**Ambrisentan**	**Macitentan**
Receptor affinity	Blocks ET_a and ET_b	Preferentially blocks ET_a	Blocks ET_a and ET_b
Half-life (hr)	5	9	16
Approved dose	62.5–125 mg BID	5–10 mg daily	10 mg daily
Outcomes	Delay clinical worsening, improves exercise capacity, improves WHO-FC, and hemodynamics	Improved symptoms, exercise capacity, hemodynamics, and time to clinical worsening	Increased exercise capacity, improved a composite end point of clinical worsening
Adverse effects	Hepatotoxicity (except ambrisentan), peripheral edema, anemia, contraindicated during pregnancy (REMS programs)		
Drug interactions	Glyburide (contraindicated) and cyclosporine (contraindicated), CYP2C9 and CYP3A4 inhibitors and inducers, hormonal contraceptives, warfarin, sildenafil and tadalafil	Caution with cyclosporine (max ambrisentan dose is 5 mg daily)	CYP2C19 and strong CYP3A4 inhibitors and inducers

BID = twice daily; ET = endothelin; REMS = risk evaluation and mitigation strategies.

Information from: Galiè N, Barberà JA, Frost AE, et al. Initial use of ambrisentan plus tadalafil in pulmonary arterial hypertension. N Engl J Med 2015;373:834-44; Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731; Pulido T, Adzerikho I, Channick RN, et al. Macitentan and morbidity and mortality in pulmonary arterial hypertension. N Engl J Med 2013;369:809-18.

c. Intermediate/High- or high-risk strata
   i. Add a parenteral prostacyclin—and/or—
   ii. Evaluate for lung transplant

E. Drug Information by Class
1. Endothelin receptor antagonists (ERAs) (Table 6)
   a. Competitively block the action of ET-1 from acting on two types of ET receptors: ET_a and ET_b. This blockade prevents endogenous ET-1 from its normal actions:
      i. ET_a = vasoconstriction and cell proliferation
      ii. ET_b = vasodilation and antiproliferation of pulmonary endothelial cells
   b. Minimal place in therapy for initiation in critically ill patients
2. PDE-5 inhibitors (Table 7)
   a. Inhibit PDE-5, which prevents the breakdown of cyclic guanosine monophosphate in pulmonary vascular smooth muscle. Increased cyclic guanosine monophosphate potentiates pulmonary vascular smooth muscle and vasodilation of the pulmonary vascular bed.
   b. Indicated for patients who continue to be symptomatic on intravenous epoprostenol.
   c. Intravenous formulation of sildenafil is available for patients who temporarily cannot ingest tablets; however, this formulation is limited by its hemodynamic effects.

Table 7. PDE5i

	Sildenafil (Revatio, Liqrev)	**Sildenafil (Revatio)**	**Tadalafil (Adcirca, Alyq, Tadliq)**
Half-life (hr)	4		35
Approved dose	20 mg oral TID	10 mg IV TID	40 mg oral daily; dose adjustment necessary for renal impairment
Outcomes	WHO FC II and III: Improve 6MWD[a] WHO FC III: Improve WHO FC[a]		WHO FC II and III: Improve 6MWD[b]
Outcomes	WHO FC III: Improve WHO FC[a]	WHO FC II and III: Improve 6MWD[b]	WHO FC II and III: Delay clinical worsening[a] WHO FC III: Improve WHO FC
Adverse effects	Headache, epistaxis, flushing, dyspepsia, hypotension, visual alterations		
Drug interactions	Contraindicated in patients receiving nitrates; avoid with strong CYP3A4 inhibitors and inducers; contraindicated with riociguat		

[a]Klinger JR, Elliott CG, Levine DJ, et al. Therapy for pulmonary arterial hypertension in adults. Chest 2019;155:565-86.

[b]Galiè N, Barberà JA, Frost AE, et al. Initial use of ambrisentan plus tadalafil in pulmonary arterial hypertension. N Engl J Med 2015;373:834-44.

IV = intravenously; TID = three times daily; WHO-FC = World Health Organization functional class; 6MWD = 6-min walk distance.

3. Soluble guanylate cyclase stimulators
   a. Riociguat exerts its effect through two mechanisms: Sensitizes endogenous soluble guanylate cyclase by stabilizing nitric oxide–soluble guanylate cyclase bonding and directly stimulates soluble guanylate cyclase through an auxiliary binding site. Both processes increase cyclic guanosine monophosphate, which influences vascular tone, proliferation, fibrosis, and inflammation.
   b. For patients on ERA/PDE5i therapy and at low-intermediate risk of death, switching from PDE5i to riociguat could be considered (Lancet Respir Med 2021:9:573-84).
   c. Elimination half-life is 12 hours. If cannot be administered for 3 days or more, retitration is required; however, a PH specialist should always be involved for issues with access and reinitiation.
   d. Approved dose is 1–2.5 mg three times daily (higher doses for smokers). In patients at risk of systemic hypotension, a starting dose of 0.5 mg three times daily may be used (2022 guidelines)

Table 8. Additional Roles in Therapy for Riociguat

Failed monotherapy	Add	Clinical outcomes	WHO FC III-IV
Bosentan	Riociguat	Improve 6MWD Improve WHO FC Delay time to clinical worsening	Recommended[a]
Ambrisentan			Recommended[a]
Prostacyclins (inhaled)			Recommended[a]

[a]Consensus-based level of evidence (Chest 2019;155:565-86)

WHO FC = World Health Organization functional class; 6MWD = 6-min walk distance.

e. Adverse effects include hypotension, hemoptysis, headache, dizziness, dyspepsia, nausea, diarrhea, vomiting, and anemia.
f. Contraindicated in patients receiving nitrates and phosphodiesterase inhibitors because of hypotension; avoid with strong CYP3A4/2C8 inhibitors and inducers and with P-glycoprotein/ breast cancer resistance protein inhibitors
g. Riociguat risk evaluation and mitigation strategies (REMS) program for embryo-fetal toxicity

4. Prostacyclins
   a. Parenteral prostacyclins (Table 9–10)
      i. Mimic the actions of PGI_2 (endogenous prostacyclin): Direct vasodilation of pulmonary and systemic arterial vascular beds, inhibition of platelet aggregation, and antiproliferative effects
      ii. Continuous-infusion epoprostenol is the most thoroughly studied medication approved for treating PAH and has been shown to prolong survival.
         • 100% survival with intravenous epoprostenol compared with 80% survival with conventional therapies at 12 weeks (p=0.003) (N Engl J Med 1996;334:296-301)
         • 88% survival with intravenous epoprostenol compared with 59% expected survival in data from historical controls at 1 year (p<0.001) (Circulation 2002;106:1477-82)
         • 55% survival with intravenous epoprostenol compared with 28% survival in historical controls at 5 years (p<0.0001) (J Am Coll Cardiol 2002;40:780-8)
      iii. Complications related to delivery include the need for a dedicated intravenous line, local catheter/bloodstream infections, and catheter-related thrombosis.
   b. Inhaled prostacyclins (Table 9 and Table 10) – Advantage over intravenous route because of selective pulmonary vasodilation with minimal systemic effects

Table 9. Additional Roles in Therapy for Prostacyclins

Failed monotherapy	Add	Clinical outcomes
1–2 classes of oral agents	IV epoprostenol	Improve WHO FC Improve 6MWD
1–2 classes of oral agents	IV treprostinil	Improve 6MWD
IV epoprostenol	Sildenafil (2022 guidelines)	Improve 6MWD
ERA	Inhaled treprostinil	Improve 6MWD
	Inhaled iloprost	Improve WHO FC Delay clinical worsening
PDE5i	Inhaled treprostinil	Improve 6MWD
	Inhaled iloprost	Improve WHO FC Delay clinical worsening

6MWD = 6-min walk distance.

Information from: Klinger JR,, Elliott CG, Levine DJ, et al. Therapy for pulmonary arterial hypertension in adults. Chest 2019;155:565-86; Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

Table 10. Prostacyclins

	Epoprostenol		Treprostinil	Treprostinil	Iloprost	Treprostinil
Brand name	**Flolan**	**Veletri**	**Remodulin**	**Tyvaso**	**Ventavis**	**Orenitram**
Route of administration	IV		SC and IV	Inhalation	Inhalation	Oral
Initial dose	1–4 ng/kg/min IV		1.25 ng/kg/min IV or SC; dose adjustments in hepatic impairment	18 mcg four times daily (solution) 16 mcg four times daily (dry powder)	2.5–5 mcg six to nine times daily	0.25 mg every 12 hr or 0.125 mg every 8 hr; caution with strong CYP2C8 inhibitors and severe; contraindicated in severe hepatic impairment
Half-life	6 min		4 hr		25 min	4 hr
Adverse effects	Flushing, headache, diarrhea nausea/ vomiting, jaw pain, thrombocytopenia			Cough, throat irritation, bronchospasm, hypotension		Headache, diarrhea, nausea, flushing, jaw pain, and abdominal discomfort
Cautions	Rebound PH with abrupt discontinuation; dosing weight should not be changed without consultation with PH specialist					

CYP = cytochrome P450; IV = intravenous; PH = pulmonary hypertension; SC = subcutaneous.

Information from: Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

c. Oral treprostinil (Table 10):
   i. FDA approved as monotherapy for PAH, improved 6MWD
   ii. The addition of oral tresprostinil to ERA or PDE5i/riociguat monotherapy is recommended to reduce the risk of morbidity and mortality in patients with idiopathic, heritable, or drug-associated PAH (Eur Heart J 2022;43:3618-731).

5. Selective prostacyclin receptor (IP receptor) agonist
   a. Selexipag is an oral nonprostanoid that targets the prostacyclin pathway.
   b. The 2019 CHEST guideline update changed their recommendation for this agent based on the included studies not achieving their prespecified definition of clinically significant improvement, defined as 33 m in the 6MWD. However, practitioners still use selexipag despite these guidelines not providing a recommendation for or against the use of this agent.
   c. The 2022 ESC/ERS guidelines support
      i. Consider the addition of selexipag to ERA/PDE5i therapy for patients who present with low/intermediate risk of death at follow-up
      ii. Add selexipag if unable to use a parenteral prostacyclin for patients at intermediate/high or high risk.
      iii. Add selexipag to ERAs and/or PDE5is to reduce the risk of morbidity and mortality events.
   d. It is not recommended to start macitentan, tadalafil, and selexipag as initial therapy, given the lack of current evidence for this strategy (J Am Coll Cardiol 2021;78:1393-403).
   e. Elimination half-life is 0.8–3 hours and, for the active metabolite, 6–14 hours. If unable to administer for 3 days or more, retitration is required; however, a PH specialist should always be involved for issues with access and reinitiation.
   f. Approved dose is 200 mcg twice daily; increase at weekly intervals to the highest tolerated dose (maximum dose 1600 mcg twice daily); adjustment necessary for moderate hepatic impairment

g. Dosing should be reduced to once daily in patients receiving moderate CYP2C8 inhibitors whereas selexipag is contraindicated in patients receiving strong CYP2C8 inhibitors (e.g., gemfibrozil); dose increase necessary with CYP2C8 inducers
h. Adverse effects include headache, diarrhea, nausea and jaw pain.

F. Managing decompensated PH
1. Decompensated PH generally presents with worsening symptoms of right heart failure such as shortness of breath, syncope, fluid retention, or even multiorgan failure due to poor perfusion.
2. Markers of imminent mortality from PH/severe right sided heart failure include a decline in central venous oxygen saturation, lactic acidosis, and decreased urine output.
3. Manage contributing factors such as infections, anemia, arrhythmias, rebound PH (nonadherence or ineffective dosing), hypoxemia, acidosis, and metabolic abnormalities.
4. Supportive therapies include optimizing RV preload through fluid optimization (diuretics or dialysis), maintaining aortic root pressure, improving RV contractility, and reducing RV afterload.
5. Temporary discontinuation of CCBs and β-blockers to eliminate negative inotropic effects
6. Hemodynamic support (Table 11)
   a. Maintaining aortic root pressure and minimizing RV ischemia can be accomplished using vasopressors, which increase the systemic vascular resistance and ultimately improve RV perfusion (Crit Care Med 2007;35:2037-50).
   b. Direct effects on the pulmonary circulation from vasopressors may increase the PVR, potentially leading to further clinical decompensation.
   c. Few studies have been published to help guide selection of the optimal vasopressor in patients with PH; recommendations are extrapolations from other patient populations.

Table 11. Agents Used for Hemodynamic Support for Decompensated PH

Drug	Site of Action (receptor activity)	Effects on PVR	Effects on CO	Comments
Dopamine	Dose-dependent dopaminergic, α_1 and β_1	↑ or ↔	↑	May not improve RV ejection fraction; arrhythmias
Norepinephrine[a]	$\alpha_1 > \beta_1$	↑	↑ or ↔	Decreased mortality in subgroup of cardiogenic shock and decreased rate of arrhythmias compared with dopamine in a randomized trial
Phenylephrine	α_1	↑	↔	Reflex bradycardia may be harmful in the setting of RV failure
Epinephrine	$\alpha_1, \beta_1 > \beta_2$	↑ or ↔	↑	Arrhythmias, hyperglycemia, increased lactate concentrations
Vasopressin	V_1 receptors	↑ or ↔	↑ or ↓	Use low dose (≤ 0.03 units/min)
Dobutamine	$\beta_1 > \beta_2$	↓	↑	Combine with peripheral vasoconstrictor to attenuate systemic vasodilation
Milrinone	PDE-3 inhibitor	↓	↑	Combine with peripheral vasoconstrictor to attenuate systemic vasodilation

[a]De Backer D, Biston P, Devriendt J, et al. Comparison of dopamine and norepinephrine in the treatment of shock. N Engl J Med 2010;362:779-89.

CO = cardiac output; PDE-3 = phosphodiesterase-3; PVR = pulmonary vascular resistance; RV = right ventricular.

   d. Inotropes are used to further augment the cardiac output of the RV and may improve PVR. Because of systemic vasodilatory properties from inotropes, expect possible systemic hypotension and need for vasopressors.
7. Unloading the RV with pulmonary vasodilators is essential to controlling decompensated PH and RV failure.
8. If a patient is not responsive to therapy and may be a transplant candidate, extracorporeal life support can be considered as a bridge to lung transplant. Candidates must be chosen prudently because RV failure must be reversible for this strategy to be successful.

Patient Case

Questions 3 and 4 pertain to the following case.
A 44-year-old man is transferred to the medical ICU for treatment of his worsening PAH. He currently receives no PAH treatment. The patient has had increased dyspnea on exertion for the past 6 months with exercise. For the past 2 months, he has had difficulty going for walks with his dog without getting fatigued, and occasionally, he has chest pain. In the past few days, he has had severe shortness of breath at rest, which prompted him to come to the hospital. His physical examination is remarkable for blood pressure 105/64 mm Hg and heart rate 85 beats/minute. Lung examination is clear, and extremities are notable for trace edema. An echocardiogram reveals an elevated pulmonary systolic pressure and a normal ejection fraction. The patient has an unfavorable response to vasodilator challenge. Pertinent laboratory data are serum urea nitrogen 10 mg/dL, SCr 0.6 mg/dL, AST 160 U/L, and ALT 100 U/L.

3. Which best describes his WHO FC and risk of mortality in the next year?
   E. WHO FC III, mortality risk 5%.
   F. WHO FC III, mortality risk 10%.
   G. WHO FC IV, mortality risk 15%.
   H. WHO FC IV, mortality risk greater than 20%.

4. Which medication therapy would be best to recommend initiating at this time?
   A. Epoprostenol infusion at 2 ng/kg/minute.
   B. Diltiazem 180 mg orally daily.
   C. Macitentan 10 mg orally daily.
   D. Sildenafil 10 mg intravenously three times daily.

III. ASTHMA EXACERBATION

A. Pathophysiology and Classification of Exacerbations
   1. Asthma is characterized by variable and recurring symptoms, airflow obstruction, bronchial hyperresponsiveness, and underlying inflammation (NAEPP 2007).
   2. Asthma exacerbations are, also termed *acute severe asthma*, characterized by decreases in expiratory airflow that can be quantified by spirometry or peak expiratory flow.
   3. Definitions and classifications of asthma exacerbations vary between experts and guidelines. Both terms below are considered life threatening exacerbations.

a. Status asthmaticus is an acute severe asthma exacerbation that does not respond to initial intensive therapy, with the potential for pulmonary compromise and death.
b. Near-fatal asthma is status asthmaticus that progresses to respiratory failure.

Table 12. Classification of Asthma Exacerbations in Acute Care Facility (GINA 2023)

	Symptoms	Initial PEF (or FEV_1)	Clinical Course
Mild/ Moderate	• Talks in phrases • Prefers sitting to laying down • Not agitated • RR increased • Accessory muscles not used • Heart rate 100–120 beats/min • SaO_2 90%–95%	> 50% of predicted or personal best	• Usually cared for at home/primary care setting • Relief from frequent inhaled SABA • Consider addition of ipratropium • Recommended course of oral CS • SaO_2 goal 93%–95%
Severe	• Talks in words • Sits hunched forward • Agitated • RR > 30 breaths/min • Accessory muscles used • Heart rate > 120 beats/min • SaO_2 < 90%	≤ 50% of predicted or personal best	• Requires ED visit and likely hospitalization • Recommended inhaled SABA + ipratropium bromide • SaO_2 goal 93%–95% • Oral or IV corticosteroids • Consider IV magnesium • Consider high dose ICS
Life threatening	• Drowsy • Confused • Silent chest (absent breath sounds)	No definition	• Requires hospitalization; possible ICU • Minimal or no relief from frequent inhaled SABA • IV corticosteroids • Adjunctive therapies are helpful

ED = emergency department; FEV_1 = forced expiratory volume in the first second of expiration; ICS = inhaled corticosteroid; ICU = intensive care unit; IV = intravenous; PEF = peak expiratory flow; RR = respiratory rate; SABA = short-acting β-agonist; SaO_2 = oxygen saturation.

Adapted from: Venkatesan P. 2023 GINA report for asthma. Lancet Respir Med 2023;11:589.

B. Common Causes of Asthma Exacerbations
   1. Viral infections
   2. Allergen exposure
   3. Food allergy
   4. Air pollution
   5. Seasonal weather changes
   6. Poor adherence to inhaled corticosteroids

C. Mortality Risk Factors
   1. History of near-fatal asthma (e.g., requiring MV)
   2. Hospitalization or ED visit for asthma in the past year
   3. Active use of oral corticosteroids or completion of recent course for asthma
   4. Not currently using inhaled corticosteroids
   5. Use of more than one canister of short-acting β-agonists (SABAs) per month
   6. Poor adherence to inhaled corticosteroid containing asthma medications and/or poor adherence to asthma action plan

7. Social history that includes major psychosocial problems or psychiatric illness
8. Food allergies
9. Comorbidities including pneumonia, diabetes, and arrhythmias

D. Social Determinants of Health in patients with asthma (Ann Allergy Asthma Immunol 2022;128:5-11)
   1. Structural racism and inequities in socioeconomic status have had profound effects on both the prevalence of asthma and frequency of exacerbations in Black and Latinx patients.
   2. Adults with a household income < $50,000 had a 1.6 times higher rate of asthma treatment failure and 2 times higher rate of asthma exacerbations compared with those with a household income >=$50,000.
   3. Long-term stress, healthcare access, environmental factors, education level, food accessibility and other factors can impact both disease prevalence and management in patients with asthma. The full extent of which of these variables has the highest effects is still under investigation.

E. COVID-19 (SARS-CoV-2) and Asthma (GINA 2023)
   1. Patients with well-controlled asthma do not appear to be at increased risk of infection or mortality due to SARS-CoV-2.
   2. However, patients who recently required oral corticosteroids for their asthma and those hospitalized with severe asthma were at increased risk of death from COVID-19.
   3. Patients on inhaled corticosteroids for their asthma had lower mortality from COVID-19 compared with patients without underlying lung disease (Lancet Respir Med 2021;9:699-711).
   4. Both GINA and the Centers for Disease Control and Prevention recommend that patients receive vaccination against COVID-19 including boosters if applicable.
   5. If patients are receiving biologics for their severe asthma, it is not recommended to give the biologic and COVID-19 vaccine on the same day.

F. Alternative Causes (mimic asthma exacerbation)
   1. Upper airway: Vocal cord dysfunction, anaphylaxis, laryngeal stenosis
   2. Central airway: Tracheomalacia, tracheal stenosis mucus plugging
   3. Lower airway: Bronchiolitis, COPD, valvular heart disease, diastolic heart dysfunction

G. ABG Assessment
   1. Patients with acute severe asthma initially experience a respiratory alkalosis.
   2. As respiratory status worsens, arterial carbon dioxide increases (patient exhaustion, inadequate alveolar ventilation and/or an increase in physiologic dead space), leading to respiratory acidosis.
   3. Metabolic (lactic) acidosis may coexist. Lactate production presumably stems from the use of high-dose β-agonists, increased work of breathing resulting in anaerobic metabolism of the ventilatory muscles, and tissue hypoxia.

H. Oxygen
   1. Oxygen therapy is important in managing acute severe asthma.
   2. Oxygen by nasal cannula or mask should be administered to patients with severe exacerbations with hypoxemia to achieve Sao_2 values greater than 93%–95% (pregnant women and patients with a cardiac history may require higher goals) (GINA 2023; Thorax 2011;66:937-41).

I. Noninvasive Ventilation: Data are insufficient for severe exacerbations. If attempted, patients should not be sedated to tolerate noninvasive ventilation.

J. Mechanical Ventilation
   1. Indications
      a. Worsening hypoxemia or hypercarbia
      b. Drowsiness or altered mental status
      c. Hemodynamic instability
      d. Increased work of breathing
   2. Low minute ventilation (by reduced tidal volume and/or respiratory rate), high inspiratory flow rate, and minimal PEEP on the ventilator will help minimize dynamic hyperinflation.

K. β-Agonists
   1. SABAs stimulate the β_2-receptors on smooth muscle cells, leading to relaxation of respiratory smooth muscle and causing bronchodilation and decreased airway obstruction.
   2. SABAs are the cornerstone in managing acute severe asthma.
   3. Patients with an asthma exacerbation should receive a SABA repeatedly by either an MDI or nebulization at presentation, and the SABA should be continued until acute symptoms have resolved (NAEPP 2007).
   4. Patients with acute severe exacerbations may benefit from continuous versus intermittent nebulization of SABAs (Cochrane Database Syst Rev 2003;4:CD001115).
      a. Intermittent dosing of albuterol: 2.5–5 mg every 20 minutes for three doses; then 2.5–10 mg every 1–4 hours as needed
      b. Continuous nebulization of albuterol: 10–15 mg/hour
   5. Systemic β-agonists can be considered if the patient does not respond to inhaled therapy after several hours (Chest 2022;162:747-56).
      a. These agents have no proven advantage over inhaled agents or over each other and are not recommended for routine use because of the high rate of adverse effects.
      b. However, if the patient's acute asthma exacerbation is caused by anaphylaxis or angioedema, epinephrine would be recommended.
      c. Epinephrine 1 mg/mL 0.3–0.5 mg intramuscularly/subcutaneously every 20 minutes up to three doses
      d. Epinephrine infusion 0.1 mcg/kg/minute
      e. Terbutaline 0.25 mg subcutaneously every 20 minutes for three doses
   6. Adverse effects include tremor, tachyarrhythmias, hypokalemia, tachyphylaxis, hyperglycemia, and type B lactic acidosis.

L. Anticholinergic Agents
   1. Inhaled anticholinergic agents selectively bind to the muscarinic receptors on smooth muscle cells in the airways and thereby reduce bronchoconstriction.
   2. Short-acting inhaled anticholinergic agents (ipratropium bromide) should be given in combination with a SABA to promote additional bronchodilation through a different pathway.
   3. Adding ipratropium to inhaled albuterol compared with using albuterol alone in patients with severe asthma improved the response; however, outcomes with this combination in status asthmaticus or near-fatal asthma remain elusive (Am J Respir Crit Care Med 2000;161:1862-8). Combination therapy consisting of inhaled short-acting anticholinergics and a SABA compared with a SABA alone has reduced hospitalization rates in severe asthma exacerbations. However, combination therapy may increase the risk of adverse drug events, including agitation and palpitations (Cochrane Database Syst Rev 2017;1:CD001284).
   4. Adverse effects include headache, flushed skin, blurred vision, tachycardia, palpitations, and urinary retention.

M. Corticosteroid Therapy
   1. Corticosteroids decrease airway obstruction during an asthma exacerbation by decreasing inflammation, increasing the number of β2-receptors and increasing their responsiveness to β-agonists, reducing airway edema, and suppressing certain proinflammatory cytokines (Respir Med 2004;98:275-84).
   2. Systemic corticosteroids should be administered to patients who have moderate or severe exacerbations or to patients who do not respond promptly and completely to SABA treatment (Am J Med 1983;74:845-51).
   3. Typically, there is a 6–8-hour delay in the response to corticosteroids in status asthmaticus or life threatening asthma; therefore, administration should be considered early in the course (within 1 hour of presentation) (Cochrane Database Syst Rev 2001;1:CD002178).
   4. Oral prednisone is considered as effective as parenteral corticosteroids in most patients; however, it may not be as effective in critically ill patients with impaired gastric absorption.
   5. GINA guidelines recommend prednisone 1 mg/kg/day up to a maximum of prednisone 50 mg daily for 5–7 days for treatment of exacerbations.
      a. Other guidelines recommend longer courses and higher doses for life threatening asthma exacerbations based on the same evidence used by GINA.
      b. For corticosteroid courses less than 2 weeks (GINA 2023), tapering is not necessary, especially if patients are concurrently using an inhaled corticosteroid.
   6. Inhaled corticosteroids can be initiated any time in the treatment of an asthma exacerbation. It is not recommended to replace systemic steroids with inhaled corticosteroids (Respir Care 2018;63:783-96).

N. Adjunctive Therapies (Respir Care 2018;63:783-96; J Clin Med 2019;8:1283)
   1. Ketamine is a phencyclidine derivative that has bronchodilatory properties that reduce airway resistance and prevent the reuptake of norepinephrine, which may stimulate β_2-receptors (Indian J Crit Care Med 2013;17:154-61). Ketamine may be used for as an induction for rapid sequence induction or sedation for MV (Chest 2022;162:747-56).
   2. Helium-oxygen (heliox) is a blended gas (mixture of about 70%–80% helium and 20%–30% oxygen) that decreases airway resistance, leading to improved airflow and ventilation. Use of heliox in status asthmaticus may delay the need for intubation by allowing other therapies to work or as a delivery mechanism for nebulized SABAs.
   3. Magnesium sulfate (2 g intravenously administered over 20–30 minutes) can be considered in patients who have life-threatening exacerbations and are unresponsive to conventional therapies after 1 hour (GINA 2023).
      a. Magnesium is thought to cause bronchodilation by inhibiting calcium channels on smooth muscle, leading to relaxation.
      b. In addition, magnesium may have anti-inflammatory properties that interfere with the activation and release of neutrophils in patients with asthma.
      c. Data for magnesium sulfate administration in severe asthma exacerbations are mixed. In addition, data exist using nebulized magnesium rather than intravenous, but evidence is currently not robust enough to recommend (Front Allergy 2023;4:1211949).
   4. Centers equipped with extracorporeal membrane oxygenation may offer this as a bridge to recovery if a patient's respiratory acidosis cannot be managed by invasive MV alone.

O. Treatments Not Recommended (GINA 2023)
   1. Methylxanthines (theophylline and aminophylline) do not improve lung function or other outcomes in hospitalized adults. Use is associated with severe adverse effects.
   2. Antimicrobials are not generally recommended for acute asthma exacerbations; however, they can be considered if there is evidence of concurrent infection.

Patient Case

Questions 5 and 6 pertain to the following case.
A 30-year-old woman (weight 115 kg) with status asthmaticus is admitted to the ICU. She has a history of severe refractory asthma that has required endotracheal intubation three times in the past 6 months. Her medical history includes hypertension, diabetes, obesity, and bipolar disorder. She has used at least three canisters of albuterol per month for the past 2 months to manage her symptoms.

5. Which best represents the patient's risk factors for higher mortality?
   A. Three hospitalizations in the past 6 months and bipolar disorder.
   B. Use of more than two canisters of SABAs in the past month and obesity.
   C. Hospitalization for asthma in the past month and diabetes.
   D. Prior episode of near-fatal asthma and hypertension.

6. The patient is endotracheally intubated and placed on MV. Which would be the most appropriate initial therapy for this patient with life-threatening asthma exacerbation?
   A. Inhaled albuterol by nebulization 2.5 mg every 4 hours.
   B. Inhaled albuterol by nebulization 2.5 mg every 4 hours and inhaled ipratropium by nebulization 0.5 mg every 6 hours.
   C. Inhaled albuterol by nebulization 2.5 mg every 4 hours, inhaled ipratropium by nebulization 0.5 mg every 6 hours, and methylprednisolone 40 mg intravenously daily.
   D. Inhaled albuterol by nebulization 2.5 mg every 4 hours, inhaled ipratropium by nebulization 0.5 mg every 6 hours, and methylprednisolone 125 mg intravenously every 6 hours.

IV. ACUTE CHRONIC OBSTRUCTIVE PULMONARY DISEASE EXACERBATION

A. Chronic Obstructive Pulmonary Disease (COPD) (GOLD 2024)
   1. Newly defined in the GOLD 2023 guidelines as a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production, exacerbations) as a result of abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction (Table 13).

Table 13. Classification of Airflow Limitation Severity in Chronic Obstructive Pulmonary Disease

Grade	Spirometric GOLD Classification
GOLD 1: Mild	$FEV_1 \geq 80\%$ predicted
GOLD 2: Moderate	$50\% \leq FEV_1 < 80\%$ predicted
GOLD 3: Severe	$30\% \leq FEV_1 < 50\%$ predicted
GOLD 4: Very severe	$FEV_1 < 30\%$ predicted

FEV_1 = forced expiratory volume in the first second of expiration; GOLD = Global Initiative for Chronic Obstructive Lung Disease.

Adapted from: Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global strategy for the diagnosis, management, and prevention of COPD. 2024.

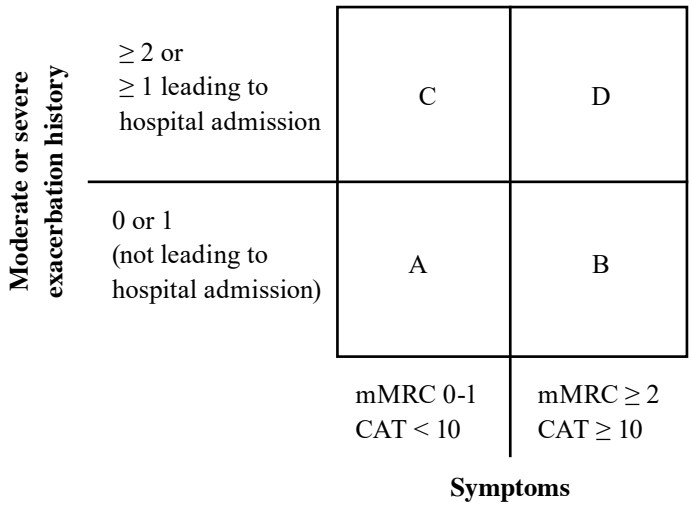


Figure 1. Assessment of Chronic Obstructive Pulmonary Disease Symptoms and Risk of Exacerbations
CAT = Chronic Obstructive Pulmonary Disease Assessment Test; mMRC = Modified British Medical Research Council Questionnaire.

2. Exacerbations of COPD (ECOPD): An event that is characterized by dyspnea and/or cough that worsens over less than 14 days, which may be accompanied by tachypnea and/or tachycardia and is often associated with local and systemic inflammation caused by airway infection, pollution, or other insults to the lungs (Am J Respir Crit Care Med 2021;204:1251-8)
3. Diagnosis of ECOPD
   a. Symptoms, severity of dyspnea using a visual analog scale, and documentation of the presence of cough should be assessed.
   b. Signs (tachypnea, tachycardia), sputum volume and color, and respiratory distress (accessory muscle use) should also be assessed.
   c. Severity should be evaluated with Spo_2, C-reactive protein (CRP), and ABG (venous blood gas to assess pH and bicarbonate is an acceptable alternative).
   d. Identify the etiology of the event (viral, bacterial, or environmental).
   e. Spirometry is not accurate during an exacerbation and is not recommended.
4. Classifications of ECOPD

Table 14. Classification of ECOPD Severity

ECOPD Severity	Description	Diagnostic Parameters
Mild	Treated with short-acting bronchodilators only	Patient must exceed three of the listed variables to move to moderate • Dyspnea VAS < 5 • RR < 24 breaths/min • HR < 95 beats/min • Resting Spo_2 ≥ 92% breathing ambient air (or patient's usual oxygen prescription) AND change ≤ 3%[a] • CRP < 10 mg/L (if available)

Table 14. Classification of ECOPD Severity (*continued*)

ECOPD Severity	Description	Diagnostic Parameters
Moderate	Treated with short-acting bronchodilators and oral corticosteroids with or without the addition of antibiotics	• Dyspnea VAS ≥ 5 • RR ≥ 24 breaths/min • HR ≥ 95 beats/min • Resting Spo_2 < 92% breathing ambient air (or patient's usual oxygen prescription) AND/OR change > 3%[a] • CRP ≥ 10 mg/L (if available) • If ABG obtained, hypoxemia and/or hypercapnia may be present (Pao_2 ≤ 60 mm Hg, $Paco_2$ > 45 mm Hg), but acidosis will be absent
Severe	Patient requires ED visit or hospitalization; patient may also experience acute respiratory failure	• Dyspnea VAS ≥ 5 • RR ≥ 24 breaths/min • HR ≥ 95 beats/min • Resting Spo_2 < 92% breathing ambient air (or patient's usual oxygen prescription) AND/OR change > 3%[a] • CRP ≥ 10 mg/L (if available) • ABG shows new hypercapnia and acidosis ($Paco_2$ > 45 mm Hg and pH < 7.35)

[a]Pulse oximetry (Spo_2) is not as accurate as an ABG, and there is a risk of overestimation of blood oxygenation saturation in patients with darker skin tones (N Engl J Med 2020;383:2477-8).

ABG = arterial blood gas; CRP = C-reactive protein; ECOPD = exacerbations of chronic obstructive pulmonary disease; ED = emergency department; HR = heart rate; RR = respiratory rate; Sao_2 = oxygen saturation; Spo_2 = functional oxygen saturation; VAS = visual analog scale.

Adapted from: The ROME Proposal (Am J Respir Crit Care Med 2021;204:1251-8); GOLD guidelines 2024.

5. Definitions of ECOPD respiratory failure (GOLD 2024)

Table 15. Definitions of ECOPD Respiratory Failure

No respiratory failure	• RR ≤ 24 breaths/min • HR < 95 beats/min • No accessory respiratory muscle use • No change in mental status • Hypoxemia improved with supplemental oxygen by Venturi mask 24%–35% Fio_2 • No increase in $Paco_2$
Acute respiratory failure – non–life threatening	• RR > 24 breaths/min • Accessory respiratory muscle use • No change in mental status • Hypoxemia improved with supplemental oxygen by Venturi mask > 35% Fio_2 • Hypercarbia; $Paco_2$ increased compared with baseline or elevated 50–60 mm Hg
Acute respiratory failure – life threatening	• RR > 24 breaths/min • Accessory respiratory muscle use • Acute changes in mental status • Hypoxemia not improved with supplemental oxygen by Venturi mask > 40% Fio_2 • Hypercarbia; $Paco_2$ increased compared with baseline or elevated > 60 mm Hg or the presence of acidosis (pH ≤ 7.25)

ECOPD = exacerbations of chronic obstructive pulmonary disease; Fio_2 = fraction of inspired oxygen; HR = heart rate; RR = respiratory rate.

B. Common Causes of ECOPD
   1. Respiratory tract infection (40%–50% of ECOPD)
      a. Bacterial (J Infect 2013;67:516-23)
         i. *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis* are the most common organisms.
         ii. *Pseudomonas aeruginosa* and other gram-negative organisms are more common in patients with frequent exacerbations (Chest 1998;113:1542-8; Chest 1999;116:40-6; Respir Med 2003;97:770-7; Chest 2007;131:44-52; Eur Respir J 2009;34:1072-8; J Infect 2013;67:516-23).
      b. Viral SARS-CoV-2, influenza, rhinovirus, parainfluenza, respiratory syncytial virus
   2. Medication nonadherence
   3. Temperature change
   4. Air pollution

C. Oxygen
   1. Oxygen therapy is important in managing ECOPD.
   2. Oxygen by nasal cannula or mask should be administered to patients with severe exacerbations to achieve an Sa_{O_2} of 88-92% (BMJ 2010;341:c5462).
   3. Caution is advised with oxygen supplementation in patients with COPD (Crit Care 2012;16:323):
      a. A slight level of hypoxemia may serve as a trigger for their respiratory drive secondary to chronic hypercapnia.
      b. May increase ventilation/perfusion mismatch and decrease respiratory rate centrally and perpetuate the Haldane effect
      c. A recent post hoc analysis of two prospective COPD trials suggests that exceeding an Sa_{O_2} of 92% is associated with increased mortality (HR 1.98 for Sa_{O_2} 93%–96%, HR 2.97 for Sa_{O_2} 97%–100%) (Emerg Med J 2021;38:170-7).

D. High-Flow Oxygen Therapy by Nasal Cannula
   1. Alternative to standard oxygen therapy or noninvasive positive pressure ventilation (GOLD 2024)
   2. Improves oxygenation and ventilation in patients with COPD and very severe underlying disease (Thorax 2016;71:759-61)

E. Noninvasive MV
   1. Indications for noninvasive MV include at least one of the following (GOLD 2024):
      a. Respiratory acidosis (Pa_{CO_2} 45 mm Hg or greater and/or arterial pH 7.35 or less)
      b. Severe dyspnea with clinical signs suggestive of respiratory muscle fatigue, increased work of breathing, or both; use of accessory muscles; paradoxical motion of the abdomen; or retraction of the intercostal spaces
      c. Persistent hypoxemia despite supplemental oxygen
   2. NIV is preferred to invasive MV as the initial management of acute respiratory failure. NIV improves gas exchange, reduces work of breathing and need for MV, decreases hospitalization duration, and improves survival (N Engl J Med 1995;333:817-22; Eur Respir J 2017;49:1600791).

F. Mechanical Ventilation: Indications for invasive MV (GOLD 2024)
   1. Unable to tolerate or failure of noninvasive ventilation
   2. Respiratory or cardiac arrest
   3. Altered level of consciousness
   4. Aspiration or vomiting
   5. Hemodynamic instability despite fluid and vasopressors
   6. Severe ventricular or supraventricular arrhythmias

7. Life-threatening hypoxemia and inability to tolerate noninvasive ventilation
8. Inability to manage respiratory secretions

G. Bronchodilators
1. Inhaled SABAs (nebulized or MDI) with or without a short-acting anticholinergic are preferred for bronchodilation in ECOPD.
2. Continue long-acting β-agonists and anticholinergic agents, although no studies have evaluated this regimen. If patients have not yet started these agents, initiate long-acting agents as soon as patient is stable and prior to discharge.
3. Currently, evidence is also lacking regarding a mode of delivery when comparing nebulizers with MDIs during ECOPD, though continuous nebulization is not recommended (GOLD 2024; Cochrane Database Syst Rev 2016;8:CD011826).
4. Treatments NOT recommended: Methylxanthines (theophylline and aminophylline) because of significant adverse effects

H. Corticosteroid Therapy
1. Corticosteroids shorten recovery time, improve lung function (FEV_1), improve oxygenation, and decrease the risk of early relapse, treatment failure, and length of hospitalization (GOLD 2024; N Engl J Med 1999;340:1941-7; Chest 2001;119:726-30). Prednisone 40 mg daily or equivalent (preferably oral) for 5 days may shorten recovery time in acute ECOPD. If oral administration is not an option, equivalent doses of intravenous methylprednisolone or nebulized budesonide can be administered (GOLD 2024).
2. Literature review
    a. Reduction in the Use of Corticosteroids in Exacerbated COPD (REDUCE) was a randomized, noninferiority trial. Patients (n=314) were randomized to prednisone 40 mg daily for either 5 days or 14 days (inhaled bronchodilators and antimicrobials in both groups). ECOPD occurred in 35.9% of patients in the 5-day group and in 36.8% of patients in the 14-day group (p=0.006) In addition, although one of the secondary endpoints notes no difference between groups regarding the need for mechanical ventilation (defined as either intubation or noninvasive ventilation), it is unclear how many patients in the study population were critically ill. (JAMA 2013;309:2223-31). This study did not include critically ill patients requiring MV.
    b. A randomized, double-blind trial including 83 adult patients with an ECOPD requiring hospitalization and ventilatory support (invasive or noninvasive) compared methylprednisolone 0.5 mg/kg intravenously every 6 hours for 72 hours, 0.5 mg/kg every 12 hours on days 4–6, and 0.5 mg/kg daily on days 7–10 with placebo. Patients receiving corticosteroids had a shorter duration of MV (p=0.04), trend toward shorter length of ICU stay (p=0.09), fewer noninvasive ventilation failures (p=0.004), and more hyperglycemia (p=0.04). This study was not powered to detect differences in length of stay or mortality (Arch Intern Med 2011;171:1939-46).
    c. An open-label, randomized trial included 217 critically ill patients 40 years and older with an ECOPD requiring MV to receive oral prednisone 1 mg/kg daily for a maximum of 10 days or usual care. No differences in ICU mortality, noninvasive ventilation failure, duration of MV, or ICU length of stay were observed. Hyperglycemia significantly increased in the steroid group (*Eur Respir J* 2014;43:717-24).
    d. A cohort study compared high-dose (greater than 240 mg/day) with low-dose (240 mg/day or less) methylprednisolone in more than 17,000 patients with an ECOPD admitted to an ICU. Patients in the high-dose group had longer ICU and hospital lengths of stay, higher hospital costs, longer durations of MV, and more hyperglycemia and fungal infections. Mortality did not differ between groups (Am J Respir Crit Care Med 2014;189:1052-64).

e. An open-label, randomized, parallel-group trial compared fixed-dosed corticosteroid treatment (prednisone 40 mg/day) with variable-dosed corticosteroid treatment for 5 days for the treatment of ECOPD in 246 hospitalized patients. Those in the variable-dosed corticosteroid group received 61.4 mg/day of prednisone on average. In-hospital treatment failure was defined as death, need for MV, administration of additional rescue steroids or aminophylline, or upgrade of antibiotic use. Those in the variable-dosed group experienced a reduced risk of in-hospital treatment failure (RR 0.37 [0.18–0.74]). Post hoc analysis did not show a significant difference in treatment failure for patients receiving less than or greater than 60 mg/day (31.3% vs. 23.4% [p=0.351]); however, treatment failure was higher in patients receiving 40 mg or less per day, 44.4%, than for those receiving greater than 40 mg/day, 22.9% (p=0.27). This study suggests a more appropriate fixed steroid dose for ECOPD is prednisolone 60 mg/day rather than 40 mg/day (Chest 2021;160:1660-9).

I. Antimicrobials (GOLD 2024; Cochrane Database Syst Rev 2018;10:CD010257)
   1. Data are conflicting on which populations benefit from antimicrobials.
      a. A 2018 systematic review comparing antibiotics with placebo for ECOPD found a treatment failure relative risk of 0.72 (n=1191) among studies in outpatient settings using only antibiotics currently in practice. No differences in all-cause mortality or treatment failure were identified (Cochrane Database Syst Rev 2018;10:CD010257).
         i. Only one trial included ICU patients (Lancet 2001;358:2020-5).
            (a) Administration of antibiotics reduced treatment failure, all-cause mortality, and hospital length of stay.
            (b) Study limitations: Conducted in 2001; generalizability to current practice may be limited; study used ofloxacin, which is no longer used
         ii. Data remain inconclusive regarding treatment failure, mortality, and repeated exacerbations for patients who are hospitalized outside the ICU with severe COPD and for outpatients.
         iii. The findings within this systematic review are further limited by the inconsistency among studies in the description of COPD severity at baseline.
   2. GOLD guidelines recommend 5–7 days of antibiotics during ECOPD in the following patient populations:
      a. Patients with all three cardinal symptoms: increase in dyspnea, increased sputum, and sputum purulence
      b. Patients with two cardinal symptoms if purulent sputum is one of the two symptoms
      c. Patients requiring invasive or noninvasive MV for their respiratory insufficiency or failure as the result of an ECOPD
   3. Choice of antimicrobial should be based on local resistance patterns.
      a. Initial therapy with amoxicillin/clavulanate, macrolide, tetracycline, or quinolone is suggested by the guidelines.
         i. Macrolide-based antimicrobial regimens may improve clinical outcomes.
            (a) A propensity score-matched cohort study found that macrolide treatment compared with non-macrolide treatment was associated with reduced 30-day readmission rates (7.3% vs. 8.8%, respectively; p<0.01) (Pharmacotherapy 2019;39:242-52).
      b. For patients with frequent exacerbations, severe airflow obstruction, and/or exacerbations requiring MV, broader-spectrum antimicrobials should be considered because of the possible presence of *P. aeruginosa*.
   4. De-escalation
      a. Antimicrobial therapy should be de-escalated as appropriate according to culture results, if available.

b. Use of procalcitonin in critically ill patients with ECOPD is controversial. A study of ECOPD in ICU patients found that using a procalcitonin-based algorithm for initiating or discontinuing antibiotics was associated with a higher mortality rate compared with patients receiving standard antibiotic regimens (Intensive Care Med 2018;44:428-37).

J. Vitamin D
   1. Mechanism of effect: Vitamin D metabolites attenuate inflammation and support immune system responses to pathogens.
   2. The GOLD guidelines recommend routine identification and supplementation of vitamin D deficiency in patients hospitalized with ECOPD with vitamin D concentrations less than 10 ng/mL.
   3. A systematic review and meta-analysis of three randomized controlled trials (469 patients with data available) showed that supplementation of vitamin D did not affect the rate of moderate/severe ECOPD (Thorax 2019;74:337-45).
      a. Prespecified subgroup analysis of 87 patients identified that vitamin D supplementation reduced the rate of moderate to severe ECOPD in patients whose concentrations were deficient (25-hydroxyvitamin D concentrations less than 10 ng/mL) at baseline.
      b. No effect on exacerbations for patients whose concentrations were not deficient at baseline
   4. In a more recent randomized controlled trial of patients with COPD investigating the effects of vitamin D supplementation on exacerbations, a prespecified subgroup analysis of patients with vitamin D concentrations of 6–10 ng/mL compared with placebo found no difference in time to first exacerbation, time to hospitalization, or rate of exacerbations (Am J Clin Nutr 2022;116:491-9).]

Patient Cases

7. A 79-year-old woman (weight 70 kg) is admitted to the ICU for the management of hypercapnic respiratory failure related to an ECOPD. She has had several admissions for ECOPD. The patient presents with fever, profound dyspnea, increased sputum production (thick and purulent), and confusion. She has a history of anaphylaxis to penicillin. Her blood pressure is 190/100 mm Hg, heart rate is 110 beats/minute, and respiratory rate is 22 breaths/minute. Her chest is hyperinflated and has poor air entry bilaterally. Her ABG values are as follows: pH 7.20, Pco_2 85 mm Hg, and Pao_2 44 mm Hg on 6 L of nasal cannula. She is intubated and placed on MV. Which would be the most appropriate group of medications to treat this patient's severe ECOPD?
   A. Methylprednisolone 1 mg/kg intravenously administered as two divided doses, inhaled albuterol by nebulization, ampicillin/sulbactam 3 g intravenously every 6 hours for 7 days, and azithromycin 500 mg intravenously daily for 5 days.
   B. Prednisone 40 mg by nasogastric tube (NGT) daily for 5 days, inhaled albuterol and ipratropium by nebulization, ampicillin/sulbactam 3 g intravenously every 6 hours for 10 days, and azithromycin 500 mg daily intravenously for 5 days.
   C. Methylprednisolone 1 mg/kg intravenously administered as two divided doses, inhaled albuterol and ipratropium by nebulization, and levofloxacin 750 mg by NGT daily for 7 days.
   D. Prednisone 40 mg by NGT daily for 5 days, inhaled albuterol and ipratropium by nebulization, and levofloxacin 750 mg intravenously daily for 5 days.

8. A 75-year-old woman presents to the ED with a productive cough and altered mental status (oriented only to person). Her medical history is significant for type 2 diabetes and severe COPD. On physical examination, she has a productive cough and is alert and oriented × 4. Her vital signs are temperature 101.8°F, heart rate 95 beats/minute, respiratory rate 33 breaths/minute, systolic blood pressure 105 mm Hg, and Spo_2 86% on her home oxygen of 2 L/minute by nasal cannula. The patient is placed on noninvasive ventilation by Venturi mask at an Fio_2 of 40%, and her Spo_2 is now 91%. A basic metabolic panel returns with SCr 1.3 mg/dL and serum urea nitrogen 55 mg/dL, and an ABG reports pH 7.37, $Paco_2$ 55 mm Hg, and Pao_2 59 mm Hg. Radiography reveals multilobar infiltrates. Which best describes the severity of the patient's ECOPD and her level of respiratory failure according to the 2024 GOLD guidelines?
   A. Moderate ECOPD with non–life-threatening acute respiratory failure.
   B. Moderate ECOPD with life-threatening acute respiratory failure.
   C. Severe ECOPD with life-threatening acute respiratory failure.
   D. Severe ECOPD with non–life-threatening acute respiratory failure.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: C**

The patient presents with a CF exacerbation, probably caused by an infection. The most likely causative organism of her infection is *P. aeruginosa*; therefore, therapy must be directed to *P. aeruginosa* (Answer B is incorrect). In addition, antibiotic treatment should include the empiric selection of two antibacterial agents, ideally a β-lactam and an aminoglycoside, dosed to effectively treat the infection (Answers A and D are incorrect). The ideal regimen should include a β-lactam dosed to treat *P. aeruginosa* – in this case, piperacillin/tazobactam at the recommended dose – and an aminoglycoside dosed once daily (Answer C is correct).

2. **Answer: B**

According to the STOP2 trial, patients who improve within 7 days of antimicrobial therapy can complete 10 days of antimicrobials. These patients' clinical outcomes are noninferior to those of patients who continue antimicrobials for 14 days (Answer B is correct). Answer A is incorrect; 7 days is too short for an exacerbation of CF. Answer C is incorrect, given the results of the STOP2 trial. Answer D is incorrect because microbiological cure is neither expected nor likely for patients with CF.

3. **Answer: D**

This patient has signs of right heart failure and rapid progression of symptoms, which place him in the high-risk category for mortality within 1 year (greater than 20%) (Answers A, B, and C are incorrect). In addition, he has symptoms at rest, which place him in WHO FC IV (Answer D is correct).

4. **Answer: A**

The patient's symptoms and physical findings place him in WHO FC IV. His unfavorable response to the vasodilator challenge makes CCBs an undesirable class of medications for him (Answer B is incorrect). Epoprostenol continuous infusion is indicated for patients with PAH presenting with WHO FC IV to improve symptoms, exercise capacity, and hemodynamics. In addition, it is the only treatment shown to reduce mortality in PAH (Answer A is correct). Macitentan and sildenafil could be considered in this patient; however, his elevated liver enzymes do not make macitentan an ideal agent (Answer C is incorrect). Intravenous sildenafil is also not ideal because of the hypotension associated with the intravenous formulation (Answer D is incorrect).

5. **Answer: C**

Risk factors for increased mortality in patients with asthma include (1) history of near-fatal asthma (e.g., requiring mechanical ventilation); (2) hospitalization or ED visit for asthma in the past year; (3) active use of oral corticosteroids or completion of recent course for asthma; (4) not currently using inhaled corticosteroids; (5) use of more than one canister of SABAs per month; (6) poor adherence to inhaled corticosteroid containing asthma medications and/or poor adherence to asthma action plan; (7) social history that includes major psychosocial problems or psychiatric illness; (8) food allergies; (9) comorbidities including pneumonia, diabetes, and arrhythmias. Two of the patient's listed characteristics are risk factors (Answer C is correct). Other answers include one risk factor or none (Answers A, B, and D are incorrect).

6. **Answer: C**

For life-threatening asthma exacerbations, SABAs, short-acting antimuscarinics, and intravenous corticosteroids are recommended (Answers A and B are incorrect). The recommended dose of intravenous corticosteroids is methylprednisolone 40 mg (equivalent to prednisone 50 mg) intravenously per day administered early in the course of the exacerbation (Answer C is correct; Answer D is incorrect).

7. **Answer: D**

The latest GOLD guidelines recommend systemic corticosteroids to shorten recovery time, improve FEV_1, and improve hypoxemia. The recommended dose is prednisone 40 mg orally once daily (or equivalent) for 5 days. Although recent data identified that some patients may have worse outcomes with fixed prednisone 40-mg doses, the data suggest prednisone 60 mg is adequate to balance the benefit of steroids with the risks of adverse effects (Answers A and C are incorrect). Adding inhaled SABAs (nebulized or MDI) with or without a short-acting anticholinergic is the preferred treatment in an ECOPD. Antibiotic treatment for 5–7 days is also indicated because the patient has signs of bacterial infection

(fever, purulent sputum) and a severe ECOPD with respiratory failure. This patient has a history of anaphylactic reaction to penicillin and repeated exacerbations that may predispose her to resistant organisms such as *P. aeruginosa* (Answers A and B are incorrect); therefore, levofloxacin would be the best recommendation (Answer D is correct).

8. **Answer: A**

The current GOLD guidelines provide specific diagnostic parameters to determine the severity of an ECOPD. A respiratory rate greater than 24 breaths/minute and an Spo_2 less than 92% on home oxygen therapy clearly identify this patient as having either a moderate or severe asthma exacerbation. Patients with a moderate ECOPD may have hypercapnia ($Paco_2$ greater than 45 mm Hg) or hypoxia (Pao_2 less than 60 mm Hg); however, they have not yet developed acidosis (pH less than 7.35), whereas those experiencing a severe ECOPD have developed acidosis (Answers C and D are incorrect). Although the degree of respiratory failure is determined using many of the same parameters as the severity of ECOPD, there is a greater focus on patient response to therapy. An improvement in hypoxemia with noninvasive ventilation by Venturi mask with an Fio_2 greater than 35% and the absence of an altered mental status are associated with non–life-threatening respiratory failure (Answer B is incorrect; Answer A is correct).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**
It is imperative to recognize that this patient has acute respiratory distress syndrome (ARDS) caused by a CF exacerbation. Therefore, an inclusive therapy plan should include appropriate treatments for ARDS and CF. For ARDS, a lung-protective ventilation strategy (tidal volume 4–6 mL/kg) and a fluid-conservative strategy (CVP less than 4 mm Hg, if not in shock) are of utmost importance (Answers A and C are incorrect because of the CVP goal; Answer D is correct). Appropriate treatment of the CF exacerbation includes empiric therapy for *P. aeruginosa* in the form of optimal doses of a β-lactam and aminoglycoside (Answer B is incorrect because of the inappropriate tobramycin dose and CVP goal of 10–14 mm Hg).

2. **Answer: D**
The highest quality of evidence for improved outcomes with CF exacerbations is with adjunctive therapy for mucous clearance. The guidelines recommend airway clearance options such as aggressive chest physical therapy, nebulized dornase alfa, and hypertonic saline as a grade B recommendation (Answer D is correct). Hypertonic saline would be appropriate, but nebulization of normal saline is not included in any of the recommendations (Answer A is incorrect). Both corticosteroid administration and concurrent administration of intravenous and inhaled antibiotics have a grade I recommendation, meaning evidence is insufficient to support them (Answers B and C are incorrect).

3. **Answer: D**
This patient presents with severe right heart failure. The primary goal is to optimize RV preload by maintaining a net negative fluid balance using gentle diuresis and blood pressure monitoring (Answer D is correct). Dopamine would increase blood pressure; however, it might worsen the patient's tachycardia, thereby worsening her already tenuous clinical status (Answer A is incorrect). Epoprostenol would help decrease pulmonary pressures; however, epoprostenol would potentially worsen the patient's blood pressure because of its peripheral vasodilating effects (Answer B is incorrect). Phenylephrine would not be optimal because this vasopressor might worsen RV function, further elevate pulmonary artery pressure by α1-receptors in the pulmonary vasculature, and potentially induce a reflex bradycardia (Answer C is incorrect).

4. **Answer: C**
Treatment goals for PAH include, but are not limited to, achieving and maintaining WHO FC I or II (Answer A is incorrect), preserving 6MWD to greater than 440 m (Answer B is incorrect), and preserving RV size and function (right atrial pressure less than 8 mm Hg and cardiac index 2.5 L/minute/m^2 or greater) (Answer C is correct). The clinician should normalize BNP to less than 50 ng/L (Answer D is incorrect).

5. **Answer: C**
This patient has shortness of breath at rest that is interfering with his conversational ability, and his FEV_1 is less than 50% of predicted; therefore, his asthma exacerbation would be classified as severe (Answer C is correct). FEV_1 would be greater than 50% in mild/moderate asthma exacerbation (Answers A and B are incorrect). In a life-threatening asthma exacerbation, the patient would have symptoms such as drowsiness, confusion, or silent chest. This patient could progress to life threatening but currently would not be classified as such (Answer D is incorrect).

6. **Answer: B**
This patient presents with near-fatal asthma that is unresponsive to initial therapy. Magnesium sulfate (2 g intravenously administered over 20–30 minutes) can be considered in patients who have life-threatening exacerbations and are unresponsive to conventional therapies after 1 hour (Answer B is correct). Antimicrobials are not routinely recommended for asthma exacerbations if the patient has no evidence of concurrent infection (Answer A is incorrect). Because of the lack of improved outcomes, risk of adverse effects, and superior bronchodilation with SABAs, methylxanthines (theophylline and aminophylline) are not recommended for acute asthma exacerbations (Answer C is incorrect). Mucolytic agents are irritating and may worsen cough and airflow obstruction (Answer D is incorrect).

7. **Answer: B**
The recommended corticosteroid dose for an ECOPD is prednisone 40 mg orally once daily (Answer D is

incorrect; Answer B is correct). Antimicrobial treatment should be initiated if (1) all three cardinal symptoms of an ECOPD (increased dyspnea, increased sputum production, and increased sputum purulence) are present; (2) two of the three cardinal signs are present, with increased sputum purulence as one of the symptoms; or (3) the patient requires noninvasive or invasive ventilation. This patient has no indication for antimicrobials (Answers A and C are incorrect).

8. **Answer: D**

Although the GOLD-recommended dose of steroids for an ECOPD is prednisone 40 mg orally (or intravenously), some studies that included populations of ICU patients have shown higher doses may be needed or may be tolerated with some increase in adverse effects. Currently, doses higher than prednisone 40 mg have not been linked to increased mortality (Answer A is incorrect). A recent investigation identified significant racial bias in the use of peripheral Sa_{O_2} because of frequent overestimation of Sa_{O_2} compared with a direct ABG measurement for patients with darker skin tones. This answer is incorrect because the publication did not investigate patient outcomes. This patient has a higher Sp_{O_2} measure than the patients in the study (92%–96%), who had a high rate of hypoxemia identified on ABG. Answer C is incorrect because no specific dose of macrolide has been linked to mortality, and azithromycin 500 mg intravenously once is the recommended loading dose. Answer D is correct because titrating oxygen therapy to peripheral Sa_{O_2} greater than 88%–92% has been linked to mortality. The correct intervention would have been to decrease the F_{IO_2} when the patient had an Sa_{O_2} greater than 92%.

Toxicology

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Learning Objectives

1. Distinguish between the common clinical toxidromes associated with acute poisonings.
2. Describe the general management of a patient with an acute overdose.
3. Determine the best options for the management of selected toxins.
4. Develop a patient care plan for a patient presenting with an acute overdose.
5. Identify the adverse effects and monitoring of the patient who is poisoned.

Abbreviations in This Chapter

ABC	Airway, breathing, and circulation
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
BP	Blood pressure
BUN	Blood urea nitrogen
cAMP	Cyclic adenosine monophosphate
CB1	Cannabinoid-1
CB2	Cannabinoid-2
CIWA	Clinical Institute Withdrawal Assessment for Alcohol Scale
CK	Creatine kinase
CNS	Central nervous system
COPD	Chronic obstructive pulmonary disease
ECG	Electrocardiogram
ED	Emergency department
GABAa	γ-aminobutyric acid
GI	Gastrointestinal
HIET	Hyperinsulinemic euglycemic therapy
HR	Heart rate
ICU	Intensive care unit
INR	International normalized ratio
NAPQI	*N*-acetyl-*p*-benzoquinoneimine
NMDA	N-methyl-d-aspartate
RR	Respiratory rate
SBP	Systolic blood pressure
SCr	Serum creatinine
SSRI	Selective serotonin reuptake inhibitor
TCA	Tricyclic antidepressant

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

1. A 38-year-old woman with type 2 diabetes is admitted for confusion and altered mental status. She has an active prescription for glipizide 10 mg by mouth twice daily, but she cannot respond to further questioning. Her vital signs are as follows: blood pressure (BP) 115/65 mm Hg, heart rate (HR) 68 beats/minute, respiratory rate (RR) 15 breaths/minute, and temperature 98.6°F (37°C). Her point-of-care glucose concentration is 45 mg/dL, and she is given 50 mL of 50% dextrose in water intravenously twice. Her follow-up point-of-care glucose concentration is 50 mg/dL after the first dose and 57 mg/dL after the second dose, with no improvement in symptoms. Which is the most appropriate intervention at this time?
   A. Dextrose.
   B. Glucagon.
   C. Octreotide.
   D. Sodium bicarbonate.

2. A 56-year-old man is admitted to the intensive care unit (ICU) after a β-blocker overdose. After administering 2 L of 0.9% sodium chloride and 3 g of calcium gluconate, his vital signs are as follows: BP 70/40 mm Hg, HR 52 beats/minute, and RR 22 breaths/minute. Which therapy is most appropriate at this time?
   A. Glucagon 5 mg intravenously.
   B. Atropine 2 mg intravenously.
   C. Insulin 0.1 unit/kg intravenously.
   D. Dopamine 2 mcg/kg/minute intravenously.

3. A 76-year-old woman is admitted to the emergency department (ED) with the chief concern of decreased mental status. Her vital signs are as follows: BP 118/72 mm Hg, HR 57 beats/minute, and RR 17 breaths/minute. She is experiencing nausea, but her physical examination is otherwise normal. Her husband is concerned that she may not be taking her medications properly. Given her presentation, which common toxidrome is the patient most likely experiencing?

A. Anticholinergic.
B. Cholinergic.
C. Opioid.
D. Sympathomimetic.

4. A 38-year-old woman is admitted to the ICU after a suspected overdose of risperidone. She was initially hypotensive, but she was stabilized after the administration of two 500-mL boluses of lactated Ringer solution. Her BP is now 118/77 mm Hg, HR 75 beats/minute, and RR 16 breaths/minute. A 12-lead electrocardiogram (ECG) shows QT prolongation (corrected QT interval [QTc] = 480 milliseconds), and her chemistry panel is significant for a bicarbonate of 24 mEq/L, potassium of 3.1 mEq/L, and magnesium of 1.8 mg/dL. Which intervention is most appropriate at this time?
   A. Potassium chloride 20 mEq every hour for two doses.
   B. Activated charcoal 50 g.
   C. Magnesium sulfate 2 g.
   D. Lorazepam 2 mg.

5. A 48-year-old man is admitted to the medical floor for community-acquired pneumonia. His medical history is significant for hypertension, hyperlipidemia, and chronic obstructive pulmonary disease (COPD), and he reports occasional alcohol use but no history of alcohol withdrawal. He is initiated on levofloxacin 750 mg intravenously daily and nebulizer treatments with albuterol and ipratropium. Twenty-four hours after admission, he is increasingly more confused and has nausea and vomiting. His vital signs are stable: BP 115/68 mm Hg, HR 122 beats/minute, RR 21 breaths/minute, and temperature 99.7°F (37.6°C). The team is concerned about possible alcohol withdrawal and asks for recommendations for initial therapy. Which is the most appropriate treatment for this patient?
   A. Lorazepam 2 mg intravenous push every 4 hours as needed according to the patient's Clinical Institute Withdrawal Assessment for Alcohol Scale (CIWA) score.
   B. Phenobarbital 65 mg by mouth every 8 hours as needed according to the patient's CIWA score.
   C. Propofol continuous infusion.
   D. Clonidine 0.1 mg by mouth every 12 hours.

6. A 57-year-old male patient on the medical floor is incorrectly administered a dose of methadone 40 mg by mouth that was written for the patient in the adjoining bed. Two hours later, the nurse finds him unresponsive with the following vital signs: BP 105/67 mm Hg, HR 61 beats/minute, RR 8 breaths/minute, and temperature 98.7°F (37.1°C). The nurse calls for the rapid response team, and, as the team pharmacist, you are asked for a recommendation. Which treatment is most appropriate at this time?
   A. Activated charcoal 50 g orally.
   B. Naloxone 0.04 mg intravenously.
   C. Whole bowel irrigation orally.
   D. 1 L of 0.9% sodium chloride intravenously.

Questions 7 and 8 pertain to the following case.
A 56-year-old female patient is admitted to the ED after an intentional overdose of 25 amlodipine 10-mg tablets. She is given activated charcoal 50 g, 2 L of 0.9% sodium chloride, and 3 g of calcium gluconate. Her current vital signs are as follows: BP 90/50 mm Hg, HR 107 beats/minute, RR 17 breaths/minute, and temperature 98.7°F (37.1°C). Serum chemistries are as follows: Na 141 mEq/L, K 2.5 mEq/L, Cl 101 mEq/L, HCO_3 24 mEq/L, blood urea nitrogen (BUN) 19 mg/dL, serum creatinine (SCr) 0.9 mg/dL, and glucose 215 mg/dL. The ED physician wants to initiate hyperinsulinemic euglycemic therapy (HIET).

7. Which is most appropriate to initiate or do first with respect to HIET?
   A. Give insulin 1 unit/kg intravenously.
   B. Give 50 mL of 50% dextrose in water intravenously.
   C. Warn the physician that full effects may take up to 30 minutes.
   D. Give 20 mEq of potassium chloride intravenously every hour for four doses.

8. The patient is not responding to HIET initiation. Her current infusion rate is 1 unit/kg/hour. Her BP remains low at 70/40 mm Hg, and her HR is now 58 beats/minute. Her repeat glucose is 186 mg/dL. Which is most appropriate to initiate at this time?
   A. Discontinue HIET, and initiate norepinephrine.
   B. Continue HIET, and increase the insulin infusion rate and initiate norepinephrine.
   C. Continue HIET, and initiate epinephrine.
   D. Discontinue HIET, and begin intravenous lipid therapy.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 3, 6
   b. Task B: 3, 4
2. Domain 2: Therapeutics and Managment
   a. Task B: 2, 3, 4, 9, 11, 15
   b. Task C: 1, 2, 3, 4
3. Domain 3: Professional Practice
   a. Task A: 1
   b. Task B: 1, 4

I. EPIDEMIOLOGY

A. Population based: The American Association of Poison Control Centers releases an annual report based on all the cases submitted by the 55 regional poison centers to the National Poison Data System to help clinicians stay abreast of the changing landscape of exposures (Clin Toxicol 2023;61:717-939).
   1. In 2022, 2,064,875 human exposures were reported. Fatalities were reported in 1491 cases (Table 1).
   2. The most common site of exposure was a residence (90.7%), followed by workplace (1.93%) and other's residence (1.81%).
   3. Most of the reported cases (1,159,170) occurred in children, defined in the report as younger than 20 years. To add perspective, 968,189 cases were reported in children 12 years and younger.
   4. The most common reasons associated with these exposures were unintentional (76.4%), intentional (18.9%), and adverse reactions (2.7%). Of note, therapeutic errors accounted for 331,539 (16.1%) of all cases. The scenarios reported for therapeutic errors included inadvertent double dosing (30.8%), incorrect dose (16.4%), incorrect medication administered or taken (15%), doses administered too close in time (11.1%), and inadvertent exposure to another person's medication (8.5%).
   5. Routes of exposure included ingestion (82.7%), inhalation/nasal (7.6%), dermal (6.9%), and ocular (4.1%). Of the 1491 exposure-related fatalities, most were by ingestion (83.4%), followed by inhalation/nasal (7.3%) and parenteral (2.7%).

B. Management based
   1. Only 30.6% of exposures were managed in a health care facility, with 3.4% managed in critical care units.
   2. Gastric decontamination was used in about half of all exposures (49.9%); however, antidotes were given in only 12.4% of the exposures. The most common gastric decontamination strategy was the use of activated charcoal (1.6% of total exposures), followed by other emetic agents (0.9%), cathartics (0.1%), whole bowel irrigation (0.05%), and gastric lavage (0.02%).

Table 1. Top 5 Most Common Medication-Related Toxic Exposures in 2022

Most Common Medication-Related Toxic Exposures	% of Total Cases
All human exposures	
Analgesics	11.51
Antidepressants	5.61
Antihistamines	4.81
Cardiovascular drugs	4.74
Sedatives/hypnotics/antipsychotics	4.60
Adult exposures (≥ 20 yr of age)	
Analgesics	11.09
Sedatives/hypnotics/antipsychotics	7.37
Antidepressants	7.30
Cardiovascular drugs	7.06
Alcohols	4.39
Anticonvulsants	3.81
Pediatric exposures (< 6 yr of age)	
Analgesics	9.54
Dietary/herbal/homeopathic supplements	6.65
Antihistamines	5.03
Vitamins	4.85
Topical preparations	4.06

Information from: Gummin DD, Mowry JB, Beuhler MC, et al. 2022 annual report of the American Association of Poison Control Centers' National Poison Data System (NPDS): 40th annual report. Clin Toxicol 2023;61:717-939.

II. EMERGENCY EVALUATION AND MANAGEMENT

A. The primary treatment strategy for managing a toxic exposure should focus on stabilizing the patient, with an emphasis on airway, breathing, and circulation (ABC). The most common factor contributing to death from a poisoning or drug overdose is the loss of the protective reflexes of the airway secondary to flaccid tongue, aspiration of gastric contents into the lungs, or respiratory compromise including arrest. Emergent endotracheal intubation should be performed when necessary. Vital signs (HR, RR, BP, temperature, and oxygen saturation) and changes in mental status should be closely monitored. After stabilization, the DEFG approach can be considered:
   D: Decontamination
   E: Enhanced elimination
   F: Focused antidote therapy
   G: Get help from a poison control center or toxicologist

B. Supportive care should be based on specific patient symptoms and may include the administration of intravenous fluids, supplemental oxygen, and advanced airway management. Other potential complications should be assessed, such as presence of rhabdomyolysis, rigidity, or dystonia. Additional tests, such as a 12-lead ECG, chest radiograph, or electroencephalogram may be required. In addition, essential laboratory tests should be conducted and assessed for the presence of an osmolar gap, anion gap acidosis, hyper/hypoglycemia, hyper/hyponatremia, hyper/hypokalemia, renal failure, and liver failure.
   1. Use of "coma cocktail" preparations is controversial and therefore not routinely recommended because they should not replace or substitute for a thorough analysis of the patient (JEMS 2002;27:54-60). Formulations vary, but they typically contain one or more of the following: dextrose, thiamine, and naloxone. The following text presents an overview of the common additives, a rationale for use, and potential controversies.
      a. Dextrose 50%, 12.5–25 g (25–50 mL) intravenously is administered to treat hypoglycemia; it is recommended to perform point-of-care blood glucose testing to confirm before administration.
      b. Thiamine 100 mg is administered intravenously to prevent Wernicke encephalopathy; often under-recognized, several doses of high-dose parenteral thiamine (e.g., 500 mg) concurrent with or immediately following intravenous dextrose before intravenous dextrose are typically required to effectively treat (J Emerg Med 2012;42:488-94).
      c. Naloxone 0.04–2 mg IV is administered in a stepwise titration to reverse respiratory depression secondary to opiate overdose.

C. Ingestions
   1. A thorough physical examination should be performed.
   2. A medication history and reconciliation should be done, including all prescription medications, over-the-counter agents, illicit substance use, and herbal products.
   3. The history of the ingestion should be determined, if possible, including the following elements (Ann Emerg Med 1999;33:735-56):
      a. Timing and route of the exposure, the possible agents involved and their strengths and amounts, and the potential intent of the patient
      b. History from the prehospital care providers, family members, or other patient advocates
      c. Onset and progression of any symptoms
   4. Some providers advocate for the use of toxidromes, which are a collection of symptoms that occur with particular classes of toxic agents. Toxidromes may help identify the toxic agent and assist in care by helping providers anticipate additional symptoms. Although they may be very useful in the care of an acute poisoning, they should be used with caution because some symptoms may overlap with other classes of toxins, may be impacted by comorbid conditions, or may be absent altogether.

5. Common Toxidromes and Presentation (Chest 2011;140:795-806) (Crit Care Clin 2012;28:180-198)
   a. Anticholinergic
      i. Mechanism of toxicity is through competitive antagonism of the effects of acetylcholine at peripheral muscarinic receptors and central receptors.
      ii. Signs and symptoms include altered mental status, delirium, hallucinations, mumbled speech, dry mucous membranes, mydriasis, anhidrosis, flushing, hyperthermia, tachycardia, hypoactive bowel sounds, and urinary retention.
      iii. Common drugs that have anticholinergic activity include antihistamines, antipsychotics, tricyclic antidepressants, and skeletal muscle relaxants.
   b. Cholinergic
      i. Mechanism of toxicity is inhibition of acetylcholinesterase causing accumulation of acetylcholine ultimately resulting in overstimulation of muscarinic and nicotinic receptors.
      ii. Signs and symptoms include confusion, central nervous system (CNS) depression, miosis, wet mucous membranes, salivation, lacrimation, diaphoresis, emesis, urination, diarrhea, muscle weakness/twitching, bronchospasm, bronchorrhea, hypertension, tachycardia, and bradycardia.
      iii. Common drugs that have cholinergic activity include organophosphates, nerve agents, nicotine, pilocarpine, and physostigmine.
   c. Opioid
      i. Mechanism of toxicity is stimulation of opioid receptors causing a decrease in autonomic activity.
      ii. Signs and symptoms include sedation, miosis, decreased bowel sounds (ileus), respiratory depression, hypotension, and bradycardia.
      iii. Common drugs that have opioid activity include heroin, morphine, codeine, synthetic opioids, loperamide, and dextromethorphan (in large quantities).
   d. Sympathomimetic
      i. Mechanism of toxicity is through an increase in sympathetic tone through release of cathecholamines, inhibition of reuptake, by direct receptor stimulation, and alterations in neurotransmitter metabolism.
      ii. Signs and symptoms include agitation, delirium, mydriasis, diaphoresis, myoclonus, hyperthermia, hypertension, and tachycardia.
      iii. Common drugs that have sympathomimetic activity include cocaine, methamphetamine, pseudoephedrine, and caffeine.
6. Drug screens are used in acute toxic ingestions, the most common of which is the qualitative urine screen. This method tests for the presence of a substance, but it cannot detect the amount of substance present. If a toxin is known, a quantitative drug screen may be used to confirm the exact amount present. Although urine drug screens may vary by institution, they may include amphetamines, barbiturates, benzodiazepines, cocaine, MDMA (ecstasy), methamphetamines, opiates, THC (marijuana), and tricyclic antidepressants (TCAs). Urine screens are not considered comprehensive; therefore, the presence of additional agents should be tested for (e.g., acetaminophen, salicylates). However, comprehensive drug screens using gas chromatography and liquid chromatography together with mass spectrometry can be used in patients presenting with severe or unexplained toxicity.
   a. A negative screen does not exclude the presence of a toxic substance, especially if the presumed agent is not present on the screen. Many agents are not identified by their designated screen; this is especially an issue with standard amphetamine, benzodiazepine, and opiate screens.

b. A positive test also does not necessarily confirm the diagnosis because another agent may be present but at concentrations below a detectable threshold. In addition, a positive test does not indicate that the patient is intoxicated on the particular substance (e.g., cocaine is positive for 3 days; however, its effects last only a few hours) or that the agent ingested is the exact agent that is screened (e.g., bupropion causes a false-positive amphetamine screen) (Am J Health Syst Pharm 2010;67:1344-50).

Patient Case

1. A 53-year-old man (height 74 inches, weight 97 kg [215 lb]) arrives in the ED confused and disoriented. He cannot provide any information about his condition or medical history. Vital signs are as follows: BP 85/50 mm Hg, HR 120 beats/minute, RR 28 breaths/minute, and temperature 99.2°F (37.3°C). On physical examination, an unmarked pill bottle is found in his pocket. Two tablets remain, and a possible drug overdose is suspected. Which is most appropriate to do first for this patient?
   A. Send a quantitative urine drug screen.
   B. Stabilize the patient's ABC.
   C. Order a coma cocktail.
   D. Try to identify the tablets in a drug database.

III. GASTRIC DECONTAMINATION/ENHANCED ELIMINATION

A. Many strategies for gastric decontamination are used to try to remove toxins or prevent further absorption. No particular strategy is preferred to another; each has certain advantages and disadvantages, and the risks and benefits must be considered before use. Consensus statements from the American Academy of Clinical Toxicology and the European Association of Poisons Centres and Clinical Toxicologists recommend against the routine use of any decontamination strategy but suggest that these strategies play a role in individualized care after a poison exposure (Clin Toxicol 2013;51:127). Table 2 lists the common dosing strategies for general decontamination and enhanced elimination.

B. Ipecac
   1. Ipecac is no longer manufactured in the United States since 2010 and is no longer recommended because of concerns for safety and ability to improve outcomes for patients who have been poisoned.
   2. Mechanism of action is to induce vomiting through irritation of the gastric mucosa and stimulation of the chemoreceptor trigger zone in the medulla.
   3. Guidelines recommend ipecac (if available) to be given only under a specific recommendation from a poison control center, ED physician, or other qualified medical personnel when all the following conditions are met (Clin Toxicol 2013;51:134-9; Clin Toxicol 2005;43:1-10):
      a. No specific contraindication exists for use.
      b. There is a substantial risk of serious toxicity of the toxin to the patient.
      c. No alternatives are available or considered effective to reduce the toxin absorption.
      d. A delay of more than 1 hour is expected before arrival to a medical facility.
      e. Use of ipecac will not adversely affect a more definitive treatment option.

C. Gastric Lavage
   1. Gastric lavage is performed by inserting a larger-bore orogastric or nasogastric catheter tube (36–40 French for adults and 24–28 French for children) with several holes at the distal end into the stomach. Aliquots of warmed tap water (200–300 mL) are then instilled until there is clearing of aspirated fluid.
   2. The efficacy is highly variable and diminishes over time; therefore, it is optimal to perform within 60 minutes of ingestion.
   3. Guidelines emphasize that gastric lavage has not been proven to decrease the severity of illness, improve recovery times, or improve outcomes (Clin Toxicol 2013;51:140-6).
   4. Should be considered only for life-threatening ingestions when it can be safely performed within 30–60 minutes of ingestion.
   5. Even in life-threatening overdoses, it may not be beneficial. Gastric lavage should not be performed routinely, if at all, for treating the patient who is poisoned. In the rare situation when it might be appropriate, clinicians should consider treatment with activated charcoal or observation and supportive care in place of gastric lavage (Clin Toxicol 2013;51:140-6).
   6. Contraindications for gastric lavage include patients with craniofacial abnormalities, concomitant head trauma, unprotected airway, increased risk of aspiration, those at risk of GI hemorrhage or perforation and caustic ingestion such as acids (e.g., boric acid) and alkalis (e.g., dishwasher detergents) because of the risk of exacerbating any esophageal or gastric injury. Patients with decreased consciousness require endotracheal or nasal intubation before the procedure.
   7. Complications associated with gastric lavage include aspiration, laryngospasm, perforation of the esophagus or stomach, arrhythmias, fluid imbalance, hyponatremia, and small conjunctival hemorrhages.

D. Cathartics
   1. Used to reduce the transit time of toxins and hence absorption, as well as in combination with activated charcoal to decrease constipating effects
   2. Cathartics have conflicting data regarding decreased transit time in the GI tract and have no data to support improved patient outcomes (J Toxicol Clin Toxicol 2004;42:243-53).
   3. Cathartic use is not recommended; if used, it should be limited to a single dose and not be used as monotherapy.
   4. Contraindications to cathartic use include absence of bowel sounds, recent GI surgery, intestinal perforation or obstruction, hypotension, electrolyte disturbances, and renal insufficiency (for magnesium-based cathartics).
   5. Complications include nausea, dehydration, hypotension, and magnesium imbalances.

E. Activated Charcoal
   1. Activated charcoal is an adsorbent that works by binding the toxin throughout the GI tract to reduce systemic absorption. Although activated charcoal binds most substances, Table 3 lists the agents for which activated charcoal is NOT recommended.
      a. Acids and alkalis should be avoided because charcoal may cause vomiting, which can be damaging in these ingestions. The black color and thickness of the activated charcoal may also cause discoloration of the stomach lining and therefore interfere with endoscopy.
      b. Alcohols bind poorly; therefore, large doses are needed, which are difficult to ingest.
      c. Cyanide will bind, but not with as much activity as other substances. Because the toxic dose of cyanide is so small, normal doses of activated charcoal may be ineffective.
      d. Hydrocarbons may lead to a significant risk of aspiration.
   2. It is optimal to administer activated charcoal within 60 minutes of the toxin ingestion to maximize efficacy.

3. The use of a cathartic (e.g., sorbitol) in combination with activated charcoal is not recommended.
4. If significant nausea occurs, it is recommended to administer an antiemetic. When choosing an antiemetic, potential drug and symptom interactions should be considered as well.
5. Complications include aspiration, accidental administration into the lung, emesis, constipation, and gastric obstruction.
6. Contraindications include an unconscious state or an inability to otherwise protect the airway without endotracheal intubation, ileus or intestinal obstruction, late presentation (more than 2 hours), and recent GI surgery.
7. Multidose activated charcoal is a method described to enhance the elimination of certain toxins. It is not more effective in reducing morbidity or mortality than single-dose charcoal, but it may be administered to enhance elimination in life-threatening ingestions caused by medications that undergo significant enterohepatic recirculation with active enterohepatic metabolites (J Toxicol Clin Toxicol 1999;37:731-51).

F. Whole Bowel Irrigation
1. Whole bowel irrigation is a strategy for cleansing the bowel to remove potential toxins by administering an osmotic polyethylene glycol solution.
2. Not recommended for routine use, but may be useful in potentially life-threatening ingestions of medications with long half-lives, sustained-release dosage forms, or enteric-coated formulations. Specifically useful for certain toxic substances not adsorbed by activated charcoal (e.g., lithium and iron). May also be beneficial for iron overdoses and for "packers" of illicit substances.
3. Concurrent administration of activated charcoal and whole bowel irrigation may decrease the efficacy of charcoal.
4. Complications of the polyethylene glycol electrolyte solutions include anaphylaxis, angioedema of the lips, aspiration, Mallory-Weiss tear, and esophageal perforation.
5. Contraindications include bowel obstruction, perforation, ileus, and use in patients with recent bowel surgery. A kidney-ureter-bladder radiograph may be used to rule out these contraindications.

G. Urine Alkalinization
1. Urine alkalinization is a strategy to improve the elimination of toxins by increasing the urine pH to levels of 7.5 or greater via administration of sodium bicarbonate or sodium acetate (J Toxicol Clin Toxicol 2004;42:1-26).
2. Specific substances that may benefit from this strategy include salicylates, phenobarbital, chlorpropamide, and other weak acids with intrinsic urinary clearance.
3. Contraindications include acute and chronic renal failure and preexisting heart failure owing to the volume of fluid required for this treatment strategy.
4. Complications include hypokalemia, hypernatremia, hypocalcemia, cerebral vasoconstriction, and coronary vasoconstriction.
5. To administer urine alkalinization, it is recommended to check baseline blood chemistries, electrolyte values, and an arterial blood gas, as well as to correct any fluid or electrolyte deficits (especially potassium because alkalemia will push potassium intracellularly). Hypokalemia will make it impossible to get the urine alkaline because of the K+-H+ exchange in the kidneys, which will excrete H+ into urine if K+ is low.
6. Guideline-recommended monitoring includes urine pH every 15–30 minutes (every 30–60 minutes is more accepted in clinical practice) until the goal pH level of 7.5–8.5 is achieved, followed by every hour; serum potassium concentrations, central venous pressure, and arterial blood gases should be measured hourly.

Table 2. Common Dosage Strategies for General Decontamination and Enhanced Elimination

Decontamination/ Elimination Strategy	Pediatric Dosing	Adult Dosing
Gastric lavage[a]	10-mL/kg aliquots, followed by return of an equal amount	200- to 300-mL aliquots, followed by return of an equal amount
Cathartics		
Magnesium citrate:	4 mL/kg	240 mL
Sorbitol:	4.3 mL/kg (35% solution)	1–2 mL/kg (70% solution)
Activated charcoal		
Single dose	Up to 1 yr of age: 0.5–1 g/kg (usually 10–25 g) 1–12 years: 0.5–1 g/kg (usually 25–50 g)	> 12 years and adults: 25–100 g (doses > 50 g may induce vomiting)[b]
Multidose	0.5–1 g/kg (25–50 g), followed by 0.25–0.5 g/kg (10–25 g) every 4 hr	50 g, followed by 25–50 g every 4 hr
Whole bowel irrigation[c]	9 mo to 6 yr: 500 mL/hr 6–12 yr: 1000 mL/hr	> 12 yr and adults: Goal is 2000 mL/hr (initiated at 500 mL/hr and doubled every 30 min)
Urine alkalinization[d]	25–50 mEq intravenously for 1 hr	250 mEq intravenously for 1 hr

[a]Sterile water or 0.9% sodium chloride; may repeat until the return fluid is clear and absent of particulate matter.

[b]Upper limit may vary depending on the capacity of the stomach.

[c]Polyethylene glycol electrolyte lavage solutions; dose until the rectal effluent is clear or the desired effect has been achieved.

[d]Sodium bicarbonate in dextrose 5% in water additional boluses can be given hourly (or begin a continuous infusion at this hourly rate) to maintain a urine pH of 7.5–8.5.

Table 3. Agents for Which Activated Charcoal Is NOT Recommended

Substance	Examples
Acids	Boric acid, mineral acids
Alcohols	Ethanol, ethylene glycol, methanol
Alkalis	Bleach, cleaning solutions, dishwasher detergents, lye
Carbamates	Insecticides, neostigmine, physostigmine
Cyanide	Cyanogen chloride, hydrogen cyanide, potassium cyanide, sodium cyanide
Hydrocarbons	Gasoline, kerosene, petroleum oils
Metals	Arsenic, iron, lead, lithium, mercury
Organic solvents	Acetic acid, acetone, ethylene glycol, glycerin, toluene
Organophosphates	Antihelminthic drugs (trichlorfon), insecticides (malathion, parathion), herbicides

IV. ACETAMINOPHEN

A. Background
   1. Acetaminophen is consistently one of the most common toxic drug exposures.
   2. Accounted for 66,710 exposures (as a single agent) and resulted in 167 deaths in 2022.
   3. In general, acute doses of 150 mg/kg or 7.5 g in adults and 200 mg/kg in children are considered toxic. It is recommended that doses exceeding this threshold be managed in a health care facility.

4. The mechanism of toxicity is caused by the active metabolite *N*-acetyl-*p*-benzoquinoneimine (NAPQI), which can lead to oxidant cell injury, hepatic failure, and death.
5. Around 90% of acetaminophen undergoes phase II conjugation to glucuronide and sulfate conjugates that are excreted in the urine. An additional 2% is excreted unchanged in the urine. The remaining amount (8%–10%) is converted by cytochrome P450 (CYP2E1) to NAPQI. NAPQI is normally converted by glutathione to cysteine conjugates, which are renally excreted. In an overdose, the sulfation and glucuronidation pathways become saturated, leading to glutathione depletion and a subsequent buildup of NAPQI (Clin Liver Dis 2013;17:587-607).

B. Clinical Presentation – Four clinical phases are associated with acetaminophen toxicity (time intervals are estimated and may vary with individual patients).
1. Phase I occurs within the first 24 hours after ingestion. Patients may present with minimal or no signs of distress. Potential signs and symptoms include nausea, vomiting, diaphoresis, and anorexia.
2. Phase II occurs 24–48 hours after exposure and is marked by initial damage to the hepatocytes. Patients may present with right upper quadrant pain, increases in liver transaminases, elevated total bilirubin concentrations, and prolonged prothrombin time.
3. Phase III occurs 72–96 hours after initial exposure and is the peak of the hepatotoxic effects. Patients may present with lactic acidosis, acute renal failure, acute pancreatitis, and fulminant hepatic failure, as evidenced by jaundice, extensive coagulopathies, hypoglycemia, and hepatic encephalopathy.
4. Phase IV occurs about 1 week after exposure and marks the recovery phase if the patient survives phase III.

C. Treatment
1. The goal of treatment is to prevent the development of hepatic toxicity and reduce mortality.
2. Gastric decontamination with a single dose of activated charcoal can be considered if the patient presents within the first hour after exposure, is not vomiting, and has no alterations in mental status. However, with the availability of highly efficacious antidotal therapy, the risks often outweigh the benefits.
3. Antidote therapy is recommended with acetylcysteine. The mechanism of action for acetylcysteine is to increase the synthesis and bioavailability of glutathione, substituting for glutathione by binding to the reduced sulfur group of NAPQI, and supplying a substrate for sulfation, thereby increasing nontoxic metabolism. Additional mechanisms of action have also been proposed, including its facilitation of reactive oxygen and nitrogen species scavenging.
4. Guidelines suggest that acetylcysteine treatment be administered to patients within the first 8 hours of exposure if they can be stratified as being at possible or probable risk of hepatotoxicity by the Rumack-Matthew nomogram (Figure 1). If patients cannot be stratified because of unknown time of ingestion, they should receive acetylcysteine if any of the following conditions apply: significantly elevated alanine aminotransferase (ALT) concentration, serum acetaminophen concentrations greater than 20 mcg/mL, or history of chronic ingestions exceeding 4 g/day with an elevated serum ALT concentration and/or presenting more than 24 hours post-ingestion with evidence of hepatotoxicity (Ann Emerg Med 2007;50:292-313).
    a. This includes patients presenting more than 24 hours postingestion with evidence of hepatotoxicity.
    b. Limitations to use of the Rumack-Matthew nomogram include the following (Ann Emerg Med 2007;50:292-313):
        i. Presentation more than 24 hours postingestion
        ii. An unknown or unreliable history of ingestion
        iii. Overdoses with extended-release formulations
        iv. Chronic or repeated supratherapeutic ingestions
        v. Patients with preexisting hepatic disease, chronic alcohol use, or concurrent medications metabolized by the CYP system or that alter gastric motility.

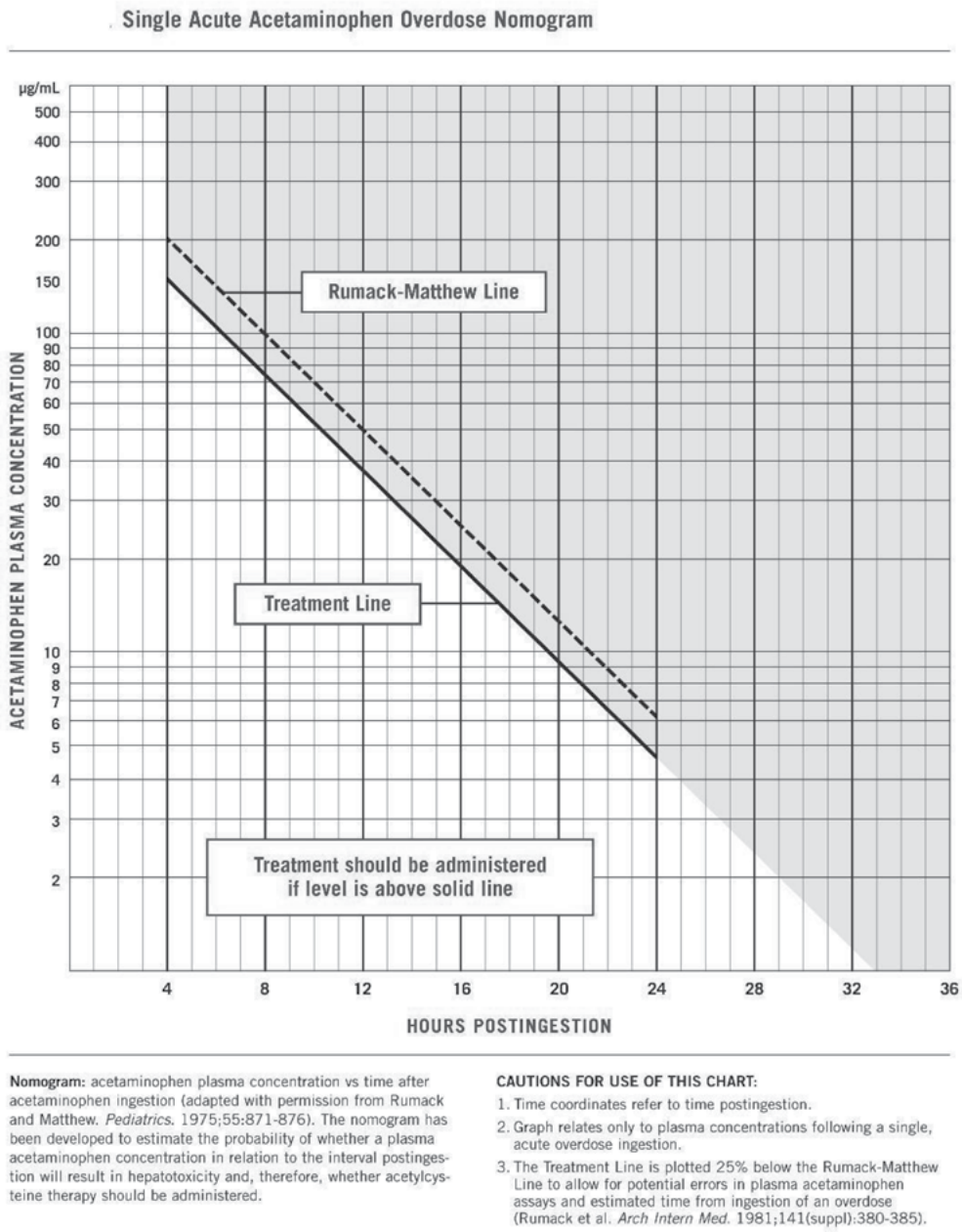


Figure 1. Rumack-Matthew nomogram.

Reprinted from: Tylenol for Healthcare Professionals. Guidelines for the Management of Acetaminophen Overdose. McNeil Consumer & Specialty Pharmaceuticals. Available at www.tylenolprofessional.com/guidelines-and-studies.html.

5. Intravenous acetylcysteine is advantageous because of its decreased overall administration time (21 hours vs. 72 hours for oral) and minimal GI adverse effects. If intravenous acetylcysteine formulation is not available and cannot be obtained in a timely fashion, poison control centers can be contacted for instructions on compounding the inhalational acetylcysteine formulation for intravenous use. It is not recommended to use this strategy except for emergencies.

6. Oral acetylcysteine is dosed for 18 total doses; doses may be repeated if emesis occurs within 1 hour of a dose. Two oral formulations exist. One preparation is an effervescent tablet that must be dissolved in water before administration, and the other is a liquid preparation that may be diluted in juice or carbonated beverages to improve palatability. Antiemetics may be administered if significant nausea or vomiting occurs. Table 4 discusses the dosing strategies for oral and intravenous administration of acetylcysteine.
    a. For exceptionally large ingestions (e.g., over 30 g), some studies have recommended alternative approaches:
        i. An increased dose of acetylcysteine (Med J Aust 2020;212:175-83).
        ii. Given that acetaminophen is dialyzable, intermittent hemodialysis is an option (Clin Toxicol 2014;52:856-67). This approach would also require an increase in the acetylcysteine dose (i.e., 2-fold) due to its concurrent increased clearance (Clin Toxicol 2016;54:519-22).
7. Although treatment guidelines recommend 18 total doses of acetylcysteine administered throughout 72 hours for oral acetylcysteine therapy and 21 hours of the intravenous infusion of acetylcysteine, many poison control centers recommend early discontinuation if the following conditions are met (or prolonged therapy if they are not met) (Ann Emerg Med 2007;50:280-1):
    a. Serum acetaminophen concentrations are undetectable or less than 10 mcg/mL.
    b. ALT concentrations are normal (60 IU/L or less) or improving. Some clinicians also advocate for an international normalized ratio (INR) of 1.3 or less.
    c. The patient is clinically improved.
8. Adverse effects (intravenous): Anaphylactoid reactions (rash, urticarial, pruritus) that are often infusion rate-associated, hyponatremia, hypervolemia, seizures (pediatric patients with unadjusted volume)
9. Adverse effects (oral): Nausea, vomiting, anaphylactoid reactions (rare)
10. Patients experiencing mild anaphylactoid reactions (rash, pruritus, flushing) can be effectively treated with diphenhydramine, and acetylcysteine therapy can be resumed.
11. Medication errors may occur because of the complex dosing regimens (Ann Pharmacother 2008;42:766-70). Common errors include delays in therapy, incorrect dosages, and incorrect infusion rates. To mitigate these errors, some institutions have developed a two-step intravenous regimen (Toxicol Commun 2018;2:81-4).

Table 4. Acetylcysteine Dosage

Route	Dose
Oral	**Loading dose:** 140 mg/kg **Maintenance doses:** 70 mg/kg every 4 hr for a total of 17 doses (72 hr)
Intravenous	**Loading dose:** 150 mg/kg (max 15 g)[a] in 200 mL of 5% dextrose in water infused for 60 min **Maintenance dose:** 50 mg/kg (max 5 g)[a] in 500 mL of 5% dextrose in water infused for 4 hr followed by 100 mg/kg (max 10 g)[a,b] in 1000 mL of 5% dextrose in water infused for 16 hr Patients weighing < 40 kg require reduced volume administration

[a] Dose limits are provided by the manufacturer; however, small retrospective studies have shown that acetylcysteine is well tolerated when using actual body weight for patients > 100 kg and exceeding the recommended max doses (Am J Ther 2016;23:e714-9.)(Am J Ther 2014;21:159-63).

[b]Higher ongoing infusion rates (e.g., 200 mg/kg over 16 hr) may be required for massive paracetamol ingestions (i.e., initial concentration that is double the standard nomogram line) and a clinical toxicologist should be consulted (Chiew AL, Reith D, Pomerleau A, et al. Updated guidelines for the management of paracetamol poisoning in Australia and New Zealand. Med J Aust 2020;212:175-183. doi: 10.5694/mja2.50428).

D. Monitoring
   1. Patients should be monitored for improvement in vital signs and mental status.
   2. The following laboratory values should be monitored periodically for improvement as well as for potential worsening.
      a. ALT, aspartate aminotransferase (AST), total bilirubin, INR, and prothrombin time
      b. Acetylcysteine may cause a dose-dependent decrease in prothrombin time, although this has no clinical effect on coagulation
      c. BUN and SCr
      d. Serum electrolytes
      e. Fulminant hepatic failure: Serum bicarbonate, serum sodium, serum lactate, arterial blood gas, serum glucose, and ammonia concentrations

Patient Case

Questions 2 and 3 pertain to the following case.
A 42-year-old woman (height 66 inches, weight 79.2 kg [176 lb]) presents to the ED with the chief concern of flu-like symptoms. Her symptoms include headache, congestion, severe nausea and vomiting, abdominal pain, and some confusion. She has been taking acetaminophen 500-mg caplets as needed for her symptoms, but she just ran out of the bottle she purchased yesterday. On presentation, she is alert and oriented. Her vital signs are as follows: BP 135/90 mm Hg, HR 83 beats/minute, RR 18 breaths/minute, and temperature 101.8°F (38.8°C). An acetaminophen concentration on admission was 100 mcg/mL, AST 560 IU/L, and ALT 310 IU/L. The physician wants to begin general management.

2. Which general management strategy is most indicated for this patient?
   A. 10% magnesium citrate 240 mL per tube once
   B. Continued stabilization of the patient and good supportive care
   C. Gastric lavage
   D. Charcoal 50 g once

3. Which is the most appropriate treatment for her acetaminophen toxicity?
   A. Give acetylcysteine 11,200 mg oral bolus, followed by 5600 mg orally every 4 hours for 17 doses.
   B. Give acetylcysteine 11,200 mg intravenous bolus, followed by 5600 mg intravenously every 4 hours for 12 doses.
   C. Give acetylcysteine 12,000 mg intravenously over 1 hour, followed by 4000 mg intravenously over 4 hours; then 8000 mg intravenously over 16 hours.
   D. Acetylcysteine therapy is not indicated in this patient.

V. SALICYLATES

A. Background
   1. Salicylates as a single agent (not in combination with other agents) accounted for 6595 overdoses and 18 deaths in 2022. These numbers, which include overdoses of both adult and pediatric formulations of acetylsalicylic acid, are often underreported because these products are not typically recognized as a potential cause.
   2. However, because of the development of child-resistant containers, aspirin overdose causing accidental death in children has reduced drastically. (Lancet 1977;2: 289-90)

B. Clinical Presentation
   1. The mechanism of toxicity for salicylates is through the interference with aerobic metabolism owing to the uncoupling of mitochondrial oxidative phosphorylation, leading to increases in anaerobic metabolism, which causes a significant lactic acidosis (Emerg Med Clin North Am 2007;25:333-46). This also leads to hypoglycemia because of glycogen depletion, gluconeogenesis, and catabolism of proteins and free fatty acids. Salicylates also directly stimulate the respiratory center, leading to hyperventilation and respiratory alkalosis. Secondary complications from hyperventilation include dehydration and compensatory metabolic acidosis.
   2. Salicylates are readily absorbed in the stomach and small intestine and are then conjugated with glycine in the liver to the active component, salicylic acid. In overdoses, the liver cannot metabolize the excess drug, and most is then excreted unchanged by the kidneys (Postgrad Med 2007;121:162-8).
   3. The most common clinical symptoms associated with a salicylate overdose are hyperventilation (causing respiratory alkalosis), tinnitus, and GI irritation. Symptoms may vary depending on the serum salicylate concentration; however, these may be low to normal early in the presentation (Postgrad Med 2007;121:162-8):
      a. Serum concentration less than 30 mg/dL: Asymptomatic
      b. Serum concentration 15–30 mg/dL: Therapeutic concentrations
      c. Serum concentration 30–50 mg/dL: Hyperventilation, nausea, vomiting, tinnitus, dizziness
      d. Serum concentration 50–70 mg/dL: Tachypnea, fever, sweating, dehydration, listlessness
      e. Serum concentration greater than 70 mg/dL: Coma, seizures, hallucinations, stupor, cerebral edema, dysrhythmias, hypotension, oliguria, renal failure
   4. Acute salicylate toxicity is typically associated more with GI symptoms; chronic toxicity is more associated with CNS-type symptoms.
   5. Absorption may be delayed up to 36 hours because of gastric pylorospasm, bezoar formation (precipitate concretions because of poor solubility), or enteric-coated formulations; therefore, these ranges should be used with caution because they may not correlate with actual symptoms (Am J Emerg Med 2010;28:383-4).

C. Treatment
   1. There is no antidote for salicylate poisoning; the goals of therapy are to limit the additional absorption of salicylates and to provide supportive care.
   2. Maintain a patent airway and assist ventilation, if necessary. Ensure adequate ventilation to prevent respiratory acidosis; however, do not mechanically ventilate patients unless clinically necessary as this will interfere with the patient's ability to appropriately compensate and maintain pH.
   3. Gastric decontamination with a single dose of activated charcoal is recommended within 60 minutes of acute ingestions and if the patient is alert with the absence of vomiting. However, there may be a role for administration after this time frame, as well as multidose administration based on the characteristics of the ingestion.
   4. Administer intravenous crystalloid fluids at a rate of 10-20 mL/kg/hr for the first 2 hours to maintain a normal BP and a urine output of 1–1.5 mL/kg/hr.
   5. Administer intravenous glucose for hypoglycemia or significant neurologic symptoms.
   6. Urine alkalinization is recommended to enhance renal elimination and increase the glomerular filtration rate.
      a. The following administration strategies have been described in the literature:
         i. Administer 250 mL of sodium bicarbonate 8.4% over 1 hour; then administer additional 50-mL boluses as needed to maintain a goal urine pH range of 7.5–8.5.
         ii. Administer a continuous infusion of 150 mL of sodium bicarbonate 8.4% in 1 L of 5% dextrose in water at 2–3 mL/kg/hour to maintain a urine output of 1–2 mL/kg/hour. Do not allow the serum pH to fall below 7.4.

   b. Oral bicarbonate is not recommended because it may enhance salicylate absorption by accelerating tablet dissolution.
   c. Discontinue bicarbonate once the serum salicylate concentrations are less than 30 mg/dL or there is a resolution of clinical symptoms.
7. Alkaline diuresis will increase potassium secretion, and hypokalemia will make it more challenging to raise the urine pH. Adequate potassium concentrations should be aggressively maintained.
8. Consider hemodialysis for any of the following (Postgrad Med 2007;121:162-8):
   a. Acute renal insufficiency
   b. End-organ damage (severe pulmonary edema, seizures, rhabdomyolysis)
   c. Altered mental status
   d. Deterioration of clinical status
   e. Severe acid-base disturbances
   f. If hemodialysis is not effective, hemoperfusion and continuous renal replacement therapies may be considered (Ann Emerg Med 2015;66:165-81).

D. Monitoring
1. Patients should be monitored for up to 24 hours because of the possibility of delayed or impaired absorption.
2. Monitor RR, and support as needed; caution is advised if intubation is required because of a requirement for an increased minute ventilation (Am J Emerg Med 2010;28:383-4).
3. During urine alkalinization, monitor for signs and symptoms of fluid overload, hypernatremia, hypokalemia, hypocalcemia, and worsening alkalemia.

Patient Case

4. A 62-year-old man presents to the ED with the chief concern of nausea, tachypnea, and flu-like symptoms. He is alert and oriented and can communicate that his symptoms have been worsening for the past 2 days. He has been taking a combination cold product, which he thinks has helped. His medical history is significant for a stroke, for which he takes aspirin 325 mg by mouth daily, and hypertension, for which he takes amlodipine 5 mg by mouth daily. His vital signs are as follows: BP 135/82 mm Hg, HR 78 beats/minute, RR 29 breaths/minute, and temperature 100.2°F (37.9°C). Arterial blood gas results are as follows: pH 7.52, Pco_2 25, and HCO_3 20 mEq/L. A salicylate concentration is sent, which is 25 mg/dL. Which treatment management strategy is most indicated for this patient?
   A. Sodium chloride infusion
   B. Urgent endotracheal intubation
   C. Sodium bicarbonate infusion
   D. Hemodialysis

VI. OPIOIDS

A. Background
   1. Opioids as a single agent (not in combination with other agents) accounted for 20,392 overdoses and 346 deaths in 2022. The Centers for Disease Control and Prevention reports that almost 15,000 deaths are caused by prescription opioid painkillers annually.
   2. The most common agents associated with a toxicologic event were fentanyl (prescription and non-prescription), heroin, oxycodone, buprenorphine, and tramadol.
   3. The most common agents associated with a toxicologic death were fentanyl (prescription and non-prescription), heroin, oxycodone, methadone, and morphine.
   4. Opioids act at the mu, delta, and kappa opioid receptors, although mu is responsible for most of the opioids' clinical effects.

B. Clinical Presentation
   1. The most common clinical symptoms associated with opioid overdose are respiratory depression (defined as fewer than 12 breaths/minute), coma, miosis, and hypoactive bowel sounds.
   2. Additional findings may include stupor, hepatotoxicity, acute renal failure, rhabdomyolysis, compartment syndrome, hypothermia, and seizures (N Engl J Med 2012;367:146-55).
   3. Diagnostic workup is as follows: (Pharmacotherapy: A Pathophysiologic Approach, 9e. New York: McGraw-Hill, 2014):
      a. 12-lead ECG to assess for QT prolongation (methadone may cause QT prolongation leading to torsades de pointes).
      b. Arterial blood gas to assess for respiratory acidosis
      c. Standard chemistry panel for electrolyte and glucose abnormalities as well as Creatine kinase (CK), BUN, and SCr for signs of rhabdomyolysis
      d. Pulse oximetry

C. Treatment
   1. Stabilize the airway, provide supplemental bag-valve mask respirations if needed, and administer supplemental oxygen. Establish an airway, if needed, by endotracheal intubation.
   2. Administer intravenous crystalloid fluids to maintain BP.
   3. Gastric decontamination with a single dose of activated charcoal may be considered only if the patient presents within the first hour after exposure and is awake with an intact airway. Whole bowel irrigation can be considered for extended-release formulations or for "packers" of illicit substances (including ingestion of fentanyl patches).
   4. Antidote therapy (N Engl J Med 2012;367:146-55; Circulation 2023;148:e149-84):
      a. Naloxone is a competitive antagonist at the opioid receptor.
      b. The intravenous route is preferred, but naloxone is also effective through the endotracheal, intramuscular, intranasal, inhalational, intraosseous, or intrapulmonary route. However, it is important to note that the bioavailability and onset of action can vary greatly between different routes.
      c. Onset of action of intravenous naloxone is 1–2 minutes, with a duration of 30–120 minutes.
      d. Dosing may be affected by the specific opioid agent and dose, affinity for the mu-receptor, and patient weight. Goal of care is reversal of ventilatory depression, not necessarily reversal of neurologic depression, as this is more likely to precipitate acute opioid withdrawal syndrome.

e. Initial dose is 0.04 mg in adult patients and 0.1 mg/kg in pediatric patients; if no response, the dose is increased every 2–3 minutes to 0.5 mg, 2 mg, 4 mg, and 10 mg, followed by 15 mg. Large doses and frequent repeat administration of naloxone are necessary for patients thought to have had an overdose of heroin laced with carfentanil.
f. It is recommended that a continuous infusion be initiated at a dose of two-thirds the effective bolus dose per hour (0.04–4 mg/hour) for patients requiring subsequent naloxone doses to sustain effect secondary to exposure to long-acting opioids (e.g., methadone) or a sustained-release product (Ann Emerg Med 1986;15:566-70).
g. Intranasal administration appears to be safe and effective when the intravenous route is not available. The dose is administered by attaching a 2 mg/2 mL naloxone prefilled syringe to an atomizer and spraying an equal portion (1 mL) of the contents into each nostril. Commercial preparations contain up to 4 mg per actuation.
h. Adverse effects are rare and may be more related to a return of sympathetic response to opioid withdrawal. Nausea and vomiting are common and can lead to aspiration. If the situation allows, providing assisted manual ventilation prior to administering naloxone can potentiate the sympathetic response to opioid reversal (Anesth Analg 1988;67:730-6).
i. If no effect is seen at the higher naloxone doses, consider other causes such as co-ingestants or alternative agents.

D. Monitoring – Observe respiratory status and vital signs for a minimum of 4 hours after the last dose of naloxone or discontinuation of the continuous infusion. The duration of a naloxone continuous infusion will vary based on the quantity, pharmacokinetics, and pharmacodynamics of the opioid exposure, in addition to patient specific factors. Closely monitor for signs and symptoms of opioid withdrawal syndrome, such as anxiety, piloerection, heightened sensation to pain, abdominal cramps, diarrhea, and insomnia.

VII. LOPERAMIDE

A. Background
1. A nonprescription antidiarrheal medication with a 91% increase in reported abuse in 2010-2015 resulting in 15 deaths (Ann Emerg Med 2017;69:73-8). Loperamide as a single agent (not in combination with other agents) accounted for 695 overdoses and 1 death in 2022.
2. Relatively safe at therapeutic doses; however, can be fatal when high doses are ingested for its euphoric effects
3. Phenylpiperidine opioid that slows intestinal transit time by stimulating mu-opioid receptors in the GI tract and blocks intestinal calcium channels
4. Toxicity of loperamide involves blockade of sodium channels and potassium channels in the cardiac tissue, causing QT prolongation and QRS interval widening and leading to life-threatening dysrhythmias and cardiac death

B. Clinical Presentation
1. Common signs/symptoms include respiratory depression, nausea, vomiting, decreased level of consciousness, miosis, decreased bowel motility/ileus, palpitations, and syncope.
2. In severe toxicity, ECG findings are abnormal, including widened QRS interval and prolonged QT interval.

3. Diagnostic workup
    a. 12-lead ECG to assess for QT prolongation and QRS interval widening and development of ventricular dysrhythmias
    b. Arterial blood gas to monitor for respiratory acidosis secondary to respiratory depression
    c. Standard chemistry panel for electrolyte and glucose abnormalities, plus CK, BUN, and SCr for signs of rhabdomyolysis
    d. Pulse oximetry

C. Treatment
1. Stabilize the airway and provide supplemental oxygen, if needed. Establish an airway if patients cannot protect their airway or have significant respiratory depression.
2. Administer intravenous crystalloid fluids to maintain BP.
3. Gastric decontamination with a single dose of activated charcoal may be considered only if patients present within 2-4 hours after a large overdose and can protect their airway.
4. Antidote therapy (Ann Emerg Med 2017;pii: S0196-0644(17)30424-9.):
    a. Naloxone
        i. Should be administered in addition to supportive care in patients with respiratory depression who cannot protect their airway
        ii. The lowest effective dose should be used; repeat doses may be necessary secondary to loperamide's slow elimination.
    b. Sodium bicarbonate
        i. May be beneficial in patients with signs and symptoms of cardiac toxicity and QRS interval widening secondary to sodium channel blockade by loperamide
        ii. Sodium bicarbonate 1-2 mEq/kg intravenously should be administered concomitantly with magnesium and potassium chloride if electrolyte abnormalities exist.

D. Monitoring
1. Patients should be closely monitored for resolution of clinical symptoms, including respiratory status, vital signs, and return to baseline mental status.
2. Monitor serum electrolytes, blood glucose, and 12-lead ECG periodically.
3. Closely monitor for signs and symptoms of opioid withdrawal.

VIII. ALCOHOLS (METHANOL AND ETHYLENE GLYCOL)

A. Background
1. Alcohol poisonings (methanol and ethylene glycol) are not as common as poisonings with other substances, accounting for <1% of all cases in 2022 (National Poison Data System), but they can be serious and potentially fatal.
2. Methanol is commonly found in products such as windshield washer fluid, antifreeze, brake and carburetor fluids, and cooking products.
3. Ethylene glycol is commonly found in products such as antifreeze, de-icing solutions, refrigerants, and brake fluids.
4. Toxicity of both agents is caused by the breakdown to toxic metabolites by alcohol dehydrogenase and aldehyde dehydrogenase.

a. Methanol is converted to formaldehyde and then to formic acid, which results in an anion gap acidosis and ocular toxicity.
b. Ethylene glycol is converted to glycoaldehyde and then to glycolic acid, followed by glyoxylic acid, and, eventually, oxalic acid. Glycolic acid results in an anion gap acidosis and CNS toxicity. Oxalic acid results in CNS toxicity and renal toxicity because of the formation of calcium oxalate crystals.

B. Clinical Presentation
1. Common symptoms include inebriation, altered mental status, nausea, vomiting, hematemesis, nystagmus, and depressed reflexes. In rare cases of ethylene glycol toxicity, patients may present with tetany caused by hypocalcemia. Symptoms typically develop over the first 24 hours; however, they can be delayed for days in the setting of ethanol coingestion.
2. Early in therapy, an osmolar gap will be present, but this will diminish as the parent compound is metabolized.
   a. As the osmolar gap declines, the anion gap will rise, resulting in a significant metabolic acidosis.
   b. Calculations:
      i. Osmolar gap (OG):
      Measured osmolality - calculated osmolality (normal OG less than 10)
      ii. Calculated osmolality:
      (sodium x 2) + (glucose/18) + (BUN/2.8)
      iii. Calculated osmolality with ethanol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (ethanol/4.6)
      iv. Calculated osmolality with methanol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (methanol/3.2)
      v. Calculated osmolality with ethylene glycol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (ethylene glycol/6.2)
      vi. Anion gap:
      $Na - (Cl + HCO_3)$
3. Methanol and ethylene glycol serum concentrations may be monitored to determine severity and to guide therapy in conjunction with an anion gap metabolic acidosis. Often, the ability to obtain these serum concentrations is not readily available and may take several hours to perform; therefore, therapy should not be delayed for lab results.

C. Treatment
1. Treatment is focused on blocking the toxic alcohol metabolism and allowing it to be excreted unchanged in the urine. Treatment is recommended when serum methanol or ethylene glycol concentrations exceed 20 mg/dL, the patient has a documented history of ingestion, there is a high clinical suspicion of ingestion combined with an osmolal gap >10 mOsm/kg, or there is a presence of an anion gap metabolic acidosis of unknown etiology.
2. Gastric decontamination is not recommended.
3. Fomepizole is the preferred antidote because of its predictable response, ease of dosing, and lack of contraindications to use.
   a. Mechanism of action is competitive inhibition of alcohol dehydrogenase.
   b. Dosing is a 15-mg/kg intravenous bolus (as a loading dose); then 10 mg/kg every 12 hours for four doses; then 15 mg/kg every 12 hours until methanol/ethylene glycol concentrations are less than 20 mg/dL.
   c. After 48 hours, fomepizole induces its own metabolism, requiring dosage increases.
   d. Oral administration is effective and may be considered if intravenous access cannot be established (Clin Toxicol 2008;46:181-6).

e. Therapy is discontinued when methanol/ethylene glycol concentrations are less than 20 mg/dL. If the patient is still symptomatic with a normal pH, further workup is warranted, and hemodialysis may be indicated.
f. Hemodialysis increases the clearance of fomepizole; therefore, doses must be administered every 4 hours during hemodialysis. Continuous renal replacement therapy has also been found to increase the clearance of fomepizole, although guidance regarding dosing adjustments is limited and requires further investigation.
g. Adverse effects may include headache, nausea, dizziness, abdominal pain, hypotension, and bradycardia.

4. Ethanol may be administered by diluting 95% alcohol for intravenous, oral, or per-tube administration.
   a. Mechanism of action is competitive inhibition of alcohol dehydrogenase.
   b. Alcohol dehydrogenase has a higher affinity for ethanol.
   c. Intravenous alcohol preparations are no longer commercially available and must be compounded.
   d. Initial dosing is 600–700 mg (7.6–8.9 mL/kg) of a 10% solution, followed by an infusion of 66 mg/kg/hour (0.83 mL/kg/hour). The infusion dose may be initiated at 154 mg/kg/hour (1.96 mL/kg/hour) in chronic drinkers. The goal is to maintain a serum ethanol concentration of 100 mg/dL (0.1%) until symptoms have diminished and methanol or ethylene glycol serum concentrations are undetectable. (Ann Emerg Med 2009;53:439-50).
   e. Disadvantages include frequent monitoring and ICU admission in some institutions.
   f. Adverse effects include CNS depression, nausea, vomiting, abdominal pain, polyuria, and hypoglycemia (especially in children).
5. Hemodialysis should be considered if the clinical condition deteriorates, as evidenced by:
   a. Methanol/ethylene glycol concentration greater than 50 mg/dL
   b. Significant metabolic acidosis
   c. Development of acute renal failure (ethylene glycol) or visual disturbances (methanol)
   d. Development of significant electrolyte abnormalities
6. Additional therapies
   a. Pyridoxine and thiamine
      i. Serve as cofactors in the metabolism of the toxic metabolites of ethylene glycol to nontoxic metabolites
      ii. Pyridoxine promotes the metabolism of glyoxylate to glycine.
         (a) Dosing is generally recommended to be one or more intravenous doses of 50 mg
      iii. Thiamine promotes the metabolism of glycolic acid to a nontoxic metabolite and is also used to prevent or treat Wernicke-Korsakoff syndrome.
         (a) Dosing is generally recommended as 100 mg intravenously, although a concomitant history of chronic alcohol use may warrant use of higher doses (e.g., 500 mg).
   b. Folinic acid
      i. Serves as a cofactor in the metabolism of the toxic metabolites of methanol to nontoxic metabolites. May reduce formate accumulation and reduce the development of metabolic acidosis (Crit Care Clin 2012;28:661-771).
         (a) Dosing is generally recommended to be 1 mg/kg (maximum of 50 mg) intravenously at 4-hour intervals
      ii. Folic acid may be used if folinic acid is unavailable.
   c. Dextrose
      i. Recommended to check a point-of-care level blood glucose concentration before administration (Crit Care Clin 2012;28: 661-771)
      ii. Administer 50 mL of 50% dextrose in water if blood glucose is 70 mg/dL or less or if testing is unavailable.

d. Magnesium – Recommended to administer 1–2 g intravenously for hypomagnesemia (more common in chronic alcohol use)
e. Antiseizure medications
    i. Benzodiazepines are the preferred emergent agents to treat seizures.
    ii. Subsequent urgent control therapy can be implemented with intravenous fosphenytoin, valproate sodium, phenobarbital, or levetiracetam.
        (a) Phenytoin would not be recommended for patients with a history of chronic ethanol ingestion due to enhanced phenytoin metabolism (Gastroenterology 1969;56:412).

D. Monitoring
1. Patient should be closely monitored for resolution of clinical symptoms and return to baseline mental status.
2. Monitor serum electrolytes and blood glucose periodically.
3. Arterial blood gases with a goal of pH greater than 7.2
4. Methanol/ethylene glycol concentrations with a goal of less than 20 mg/dL
5. Monitor the osmolar and anion gaps to ensure the toxic alcohol and metabolites are being cleared appropriately.

Patient Cases

5. A 35-year-old man is admitted to the ED appearing inebriated. He is alert but oriented only to person. His vital signs are BP 122/80 mm Hg, HR 82 beats/minute, and RR 25 breaths/minute. His serum ethanol concentration is 20 mg/dL, and his ethylene glycol concentration is 100 mg/dL. Which is the most appropriate therapy at this time?

A. Fomepizole
B. Ethanol infusion
C. Thiamine
D. Activated charcoal

6. A patient with methanol intoxication is initiated on fomepizole treatment together with hemodialysis. After the 15-mg/kg bolus dose is given, which would be best for adjusting the maintenance fomepizole doses during dialysis?

A. 10 mg/kg every 12 hours
B. 20 mg/kg every 12 hours
C. 10 mg/kg every 4 hours
D. 20 mg/kg every 4 hours

IX. ALCOHOL WITHDRAWAL

A. Background
   1. Alcohol withdrawal is a relatively common consequence of hospital admission.
   2. The strongest risk factor is a history of alcohol withdrawal.
      a. Additional tools have been developed, such as the Prediction of Alcohol Withdrawal Severity Scale, to help identify patients at risk for alcohol withdrawal before development of severe symptoms (Alcohol 2014;48:375-90).

B. Clinical Presentation
   1. Withdrawal symptoms typically occur within 8 hours after blood alcohol concentrations decrease, peak at 72 hours, and are markedly reduced at 5–7 days (N Engl J Med 2014;371:2109-13).
   2. Common symptoms include tremors, diaphoresis, nausea, vomiting, and abnormal vital signs, including hypertension, tachycardia, hyperthermia, and tachypnea (Am J Emerg Med 2013;31:734-42).
   3. Additional symptoms categorized as moderate to severe withdrawal include:
      a. Alcoholic hallucinations – Auditory, visual, or tactile; may last up to 6 days
      b. Alcohol withdrawal seizures (tonic-clonic) – Occur within 72 hours; however, it is highest in the first 24 hours after the last drink.
      c. Delirium tremens – Severe and potentially life-threatening symptom that may develop within 72 hours; presentation includes autonomic hyperactivity, confusion, delirium, psychosis, hallucinations, and seizures.

C. Treatment
   1. The goal of therapy is to keep the patient safe, alleviate and prevent the progression of symptoms, and treat comorbidities (Crit Care Med 2010;38(suppl):S494-S501). Agents for the treatment of alcohol withdrawal are listed in Table 5.
   2. Benzodiazepines are the primary agents used in treatment; they bind to the γ-aminobutyric acid ($GABA_A$) receptor, resulting in hyperpolarization and membrane stabilization.
      a. Lorazepam and diazepam are preferred because of their more predictable effects.
      b. Chlordiazepoxide is not recommended in the acute stabilization setting.
      c. Symptom-triggered therapy is preferred because it reduces benzodiazepine use, duration of mechanical ventilation, and duration of ICU stay.
         i. A front-loading regimen (i.e., a single dose of preventative medication) is recommended for patients at high risk for severe withdrawal (J Addict Med 2020;14:1-72).
      d. Scheduled treatment may be necessary if symptoms are severe or difficult to control.
   3. Use of ethanol to control alcohol withdrawal is controversial and is not routinely recommended.
   4. Phenobarbital
      a. Barbiturate with sedative, hypnotic, and antiseizure activity. Increases the binding of GABA to $GABA_A$ receptor and prolonging the chloride channel opening. It may also concomitantly mitigate neuronal stimulation through NMDA receptor antagonism. Potential advantage over benzodiazepines in alcohol withdrawal because it does not require GABA to be effective (GABA may be depleted), and a longer duration of action.
      b. Historically considered a second-line agent if benzodiazepines fail to adequately control symptoms. Current data suggests that phenobarbital appears to have equivalent outcomes as that of benzodiazepines, and may be more advantageous for certain patients based on patient-specific factors (Pourmand A, AlRemeithi R, Kartiko S, Bronstein D, Tran QK. Evaluation of phenobarbital based approach in treating patient with alcohol withdrawal syndrome: A systematic review and meta-analysis. Am J Emerg Med. 2023 Jul;69:65-75. doi: 10.1016/j.ajem.2023.04.002. Epub 2023 Apr 5. PMID: 37060631.)

   c. May increase the efficacy of benzodiazepines when used in combination by increasing the binding to the $GABA_A$ receptor.
5. Clonidine: α_2-receptor agonist that helps control the catecholamine surge associated with withdrawal (responsible for elevations in BP and HR); however, it will not prevent seizures.
6. Baclofen: Selective $GABA_B$ receptor agonist that reduces the signs and symptoms of alcohol withdrawal; however, it will not prevent seizures.
7. Gabapentin: Increases GABA syntheses and concentrations; may reduce benzodiazepine requirements; however, it may not effectively control seizures when used as monotherapy (Ann Pharmacother 2015;49:897-906).
8. Propofol
   a. General anesthetic acting through $GABA_A$ receptor agonism and N-methyl-D-aspartate (NMDA) receptor antagonism; chronic alcohol use is associated with an up-regulation of NMDA.
   b. Useful for controlling delirium and preventing seizures; patients must be intubated to receive propofol for this indication
9. Dexmedetomidine
   a. α_2-receptor agonist, which may help control BP and HR; however, it will not prevent seizures.
   b. May reduce overall benzodiazepine requirements; however, it lacks the activity necessary to prevent seizures.
   c. Recommended when clonidine cannot be administered and as adjunct therapy
10. Ketamine: NMDA antagonist; may reduce overall benzodiazepine use and can be considered in patients refractory to other therapies (Crit Care Med 2018;46:e768-71).
11. Supportive care: Alcohol-dependent patients are often nutritionally deficient and at risk of Wernicke encephalopathy and hypomagnesemia. High dose intravenous folic acid, thiamine, and magnesium should be administered to nutrient-deficient patients. Dextrose containing fluids may also be considered (Crit Care Med 2016;44:1545-52).

D. Monitoring
1. Clinical Institute Withdrawal Assessment for Alcohol Scale (revised version) (CIWA-Ar) or the Minnesota Detoxification Scale (MINDS) to determine the severity of symptoms and treatment
2. A sedation score, such as the Richmond Agitation-Sedation Scale, should be used in place of the CIWA in intubated patients to guide benzodiazepine dosing.
3. Vital signs every 2–4 hours
4. Electroencephalogram for sustained seizure-related activity

Table 5. Agents for Treatment of Alcohol Withdrawal

Agent	Suggested Starting Dose	Suggested Interval/Infusion Dose Range
Diazepam	5–20 mg PO/IV	Every 6–8 hr
Lorazepam	2–4 mg PO/IV	Every 4–6 hr
Phenobarbital	65– 260 mg or 10 mg/kg IV	Every 15–20 min until symptoms are controlled
Clonidine	0.1–0.3 mg PO	Every 8–12 hr
Baclofen	5–10 mg PO	Every 8–12 hr
Gabapentin	600–800 mg PO	Every 8 hr
Propofol	10–20 mcg/kg/min IV	20–70 mcg/kg/min
Dexmedetomidine	0.2–0.4 mcg/kg/hr IV	0.4–1.5 mcg/kg/hr
Ketamine	0.15–0.2 mg/kg/hr IV	0.2–0.3 mg/kg/hr
Thiamine	100–500 mg IV	Every 8–24 hr for 3–5 days
Folic acid	1–5 mg IV	Once daily for 3–5 days
Magnesium	1–4 g IV	Once daily for 3–5 days

IV = intravenously; PO = orally or per tube.

X. β-BLOCKERS AND CALCIUM CHANNEL BLOCKERS

A. Background
   1. Cardiovascular agents accounted for a little more than 110,000 toxic exposures in 2022 and were a leading cause of death secondary to pharmaceutical exposure.
   2. Two of the most common cardiovascular agents involved in single-agent toxic exposures were β-blockers (28,547 cases and 23 deaths in 2022) and calcium channel blockers (15,718 cases and 30 deaths in 2022).

B. Clinical Presentation
   1. β-Blocker overdoses are characterized by hypotension, bradycardia, and prolonged atrioventricular conduction.
   2. Calcium channel blocker overdoses with the non-dihydropyridine agents are characterized by hypotension, prolonged atrioventricular conduction, bradycardia, lethargy, hyperglycemia, and depressed consciousness. The dihydropyridine agents act peripherally and are primarily associated with vasodilation, hypotension, and reflex tachycardia. However, this receptor selectivity is lost at high doses for both categories (Ann Emerg Med 1993;22:196-200).

C. Treatment
   1. Consider gastric lavage or activated charcoal if patients present within 1–2 hours of overdose. Whole bowel irrigation is recommended for delayed presentation or for sustained- or extended-release formulations.
   2. Maintenance of hemodynamic stability
      a. Goal of therapy is a mean arterial pressure greater than 65 mm Hg or systolic blood pressure (SBP) greater than 90 mm Hg.
      b. Administer isotonic fluids (0.9% sodium chloride or lactated Ringer solution) at 20–30 mL/kg (preferred) or colloidal solutions (e.g., albumin 5% 250 mL).
   3. Administer intravenous calcium chloride or calcium gluconate (first line therapy for calcium channel blocker toxicity).
      a. Calcium chloride 1–2 g (central line is preferred; however, bolus doses may be administered in a peripheral line if needed)
      b. Calcium gluconate 3–6 g. Repeat dose may be given every 15–30 minutes; may also consider continuous infusion at 0.3–0.7 mEq/kg/hour.
   4. Consider sodium bicarbonate for QRS widening, severe acidosis, or dysrhythmia: 1– 2 mEq/kg bolus; may consider a continuous infusion if symptoms persist.
   5. Treat symptomatic bradycardia and hypotension as follows:
      a. Atropine 0.5–1 mg intravenously; if no response, continue to the following options:
      b. Glucagon 5–10 mg (100 mcg/kg) intravenous push for 1 minute (consider for β-blocker toxicity only). If symptom response is achieved from the bolus, may consider a continuous intravenous infusion initiated at the same rate (milligrams per hour) as the bolus dose that achieved response.
         i. Stimulates adenylate cyclase, which increases intracellular cAMP (cyclic adenosine monophosphate), leading to increased inotropy, chronotropy, and cardiac conduction
         ii. Adverse effects include nausea, vomiting, and hyperglycemia. Consider premedicating with an antiemetic intravenously before bolus dosing of glucagon to decrease the risk of nausea and vomiting.
         iii. Use with caution with decreased mental status because of possible aspiration or airway obstruction (also causes lower esophageal sphincter relaxation).
         iv. Not recommended as a preferred treatment option for calcium channel blockers because of limited efficacy and cost

c. Norepinephrine continuous infusion initiated at 2–5 mcg/minute (0.1 mcg/kg/minute) and titrated to target blood pressure in the setting of vasodilatory shock
d. In the presence of cardiogenic shock, epinephrine and dobutamine are the agents of choice. Epinephrine continuous infusion should be initiated at 1 mcg/minute (0.01–0.1 mcg/kg/minute) and titrated to target hemodynamic parameters. Dobutamine continuous infusion should be initiated at 2–20 mcg/kg/min and titrated to target hemodynamic parameters. Dopamine is not recommended in this setting.
e. Transcutaneous or transvenous pacing or intra-aortic balloon pumps may be required in refractory cases

6. Hyperinsulinemic euglycemic therapy (HIET)
   a. Mechanism of action:
      i. Insulin increases the plasma concentrations of ionized calcium, improves the hyperglycemic acidotic state, improves the myocardial use of carbohydrates, and exerts an independent inotropic effect (Am J Crit Care Med 2007;16:498-503).
      ii. Dextrose prevents the development of hypoglycemia after insulin administration.
      iii. Potassium prevents the development of hypokalemia after insulin administration.
      iv. Onset of action is as soon as 5 minutes; however, it may take up to 30 minutes for full effects to be seen.
   b. HIET administration (Clin Toxicol 2011;49:277-83; Am J Health Syst Pharm 2006;63:1828-35; Circulation 2023;148:e149-84):
      i. If baseline glucose is less than 200 mg/dL, administer 50 mL of 50% dextrose in water before insulin administration; may consider an infusion of 10%–20% dextrose to maintain a serum glucose concentration greater than 100 mg/dL
      ii. Correct hypokalemia before initiating insulin therapy. Maintain normokalemia with potassium replacement during insulin infusion and after withdrawal.
      iii. Bolus 1 unit/kg of regular insulin intravenously, followed by a continuous intravenous infusion at 0.5–1 unit/kg/hour; increase rate every 10 minutes to a maximum of 10 units/kg/hour.
         (a) HIET should be continued until existing vasopressor therapy is weaned and hemodynamic stability is obtained. Then, HIET should be tapered over several hours while monitoring for hemodynamic deterioration. Dextrose therapy likely must be continued after discontinuation of HIET to prevent hypoglycemia.
   c. Adverse effects: Hypoglycemia, hypomagnesemia, and hypokalemia
   d. Monitoring:
      i. Vital signs every 15–60 minutes with a goal mean arterial pressure greater than 65 mm Hg and a HR greater than 50 beats/minute
      ii. Serum glucose every 15 minutes; then every 30–60 minutes once stable to target serum concentrations greater than 100 mg/dL
      iii. Serum potassium every hour during the insulin infusion; then every 6 hours to maintain concentrations above 2.8 mEq/L
      iv. Monitor for fluid overload and iatrogenic hyponatremia. Utilization of more concentrated dextrose infusions (e.g., 50% dextrose) and more concentrated insulin infusions (e.g., 16 units/mL) may be necessary.

7. Intravenous lipid emulsion (J Emerg Med 2015;48:387-97; J Med Toxicol 2017;13:124-5)
   a. Mechanism of action is not well known; however, it is thought to be owing to a combination of binding lipid-soluble agents and the provision of free fatty acids that increase cardiac energy and intracellular calcium.
   b. Improves HR and may reduce mortality as an individual treatment or in combination with other therapies.
   c. Evidence is limited to animal models and human case reports, and its role in therapy is controversial.

d. Administration:
   i. Bolus of 1.5 mL/kg of 20% lipid emulsion (such as Intralipid) over 2-3 minutes (typical dose is usually 100 mL)
   ii. A repeat bolus may be considered if no response after initial bolus
   iii. The bolus may be followed immediately by an infusion of 20% lipid emulsion at a rate of 0.25 mL/kg/minute. After 3 minutes at this infusion rate, the infusion rate may be adjusted to 0.025 mL/kg/minute (i.e., 1/10 the initial rate) if there has been a significant response; total duration is not to exceed 6 hours.
   iv. BP, HR, and other available hemodynamic parameters should be recorded at least every 15 minutes during the infusion.
   v. Adverse effects may include pancreatitis, jaundice, coagulopathies, interference with laboratory results, and fat embolism.
   vi. Drug interactions are not well known.

Patient Case

Questions 7 and 8 pertain to the following case.
A 52-year-old man is admitted to the ED with concerns about dizziness and headache. His vital signs are as follows: temperature 98.9°F (37.2°C), BP 87/50 mm Hg, and HR 58 beats/minute. His wife reports that he has a history of hypertension and that he was recently given a diagnosis of being in the early stages of Alzheimer disease. She has brought his medications with her; the 1-month supply was refilled 2 days ago: a bottle of diltiazem CD 120 mg/day (7 tablets remaining) and a bottle of donepezil 5 mg once daily (28 tablets remaining).

7. Which decontamination strategy would provide the most benefit?
   A. Charcoal 25 g every hour until his BP improves
   B. Ipecac 30 mL, followed by 240 mL of water
   C. Polyethylene glycol-electrolyte solution 1500 mL/hour until the rectal effluent is clear
   D. Magnesium citrate 240 mL, followed by 240 mL of water

8. Which antidote would be best to administer first?
   A. Calcium chloride 1 g intravenously over 1 minute
   B. Glucagon 5 mg intravenously over 1 minute
   C. Atropine 2 mg intravenously over 1 minute
   D. Epinephrine 1 mg intravenously over 1 minute

XI. DIGOXIN

A. Background
   1. The cardiac glycosides accounted for 1051 single-agent toxic exposures and 29 deaths in 2022.
   2. Mechanism of action is inhibition of the sodium-potassium adenosine triphosphatase pump and suppression of the atrioventricular node.
   3. Because of its narrow therapeutic index, toxicity has been reported in as many as 35% of patients receiving digoxin (Postgrad Med 1993;69:337-9).

a. The normal therapeutic range is 0.8–2.1 ng/mL.
b. Toxicity may be related to an acute ingestion or may be an issue with chronic use in renal dysfunction.

4. Risk factors for digoxin toxicity include renal failure, advanced age, ischemic heart disease, left ventricular dysfunction, electrolyte imbalances (hypokalemia, hypomagnesemia, hypercalcemia), and hypothyroidism (Postgrad Med 1993;69:337-9).

B. Clinical Presentation
1. Cardiac effects associated with digoxin toxicity include second- and third-degree heart block, tachyarrhythmias, and bradyarrhythmias. More specific examples include fascicular tachycardia, ventricular bigeminy, and ventricular tachycardia (Am J Cardiol 1992;69:108G-119G).
2. Noncardiac effects associated with digoxin toxicity include nausea and vomiting, lethargy, headaches, confusion, and visual disturbances.

C. Treatment
1. Consider decontamination strategies if patients present within 2 hours of overdose.
a. Multidose activated charcoal is beneficial because of the enterohepatic recirculation of digoxin. Load 50–100 g; then 10 g/hour, 10–20 g every 2 hours, or 40 g every 4 hours (Postgrad Med 1993;69:337-9).
b. Colestipol or cholestyramine is an effective drug-binding alternative to charcoal, but it may not be useful in acute toxicity (Am J Cardiol 1992;69:108G-119G).
c. Hemodialysis is not considered effective.
2. Correct serum electrolyte abnormalities.
a. Correct serum potassium concentration to a goal of 3.5–4 mEq/L.
b. Correct serum magnesium concentration to a goal of 1.5–2.2 mg/dL.
c. Correct serum calcium concentration to a goal of 8.5–10.5 mg/dL.
3. Treat symptomatic bradyarrhythmias with atropine 0.5 mg intravenously.
4. Digoxin immune antigen-binding fragments (Fab)
a. Antibodies that bind to digoxin molecules that are then renally excreted
b. Indications for use in acute intoxications (Crit Care Clin 2012;28:527-35; Circulation 2023;148:e149-84):
i. Life-threatening arrhythmias: asystole, ventricular fibrillation or tachycardia, complete heart block, symptomatic bradycardia
ii. Evidence of end-organ damage (e.g., renal failure, altered mental status)
iii. Hyperkalemia (greater than 5–5.5 mEq/L)
c. Products:
i. Digibind (Digoxin Immune Fab): 38 mg per vial
ii. DigiFab (Digoxin Immune Fab): 40 mg per vial
d. Dosing:
i. If amount of digoxin ingested is unknown: 10–20 vials for acute toxicity or 6 vials for chronic toxicity
ii. If the amount of digoxin ingested is known:
Dose (vials) = total body load (0.8 × mg of digoxin ingested)/0.5
iii. If digoxin concentration is known:
Dose (vials) = [serum digoxin concentration (ng/mL) × weight (kg)]/100
iv. May consider lower doses of 1 or 2 vials (40–80 mg) for acute ingestions with repeat doses if necessary
e. Adverse effects include heart failure exacerbation, atrial fibrillation, orthostatic hypotension, hypokalemia, and phlebitis.

D. Monitoring
   1. Monitor vital signs every 30–60 minutes initially. Goal HR of greater than 60 beats/minute and asymptomatic
   2. Monitor serum potassium concentrations hourly for at least the first 6 hours.
   3. Additional serum digoxin concentrations are not recommended after the administration of digoxin immune Fab. Minimal change in serum concentrations will be expected as the digoxin laboratory test will still measure the bound, inactive digoxin molecules. Fab-digoxin complexes may stay in the serum for several days. Repeat serum digoxin concentrations may be checked 24 hours after the initial treatment if Fab is not administered.

XII. ANTIDEPRESSANTS

A. Background
   1. Antidepressants accounted for more than 63,974 single agent toxic exposures and 56 deaths in 2022.
   2. Some of the most common agents involved in toxic exposures were the selective serotonin reuptake inhibitors (SSRIs) and the tricyclic antidepressants (TCAs).
   3. SSRIs block the reuptake of serotonin at the presynaptic neuron.
   4. Patients with SSRI overdoses are often asymptomatic with self-limiting effects (Emerg Med Clin North Am 2007;25:477-97). The most common adverse effects may include drowsiness, tremor, altered mental status, nausea and vomiting, tachycardia, hypotension, seizures, and QRS- or QT-interval prolongation.
   5. TCAs exert many effects, including blocking the reuptake of norepinephrine and serotonin at the presynaptic neuron, blocking muscarinic cholinergic receptors, blocking antihistamine effect, blocking the sodium channel, and, to a lesser degree, blocking α-adrenergic receptors.
   6. Individuals with TCA overdoses may present with the following (Emerg Med Clin North Am 1994;12:533-47):
      a. Cardiovascular: Hypo- or hypertension, tachy- or bradycardia, increased QRS or QT interval, atrioventricular-conduction block, complete heart block
      b. Respiratory: Hypoventilation, crackles, hypoxia
      c. Neurologic: Delirium, lethargy, seizures, coma
      d. Other: Hyperthermia, dry mucous membranes, urinary retention, blurred vision

B. Treatment
   1. There are no specific antidotes for antidepressant overdoses; general supportive care is recommended, with a focus on ABC.
   2. Gastric decontamination is not typically recommended; however, single-dose activated charcoal may be administered within the first hour of exposure (Emerg Med Clin North Am 2000;18:637-54).
   3. Administer crystalloid or colloid fluids to maintain BP and HR, with the goal of a mean arterial pressure >65 mmHg, a SBP greater than 90 mm Hg, and a HR greater than 60 beats/minute.
      a. Norepinephrine or epinephrine may be used if fluid resuscitation alone is unsuccessful.
      b. Dopamine may not be an effective agent because endogenous norepinephrine stores are depleted in an overdose.
   4. Sodium channel blockade
      a. Alkalinization of blood to a pH of 7.45–7.55 is recommended for the TCAs to resolve metabolic acidosis and improve cardiac symptoms. Requires frequent monitoring of arterial pH (varies by effect, but as often as every 15–30 minutes). Sodium load may overcome TCA blockade of sodium channels by increasing the electrochemical gradient.

b. Administer sodium bicarbonate
   i. Recommended bolus dose of 1 mEq/kg (minimum: 50 mEq) intravenously
   ii. May repeat bolus every 15 minutes until ECG stabilized or arterial pH goal achieved.
   iii. May consider a continuous infusion of hypertonic saline in patients with refractory cardiac conduction despite optimal serum alkalinization by increasing sodium load.
c. Proposed indications for sodium bicarbonate include the following (Chest 2008;133:1006-13; Circulation 2023;148:e149-84):
   i. QRS greater than 100–120 milliseconds
   ii. Wide complex tachycardia
   iii. Cardiac arrest
   iv. Right bundle-branch block
   v. Refractory hypotension
d. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation is present. Magnesium replacement is recommended for membrane stabilization in patients with a QTc greater than 500 milliseconds and normal serum magnesium concentrations.
e. Seizures should be managed with benzodiazepines. Phenobarbital and propofol may be considered if the patient is refractory to benzodiazepines and has a stable BP.
   i. Phenytoin and lacosamide are not recommended because of their effects on sodium channels and cardiac adverse effects, respectively.
f. Intravenous fat emulsion
   i. Many case reports for use in amitriptyline and SSRI overdose
   ii. Administration (use lean body mass):
      (a) Bolus of 1.5 mL/kg of 20% lipid emulsion (Intralipid) over 1–5 minutes (typical dose is usually 100 mL for patients weighing 70 kg and more)
      (b) May repeat up to two times for persistent cardiovascular collapse
      (c) Bolus doses should be followed by an intravenous infusion of 0.25–0.5 mL/kg/minute for 60 minutes (typical dose is 18 mL/minute for a 70 kg patient)
      (d) Continue for up to 10 minutes after cardiovascular recovery.

C. Monitoring – Patients should be monitored for clinical improvement for at least 6–8 hours and for a minimum of 24 hours for more severe adverse effects or with citalopram or escitalopram (because of the longer half-lives of these agents).
   a. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, troponins, BP, and HR.
   b. Monitor for signs and symptoms of respiratory depression with RR and pulse oximetry.

D. Serotonin Syndrome
   1. Excessive serotonin concentrations lead to overstimulation of serotonin-1A and serotonin-2A receptors in the central and peripheral nervous systems (Emerg Med Clin North Am 2007;25:477-97).
   2. Adverse effects include altered mental status, autonomic instability (hyperthermia, tachycardia, hypertension, arrhythmias), and neuromuscular changes (hyperreflexia, increased rigidity).
   3. Diagnosis is made according to clinical findings; many clinicians support the use of the Hunter Serotonin Toxicity Criteria (QJM 2003;96:635-42). By this method, patients are likely to have serotonin toxicity if they have taken a serotonergic agent and one of the following criteria are present:
      a. Spontaneous clonus
      b. Inducible clonus PLUS agitation or diaphoresis
      c. Ocular clonus PLUS agitation or diaphoresis
      d. Tremor PLUS hyperreflexia
      e. Hypertonia PLUS temperature above 100.4°F (38°C) PLUS ocular clonus or inducible clonus

4. Treatment should focus on supportive care with intravenous fluids; symptoms typically resolve within 24–48 hours.
    a. Discontinue the offending agent.
    b. Benzodiazepines should be administered as first line for agitation and muscle rigidity.
    c. Cyproheptadine is a histamine-1 receptor antagonist and nonspecific serotonin receptor antagonist. A single dose of 8–12 mg by mouth should be administered for agitation and muscle rigidity as an adjunct to benzodiazepines. A second dose may be administered in 6–8 hours if symptoms persist.
    d. If condition worsens, may require intubation with continuous infusion benzodiazepines
    e. Although not well studied, case reports have shown the efficacy of dexmedetomidine at doses of 0.05–0.8 mcg/kg/hour (in pediatric patients).

Patient Case

9. A 21-year-old man is admitted to the ED after taking 30 citalopram 20-mg tablets about 2 hours ago. His vital signs are as follows: BP 125/85 mm Hg, HR 77 beats/minute, RR 15 breaths/minute, and temperature 98.7°F (37.1°C). Which is the best intervention for this patient?
    A. Administer lorazepam 2 mg intravenously to prevent seizure activity.
    B. Administer cyproheptadine 8 mg by mouth to prevent muscle rigidity.
    C. Recommend a cooling blanket to prevent serotonin syndrome–related hyperthermia.
    D. Order a 12-lead ECG to monitor for cardiac conduction disturbances.

XIII. ATYPICAL ANTIPSYCHOTICS

A. Background
    1. The atypical antipsychotic agents accounted for about 16,541 toxic exposures and 17 deaths in 2022.
    2. These agents are classified primarily as having D2-dopaminergic receptor and serotonin-2A receptor antagonism. Additional effects include antagonism of the α_1- and histamine-1 receptors.
    3. Adverse effects associated with the atypical antipsychotics are typically self-limiting.
        a. More severe adverse effects may include CNS depression, tachycardia, hypotension, and QT prolongation.
        b. Less severe adverse effects include dizziness, drowsiness, miosis, blurred vision, urinary retention, and CNS excitation.

B. Treatment
    1. There are no specific antidotes for the atypical antipsychotics; general supportive care is recommended, focusing on ABC.
    2. Gastric decontamination is not typically recommended; however, single-dose activated charcoal may be administered within the first hour of exposure if no contraindications exist (J Emerg Med 2012;43:906-13).
    3. Administer crystalloid to maintain BP with a goal mean arterial blood pressure >65 mm Hg, an SBP greater than 90 mm Hg, and an HR greater than 60 beats/minute.
        a. Consider vasopressors if fluid resuscitation is inadequate.
        b. Because of the α-receptor antagonist activity of these agents, norepinephrine or phenylephrine is preferred if vasopressors are needed.
    4. Administer sodium bicarbonate if QRS prolongation is present (quetiapine overdose only)

5. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation. Magnesium replacement is recommended for membrane stabilization in patients with a QTc greater than 500 milliseconds and normal serum magnesium concentrations.
6. Seizure activity should be managed with benzodiazepines, barbiturates, or propofol.
7. Anticholinergic symptoms can potentially be treated with IV physostigmine in select patients (e.g., normal ECG); however, substantial controversy exists with this approach, particularly in the setting of a concomitant TCA overdose and its overall functional use considering its short half-life (Ann Emerg Med 2000;35:374-81). Currently, it is not frequently used and further investigations into its safety and efficacy are necessary (J Med Toxicol 2015;11:179-84).
8. Lipid emulsion therapy may be effective because of the high lipophilicity of these agents and may be considered if more traditional treatment means do not improve the cardiovascular complications of decreased HR and/or BP (J Emerg Med 2012;43:906-13). Administer an intravenous bolus dose of 1.5 mL/kg of 20% lipid emulsion over 2–3 minutes, followed by a continuous intravenous infusion of 0.25–0.5 mL/kg/min for 60 minutes, if necessary.

C. Monitoring: Patients should be monitored for clinical improvement for at least 8–12 hours.
   1. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, and troponins.
   2. Monitor for respiratory depression with RR and pulse oximetry.

XIV. LITHIUM

A. Background
   1. Lithium was associated with almost 3341 toxic exposures and seven deaths in 2022.
   2. Mechanism of action is through an influence on serotonin and norepinephrine reuptake, inhibition of the phosphatidylinositol cycle, and inhibition of the post-synaptic D2 receptor.
   3. Adverse effects associated with lithium include:
      a. Acute overdose:
         i. GI: Nausea, vomiting, diarrhea
         ii. CNS: Confusion, tremor, myoclonus, seizures, coma
         iii. Cardiovascular: T-wave inversion, ventricular arrhythmias
      b. Chronic adverse effects:
         i. Endocrine: Hypothyroidism, myxedema coma
         ii. Nephrogenic diabetes insipidus

B. Treatment
   1. There are no specific antidotes for lithium; general supportive care is recommended, focusing on ABC.
   2. Gastric decontamination is not typically recommended in toxic acute ingestions. Single-dose activated charcoal is not effective for lithium overdoses; whole bowel irrigation may be beneficial.
   3. Administer crystalloid to maintain BP, with a goal SBP greater than 90 mm Hg. Consider vasopressors if fluid resuscitation is not adequate.
   4. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation is present.
   5. Seizure activity should be managed with benzodiazepines, barbiturates, or propofol.
   6. Lithium overdoses are primarily managed with fluid resuscitation and renal replacement therapy.
      a. Saline infusions may be administered if there are no contraindications to fluid therapy (goal is a serum sodium concentration of 140–145 mEq/L). Lithium clearance is reduced in hyponatremia.
      b. Intermittent hemodialysis may require several sessions to fully remove lithium concentrations because of the rebound of lithium concentrations that occurs after dialysis sessions. Consider continuous renal replacement therapy (CRRT) for hemodynamically unstable patients.

c. Suggested indications for the use of hemodialysis include (Chest 2008;133:1006-13):
   i. Severe toxicity (severe altered mental status or seizures)
   ii. Renal failure (cannot eliminate lithium)
   iii. Lithium concentrations greater than 2.5 mmol/L in chronic exposures
   iv. Lithium concentrations greater than 4 mmol/L in acute exposures

C. Monitoring – Patients should be monitored for clinical improvement for at least 8–12 hours.
   1. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, and troponins.
   2. Monitor for respiratory depression with RR and pulse oximetry.
   3. Monitor renal function with urine output, BUN, and SCr.
   4. Monitor baseline lithium concentrations and then every 6 hours after until concentrations have decreased to less than 1.5 mmol/L (normal 0.6–1.2 mmol/L).

Patient Case

10. A 24-year-old woman is brought to the ED by her roommate. She has been in a normal state of health, but the roommate is concerned because she "seems really out of it." According to the roommate, the patient had an appointment with the physician today, and she had been given a prescription to refill olanzapine 5 mg by mouth daily, but the bottle is empty. On physical examination, she is alert and oriented person, place and time but she dozes off several times. Her vital signs are stable, and a 12-lead ECG shows sinus tachycardia. Which intervention is most appropriate for this patient?
    A. Lactated Ringer solution 500 mL intravenously
    B. 8.4% sodium bicarbonate 50 mL intravenously
    C. Lorazepam 2 mg intravenously
    D. Clinical monitoring for 6 hours

XV. ORAL HYPOGLYCEMICS

A. Background
   1. Oral hypoglycemics accounted for 5944 single-agent exposures and 29 deaths in 2022.
   2. The most common oral hypoglycemic involved in toxic exposures was metformin, followed by the sulfonylureas and the glucagon-like peptide-1 (GLP-1) receptor agonists.
   3. The most serious adverse effects were reported with the sulfonylureas; however, most fatalities were associated with metformin.

B. Clinical Presentation
   1. Clinical signs and symptoms include hypoglycemia (not with metformin), nausea, vomiting, dizziness, tachycardia, and diaphoresis.
   2. More severe adverse effects include seizures, palpitations, tachyarrhythmias, electrolyte abnormalities, and metabolic (lactic) acidosis.

C. Treatment
   1. Stabilization of the ABC
   2. Identifying the causative agent is important because specific treatment will vary by the agent involved.
   3. Consider gastric decontamination with single-dose activated charcoal if patients present within 1 hour of overdose.
   4. Observe clinically asymptomatic patients for a minimum of 8 hours (Am J Health Syst Pharm 2006;63:929-38).

5. For symptomatic patients or blood glucose less than 70 mg/dL, treat with glucose:
   a. Conscious patients: Administer 8 oz of an oral carbohydrate (such as juice, non-diet sodas, or milk) or oral glucose tablets or gels.
   b. Unconscious patients: Administer intravenous dextrose, 0.5–1 g/kg
   c. Repeat doses may be required; consider a continuous infusion of dextrose if needed. Glucose concentrations should be monitored often (every 15–60 minutes) until stable.
   d. Use caution to avoid overcorrection of serum glucose.
6. Octreotide
   a. Mechanism of action is a somatostatin analog that inhibits the secretion of insulin.
   b. Primarily studied in sulfonylurea overdose, but considered a treatment option for all oral hypoglycemic toxic exposures
   c. Administer 50 mcg subcutaneous or intravenous, followed by 50-mcg doses every 6 hours. Intravenous dextrose infusion should be slowly tapered off.
   d. Adverse effects include headache, dizziness, nausea, abdominal pain, and sinus bradycardia.
7. Sodium bicarbonate
   a. Indicated for severe metformin-associated lactic acidosis
   b. 1–2 mEq/kg or 50–200 mEq of 8.4% sodium bicarbonate intravenously
8. Glucagon
   a. Mechanism of action is stimulation of gluconeogenesis.
   b. May trigger additional insulin secretion, leading to a secondary hypoglycemia
   c. May provide benefit in prehospital settings when oral or intravenous options are not available, but is not routinely recommended
   d. Not recommended in pediatric patients, in malnourished patients, or for sulfonylurea toxic exposures
9. Hemodialysis or continuous renal replacement therapy may be necessary to enhance metformin clearance in severe cases.

D. Monitoring
   1. Regular assessment of vital signs and mental status (Emerg Med J 2006;23:565-7)
   2. Measure capillary blood glucose at a minimum of every hour for 24 hours with a goal of greater than 70 mg/dL.
   3. Measure BP hourly, especially after octreotide administration.

Patient Case

11. As the pharmacist in the ICU satellite, you receive a call from a distressed nurse about a patient in the cardiac step-down unit. The patient was found unconscious, and, on investigation, it was discovered that he had received a glyburide 20-mg tablet 1 hour earlier that was meant for another patient. The patient has stable vital signs, but his point-of-care blood glucose concentration is 37 mg/dL. Which intervention is most appropriate at this time?
    A. 8 oz of milk by mouth
    B. 50 mL of 50% dextrose in water intravenously
    C. Octreotide 100 mcg subcutaneously
    D. Glucagon 1 mg intramuscularly

XVI. DRUGS OF ABUSE

A. Background
   1. Miscellaneous stimulants and street drugs accounted for 46,753 toxic exposures and 186 deaths in 2022.
   2. The most common drugs of abuse involved in toxic exposures were amphetamines, marijuana (and derivatives), cocaine, methamphetamines, and heroin.
   3. Total numbers are difficult to determine because few of these agents can be detected with current techniques.

B. Amphetamines/Methamphetamines/MDMA (ecstasy)
   1. Mechanism of action/toxicity: stimulation of the CNS, peripheral release of catecholamines, inhibition of catecholamine reuptake, and inhibition of monoamine oxidase (Intensive Care Med 2004;30:1526-36)
   2. Clinical presentation (Intensive Care Med 2004;30:1526-36)
      a. Common: Confusion, tremor, anxiety, agitation, irritability, mydriasis, tachyarrhythmias
      b. Severe: Hepatocellular necrosis, acute hepatitis, myocardial ischemia, hypertension, cerebral hemorrhage, seizures, hyponatremia
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration and airway maintenance
      b. Gastric lavage or activated charcoal if within 1 hour of ingestion
      c. BP control with benzodiazepines, α-blocker, or vasodilator (e.g., nitroglycerin, nitroprusside, nicardipine). Avoid the use of β-receptor blocking agents because of risk of unopposed α-receptor activity leading to an increase in BP.
      d. Benzodiazepines for agitation, anxiety, psychosis, and/or uncontrolled hypertension titrated to effect. Haloperidol or phenothiazines may be considered for use (with caution) in patients with primary psychiatric disorders or dopamine-mediated movement disorders.
      e. Cooling therapy if hyperthermia present
      f. Monitor for serotonin syndrome.

C. Synthetic Cannabinoids: K2/Spice
   1. Mechanism of action/toxicity: stimulation of the cannabinoid-1 (CB1) and cannabinoid-2 (CB2) receptors (Pharmacotherapy 2015;35:189-97)
      a. CB1 receptors modulate glutamate and GABA and are found both peripherally and centrally.
      b. CB2 receptors are located in immune tissue and the CNS and modulate pain and emesis.
   2. Clinical presentation (J Pharm Pract 2015;28:50-65)
      a. Predominantly euphoria or excited delirium
      b. Common adverse effects include:
         i. Psychiatric: Agitation, anxiety, hallucinations, paranoia, catatonia
         ii. Neurologic: Cognitive impairment, ataxia, dizziness, headache, seizures
         iii. Cardiovascular: Tachycardia, palpitations, hypertension
         iv. GI: Nausea, vomiting
         v. Renal: Acute kidney injury, rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines for agitation or seizure activity. Antipsychotics for agitation may be considered for use (with caution) in patients with primary psychiatric disorders or dopamine-mediated movement disorders.

D. Cocaine
   1. Mechanism of action/toxicity (Intensive Care Med 2004;30:1526-36):
      a. Potent sympathetic nervous system stimulant
      b. Inhibits the presynaptic reuptake of epinephrine and norepinephrine; also stimulates norepinephrine release
      c. Acts as a reuptake inhibitor of dopamine, norepinephrine, and serotonin
   2. Clinical presentation (Intensive Care Med 2004;30:1526-36)
      a. Predominantly euphoria or excited delirium
      b. Common/severe adverse effects include:
         i. Psychiatric: Agitation, anxiety, psychosis, delirium
         ii. Neurologic: Stroke, subarachnoid or intracranial hemorrhage, seizures
         iii. Cardiovascular: Tachycardia, hypertension, palpitations, arrhythmias, heart failure, aortic dissection
         iv. GI: Gastric ulcers, gastric perforation, bowel ischemia
         v. Respiratory: Status asthmaticus, pulmonary hypertension, pulmonary edema, alveolar hemorrhage
         vi. Renal: Acute kidney injury, rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Gastric decontamination is typically not recommended. Activated charcoal may provide benefit if given within 1 hour of oral ingestion.
      c. Benzodiazepines are administered for agitation or seizure activity and titrated to relaxation. Because of their ability to block the CNS stimulant effect of cocaine, benzodiazepines are also an effective treatment for hypertension and tachycardia.
      d. BP control with benzodiazepines, α-blocker, or vasodilator (e.g., nitroglycerin, nitroprusside, nicardipine). Avoid the use of β-receptor blocking agents because of risk of unopposed α-receptor activity leading to an increase in BP.

E. Synthetic Cathinones: Bath Salts
   1. Mechanism of action/toxicity: increases presynaptic concentrations of serotonin, dopamine, and norepinephrine by stimulating release and antagonizing monoamine reuptake (Pharmacotherapy 2014;34:745-57)
   2. Clinical presentation (J Pharm Pract 2015;28:50-65):
      a. Predominantly euphoria, increased energy, and alertness
      b. Common adverse effects include:
         i. Psychiatric: Agitation, aggression, anxiety, hallucinations, paranoia
         ii. Neurologic: Amnesia, confusion, insomnia, seizures, dizziness, headache
         iii. Cardiovascular: Angina, hypertension, tachycardia, palpitations
         iv. GI: Abdominal pain, nausea, vomiting, anorexia, hepatic failure
         v. Renal: Acute kidney injury, increased SCr, and rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines or antipsychotics for agitation or anxiety
         i. Haloperidol is an alternative option, but it should be used with caution and close monitoring because of the potential to worsen hyperthermia.
         ii. Consider propofol or dexmedetomidine for severe symptoms.
      c. Antiemetics for nausea and vomiting
      d. Monitor for serotonin syndrome.

F. Piperazines
   1. Mechanism of action/toxicity: enhances neurotransmitter release and reuptake inhibition of dopamine, serotonin, and norepinephrine release (Emerg Med Clin North Am 2014;32:1-28)
   2. Clinical presentation (J Pharm Pract 2015;28:50-65):
      a. Predominantly euphoria and increased energy
      b. Common adverse effects include:
         i. Psychiatric: Agitation, anxiety, hallucinations, psychosis, depressed mood or mood swings, paranoia
         ii. Neurologic: Confusion, insomnia, tremor, seizures, dizziness, headache
         iii. Cardiovascular: Angina, hypertension, tachycardia, palpitations, QT prolongation
         iv. GI: Abdominal pain, nausea, vomiting
         v. Renal: Urinary retention
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines for agitation or seizures
      c. Avoid antipsychotics because of the risk of worsening hyperthermia, extrapyramidal effects, and hypotension or arrhythmias.
      d. Treat hypertension with parenteral antihypertensives or clonidine.
      e. Hyperthermia treatment if above 104°F (40°C)
      f. Monitor for serotonin syndrome.

G. Ketamine
   1. Mechanism of action/toxicity (Lancet 2005;365:2137-45):
      a. Noncompetitive NMDA receptor antagonist (blocks glutamate and aspartate)
      b. Mild to moderate blockade of catecholamine reuptake
   2. Clinical presentation:
      a. Predominantly hallucinations and vivid dreams
      b. Common adverse effects include:
         i. Psychiatric: Impaired memory, cognitive dysfunction, severe agitation
         ii. Cardiovascular: Hypertension, tachycardia, cardiac arrhythmias
         iii. Respiratory: Laryngospasm, apnea, respiratory depression
         iv. GI: Anorexia, nausea, vomiting
         v. Genitourinary: Cystitis, irritable bladder, urethritis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
         i. Monitor for rhabdomyolysis
         ii. Aspiration precautions are recommended in comatose patients
         iii. Urinalysis and serum chemistries if symptomatic for cystitis
      b. Activated charcoal may provide benefit if given within 1 hour of oral ingestion. Additional doses every 4 hours may be considered.
      c. Benzodiazepines are recommended for agitation or seizures.
      d. Haloperidol may be used if benzodiazepines are not effective. Monitor closely because of the potential of lowering the seizure threshold and worsening of dystonia, hypotension, neuroleptic malignant syndrome, and/or myoglobinuria.

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ANSWERS AND EXPLANATION TO PATIENT CASES

1. **Answer: B**

The most important first step in all drug overdose cases is to try to stabilize the patient's ABC (Answer B is correct). This may involve the use of supplemental oxygen or advanced airway management, establishment of intravenous access, and administration of intravenous fluids. Once the patient is stable, the process of identifying the suspected toxin can begin (Answer D is incorrect). This may include thoroughly examining the patient, speaking with family or first responders, and communicating with the patient's physicians and pharmacies. Blood and urine samples may be sent for quantitative or qualitative toxicologic assays (Answer A is incorrect). A coma cocktail may provide some benefit, but a clear cause should be established before considering its use (Answer C is incorrect).

2. **Answer: B**

The patient has symptoms of an acute acetaminophen overdose, and stabilizing the patient, together with providing good supportive care, is indicated until a determination for additional therapy can be made (Answer B is correct). Typical decontamination strategies may provide benefit, but they do not definitively improve patient outcome. Magnesium citrate (and cathartics as a whole) is not considered an effective decontamination strategy (Answer A is incorrect). Gastric lavage is of most benefit within the first 60 minutes of exposure, and the potential adverse effects outweigh any potential benefit (Answer C is incorrect). Similarly, single-dose charcoal requires more rapid administration (Answer D is incorrect).

3. **Answer: C**

The patient is considered at high risk of developing hepatic damage from acetaminophen and requires therapy with intravenous acetylcysteine (Answer D is incorrect). The dose of intravenous acetylcysteine is as follows (doses are calculated using actual body weight): loading dose: 150 mg/kg in 200 mL of 5% dextrose in water for 60 minutes; maintenance dose: 50 mg/kg in 500 mL of 5% dextrose in water for 4 hours, followed by 100 mg/kg in 1000 mL of 5% dextrose in water for 16 hours (Answer C is correct and Answer B is incorrect). Oral dosing of acetylcysteine is not a viable option because of the patient's severe nausea and vomiting (Answer A is incorrect).

4. **Answer: C**

This patient has an acute salicylate overdose. Although his serum salicylate concentrations are in the therapeutic range, he has symptoms consistent with salicylate toxicity, as evidenced by his nausea, tachycardia, and respiratory alkalosis. He is currently stable, but his serum salicylate concentrations may continue to rise; therefore, enhanced elimination with serum bicarbonate is the best option (Answer C is correct). His vital signs, which are stable, should be monitored for changes; however, although he is not experiencing signs of significant dehydration, he would benefit from the fluid administration of sodium bicarbonate. Sodium chloride would be more beneficial if his vital signs were more unstable (Answer A is incorrect). His RR, which is elevated, should be monitored; however, he is alert and able to communicate and therefore does not need intubation at this time (Answer B is incorrect). He is also not indicated for hemodialysis because of his moderate symptoms, but this could be considered if his condition deteriorates (Answer D is incorrect).

5. **Answer: A**

The most appropriate therapy for an ethylene glycol intoxication is fomepizole (Answer A is correct). An ethanol infusion is a possible treatment option, but it is not preferred because of the difficulties in dosing and adverse effects (Answer B is incorrect). Thiamine is a cofactor in the metabolism of ethylene glycol, but it would not be preferred to administer thiamine before fomepizole (Answer C is incorrect). Activated charcoal is not an option for gastric decontamination because it is not effective for alcohols (Answer D is incorrect).

6. **Answer: C**

After the initial bolus of fomepizole, 10 mg/kg should be administered every 12 hours. Because of the increased clearance of fomepizole during hemodialysis, the frequency is changed to every 4 hours during dialysis (Answer C is correct and Answer A is incorrect). When dialysis is completed, the dose returns to 10 mg/kg administered every 12 hours; and once the patient has been on 48 hours of therapy, dosing increases to 15 mg/kg because of self-induction. There is no indication for a dose increase (Answer B and D are incorrect).

7. **Answer: C**
The best treatment option for this patient is whole bowel irrigation because of the extended-release formulation of diltiazem (Answer C is correct). Activated charcoal may provide some benefit, but, similar to ipecac, the time interval is not known, and diltiazem does not undergo enterohepatic recirculation (Answer A is incorrect). Ipecac is not recommended because it may impede treatment with more effective treatment options and because it is no longer manufactured in the United States (Answer B is incorrect). A cathartic would not be useful in this situation; guidelines recommend its use only in combination with other decontamination strategies, not as a single agent (Answer D is incorrect).

8. **Answer: A**
There are several potential antidotes for a calcium channel blocker overdose. Calcium is the most effective, and it should be given by bolus, followed by continuous infusion if needed (Answer A is correct). Glucagon is not an effective antidote and is therefore not an option for this patient (Answer B is incorrect). Atropine is effective for symptomatic bradycardia caused by the calcium channel blocker, but the dose should be 0.5–1 mg (Answer C is incorrect). Epinephrine is an alternative to glucagon, but this dose is excessive for a patient not experiencing cardiac arrest (Answer D is incorrect).

9. **Answer: D**
Most of the SSRIs are relatively safe, and many patients will present as asymptomatic after an overdose. However, there is a potential for a patient to develop serious adverse effects, such as serotonin syndrome, seizures, and cardiac toxicity. Although this patient is stable and has no specific concerns, it is recommended to check a 12-lead ECG to measure for QT-interval prolongation and treat with sodium bicarbonate, if necessary (Answer D is correct). A benzodiazepine should be administered if muscle rigidity develops, but it should not be used as a prophylactic measure (Answer A is incorrect). It is recommended that the patient be observed for at least 6–8 hours. Cyproheptadine is only indicated for symptomatic patients (Answer B is incorrect). Measures should be performed to reduce hyperthermia if a serotonergic syndrome develops, but this should be treated with measures to reduce muscle activity (i.e., sedation or chemical paralysis), not by applying measures to enhance surface cooling (Answer C is incorrect).

10. **Answer: D**
Although the patient appears to have taken an overdose of olanzapine, she is experiencing only mild symptoms. The best intervention would be to monitor her for 6 hours for the progression of her symptoms or development of additional complications (Answer D is correct). Intravenous fluids would be appropriate if the patient has dehydration or hypotension (Answer A is incorrect). Sodium bicarbonate is indicated for QRS prolongation and is not warranted at this time (Answer B is incorrect). Olanzapine does not cause seizures; therefore, lorazepam would not be indicated (Answer C is incorrect).

11. **Answer: B**
The most appropriate intervention at this time is to give the patient intravenous dextrose (Answer B is correct). Oral glucose is a viable option, but it cannot be administered to an unconscious patient without oral access (Answer A is incorrect). Octreotide should be reserved for use if the administration of a glucose solution fails to raise the blood glucose above 70 mg/dL for two consecutive readings (Answer C is incorrect). Glucagon is a potential option for treatment, but because the patient has intravenous access, the intramuscular route would not be preferred (Answer D is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**

The best option for this patient right now is to administer octreotide 50–100 mcg subcutaneously (Answer C is correct). The patient has not responded to two doses of intravenous dextrose, as evidenced by point-of-care glucose concentrations less than 70 mg/dL; therefore, additional doses of dextrose are not indicated (Answer A is incorrect). Although glucagon is also a potential option, it is not recommended for sulfonylurea exposures (Answer B is incorrect). Sodium bicarbonate intravenously may be indicated in this scenario if the patient had metformin-associated lactic acidosis; however, the patient is not on metformin and has no signs of lactic acidosis (Answer D is incorrect).

2. **Answer: A**

Any of the options listed in this question are possible treatments for a patient with a β-blocker overdose who is not responding to the administration of intravenous fluids and calcium. The optimal choice ultimately involves efficacy and appropriate dosing. Glucagon is an option, and it should be dosed at 5–10 mg intravenously initially (Answer A is correct). Atropine is an option for the patient's bradycardia, but the initial recommended dose is 0.5–1 mg intravenously (Answer B is incorrect). Hyperinsulinemic euglycemic therapy may be preferred in this setting; however, the correct bolus dose is 1 unit/kg intravenously (Answer C is incorrect). Dopamine is an option for the treatment of hypotension and bradycardia, but the correct dose would be the initiation of an infusion at 5–10 mcg/kg/minute titrated to effect (Answer D is incorrect).

3. **Answer: B**

Given the patient's presentation and the common toxidromes, the most likely scenario is a cholinergic agent (Answer B is correct). The patient is experiencing bradycardia with a normal BP and RR, has a decrease in mental status, and is experiencing nausea. Although not an absolute, anticholinergics and sympathomimetics are more commonly associated with tachycardia (Answers A and D are incorrect). Similarly, opioids are typically associated with a decrease in respirations (Answer C is incorrect).

4. **Answer: A**

The patient is experiencing QT prolongation after an atypical antipsychotic overdose. It is important to stabilize the patient by administering intravenous sodium bicarbonate and electrolyte replacement (Answer A is correct). Her potassium concentration is low, requiring replacement. Because the time interval of the overdose is not known, there is limited benefit for activated charcoal (Answer B is incorrect). Although her magnesium concentration is normal, it should be monitored; however, her magnesium concentration does not require replacement at this time because her QTc is less than 500 milliseconds (Answer C is incorrect). Lorazepam is not indicated for prophylaxis of seizure activity (Answer D is incorrect).

5. **Answer: A**

This patient has the clinical signs and symptoms of alcohol withdrawal. Management should focus on the patient's safety and controlling his symptoms, and treatment should be administered using a symptom-triggered therapy strategy. The primary agents used to control symptoms are the benzodiazepines, and lorazepam is a good option (Answer A is correct). Barbiturates such as phenobarbital are typically reserved for patients who do not respond to benzodiazepine therapy because of benzodiazepine's long elimination half-life and stronger sedative effects and oral dosing may be difficult with his level of confusion (Answer B is incorrect). Propofol should be avoided in non-intubated patients (Answer C is incorrect). Clonidine is a potential option, especially because this patient has borderline hypertension, but oral dosing may be difficult with his level of confusion (Answer D is incorrect).

6. **Answer: B**

The patient is experiencing an unintended opioid overdose, as evidenced by the decreased RR and decreased consciousness. Administration of the antidote, naloxone, is the best option (Answer B is correct). Because 2 hours have passed since the methadone dose was given, there is limited usefulness for activated charcoal at this time, and it would not be advisable to administer it to an unconscious patient without an established airway (Answer A is incorrect). Whole bowel irrigation is also not useful in this situation because it is too late to prevent

drug absorption together with the airway safety concern (Answer C is incorrect). Administration of intravenous fluids would be beneficial to improve BP but should not be administered in this case before naloxone (Answer D is incorrect).

7. **Answer: D**

The patient is not responding to the initiation of intravenous fluids and calcium gluconate, so HIET is warranted. Because of the patient's low serum potassium concentrations, it is critical to replace this before administering insulin (Answer A is incorrect and Answer D is correct). The patient's glucose concentration is greater than 200 mg/dL, so additional glucose need not be given at this time (Answer B is incorrect). Full effects may take up to 30 minutes to be seen, but this should not prevent the initiation of HIET (Answer C is incorrect).

8. **Answer: B**

The patient is not responding to the initiation of intravenous fluids, calcium, and HIET. The most appropriate option at this time would be to increase the rate of the insulin infusion and initiate vasopressor therapy to improve hemodynamic stability in the interim (Answer B is correct). The initial infusion rate is 0.5–1 unit/kg/hour and is titrated every 15–20 minutes until hemodynamically stable. The next option would be to initiate a vasopressor agent (Answers A and C are incorrect). From the choices listed, the best first option is norepinephrine initiated at 4 mcg/minute and titrated to the desired effect. Epinephrine is also a possible option, but it would be recommended if the patient were not responding to increasing doses of norepinephrine. Intravenous lipid emulsion is a potential therapy, but it is typically administered in a patient with severe decompensation caused by a lipophilic medication who is not responding to fluids or vasopressors (Answer D is incorrect).

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